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CHAPTER 4. ACADEMIC STANDARDS AND ASSESSMENT

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Authority

The provisions of this Chapter 4 issued under the Public School Code of 1949 (24 P.S. § § 1-101—27-2702), unless otherwise noted.

Source

The provisions of this Chapter 4 adopted January 15, 1999, effective January 16, 1999, 29 Pa.B. 399, unless otherwise noted.

Cross References

This chapter cited in 22 Pa. Code § 11.27 (relating to graduation); 22 Pa. Code § 11.31 (relating to students not enrolled in public schools due to private tutoring); 22 Pa. Code § 16.1 (relating to definitions); 22 Pa. Code § 16.22 (relating to gifted multidisciplinary evaluation); 22 Pa. Code § 49.14 (relating to approval of institutions); 22 Pa. Code § 49.42 (relating to letter of eligibility); 22 Pa. Code § 49.81 (relating to general); 22 Pa. Code § 49.101 (relating to general); 22 Pa. Code § 49.111 (relating to Supervisory Certificate); 22 Pa. Code § 49.121 (relating to Administrative Certificate); 22 Pa. Code § 49.141 (relating to general); and 22 Pa. Code § 339.2 (relating to operation).

GENERAL PROVISIONS

§ 4.1. Statutory authority.

The statutory authority for this chapter is the School Code.

§ 4.2. Purpose.

The purpose of this chapter is to establish rigorous academic standards and assessments, applicable only to the public schools in this Commonwealth, to facilitate the improvement of student achievement and to provide parents and communities a measure by which school performance can be determined.

Authority

The provisions of this § 4.2 amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.2 amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131. Immediately preceding text appears at serial page (334860).

§ 4.3. Definitions.

The following words and terms, when used in this chapter, have the following meanings, unless the context clearly indicates otherwise:

ACTS—Area career and technical school—A public school that provides career and technical education to secondary school students, out-of-school youth and adults in a geographical area comprised and operated by one or more school districts and established under sections 1840—1853 of the School Code (24 P.S. § § 18-1840—18-1853).

Academic standard—What a student should know and be able to do at a specified grade level.

Apprenticeship program—A competency-based program that coordinates and integrates classroom instruction with a structured work-based employment experience designed for students.

Assessment—A valid and reliable measurement of student performance on a set of academic standards in a subject area that captures student understanding of the set as a whole and the central concepts, knowledge and skills of each content area.

Board—The State Board of Education established under sections 2601-B—2606-B of the School Code (24 P.S. § § 26-2601-B—26-2606-B).

Career and technical education—Programs under public supervision and control which provide an organized process of learning experiences designed to develop integrated academic and occupational skills, knowledge, attitudes, work habits and leadership ability for entry into and advancement within various levels of employment in occupational areas of agriculture, business, marketing and distribution, health, home economics and trade and industry and for participation in postsecondary education and training.

Chief school administrator—The superintendent of a school district, the superintendent of an ACTS or the chief executive officer of a charter school.

Cooperative career and technical education—A planned method of instruction developed through a signed cooperative arrangement among school representatives, students, parents and employers in the community to provide students with an opportunity to alternate in-school academic and career and technical instruction in entry-level paid employment in an occupational field, in which the student's total occupational work experience is planned, coordinated and supervised by the school in close cooperation with the employer.

Curriculum—A series of planned instruction aligned with the academic standards in each subject that is coordinated and articulated and implemented in a manner designed to result in the achievement at the proficient level by all students.

Department—The Department of Education of the Commonwealth.

ESOL—English to speakers of other languages.

Employment area—A geographic area where career and technical education program completers are most likely to be employed.

Individuals with Disabilities Education Act—20 U.S.C.A. § § 1400—1482.

Intermediate unit—A regional educational service agency established under sections 951—974 of the School Code (24 P. S. § § 9-951—9-974), which provides educational services to participating school districts as part of the public school system of this Commonwealth.

Keystone Exams—State-developed end-of-course exams. Designated exams will be used to determine, in part, a student's eligibility for high school graduation.

Local Assessment Validation Advisory Committee—An advisory committee established by the Department composed of up to two representatives each from the Department and Board, four representatives from the Pennsylvania School Boards Association and up to four additional members who are jointly selected by the Committee. The purpose of the Committee is to develop the criteria for the local validation process and criteria for selection of approved validation entities.

NOCTI—National Occupational Competency Testing Institute.

PSSA—Pennsylvania System of School Assessment.

Parent or *guardian*—A person legally responsible for a student's care.

Pennsylvania Core Standards—Academic standards for English language arts and mathematics based upon a Nationwide, state-led process coordinated by the National Governors Association and the Council of Chief State School Officers and in collaboration with teachers, content experts and other education stakeholders. The standards define the knowledge and skills students should have within their K-12 education careers so that they will graduate high school able to succeed in entry-level, credit-bearing academic college courses and in work-force training programs.

Performance Level Advisory Committee—An advisory committee established by the Department to assist the Department in developing Keystone Exam performance level descriptors and performance level cut scores. The Committee includes teachers, principals, school administrators, school board members, higher education officials, representatives of the United States Armed Forces, employers and others with at least 1/2 of its members selected from nominations made by Statewide teachers' unions and other education stakeholder organizations.

Planned instruction—Instruction offered by a school entity based upon a written plan to enable students to achieve the academic standards under § 4.12 (relating to academic standards) and any additional academic standards as determined by the school entity.

Prekindergarten—A program operated by a school district or by a community agency under contract from a school district that is open to children who are at least 3 years of age and completed prior to the school district's entry age for kindergarten.

School Code—The Public School Code of 1949 (24 P. S. § § 1-101— 27-2702).

School entity—A local public education provider (for example, public school district, charter school, cyber charter school, ACTS or intermediate unit).

School organization—The organization of a school district's programs into kindergarten, primary, intermediate level, middle level and high school programs, including programs operated at ACTSs.

Secretary—The Secretary of Education of the Commonwealth.

State assessment—A valid and reliable measurement of student performance on a set of academic standards as measured by the Pennsylvania System of School Assessment or the Keystone Exams.

State Assessment Validation Advisory Committee—An advisory committee established by the Department to advise it on its plans to conduct a validity study of the Keystone Exams and review and provide feedback on study findings. The Committee is composed of up to two representatives each from the Department, Board, Pennsylvania State Education Association, American Federation of Teachers-Pennsylvania and up to four additional members who are jointly selected by the Committee.

Tech-prep program—A combined secondary and postsecondary program which leads to an associate degree or certificate and employment by providing technical preparation in engineering technology, applied science, mechanical, industrial or practical art or trade, agriculture, health or business, including development of competence in mathematics, science and communications through a sequential course of study.

Authority

The provisions of this § 4.3 amended under the Public School Code of 1949 (24 P.S. § § 1-101—27-2702).

Source

The provisions of this § 4.3 amended December 15, 2006, effective December 16, 2006, 36 Pa.B. 7542; amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended January 8, 2010, effective January 9, 2010, 40 Pa.B. 240; amended October 15, 2010, effective October 16, 2010, 40 Pa.B. 5903; corrected April 6, 2012, effective January 1, 2011, 42 Pa.B. 1859; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371071) to (371073).

Cross References

This section cited in 22 Pa. Code § 4.23 (relating to high school education); 22 Pa. Code § 4.28 (relating to special education); 22 Pa. Code § 338.2 (relating to definitions); and 22 Pa. Code § 339.1a (relating to definitions).

§ 4.4. General policies.

(a) It is the policy of the Board that the local curriculum be designed by school entities to achieve the academic standards under § 4.12 (relating to academic standards) and any additional academic standards as determined by the school entity.

(b) It is the policy of the Board that local school entities have the greatest possible flexibility in curriculum planning consistent with providing quality education and in compliance with the School Code, including requirements for courses to be taught (24 P. S. § § 15-1501 and 16-1605); subjects to be taught in the English language (24 P. S. § 15-1511); courses adapted to the age, development and needs of the pupils (24 P. S. § 15-1512); minimum school year of 180 days and minimum of 900 hours of instruction at the elementary level and 990 hours of instruction at the secondary level (24 P. S. § § 15-1501 and 15-1504); employment of sufficient numbers of qualified professional employees (24 P. S. § 11-1106) and superintendents to enforce the curriculum requirements of State law (24 P. S. § 10-1005); and this part.

(c) Access to educational programs shall be provided without discrimination on the basis of a student's race, sex, color, religion, disability, sexual orientation or national origin.

(d) School entities shall adopt policies to assure that parents or guardians have the following:

(1) Access to information about the curriculum, including academic standards to be achieved, instructional materials and assessment techniques.

(2) A process for the review of instructional materials.

(3) The right to have their children excused from specific instruction that conflicts with their religious beliefs, upon receipt by the school entity of a written request from the parent or guardians.

(4) The right to review a State assessment in the school entity during convenient hours for parents and guardians, at least 2 weeks prior to their administration, to determine whether a State assessment conflicts with their religious belief. To protect the validity and integrity of the State assessments, each school entity shall have in place procedures to be followed when parents or guardians request to view any State assessment. Procedures must be consistent with guidance provided by the Department in its assessment administration instructions. If upon inspection of a State assessment parents or guardians find the assessment to be in conflict with their religious belief and wish their students to be excused from the assessment, the right of the parents or guardians will not be denied upon written request that states the objection to the applicable school district superintendent, charter school chief executive officer or AVTS director.

(5) The right to have their children excluded from research studies or surveys conducted by entities other than a school entity unless prior written consent has been obtained.

(e) The Department will provide support to school districts, ACTSs and charter schools, including cyber charter schools, in developing educational programs that enable students to attain academic standards under § 4.12. Department support will include:

(1) Establishment of a voluntary model curriculum and diagnostic supports aligned with State academic standards in each of the content areas assessed by the Keystone Exams under § 4.51b(i) (relating to Keystone Exams).

(2) Assistance in the development of effective student tutoring, remediation and extended instructional time programs.

(3) Opportunities for continuing professional education designed to improve instruction in each of the content areas assessed by the Keystone Exams under § 4.51b(i).

(4) Technical guidance in developing local assessments that meet the requirements of § 4.24(c)(1)(iii)(B) (relating to high school graduation requirements), upon request.

(f) The Department may not, and the Board will not, require school entities to utilize a Statewide curriculum or Statewide reading lists.

Authority

The provisions of this § 4.4 amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.4 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended January 8, 2010, effective January 9, 2010, 40 Pa.B. 240; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371074) to (371075).

Cross Refenences

This section cited in 22 Pa. Code § 11.7 (relating to religious objections).

ACADEMIC STANDARDS AND PLANNING

§ 4.11. Purpose of public education.

(a) This section and § 4.12 (relating to academic standards) describe the purpose of public education and its relationship with the academic standards.

(b) Public education prepares students for adult life by attending to their intellectual and developmental needs and challenging them to achieve at their highest level possible. In conjunction with families and other community institutions, public education prepares students to become self-directed, life-long learners and responsible, involved citizens.

(c) Together with parents, families and community institutions, public education provides opportunities for students to:

(1) Acquire knowledge and skills.

(2) Develop integrity.

(3) Process information.

(4) Think critically.

(5) Work independently.

(6) Collaborate with others.

(7) Adapt to change.

(d) The academic standards describe the knowledge and skills that students will be expected to demonstrate before graduating from a public school.

(e) Achievement of high academic standards in public education is dependent upon the quality of instruction in schools and student effort supported by the involvement of family and community.

(f) Assessment in public education is designed to determine student attainment of State and local academic standards.

(g) Public schools provide instruction throughout the curriculum so that students may develop knowledge and skills in the following areas:

(1) English language arts.

(2) Mathematics.

(3) Science and environment and ecology.

(4) Technology and engineering.

(5) Social studies (civics and government, geography, economics and history).

(6) Arts and humanities.

(7) Career education and work.

(8) Health, safety and physical education.

(9) Family and consumer sciences.

(10) Personal finance.

(h) Public education provides planned instruction to enable students to attain academic standards under § 4.12. Planned instruction consists of at least the following elements:

(1) Objectives of a planned course, instructional unit or interdisciplinary studies to be achieved by all students.

(2) Content, including materials and activities, and estimated instructional time to be devoted to achieving the academic standards. Courses, instructional units or interdisciplinary studies of varying lengths of time may be taught.

(3) The relationship between the objectives of a planned course, instructional unit or interdisciplinary studies and academic standards specified under § 4.12 and any additional academic standards as determined by the school entity.

(4) Procedures for measurement of the objectives of a planned course, instructional unit or interdisciplinary studies.

Authority

The provisions of this § 4.11 amended under the Public School Code of 1949 (24 P.S. § § 1-101—27-2702).

Source

The provisions of this § 4.11 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended October 15, 2010, effective October 16, 2010, 40 Pa.B. 5903; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended June 15, 2022, effective July 16, 2022, 52 Pa.B. 3946; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410340) to (410341).

§ 4.12. Academic standards.

(a) School entities may develop, expand or improve existing academic standards in the following content areas:

(1) The following apply:

(i) *Through June 30, 2025*: *Science and technology*. Study of the natural world and facts, principles, theories and laws in the areas of biology, chemistry, physics and earth sciences. Technology is the application of science to enable societal development, including food and fiber production, manufacturing, building, transportation and communication. Science and technology share the use of the senses, science processes, inquiry, investigation, analysis and problem solving strategies. The Pennsylvania Core Standards for Reading in Science and Technology and the Pennsylvania Core Standards for Writing in Science and Technology will be an appendix to the Commonwealth's academic standards for Science and Technology upon publication in the *Pennsylvania Bulletin*.

(ii) *Effective July 1, 2025*: *Pennsylvania Integrated Standards for Science, Environment, Ecology, Technology and Engineering (Grades K—5)*. Guide the elementary-level study of the natural and human-made world through inquiry, problem-solving, critical thinking and authentic exploration. The integration of these disciplines in the elementary grades highlights the interconnectedness of scientific study and the integral relationship between humans and the environment.

(2) The following apply:

(i) *Through June 30, 2025*: *Environment and ecology*. Understanding the components of ecological systems and their interrelationships with social systems and technologies. These components incorporate the disciplines of resource management, agricultural diversity, government and the impact of human actions on natural systems. This interaction leads to the study of watersheds, threatened and endangered species, pest management and the development of laws and regulations.

(ii) *Effective July 1, 2025*: *Pennsylvania Integrated Standards for Science, Environment and Ecology (Grades 6—12)*. Utilize a three-dimensional approach to guide the study of physical sciences, life sciences, and earth and space sciences at the middle and high school levels. The standards highlight the critical intersections of these disciplines with environmental science, ecology, and agriculture. These three-dimensional standards integrate disciplinary core ideas, practices in science and engineering, and crosscutting concepts into coherent learning progressions across the grade bands.

(iii) *Effective July 1, 2025*: *Pennsylvania Technology and Engineering Standards (Grades 6—12)*. Define the knowledge, skills, and abilities necessary for establishing literacy in technology and engineering literacy. Students develop a practical understanding of how humans are influenced by science and technology. Four core disciplinary standards describe practices in technology and engineering and include specific benchmarks for students in grade bands 6—8 and 9—12.

(2.1) Appendix B (relating to academic standards for science and technology and environment and ecology) shall expire July 1, 2025. The academic standards under paragraphs (1)(ii) and (2)(ii) and (iii) shall be based on the provisions of Appendix B-1 (relating to Pennsylvania integrated standards for science, environment, ecology, technology and engineering (Grades K—5), Pennsylvania integrated standards for science, environment and ecology (Grades 6—12), and Pennsylvania technology and engineering standards (Grades 6—12), on and after July 1, 2025.

(3) *Social studies*.

(i) *History*. Study of the record of human experience including important events; interactions of culture, race and ideas; the nature of prejudice; change and continuity in political systems; effects of technology; importance of global-international perspectives; and the integration of geography, economics and civics studies on major developments in the history of the Commonwealth, the United States and the world.

(ii) *Geography*. Study of relationships among people, places and environments, of geographic tools and methods, characteristics of place, concept of region and physical processes.

(iii) *Civics and government*. Study of United States constitutional democracy, its values and principles, study of the Constitution of the Commonwealth and government including the study of principles, operations and documents of government, the rights and responsibilities of citizenship, how governments work and international relations.

(iv) *Economics*. Study of how individuals and societies choose to use resources to produce, distribute and consume goods and services. Knowledge of how economies work, economic reasoning and basic economic concepts, economic decision making, economic systems, the Commonwealth and the United States economy and international trade. The academic standards for Economics in Appendix C (relating to academic standards for civics and government and economics and geography and history and the Pennsylvania core standards for reading in history and social studies and the Pennsylvania core standards for writing in history and social studies) shall expire July 1, 2026. The academic standards under this subparagraph shall be based on the provisions of Appendix C-1 (relating to academic standards for economics) on and after July 1, 2026.

(v) *Appendix*. The Pennsylvania Core Standards for Reading in History and Social Studies and the Pennsylvania Core Standards in Writing for History and Social Studies will be an appendix to the Commonwealth's academic standards for History upon publication in the *Pennsylvania Bulletin*.

(4) *Arts and humanities*. Study of dance, theatre, music, visual arts, language and literature including forms of expression, historical and cultural context, critical and aesthetic judgment and production, performance or exhibition of work.

(5) *Career education and work*. Understanding career options in relationship to individual interests, aptitudes and skills including the relationship between changes in society, technology, government and economy and their effect on individuals and careers. Development of knowledge and skill in job-seeking and job-retaining skills and, for students completing career and technical education programs, the skills to succeed in the occupation for which they are prepared. Appendix E (relating to academic standards for career education and work) shall expire July 1, 2026. The academic standards under this paragraph shall be based on the provisions of Appendix E-1 (relating to academic standards for career education and work) on and after July 1, 2026.

(6) *Health, safety and physical education*. Study of concepts and skills which affect personal, family and community health and safety, nutrition, physical fitness, movement concepts and strategies, safety in physical activity settings, and leadership and cooperation in physical activities.

(7) *Family and consumer sciences*. Understanding the role of consumers as a foundation for managing available resources to provide for personal and family needs and to provide basic knowledge of child health and child care skills. The academic standards for Family and Consumer Sciences in Appendix D (relating to academic standards for the arts and humanities and health, safety and physical education and family and consumer sciences) shall expire July 1, 2026. The academic standards under this paragraph shall be based on the provisions of Appendix D-1 (relating to academic standards for family and consumer sciences) on and after July 1, 2026.

(8) *Pennsylvania Core Standards for English Language Arts*. The English Language Arts standards describe what students should know and be able to do with the English language as follows:

(i) *Reading*. The application of phonemic awareness, phonics and word study, vocabulary, fluency and text comprehension in reading critically across subject areas; the interpretation and analysis of literary expression with analysis of the origins and structures of the English language and learning how to

search a variety of texts to conduct research.

(ii) *Writing*. Narrative, informational and persuasive formal writing for an audience, including spelling and editing skills; and informal writing to capture and organize information for individual use.

(iii) *Speaking and listening*. Participation in conversation and formal speaking presentations.

(iv) Academic standards are based on the Pennsylvania Core Standards for English Language Arts in Appendix A-2 (relating to Pennsylvania core standards for English language arts and mathematics).

(9) *Pennsylvania Core Standards for Mathematics*. The understanding of fundamental ideas and the development of proficient mathematical skills in numbers, computation, measurement, statistics and data analysis, probability and predictions, algebra and functions, geometry, trigonometry and concepts of calculus. Using this content, students will learn to think, reason and communicate mathematically. Students will learn to model real-world situations by creating appropriate representations of numerical quantities and plan and implement problem-solving strategies to answer the question in the context of the situation. Academic standards are based on the Pennsylvania Core Standards for Mathematics in Appendix A-2.

(10) *Personal Finance*. Understanding the process of planning and managing personal financial activities such as income generation, spending, saving, investing and risk protection. Development of knowledge of personal finance foundations, income, spending, saving and investing, risk and insurance, and credit. Personal finance is also known as personal financial literacy and financial literacy. On and after July 1, 2026, the academic standards under this paragraph shall be based on the provisions of Appendix F (relating to academic standards for personal finance).

(b) In designing educational programs, school entities shall provide for the attainment of the academic standards under subsections (a) and (c) and any additional academic standards as determined by the school entity. Attaining the academic standards in this section requires students to demonstrate the acquisition and application of knowledge.

(c) School entities shall prepare students to attain academic standards in mathematics and English Language Arts in Appendix A-2 and incorporated here by reference and additional standards as may be adopted by the Board and promulgated as amendments to this chapter.

(d) A school entity's curriculum shall be designed to provide students with planned instruction needed to attain these academic standards.

(e) School entities shall apply academic standards for students in all areas described under subsections (a) and (c). The local assessment plan under § 4.52 (relating to local assessment system) must include a description of how the academic standards will be measured and how information from the assessments is used to assist students having difficulty meeting the academic standards.

(f) School entities shall assess the attainment of academic standards developed under subsections (a) and (c) and any other academic standards that they develop under § 4.52(c) for purposes of high school graduation and strategies for assisting students to attain them. Plans for assessment developed by school entities must take into account that academic standards in subsections (a) and (c) may be attained by students in various ways and shall be assessed in various ways. Children with disabilities may attain the academic standards by completion of their individualized education programs under the Individuals with Disabilities Education Act and this part.

(g) In planning any revision of the academic standards in subsection (a) content areas, the Secretary will consult with educators, business and community leaders and parents.

(h) School entities are responsible under subsections (a), (c) and (f) for assessing individual student attainment of academic standards and for assisting those students having difficulty attaining them. Upon request by a school entity, the Department will provide the requestor with technical assistance in the development of academic standards and assessments that are sufficient to assure that students are making progress toward the attainment of standards required for high school graduation under subsection (f).

(i) No sooner than every 5 years and no later than every 10 years, the Board will review the State academic standards and State assessments under this section to determine if they are appropriate, clear, specific and challenging, and will make revisions as necessary by revising this chapter.

(j) The Department may not expand the collection of student data and, in accordance with section 444 of the Family Educational Rights and Privacy Act of 1974 (20 U.S.C.A. § 1232g), regarding family educational and privacy rights, may not collect personal family data due to the implementation of Pennsylvania Core Standards in Appendix A-2.

Authority

The provisions of this § 4.12 amended under the Public School Code of 1949 (24 P.S. § § 1-101—27-2702).

Source

The provisions of this § 4.12 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended October 15, 2010, effective October 16, 2010, 40 Pa.B. 5903; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410341) to (410345).

Cross References

This section cited in 22 Pa. Code § 4.3 (relating to definitions); 22 Pa. Code § 4.4 (relating to general policies); 22 Pa. Code § 4.11 (relating to purpose of public education); 22 Pa. Code § 4.26 (relating to ESOL); 22 Pa. Code § 4.31 (relating to career and technical education); 22 Pa. Code § 4.51 (relating to State assessment system); 22 Pa. Code § 4.51a (relating to Pennsylvania System of School Assessment); 22 Pa. Code § 4.52 (relating to local assessment system); and 22 Pa. Code § 4.82 (relating to exceptions).

STRATEGIC PLANS

§ 4.13. Strategic plans.

(a) Each school entity shall submit to the Secretary for approval a professional education plan every 3 years as required under § 49.17(a) (relating to continuing professional education). A school entity shall make its professional education plan available for public inspection and comment for a minimum of 28 days prior to approval of the plan by the school entity's governing board andsubmission of the plan to the Secretary. A school entity shall post the plan approved by its governing board on its publicly accessible website.

(b) Each school entity shall submit to the Department for approval an induction plan every 6 years as required under § 49.16(a) (relating to approval of induction plans). A school entity shall make its induction plan available for public inspection and comment for a minimum of 28 days prior to approval of the plan by the school entity's governing board and submission of the plan to the Department. A school entity shall post the plan approved by its governing board on its publicly accessible website.

(c) Each school entity shall develop and implement a comprehensive and integrated K—12 program of student services based on the needs of its students every 6 years as provided in § 12.41(a) (relating to student services). A school entity shall make its student services plan available for public inspection and comment for a minimum of 28 days prior to approval of the plan by the school entity's governing board. A school entity shall post the plan approved by its governing board on its publicly accessible website.

(d) Each school district shall develop, submit to the Department for approval and implement a special education plan every 3 years as required under § 14.104 (relating to special education plans). A school district shall make its special education plan available for public inspection and comment for a minimum of 28 days prior to approval of the plan by the school district's board of directors and submission of the plan to the Department. A school district shall post the plan approved by its governing board on its publicly accessible website.

(e) Each school district shall develop and implement a gifted education plan every 6 years as required under § 16.4 (relating to gifted education plans). A school district shall make its gifted education plan available for public inspection and comment for a minimum of 28 days prior to approval of the plan by the school district's board of directors. A school district shall post the plan approved by its governing board on its publicly accessible website.

Authority

The provisions of this § 4.13 amended under sections 121, 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.13 amended December 15, 2006, effective December 16, 2006, 36 Pa.B. 7542; amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended March 7, 2008, effective February 16, 2008; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410345) to (410346).

Cross References

This section is cited in 22 Pa. Code § 12.41 (relating to student services); 22 Pa. Code § 14.104 (relating to special education plans); 22 Pa. Code § 16.4 (relating to gifted education plans); 22 Pa. Code § 49.16 (relating to approval of induction plans); and 22 Pa. Code § 49.17 (relating to continuing professional education).

CURRICULUM AND INSTRUCTION

§ 4.20. Prekindergarten education.

School districts are not required to offer a prekindergarten program, and parents are not required to enroll their children in those programs if offered. Prekindergarten programs shall be designed so that students complete the program prior to their reaching the school district's entry age for kindergarten. The program, when offered, must provide a comprehensive program appropriate for the age and varying developmental levels of the students; be based on how young children develop and learn; include instruction to support each child's development in the areas of approaches to learning—creative expression, language and literacy, math, logic and science, social-personal development and physical development and health—and must be open to children with disabilities.

(1) The Secretary will provide academic standards, appropriate for early learning at the prekindergarten level, as guidance for the use of school districts that offer prekindergarten programs.

(2) Curriculum and instruction in the prekindergarten program must be standards-based.

(3) Prekindergarten programs may be offered to all 3 and 4 year olds or may be targeted to children who are most in need of prekindergarten services who reside in the district. Targeted programs may serve children who are at risk of school failure because of limited English proficiency, community factors, economic disadvantage, but may not exclude or be limited exclusively to children with disabilities. If a program is limited to an attendance area, children with disabilities must live in that attendance area to participate in the program. An attendance area is the geographic area within a school district designated by the school board for the purpose of assigning students to a school.

(4) The Secretary will issue guidance to school districts on developmentally appropriate curriculum, instruction and assessments for prekindergarten.

(5) Each school district that provides prekindergarten shall design an assessment system that includes prekindergarten and uses a variety of assessment strategies, which may include those listed in § 4.52(b) (relating to local assessment system), as appropriate.

(6) Prekindergarten programs must have a student/teacher ratio of no more than 20 students for one teacher and one teacher aide in a classroom (2 adults in a classroom for every 20 students). Programs of high quality ordinarily have a student/teacher ratio of 17 students for one teacher and one teacher aide in a classroom (2 adults for every 17 students). Programs operating under contract with community providers must comply with staffing qualifications as required by § 49.85(e) (relating to limitations).

(7) Beginning in the 2009-2010 school year, a teacher aide in a prekindergarten program shall meet one of the following criteria:

(i) Completion of at least 2 years of postsecondary study.

(ii) Possession of an associate's degree or higher.

(iii) Ability to meet a rigorous standard of quality and demonstration through a formal State or local academic assessment of knowledge in and ability to assist in instructing reading, writing and mathematics. A rigorous standard of quality includes a demonstration of competence in basic literacy skills, including the ability to speak and write standard English and instruction of prekindergarten students in the acquisition of the knowledge, skills and abilities described in the early learning standards issued under paragraph (1).

(8) The Secretary may approve a meritorious prekindergarten program that does not meet all regulatory requirements for the program when, in the Secretary's judgment, the program provides high quality learning opportunities for students and meets the following conditions:

(i) The school district has submitted to the Secretary a written request that provides justification for the waiver and includes a description of how the meritorious program will provide high quality learning opportunities for students.

(ii) The approval of the meritorious prekindergarten program is valid only for 1 school year.

(iii) Requests for renewals include evidence of positive student outcomes.

(9) A school district may make individual exceptions to the age of prekindergarten students based upon local policy to permit the enrollment of children under 3 years of age and 5 years of age or older.

(10) A school district planning to offer or contract with a community agency to offer a prekindergarten program shall develop an implementation plan that describes the program and its target population. The plan must identify the facilities, staffing needs and other resources that it will use to deliver the program. The school district shall consult with parents, community agencies and organizations, and child care, early intervention and head start representatives when developing the implementation plan. In years subsequent to the initial year of the program, the implementation plan must be submitted to the Department every 3 years or when the plan is amended, whichever is sooner. A school district shall make the implementation plan available for public inspection and comment for a minimum of 28 days prior to approval of the plan by the school district's board of directors and submission of the plan to the Department.

(11) School district contracted prekindergarten programs operated by a community provider shall provide a lead teacher for each classroom who meets the following minimum qualifications:

(i) An associate's degree or greater in early childhood education or child development.

(ii) For programs operating before December 16, 2006, lead teachers shall possess a bachelor's degree and early childhood certificate as provided in § 49.85(a) on or before December 16, 2011.

(iii) For programs contracted after December 16, 2006, lead teachers shall possess a bachelor's degree and early childhood certificate as provided in § 49.85 within 5 years from the date students first attend the prekindergarten program.

Authority

The provisions of this § 4.20 issued under section 2603-B of the Public School Code of 1949 (24 P.S. § 26-2603-B); amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.20 adopted December 15, 2006, effective December 16, 2006, 36 Pa.B. 7542; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131. Immediately preceding text appears at serial pages (332269) to (332270) and (350149).

Cross References

This section cited in 22 Pa. Code § 405.42 (relating to program day and developmentally appropriate instructional practices and activities).

§ 4.21. Elementary education: primary and intermediate levels.

(a) The primary program shall ordinarily be completed by children who are approximately 6 years of age. School districts, including charter schools, shall provide opportunities for individualized rates of learning and social and emotional development that reflect differing rates of development and learning styles of young children.

(b) Curriculum and instruction in the primary program must be standards-based and focus on introducing young children to formal education, developing an awareness of the self in relation to others and the environment, and developing skills of communication, thinking and learning. Literacy skills, including phonemic awareness, phonological awareness, fluency, vocabulary and comprehension and developmental writing will begin in prekindergarten and kindergarten, if offered, and developed appropriately for the primary grade level.

(c) The intermediate level program shall ordinarily be completed by children who are approximately 11 years of age.

(d) Standards-based curriculum and instruction in the intermediate level must enable all students to reach the proficient level on the local assessment system and the Statewide assessment system. Academic standards will guide the focus on learning specific subject matter content.

(e) Planned instruction aligned with academic standards in the following areas shall be provided to every student every year in the primary program. Planned instruction may be provided as a separate course or other interdisciplinary activity.

(1) English Language Arts, integrating foundational skills, reading informational text, reading literature, writing, speaking and listening, and information management, including library skills.

(2) Mathematics, including problem-solving and computation skills.

(3) Science, environment and ecology, involving active learning experiences for students.

(4) Technology and engineering education, involving active learning experiences for students.

(5) Social studies (civics and government, economics, geography and history).

(6) Health, safety and physical education, including instruction in concepts and skills which affect personal, family and community health and safety, nutrition, the prevention of alcohol, chemical and tobacco abuse, knowledge and practice of lifetime physical activities, personal fitness, basic movement skills and concepts, motor skill development, principles and strategies of movement, and safety practices in physical activity settings.

(7) The arts, including active learning experiences in art, music, dance and theatre.

(8) Career education, including exposure to various interpersonal skills and employability skills, career awareness and learning experiences and the educational preparation necessary to achieve those experiences.

(9) Personal finance.

(f) Planned instruction in the following areas shall be provided to every student every year in the intermediate level program. Planned instruction may be provided as a separate course or as an instructional unit within another course or other interdisciplinary instructional activity:

(1) English Language Arts, integrating foundational skills, reading informational text, reading literature, writing, and speaking and listening.

(2) Mathematics, including problem-solving and computation skills.

(3) Science, environment and ecology, including instruction about agriculture and agricultural science.

(4) Technology and engineering.

(5) Social studies (civics and government, economics, geography and history).

(6) The arts, including art, music, dance and theatre.

(7) Understanding and use of library and other information sources.

(8) Health, safety and physical education, including instruction in concepts and skills which affect personal, family and community health and safety, nutrition, the prevention of alcohol, chemical and tobacco abuse, knowledge and practice of lifetime physical activities, personal fitness, basic movement skills and concepts, motor skill development, principles and strategies of movement and safety practices in physical activity settings.

(9) Personal finance.

(10) Career education, including exposure to various interpersonal skills and employability skills, career awareness and learning experiences and the educational preparation necessary to achieve those experiences.

(g) Planned instruction aligned with academic standards in the following areas shall be provided to every student at least once by the end of elementary school. Planned instruction may be provided as a separate course or as an instructional unit within another course or other interdisciplinary instructional activity. See section 1511 of the School Code (24 P.S. § 15-1511).

(1) History of the United States.

(2) History of the Commonwealth.

(3) Geography.

(4) Civics.

(h) This section does not preclude the teaching of other planned instruction designed to achieve a school entity's mission, goals and academic standards.

(i) School districts, including charter schools, shall determine the most appropriate way to operate their primary and intermediate level elementary programs to achieve the purposes under subsections (b) and (d) and any mission, goals and academic standards as determined by the school entity.

(j) Students who have not achieved proficiency in reading and mathematics during their primary grades (K-3), as determined by the school entity, shall be afforded additional instructional opportunities through a grade-level learning plan developed by the school entity. The plan will assist the student in acquiring the knowledge and skills necessary to achieve at the proficient level. Assessments to measure proficiency shall be described in the local assessment system under § 4.52 (relating to local assessment system).

(k) Students who have not achieved proficiency in reading and mathematics by the end of grade 5 as determined on State assessments under § 4.51a (relating to Pennsylvania System of School Assessment) shall be afforded instructional opportunities to develop knowledge and skills necessary to achieve the proficient level.

Authority

The provisions of this § 4.21 amended under sections 121, 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.21 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410348) to (410350).

Cross References

This section cited in 22 Pa. Code § 4.27 (relating to physical education and athletics).

§ 4.22. Middle level education.

(a) The middle level planned instruction aligned with academic standards serves children who are approximately 11—14 years of age. School entities may modify the grouping of students based upon student needs identified by the school entity.

(b) Curriculum and instruction in the middle level program must be standards-based and focus on mastery of academic subjects, the development of critical and creative thinking, information literacy, good health and encourage active participation in the school and community.

(c) Planned instruction aligned with academic standards in the following areas shall be provided to every student in the middle level program. Planned instruction may be provided as a separate course or as an instructional unit within a course or other interdisciplinary instructional activity:

(1) English Language Arts, integrating reading informational text, reading literature, writing, and speaking and listening.

(2) Mathematics, including mathematical reasoning, algebra and problem-solving.

(3) Science, environment and ecology, which involves active learning experiences and which may include laboratory experiments, instruction in agriculture and agricultural science, and political and economic aspects of ecology.

(4) Social studies (civics and government, economics, geography and history, including the history and cultures of the United States, the Commonwealth, and the world).

(5) {Reserved}.

(6) Information skills, including access to traditional and electronic information sources, computer use and research.

(7) Health, safety and physical education, including instruction in concepts and skills which affect personal, family and community health and safety, nutrition, physical fitness, movement concepts, motor skill development, safety in physical activity settings, and the prevention of alcohol, chemical and tobacco abuse.

(8) The arts, including art, music, dance and theatre.

(9) Career education, including exposure to various employability skills, career awareness and work-based learning experiences and the educational preparation necessary to achieve those experiences.

(10) Technology and engineering, focusing on the interactions among technology, engineering, society, the environment, and other disciplines, with a goal of developing students that can create, utilize, and assess current and emerging technologies.

(11) Family and consumer sciences, including principles of food science and nutrition, consumer behavior, and basic knowledge of child development, child health and child care skills.

(12) Personal finance.

(d) This section does not preclude the teaching of other planned instruction designed to achieve a school entity's academic standards.

(e) School entities shall determine the most appropriate way to operate their middle level programs to achieve the purposes under subsection (b) and any additional academic standards as determined by the school entity.

Authority

The provisions of this § 4.22 amended under sections 121, 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.22 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410350) to (410351).

Cross References

This section cited in 22 Pa. Code § 4.27 (relating to physical education and athletics).

§ 4.23. High school education.

(a) Instruction in the high school program must focus on the development of abilities needed to succeed in work and advanced education through planned instruction.

(b) Curriculum and instruction in the high school must be standards-based and provide all students opportunities to develop the skills of analysis, synthesis, evaluation and problem-solving and information literacy.

(c) Planned instruction aligned with academic standards in the following areas shall be provided to every student in the high school program. Planned instruction may be provided as a separate course or as an instructional unit within a course or other interdisciplinary instructional activity:

(1) English Language Arts, integrating reading informational text, reading literature, writing, and speaking and listening.

(2) Mathematics, including problem-solving, mathematical reasoning, algebra, geometry and concepts of calculus.

(3) Science, environment and ecology, including scientific, social, political and economic aspects of ecology, participation in hands-on experiments and at least one laboratory science chosen from life sciences, earth and space sciences, chemical sciences, physical sciences and agricultural sciences.

(4) Social studies (civics and government, economics, geography and history, including the history and cultures of the United States, the Commonwealth and the world).

(5) {Reserved}.

(6) The arts, including art, music, dance, theatre and humanities.

(7) Use of applications of microcomputers and software, including word processing, database, spreadsheets and telecommunications; and information skills, including access to traditional and electronic information sources, computer use and research.

(8) Health, safety and physical education, including instruction in concepts and skills which affect personal, family and community health and safety, nutrition, physical fitness, movement concepts, motor skill development, safety in physical activity settings, and the prevention of alcohol, chemical and tobacco abuse.

(9) Family and consumer sciences, including principles of food science and nutrition, consumer behavior, and basic knowledge of child development and health, child care and early literacy skill development.

(10) Career education, including exposure to various employability skills, career awareness and work-based learning experiences, and the educational preparation necessary to achieve those experiences.

(11) Personal finance.

(d) The following planned instruction shall be made available to every student in the high school program:

(1) Career and technical education under § § 4.3 and 4.31—4.35.

(2) Business education, including courses to assist students in developing business and information technology skills.

(3) World languages under § 4.25 (relating to languages).

(4) Technology and engineering, focusing on the interactions among technology, engineering, society, the environment, and other disciplines, with a goal of developing individuals that can create, utilize, and assess current and emerging technologies.

(e) College-level advanced placement courses may be offered as planned instruction in the high school curriculum.

(f) This section does not preclude the teaching of other planned instruction designed to achieve a school district's, including a charter school's, academic standards.

(g) School districts, including a charter school, shall determine the most appropriate way to operate their high school programs to achieve the purposes under subsection (a) and any additional academic standards as determined by the school entity.

Authority

The provisions of this § 4.23 amended under sections 121, 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.23 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410351) to (410353).

Cross References

This section cited in 22 Pa. Code § 4.27 (relating to physical education and athletics).

§ 4.24. High school graduation requirements.

(a) *Approval.* High school graduation requirements and revisions to them shall be approved by a school entity's governing board no later than the beginning of the 2020-2021 school year, and a copy of the requirements shall be published and distributed to students, parents and guardians. Copies of the requirements also shall be available in each school building or on each school entity's publicly accessible web site. Changes to high school graduation requirements shall be published and distributed to students, parents and guardians and made available in each school building or on each school entity's publicly accessible web site immediately following approval by the governing board.

(b) *Requirements through the 2021-2022 school year.* Each school district, charter school (including a cyber charter school) and ACTS, if applicable, shall specify requirements for graduation. Requirements through the 2021-2022 school year must include course completion and grades, completion of a culminating project, results of local assessments aligned with the academic standards and a demonstration of proficiency in English Language Arts and Mathematics on either the State assessments administered in grade 11 or 12 or local assessments aligned with academic standards and State assessments under § 4.52 (relating to local assessment system) at the proficient level or better to graduate. The purpose of the culminating project is to assure that students are able to apply, analyze, synthesize and evaluate information and communicate significant knowledge and understanding.

(c) *Requirements beginning in the 2022-2023 school year.*

(1) *General.* Beginning in the 2022-2023 school year, each school district, charter school (including a cyber charter school) and ACTS, if applicable, shall adopt and implement requirements for high school graduation that, at minimum, include:

(i) Course completion and grades.

(ii) Demonstration of proficiency as determined by the school district, charter school (including a cyber charter school) or ACTS, if applicable, in each of the State academic standards not assessed by a State assessment under § 4.51, § 4.51a or § 4.51b (relating to State assessment system; Pennsylvania System of School Assessment; and Keystone Exams).

(iii) Demonstration of proficiency or above in each of the following State academic standards: English Language Arts and Mathematics (Appendix A-2); through June 30, 2025, Science and Technology and Environment and Ecology (Appendix B), and, beginning July 1, 2025, Science, Environment and Ecology (Appendix B-1), as determined through any one or a combination of the following:

(A) Completion of secondary level coursework in English Language Arts (Literature), Algebra I and Biology in which a student demonstrates proficiency on the associated Keystone Exam or through a pathway established in section 121(c) or (c.1) of the School Code (24 P.S. § § 1-121(c) or (c.1)).

(I) A school district, ACTS or charter school, including a cyber charter school, shall allow a student to take a Keystone Exam prior to taking the course associated with the exam's content provided that the student achieved a score of advanced on the most recent associated PSSA assessment administered to the student.

(II) A school district, ACTS or charter school, including a cyber charter school, shall allow a student who transfers from another state to take a Keystone Exam prior to taking the course associated with the exam's content, provided that the student achieved a score comparable to the PSSA's advanced performance level on a comparable assessment administered by another state.

(III) A school district, ACTS or charter school, including a cyber charter school, may allow a student who scores at the advanced level on a particular Keystone Exam prior to taking the course to be granted course credit for the course without having to complete the course.

(B) Locally approved and administered assessments, which shall be independently and objectively validated once every 6 years. Local assessments may be designed to include a variety of assessment strategies listed in § 4.52(c) and may include the use of one or more Keystone Exams. Except for replacement of individual test items that have a similar level of difficulty, a new validation is required for any material changes to the assessment. Validated local assessments must meet the following standards:

(I) Alignment with the following State academic standards: English Language Arts (Literature and Composition); Mathematics (Algebra I), and Science and Environment and Ecology (Biology).

(II) Performance level expectations and descriptors that describe the level of performance required to achieve proficiency comparable to that used for the Keystone Exams.

(III) Administration of the local assessment to all students, as a requirement for graduation, except for those exempted by their individualized education program under subsection (d), regarding special education students, or gifted individualized education plan as provided in § 16.32 (relating to GIEP).

(IV) Subject to appropriations provided by law, the cost to validate local assessments shall be evenly divided between the school district, ACTS or charter school, including a cyber charter school, and the Department. If the Department does not provide sufficient funding to meet its share, local assessments submitted for validation shall be deemed valid until a new validation is due to the Department.

(V) The Department will establish a list of entities approved to perform independent validations of local assessments in consultation with the Local Assessment Validation Advisory Committee as provided in § 4.52(f).

(VI) School boards shall only approve assessments that have been determined to meet the requirements of this subsection by an approved entity performing the independent validation. If a school district, ACTS or charter school, including a cyber charter school, uses a local assessment that has not been independently validated, the Secretary will direct the school entity to discontinue its use until the local assessment is approved through independent validation by an approved entity.

(d) *Special education students*. Children with disabilities who satisfactorily complete a special education program developed by an Individualized Education Program team under the Individuals with Disabilities Education Act and this part shall be granted and issued a regular high school diploma by the school district of residence, charter school (including cyber charter school) or ACTS, if applicable. This subsection applies if the special education program of a child with a disability does not otherwise meet the requirements of this chapter.

(e) *Demonstration of proficiency*. For purposes of this section, a student shall be deemed proficient in the State-assessed standards whenever the student demonstrates proficiency through any of the options in subsection (c)(1)(iii), regardless of the student's grade level or age.

(f) *Transcripts*. The performance level demonstrated by a student in each of the state academic standards, including the highest performance level demonstrated by a student on the associated Keystone Exam, may be included on a student's transcript as determined by each school entity.

(g) *Release of scores*. This section does not allow for the release of individual student PSSA or Keystone Exam scores to the Department or other Commonwealth entities in accordance with § 4.51(f) and (g).

(h) *Supplemental instruction*. A student who does not demonstrate proficiency on a Keystone Exam or a locally validated assessment specified in subsection (c) may be offered supplemental instructional support by the student's school district, ACTS or charter school, including a cyber charter school consistent with section 121(c.4) of the School Code.

(i) *Out-of-state transfers*. A school district, ACTS or charter school, including a cyber charter school, shall determine whether a student who transfers from an out-of-State school having demonstrated proficiency in coursework and assessments aligned with the academic standards assessed by each Keystone Exam may satisfy the requirements of subsection (c).

(j) *Waiver of testing and accountability requirements*. In any year in which the Federal government has waived the testing and accountability requirements of the Elementary and Secondary Education Act, as amended by the Every Student Succeeds Act (20 U.S.C.A. § § 6301—7981), a student shall not be required to take a Keystone Exam for the purpose established in this section and shall be deemed proficient for purposes of this section by meeting the requirements set forth in section 121.1 of the School Code (24 P.S. § 1-121.1).

Authority

The provisions of this § 4.24 amended under section 32 of the act of June 29, 2002 (P.L. 524, No. 88); and sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.24 amended October 25, 2002, effective October 26, 2002, 32 Pa.B. 5266; amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended January 8, 2010, effective January 9, 2010, 40 Pa.B. 240; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371088) to (371092).

Notes of Decisions

IEP Compliance

Court determined that hearing officer, when reaching the conclusion that the student had completed the graduation requirements, failed to consider whether the requirements of the student's individualized education program (IEP) were fulfilled, as required by § 4.24(e). Because school district did not fulfill IEP obligation, it was required to pay tuition and fees for a transitional program after high school for one year, which the court found fulfilled the graduation requirements of § 4.24(e), *Susquehanna Township School District v. Frances J.*, 823 A.2d 249, 255-56 (Pa.Cmwlth. 2003).

Cross References

This section cited in 22 Pa. Code § 4.4 (relating to general policies); 22 Pa. Code § 4.31 (relating to career and technical education); 22 Pa. Code § 4.51b (relating to Keystone Exams); 22 Pa. Code § 4.51d (relating to waivers); and 22 Pa. Code § 4.52 (relating to local assessment system).

§ 4.25. Languages.

(a) World language programs must prepare students to be proficient in meeting the World Language Standards issued by the Department and available on its web site. Every school district shall provide planned instruction in at least two languages in addition to English, at least one of which shall be a modern language, and at least one of which shall be offered in a minimum 4-year sequence in the secondary program (middle level and high school).

(b) World language planned instruction under subsection (a) may be offered beginning at any grade level, including the elementary grades.

(c) World Language Standards issued by the Department will address the ability of students to communicate in a language other than English, including the ability to understand and interpret written and spoken language on a variety of topics and to develop knowledge and understanding of other cultures.

(d) As used in this section, the term ''world language'' means the study of the language, cultures, traditions and histories of different communities of people who communicate in languages other than English. American sign language is a world language.

Authority

The provisions of this § 4.25 amended under section 2603-B of the Public School Code of 1949 (24 P.S. § 26-2603-B).

Source

The provisions of this § 4.25 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872. Immediately preceding text appears at serial page (304974).

Cross References

This section cited in 22 Pa. Code § 4.23 (relating to high school education).

§ 4.26. ESOL.

Every school district shall provide a program for each student whose dominant language is not English for the purpose of facilitating the student's achievement of English proficiency and the academic standards under § 4.12 (relating to academic standards). Programs under this section shall include appropriate bilingual-bicultural or English as a second language (ESL) instruction.

§ 4.27. Physical education and athletics.

(a) Physical education shall be taught as required under § § 4.21(e)(6) and (f)(8), 4.22(c)(7) and 4.23(c)(8) (relating to elementary education: primary and intermediate levels; middle level education; and high school education).

(b) The physical education program must be adapted for students who are medically unable to participate in the regular physical education program.

(c) The physical education program shall provide coeducational instruction, except that separation by sex may be permitted in courses involving contact sports. Separation by sex may not be used to exclude students of either sex from participating in any physical education instruction.

(d) In addition to physical education instruction under subsections (a)—(c), students of both sexes shall have equal access in interscholastic and intramural athletic programs to all of the following:

(1) School facilities.

(2) Coaching and instruction.

(3) Scheduling of practice time and games.

(4) Number of activities at each level of competition.

(5) Equipment, supplies and services.

(6) Funding appropriate to the sport.

(e) School districts may sponsor coeducational teams in interscholastic and intramural sports programs.

(f) Interscholastic and intramural teams playing contact sports may be separated by sex, but this subsection may not be used to exclude students of either sex from participating in a sport.

Authority

The provisions of this § 4.27 amended under section 2603-B of the Public School Code of 1949 (24 P.S. § 26-2603-B).

Source

The provisions of this § 4.27 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872. Immediately preceding text appears at serial page (293053).

§ 4.28. Special education.

(a) Under the Individuals with Disabilities Education Act and this part, children with disabilities shall be provided an education which enables them to be involved in and progress in the general curriculum under this chapter.

(b) Students who are gifted as defined in this part shall be provided an education that enables them to participate in acceleration or enrichment, or both, as appropriate.

(c) The educational program provided to children with disabilities shall be in accordance with their Individualized Education Programs under the Individuals with Disabilities Education Act and this part, even if the Individualized Education Program does not otherwise meet all requirements of this chapter.

(d) Planned instruction for children with disabilities shall conform to the requirements established for planned courses in § 4.3 (relating to definitions) as it relates to planned instruction.

§ 4.29. HIV/AIDS and other life-threatening and communicable diseases.

(a) Instruction regarding prevention of human immunodeficiency virus (HIV) infection/acquired immunodeficiency syndrome (AIDS) and other life-threatening and communicable diseases shall be given for primary, intermediate, middle school and high school education and shall follow the requirements of subsections (b) and (c).

(b) Educational materials and instruction shall be determined by the local school district and be appropriate to the age group being taught. The program of instruction must include information about the nature of the diseases, treatments and cures, methods of transmission and how infection can be prevented. The school district may omit instruction in the elementary grades on transmission of disease through sexual activity. Programs discussing transmission through sexual activity must stress that abstinence from sexual activity is the only completely reliable means of preventing sexual transmission. Programs must stress that avoidance of illegal drug use is the only completely reliable means of preventing transmission of disease through shared drug paraphernalia.

(c) A school entity shall excuse a pupil from HIV/AIDS instruction when the instruction conflicts with the religious beliefs or principles of the pupil or parent or guardian of the pupil and when excusal is requested in writing. Prior to the commencement of instruction, a school district shall publicize that detailed curriculum outlines and curricular materials used in conjunction with the instruction are available to parents and guardians during normal school hours or at teacher-parent conferences. Curricular materials, if practical, shall be made available by the school entity for home instructional use by a parent or guardian if the student has been excused from the school entity's HIV/AIDS instruction.

Authority

The provisions of this § 4.29 amended under section 2603-B of the Public School Code of 1949 (24 P.S. § 26-2603-B).

Source

The provisions of this § 4.29 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872. Immediately preceding text appears at serial page (252333).

CAREER AND TECHNICAL EDUCATION

§ 4.31. Career and technical education.

(a) Career and technical education courses shall be developed in the planned instruction format and be accessible to all high school students attending those grades in which career and technical education courses are offered. All students and their parents or guardians shall be informed of the students' rights to participate in career and technical education programs and courses and that students with disabilities enrolled in the programs are entitled to services under Chapter 14 (relating to special education services and programs). Students who complete approved career and technical education programs shall have their occupational competency assessed by completion of the appropriate assessment under the Pennsylvania Skills Certificate Program or by completion of another occupational competency assessment approved by the Department. A student with a disability shall be provided appropriate accommodations when provided for in the student's individualized education program. Students shall also demonstrate proficiency in meeting academic standards as required under § 4.24 (relating to high school graduation requirements), including § 4.12(f) (relating to academic standards) and § 4.24(d) for students with disabilities with an individualized education program.

(b) Career and technical education courses may be taught at ACTSs or other high schools.

(c) Career and technical education programs must consist of a series of planned academic and career and technical education courses that are articulated with one another so that knowledge and skills are taught in a systematic manner. When appropriate, career and technical education programs must adopt, in program areas for which they are available, industry recognized skills standards and may also include cooperative career and technical education and participation in career and technical student organizations to develop leadership skills.

(d) Career and technical education courses must include content based upon occupational analysis, clearly stated performance objectives deemed critical to successful employment and assessment of student competencies based upon performance standards.

(e) The record of a student enrolled in a career and technical education program must include the student's educational and occupational objectives and the results of the assessment of student competencies under subsection (d).

(f) Safety education, consisting of safety practices, accident prevention, occupational health habits and environmental concerns shall be integrated into the instruction and practices in career and technical education programs.

(g) School districts and ACTSs administering career and technical education programs shall develop written policies regarding admissions. Course announcements, guidance materials and other communications must convey the philosophy of equal access to students considering enrolling in ACTSs and include a description of admissions policies. The policies must assure that when admissions to ACTSs must be limited, the admissions shall be on a nondiscriminatory basis.

Authority

The provisions of this § 4.31 amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.31 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371094) to (371095).

Cross References

This section cited in 22 Pa. Code § 4.23 (relating to high school education); 22 Pa. Code § 4.32 (relating to standards and reports).

§ 4.32. Standards and reports.

(a) The Secretary is responsible for the promulgation of standards appropriate for implementing § 4.31 (relating to career and technical education). Present standards, to the extent that they are inconsistent, are superseded by this chapter.

(b) The Secretary will report annually to the Board on the status of career and technical education programs, including tech-prep and apprenticeship programs. Reports will include numbers and types of programs, numbers of students, post-program status of students, Statewide competency standards and assessment information.

Authority

The provisions of this § 4.32 amended under sections 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.32 amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial page (371096).

Cross References

This section cited in 22 Pa. Code § 4.23 (relating to high school education).

§ 4.33. Advisory committees.

(a) A school district or ACTS administering or planning to administer career and technical education programs shall appoint a local advisory committee. Membership on the committee shall consist of business and industry representatives, public sector employers, agriculture, labor organizations, community organizations, postsecondary education institutions and the general public. The appointed advisory committee shall meet at least once each year and give advice to the board and the administration concerning the program of the school, including its general philosophy, academic and other standards, course offerings, support services, safety requirements and the skill needs of employers. An advisory committee may serve multiple institutions where employment areas overlap.

(b) An administrative committee, composed of chief school administrators representing participating school districts, shall be included in the organization of each ACTS. The committee shall advise the ACTS board and the administration concerning the educational program and policies of the school.

(c) An occupational advisory committee shall be established for each career and technical education program or cluster of related programs offered by a school district or ACTS. The committee shall be appointed by the board of directors, and a majority of the members of the committee shall be employees and employers in the occupation for which training is provided. The committee shall meet at least twice each year to advise the board, administration and staff on curriculum, equipment, instructional materials, safety requirements, program evaluation and other related matters and to verify that the programs meet industry standards and, if appropriate, licensing board criteria and that they prepare students with occupation related competencies.

Authority

The provisions of this § 4.33 amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.33 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371096) to (371097).

Cross References

This section cited in 22 Pa. Code § 4.23 (relating to high school education); 22 Pa. Code § 339.13 (relating to local advisory committee); and 22 Pa. Code § 339.14 (relating to occupational advisory committee).

§ 4.34. Programs and equipment.

(a) A satellite career and technical education program may be operated by an ACTS board in conformity with a memorandum of understanding adopted with the participating school district's board of school directors.

(b) Certified guidance personnel in each secondary school and ACTS shall be assigned responsibility to provide pupils with career and technical education guidance services.

(c) Equipment will be deemed appropriate if it is compatible, insofar as practical, to that used in occupations or households for which career and technical education is provided.

Authority

The provisions of this § 4.34 amended under sections 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.34 amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial page (371097).

Cross References

This section cited in 22 Pa. Code § 4.23 (relating to high school education).

§ 4.35. ACTSs.

(a) ACTS attendance areas shall conform to the plan of the State Board of Career and Technical Education. Boards of school directors may petition the State Board of Career and Technical Education for attendance area assignment or reassignment.

(b) The following provisions apply to the establishment of ACTSs:

(1) Where more than one district constitutes an attendance area, the appropriate intermediate unit may, and upon the request of any school district shall, call for an election by the boards of school directors within the attendance area to determine if an ACTS shall be established.

(2) A school district within the attendance area may elect to participate in the establishment of the ACTS.

(3) Where a single school district constitutes an attendance area, the board of school directors of that district may establish and operate ACTSs and be considered an ACTS board.

(c) The following provisions apply to articles of agreement for the establishment and operation of ACTSs:

(1) The boards of school directors of the school districts electing to participate in the ACTS shall enter into a written agreement setting forth rights and obligations of the participating school districts.

(2) No change will be made in the articles of agreement under paragraph (1) without the consent of each participating school district by the affirmative vote of each board of school directors.

(3) No school district may withdraw from the articles of agreement under paragraph (1) without the consent of each participating school district.

Authority

The provisions of this § 4.35 amended under sections 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.35 amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial page (371097).

Cross References

This section cited in 22 Pa. Code § 4.23 (relating to high school education).

SCHEDULING AND LEARNING OPTIONS

§ 4.41. Scheduling.

(a) Kindergarten programs shall provide each kindergarten student with at least 2 1/2 hours of instruction each day for the full school term unless the school district, including charter schools, obtains prior Department approval for an alternative kindergarten program.

(b) A school district, including charter schools, shall obtain approval of the Department prior to scheduling 1/2-day sessions other than in kindergarten under subsection (a). A school district is not required to obtain approval of the Department prior to scheduling 1/2 day sessions for prekindergarten under subsection (e).

(c) A school district shall obtain approval of the Department prior to establishing a new school or changing school organization.

(d) Planned instruction offered in summer school may be designed as credit or noncredit offerings.

(e) School districts with prekindergarten programs shall provide prekindergarten students with at least 2 1/2 hours of instruction each day for the full school term unless the school district obtains prior Department approval for an alternative prekindergarten program.

Authority

The provisions of this § 4.41 amended under section 2603-B of the Public School Code of 1949 (24 P.S. § 26-2603-B).

Source

The provisions of this § 4.41 amended December 15, 2006, effective December 16, 2006, 36 Pa.B. 7542. Immediately preceding text appears at serial pages (252336) to (252337).

§ 4.42. Grade structure.

This chapter does not require educational programs to be organized in traditional grades according to students' chronological ages or academic achievement levels.

ASSESSMENT

§ 4.51. State assessment system.

(a) The State assessment system shall be designed to serve the following purposes:

(1) Provide students, parents, educators and citizens with an understanding of student and school performance consistent with the Every Student Succeeds Act (Pub.L. No. 114-95).

(2) Determine the degree to which school programs enable students to attain proficiency of academic standards under § 4.12 (relating to academic standards).

(3) Provide information to State policymakers, including the General Assembly and the Board, on how effective schools are in promoting and demonstrating student proficiency of academic standards.

(4) Provide information to the general public on school performance.

(5) Provide results to school entities based upon the aggregate performance of all students, for students with an Individualized Education Program (IEP) and for those without an IEP.

(6) Assess student proficiency in the Academic Standards for English Language Arts (Appendix A-2), Mathematics (Appendix A-2), and, through June 30, 2025, Science and Technology and Environment and Ecology (Appendix B), and, beginning July 1, 2025, Science, Environment, Ecology and Technology

and Engineering (Appendix B-1) for the purpose of determining, in part, a student's eligibility for high school graduation.

(b) The State assessment system must include PSSA assessments and Keystone Exams.

(c) Neither State assessments nor academic standards under § 4.12 may require students to hold or express particular attitudes, values or beliefs.

(d) The Department will make samples of State assessment questions, assessment formats and scoring guides available to the public after each administration of State assessments.

(e) To ensure that information regarding student performance is available to parents and teachers, State assessments developed under this section must include student names.

(f) Individual assessment results shall be used in planning instruction only by parents, teachers, administrators and guidance counselors with a need to know based upon local board policy on testing and in reporting academic progress.

(g) The Department and other Commonwealth entities are prohibited from collecting individual student test scores and may collect only aggregate test scores by school and district.

(h) The Board will authorize the expansion of the State assessment system through a revision of this chapter.

(1) The Board will not include National assessments as part of the State assessment system unless, upon consultation with teachers, counselors and parents representing students who have been identified under Chapter 14 (relating to special education services and programs), the Board determines the assessment is an appropriate means of assessing the academic progress of students identified under Chapter 14, or unless the General Assembly authorizes the use of a National assessment.

(2) Subject to paragraph (3), the Board will not, and the Department may not, be a governing state in any consortium for the development of a National assessment for the purpose of utilization as part of the State assessment system.

(3) The Department may continue to participate in a consortium to develop an alternate assessment to measure the academic progress of students identified under Chapter 14.

(i) The Department will implement provisions for security of the State assessment system, including the following:

(1) Action by a professional employee or commissioned officer that is willfully designed to divulge test questions, falsify student scores or in some other fashion compromise the integrity of the State assessment system as determined by the school district, ACTS or charter school, including a cyber charter school, shall be subject to disciplinary action under the Educator Discipline Act (24 P.S. § § 2070.1a—2070.18c).

(2) Cheating by students or employees other than those covered in paragraph (1) shall be subject to disciplinary action by the school district, ACTS or charter school, including a cyber charter school.

(3) Cheating or breaches of assessment security shall be reported to the Secretary as soon as detected.

(j) The Secretary is authorized to establish guidelines for the administration of the State assessment system.

(k) The Secretary will report each September to the Board and the General Assembly information and pertinent data regarding the State assessment system. The Secretary also will provide each school entity information and pertinent data for the school entity and its students.

(l) Children with disabilities and children with limited English proficiency shall be included in the State assessment system as required by Federal law, with appropriate accommodations when necessary. As appropriate, the Commonwealth will develop guidelines for the participation of children with disabilities in alternate assessments for those children who cannot participate in the PSSA or Keystone Exams as determined by each child's individualized education program team under the Individuals with Disabilities Education Act and this part.

Authority

The provisions of this § 4.51 amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.51 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended January 8, 2010, effective January 9, 2010, 40 Pa.B. 240; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371098) and (404271) to (404272).

Cross References

This section cited in 22 Pa. Code § 4.24 (relating to high school graduating requirements); 22 Pa. Code § 4.51b (relating to Keystone Exams); 22 Pa. Code § 14.105 (relating to personnel); and 22 Pa. Code § 19.1a (relating to general provisions).

§ 4.51a. Pennsylvania System of School Assessment.

(a) All PSSA assessments administered in English Language Arts, Mathematics, and Science, Environment, Ecology, Technology and Engineering will be standards-based and criterion referenced and include essay or open-ended response items in addition to other item formats. The proportion of type of items will vary by grade level. The criteria for judging performance on PSSA assessments are as follows:

(1) Performance on PSSA English Language Arts assessments shall be demonstrated by students' responses to comprehension questions about age-appropriate reading passages, by their written responses to in-depth comprehension questions about the passages and by the quality of their written compositions on a variety of topics and modes of writing.

(2) Performance on PSSA mathematics assessments shall be demonstrated by students' responses to questions about grade-appropriate content and by the quality of their responses to questions that require a written solution to a problem.

(3) Performance on PSSA science assessments shall be demonstrated by students' responses to grade appropriate content and by the quality of their responses to questions that demonstrate knowledge of each category of the standards for science, environment, ecology, technology and engineering.

(4) Performance levels shall be advanced, proficient, basic and below basic. In consultation with educators, students, parents and citizens, the Department will develop and recommend to the Board for its approval specific criteria for advanced, proficient, basic and below basic levels of performance.

(b) The Department will develop or cause to be developed PSSA assessments based on Pennsylvania Core Standards in Mathematics and English Language Arts under § 4.12 (relating to academic standards) and contained in Appendix A-2. Through June 30, 2025, the Department will develop or cause to be developed PSSA assessments based on academic standards in Science, Technology, Environment and Ecology under § 4.12 and contained in Appendix B. Beginning July 1, 2025, the Department will develop or cause to be developed PSSA assessments based on academic standards in Science, Environment, Ecology, Technology and Engineering under § 4.12 and contained in Appendix B-1. In developing PSSA assessments, the Department will consult with educators, students, parents and citizens regarding the specific methods of assessment.

(c) The PSSA assessments shall be administered annually and include assessments of the State academic standards in Mathematics and English Language Arts at grades 3 through 8, and in Science, Environment, Ecology, Technology and Engineering at grades 4 and 8. Beginning in the 2025-2026 school year, the PSSA that assesses State academic standards in Science, Environment, Ecology, Technology and Engineering shall be administered at grade 5 and grade 8 and the PSSA assessment of these standards administered at grade 4 shall be discontinued.

Authority

The provisions of this § 4.51a issued under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B); amended under sections 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.51a adopted February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410365) to (410366).

Cross References

This section cited in 22 Pa. Code § 4.21 (relating to elementary education: primary and intermediate levels); and 22 Pa. Code § 4.24 (relating to high school graduation requirements).

§ 4.51b. Keystone Exams.

(a) The Department will develop or cause to be developed Keystone Exams as provided in this subsection. (This subsection is intended by the Board to be a continuation of § 4.51(f) (relating to State assessment system) as published at 40 Pa.B. 240 (January 9, 2010) and referenced in section 102 of the School Code (24 P.S. § 1-102).)

(1) One assessment aligned with the Mathematics standards, contained in Appendix A-2, that assesses the academic content traditionally included in an Algebra I course.

(2) One assessment aligned with select English Language Arts standards, contained in Appendix A-2, that assesses academic content traditionally included in a high school literature course.

(3) Through June 30, 2025, one assessment aligned with select standards for Science, Technology, Environment and Ecology, contained in Appendix B, that assesses academic content traditionally included in a high school level Biology course. Beginning July 1, 2025, one assessment aligned with select standards for Science, Environment and Ecology, contained in Appendix B-1, that assesses academic content traditionally included in a high school level Biology course.

(b) Keystone Exams shall be offered at least three times each year: once each in the fall, spring and summer.

(c) Keystone Exams shall be administered, reviewed and scored so that scores for candidates for graduation are provided to schools no later than 10 calendar days prior to graduation. A school district, ACTS or charter school, including a cyber charter school, may request the Department to approve alternative test administration and scoring time frames. The Department will publish guidelines and procedures for approving alternative test administration and scoring time frames on its web site. The guidelines will provide for approval of all requests unless the approval is contrary to standards of test validity and scoring.

(d) A student shall be permitted to retake any Keystone Exam, or Keystone Exam module, in which the student did not score proficient or above at the next available testing date. There is not a limit on the number of times a student who did not score proficient on a Keystone Exam is permitted to retake the Keystone Exam or Keystone Exam module. A student who has achieved a score of proficient or advanced on a Keystone Exam shall be permitted to retake a Keystone Exam only if the student or parent submits a request in writing to the school entity.

(e) Each Keystone Exam will be designed in modules that reflect distinct, related academic content that is common to the traditional progression of coursework to allow students who do not score proficient or above to retake those portions of the test in which they did not score proficient or above.

(f) A student taking Keystone Exams, or Keystone Exam modules, who did not score proficient on a Keystone Exam, or Keystone Exam module, may be provided supplemental instruction consistent with the student's educational program by the student's school district, ACTS or charter school, including a cyber charter school, consistent with section 121(c.4) of the School Code (24 P.S. § 1-121(c.4)).

(g) Performance levels for Keystone Exams shall be set at the advanced, proficient, basic and below basic levels. In consultation with the Performance Level Advisory Committee, the Department will develop and recommend to the Board for its approval performance level descriptors and performance level cut scores for the Keystone Exams and any alternative assessments developed to assess students with disabilities as permitted by the Every Student Succeeds Act (Pub.L. No. 114-95). The Department will use widely-accepted psychometric procedures to establish the cut scores. Cut scores shall be presented at a public meeting of the Board for its review at least 2 weeks prior to scheduled Board action on the cut scores.

(h) The Department will provide guidance to school districts, ACTSs and charter schools, including cyber charter schools, as to the appropriate accommodations school entities shall provide to students with disabilities, students who are gifted and English language learners, when appropriate.

(i) Beginning in the 2012-2013 school year, Keystone Exams in the following subjects will be developed by the Department and made available for use by school districts, ACTSs and charter schools, including cyber charter schools, for the purpose of assessing high school graduation requirements in § 4.24(c)(1)(iii) (relating to high school graduation requirements):

Algebra I

Literature

Biology

(j) The Department will seek to have the Keystone Exams approved as the high school level single accountability system under the Every Student Succeeds Act (20 U.S.C.A. § § 6301—7981) or its successor Federal statute. If the Keystone Exams receive approval as the high school level accountability measure, school districts, ACTSs and charter schools, including cyber charter schools, shall administer the Literature, Algebra I and Biology exams as end-of-course tests in the grade level in which students complete the relevant coursework.

(k) The 11th grade PSSA exams in Reading, Writing, Math and Science shall be discontinued upon implementation of the Keystone Exams as the approved assessment system under section 1111(b)(2)(C) of the No Child Left Behind Act of 2001 (20 U.S.C.A. § 6311(b)(2)(C)).

(l) At least once every 5 years, the Department will contract with a qualified, independent research organization to perform a validity study of the Keystone Exams using generally accepted education research standards. These studies will determine, at a minimum, the degree to which the Keystone Exams and performance level cut scores are valid for the purposes for which they are used; aligned with State academic standards; aligned with performance levels of other states; internationally benchmarked; and predict college and career success. In addition, all Keystone Exams, performance level descriptors and cut scores will be subject to the best available forms of content, criterion and consequential validation.

(m) The Department will establish a State Assessment Validation Advisory Committee (Committee). The Committee will advise the Department on its plans to conduct the validity study and review and provide feedback on its findings.

(n) The Department and the Committee will investigate the use of a certificate based on industry approved standards and performance on an NOCTI exam as an alternative pathway to graduation and will make a report and recommendation to the Board by January 10, 2011.

Authority

The provisions of this § 4.51b issued under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B); amended under sections 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.51b adopted February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371101) to (371104).

Cross References

This section cited in 22 Pa. Code § 4.4 (relating to general policies); 22 Pa. Code § 4.24 (relating to high school graduation requirements); 22 Pa. Code § 4.51c (relating to project-based assessment); and 22 Pa. Code § 4.51d (relating to waivers).

§ 4.51c. Project-based assessment.

The Department will develop a project-based assessment system that is aligned with the modules for the Keystone Exams in Literature, Algebra I and Biology. School entities may utilize project-based assessments for students consistent with section 121(c.5) of the School Code (24 P.S. § 1-121(c.5)).

Authority

The provisions of this § 4.51c issued under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B); amended under sections 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.51c adopted February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371104) to (371105).

§ 4.51d. Waivers.

A chief school administrator, in his sole discretion, may waive the requirements in § 4.24 (relating to high school graduation requirements) consistent with section 121(c.3) of the School Code (24 P.S. § 1-121(c.3)). The following apply:

(1) The chief school administrator of each school district, ACTS and charter school, including a cyber charter school, shall annually report to the Department the number of waivers granted to students in the most recent graduating class consistent with section 121(c.11)(6) of the School Code, and the Department will annually report to the Board the number of waivers granted by each school district, ACTS and charter school, including a cyber charter school.

(2) The waiver process described in this section does not confer an individual right on any student.

(3) The decision of a chief school administrator concerning a waiver request is not an adjudication.

Authority

The provisions of this § 4.51d issued under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B); amended under sections 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.51d adopted February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371105) to (371106).

§ 4.52. Local assessment system.

(a) Each school entity shall design an assessment system to do the following:

(1) Determine the degree to which students are achieving academic standards under § 4.12 (relating to academic standards). The school entity shall provide assistance to students not attaining academic standards at the proficient level or better.

(2) Use assessment results to improve curriculum and instructional practices and to guide instructional strategies.

(3) Provide information requested by the Department regarding the achievement of academic standards, but which does not include student names, identification numbers or individually identifiable information.

(4) Provide summary information, including results of assessments under this section, to the general public regarding the achievement of students, but which does not include student names, identification numbers or individually identifiable information.

(b) The local assessment system shall be approved by the board of school directors, at minimum, once every 6 years and implemented no later than 1 year after the approval date.

(c) The local assessment system shall be designed to include a variety of assessment strategies which may include the following:

(1) Written work by students.

(2) Scientific experiments conducted by students.

(3) Works of art or musical, theatrical or dance performances by students.

(4) Other demonstrations, performances, products or projects by students related to specific academic standards.

(5) Examinations developed by teachers to assess specific academic standards.

(6) Nationally-available achievement tests.

(7) Diagnostic assessments.

(8) Evaluations of portfolios of student work related to achievement of academic standards.

(9) Other measures as appropriate, which may include standardized tests.

(d) Individual test information shall be maintained in a student's educational record in a manner consistent with section 444 of the Family Educational Rights and Privacy Act of 1974 (20 U.S.C.A. § 1232g) and 34 CFR Part 99 (relating to family educational rights and privacy).

(e) Children with disabilities shall be included in the local assessment system, with appropriate accommodations, when necessary. As appropriate, the school district, including a charter school, including a cyber charter school, or ACTS shall develop guidelines for the participation of children with disabilities in alternate assessments for those children who cannot participate in the local assessment as determined by each child's Individualized Education Program team under the Individuals with Disabilities Education Act and this part.

(f) The Department will establish a Local Assessment Validation Advisory Committee (Committee). The Committee will develop the criteria for the local validation process and criteria for selection of approved validation entities as provided in § 4.24(c)(1)(iii)(B) (relating to high school graduation requirements). The Department, in consultation with the Committee, will establish a list of entities approved to perform independent validations of local assessments. The Committee will submit its recommendations for approval or disapproval to the Board. The Department will post the approved criteria, selection criteria and list of approved entities on its web site.

Authority

The provisions of this § 4.52 amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.52 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended January 8, 2010, effective January 9, 2010, 40 Pa.B. 240; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; amended July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946. Immediately preceding text appears at serial pages (371106) to (371108).

Cross References

This section cited in 22 Pa. Code § 4.12 (relating to academic standards); 22 Pa. Code § 4.20 (relating to prekindergarten education); 22 Pa. Code § 4.21 (relating to elementary education: primary and intermediate levels); 22 Pa. Code § 4.24 (relating to high school graduation requirements); and 22 Pa. Code § 14.105 (relating to personnel).

SCHOOL PROFILES

§ 4.61. School profiles.

(a) School profiles developed by the Secretary will include information as required under section 220 of the School Code (24 P. S. § 2-220).

(b) The Secretary will prescribe procedures for reporting State assessment data to schools and communities.

(c) The Secretary will make available to the public, and report to the public with the same frequency and in the same detail as for children who are nondisabled, all data as required under the Individuals with Disabilities Education Act.

Authority

The provisions of this § 4.61 amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P. S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this § 4.61 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131. Immediately preceding text appears at serial pages (346993) to (346994).

PROVISIONS RELATING TO OTHER THAN PUBLIC SCHOOLS

§ 4.71. Certification by principal of nonpublic nonlicensed school.

Elementary or secondary nonpublic nonlicensed schools, shall, within 30 days of beginning classes, file a notarized certificate with the Secretary as required by section 1327(b)(1) and (2) of the School Code (24 P. S. § 13-1327(b)(1) and (2)) in the form prescribed by the Secretary.

§ 4.72. Credentials other than the high school diploma.

The requirements for a Commonwealth secondary school diploma are as follows:

(1) The Commonwealth secondary school diploma may be issued to an applicant who is a resident of this Commonwealth and does not possess a secondary school diploma upon presentation of evidence of full matriculation and the satisfactory completion of a minimum of 1 full year or 30 semester hours of study at an accredited institution of postsecondary education.

(2) In addition to the provisions of paragraph (1), the Commonwealth secondary school diploma may be issued to an applicant who is a resident of this Commonwealth, does not possess a secondary school diploma and is not enrolled in a public, licensed private, registered accredited or licensed nonpublic secondary school upon earning a passing score as determined by the Department on the high school level tests of General Educational Development (GED). A person 18 years of age or older may qualify for GED testing upon request. A person between 16 and 18 years of age may qualify for GED testing upon the issuance of a court order or at the written request of one of the following:

(i) An employer who requires a high school equivalency credential for job opportunities.

(ii) An official of an accredited institution of postsecondary education which accepts applicants on the basis of GED test scores.

(iii) A recruiting officer of a branch of the armed forces that requires a high school equivalency credential for entry of new recruits.

(iv) The director of a State institution on behalf of residents, patients or inmates.

(3) The Department will not ordinarily issue a diploma until after the high school class of which the applicant was a member has been graduated. This restriction may be waived by the Department upon the recommendation of the school district for persons between 16 and 18 years of age who meet the higher education or GED requirements for the secondary school diploma.

Cross References

This section cited in 22 Pa. Code § 4.74 (relating to students in special situations).

§ 4.73. Correspondence schools.

An applicant 18 years of age or older will be issued a Certificate of Preliminary Education upon presentation to the Department of evidence of the issuance of a high school diploma by an accredited private correspondence school licensed or approved by the State Board of Private Licensed Schools.

§ 4.74. Students in special situations.

(a) A foreign student without educational credentials may earn the Commonwealth secondary school diploma by meeting the requirements under § 4.72 (relating to credentials other than the high school diploma).

(b) A graduate of a secondary school in another state which is not on an approved list of secondary schools may earn an appropriate credential by passing an examination administered by the education agency of that state or by its designee or by meeting the requirements for the Commonwealth secondary school diploma under § 4.72.

(c) Credit granted by a public school in this Commonwealth shall be accepted by all public schools and institutions in this Commonwealth upon the transfer of a student.

ENFORCEMENT AND IMPLEMENTATION

§ 4.81. Allegations of deficiencies.

(a) The Secretary will receive and investigate allegations of curriculum deficiencies from professional employees, commissioned officers, parents of students or other residents of a school entity.

(b) The Secretary will notify the school entity's superintendent or chief executive of allegations and may require the superintendent or chief executive to submit one or more of the following:

(1) Relevant descriptions of planned instruction.

(2) A series of written articulated courses of instructional units.

(3) Relevant student assessment information.

(4) Information on staff assignments.

(5) Other information pertinent to investigating a specific allegation.

(c) If the Secretary determines that a curriculum deficiency exists, the school entity shall be required to submit to the Secretary for approval a plan to correct the deficiency.

(d) Within 1 year of the implementation of a corrective action plan under subsection (c), the Secretary will review the actions taken to correct the deficiency. If the deficiency remains uncorrected, the Secretary will send a formal notice of deficiency to the governing board of the school entity, and the notice shall be announced at the meeting of the school entity's governing board immediately following its receipt.

(e) If the school entity does not take appropriate actions to correct the deficiency after the notice of deficiency is announced, the Secretary will take action under State law.

Authority

The provisions of this § 4.81 amended under the Public School Code of 1949 (24 P. S. 26-2603-B).

Source

The provisions of this § 4.81 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872. Immediately preceding text appears at serial page (286559).

Notes of Decisions

Inapplicable Offense

Educator's argument that the offensive conduct of manufacturing of grades was a curriculum deficiency that should be resolved under the Academic Standards and Assessment Chapter of the Administrative Code is misplaced. The conduct is properly prosecuted under the Teacher Certification Law. *Seltzer v. Department of Education*, 782 A.2d 48 (Pa. Cmwlth. 2001).

§ 4.82. Exceptions.

(a) The Secretary may grant exceptions to specific provisions of this chapter when it is necessary to adapt them to the curriculum needs of individual school entities or to facilitate transition to the revised provisions of this chapter. Specific exception may be made for school entities that develop or implement academic standards that are comparable to or exceed those found in § 4.12 (relating to academic standards). Exceptions may be granted under the following conditions:

(1) The request for an exception must be in writing and include relevant information supporting the need for the exception.

(2) The exception will be valid for a limited term not to exceed 2 years.

(3) The request shall be made prior to initiating the action requiring approval and shall have the prior approval of the board of school directors.

(b) The Secretary will report annually to the Board on the nature and status of requests for exceptions under this section.

Authority

The provisions of this § 4.82 amended under the Public School Code of 1949 (24 P.S. 26-2603-B).

Source

The provisions of this § 4.82 amended February 15, 2008, effective February 16, 2008, 38 Pa.B. 872. Immediately preceding text appears at serial pages (286559) to (286560).

§ 4.83. {Reserved}.

Source

The provisions of this § 4.83 reserved February 15, 2008, effective February 16, 2008, 38 Pa.B. 872. Immediately preceding text appears at serial page (286560).

ACADEMIC STANDARDS

APPENDIX A

{Expired}

Source

The provisions of this Appendix A adopted January 15, 1999, effective January 16, 1999, 29 Pa.B. 399; reserved by correction July 19, 2013, effective June 30, 2013, replaced by Appendix A-1, 43 Pa.B. 4079. Immediately preceding text appears at serial pages (252345) to (252422) and (286561) to (286562).

APPENDIX A-1

{Expired}

Source

The provisions of this Appendix A-1 renumbered from Appendix B adopted October 15, 2010, effective July 1, 2013, 40 Pa.B. 5903; correction published at 43 Pa.B. 4079; reserved February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131. Immediately preceding text appears at serial pages (367438) and (353099) to (353316).

APPENDIX A-2
Pennsylvania Core Standards for English Language Arts and Mathematics

Authority

The provisions of this Appendix A-2 issued under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix A-2 adopted February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131.

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards); 22 Pa. Code § 4.24 (relating to high school graduation requirements); 22 Pa. Code § 4.51 (relating to state assessment system); 22 Pa. Code § 4.51a (relating to Pennsylvania System of School Assessment); and 22 Pa. Code § 4.51b (relating to Keystone Exams).

Pennsylvania Core Standards for English Language Arts
Grades Pre K—5

INTRODUCTION

These standards describe what students should know and be able to do with the English language, prekindergarten through Grade 12. The standards provide the targets for instruction and student learning essential for success in all academic areas, not just language arts classrooms. Although the standards are not a curriculum or a prescribed series of activities, school entities will use them to develop a local school curriculum that will meet local students' needs.

Five standard categories are designed to provide a Pre K-12 continuum to reflect the demands of a college- and career-ready graduate:

Standard 1:	*Foundational Skills* begin at prekindergarten and focus on early childhood, with some standards reflected through Grade 5. These foundational skills are a necessary and important component of an effective, comprehensive reading program designed to develop proficient readers with the capacity to comprehend text, both literary and informational, across disciplines.
Standard 2:	*Reading Informational Text* enables students to read, understand, and respond to informational text.
Standard 3:	*Reading Literature* enables students to read, understand, and respond to works of literature.
Standard 4:	*Writing* develops the skills of informational, argumentative, and narrative writing, as well as the ability to engage in evidence-based analysis of text and research.
Standard 5:	*Speaking and Listening* focuses students on communication skills that enable critical listening and effective presentation of ideas.

With a focus on college and career readiness, the instructional shifts as reflected in Common Core are evident throughout the PA Core Standards:

• Balancing the reading of informational and literary texts so that students can access nonfiction and authentic texts, as well as literature

• Focusing on close and careful reading of text so that students are learning from the text

• Building a staircase of complexity (i.e., each grade level requires a ''step'' of growth on the ''staircase'') so that students graduate college or career ready

• Supporting writing from sources (i.e., using evidence from text to inform or make an argument) so that students use evidence and respond to the ideas, events, facts, and arguments presented in the texts they read

• Stressing an academically focused vocabulary so that students can access more complex texts

The English Language Arts Standards also provide parents and community members with information about what students should know and be able to do as they progress through the educational program and at graduation. With a clearly defined target provided by the standards, parents, students, educators, and community members become partners in learning. Each standard implies an end-of-year goal—with the understanding that exceeding the standard is an even more desirable end goal.

Note: The Aligned Eligible Content is displayed with the standard statement. On the Standard Aligned System portal, it is a live link.

TABLE OF CONTENTS

Foundational Skills (Pre K-5)… **1.1**

Students gain a working knowledge of concepts of print, alphabetic principle, and other basic conventions. These foundational skills are not an end in and of themselves; rather, students apply them as effective readers.

• Book Handling

• Print Concepts

• Phonological Awareness

• Phonics and Word Recognition

• Fluency

Reading Informational Text… **1.2**

Students read, understand, and respond to informational text—with an emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with a focus on textual evidence.

• Key Ideas and Details

• Craft and Structure

• Integration of Knowledge and Ideas

• Vocabulary Acquisition and Use

• Range of Reading

Reading Literature… **1.3**

Students read and respond to works of literature—with emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with focus on textual evidence.

• Key Ideas and Details

• Craft and Structure

• Integration of Knowledge and Ideas

• Vocabulary Acquisition and Use

• Range of Reading

Writing… **1.4**

Students write for different purposes and audiences. Students write clear and focused text to convey a well-defined perspective and appropriate content.

• Informative/Explanatory

• Opinion/Argumentative

• Narrative

• Response to Literature

• Production and Distribution of Writing

• Technology and Publication

• Conducting Research

• Credibility, Reliability, and Validity of Sources

• Range of Writing

Speaking and Listening… **1.5**

Students present appropriately in formal speaking situations, listen critically, and respond intelligently as individuals or in group discussions.

• Comprehension and Collaboration

• Presentation of Knowledge and Ideas

• Integration of Knowledge and Ideas

• Conventions of Standard English

1.1	**Foundational Skills** **Students gain a working knowledge of concepts of print, alphabetic principle, and other basic conventions.**						
	Grade Pre K	**Grade K**	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Grade 5**
Book Handling	**CC.1.1.PK.A** Practice appropriate book handling skills.	**CC.1.1.K.A** Utilize book handing skills.	Intentionally Blank	Intentionally Blank	Intentionally Blank	Intentionally Blank	Intentionally Blank
Print Concepts	**CC.1.1.PK.B** Identify basic features of print. • Differentiate between numbers and letters and letters and words. • Recognize and name some uppercase and lowercase letters of the alphabet.	**CC.1.1.K.B** Demonstrate understanding of the organization and basic features of print. • Follow words left to right, top to bottom, and page by page. • Recognize that spoken words are represented in written language by specific sequences of letters. • Understand that words are separated by spaces in print. • Recognize and name all uppercase and lowercase letters of the alphabet.	**CC.1.1.1.B** Demonstrate understanding of the organization and basic features of print. • Recognize the distinguishing features of a sentence.	Intentionally Blank	Intentionally Blank	Intentionally Blank	Intentionally Blank
Phonological Awareness	**CC.1.1.PK.C** Demonstrate understanding of spoken words, syllables, and sounds (phonemes). • Recognize rhyming words and when two or more words begin with the same sound (alliteration). • Count syllables in spoken words. • Segment single-syllable spoken words. • Isolate and pronounce initial sounds.	**CC.1.1.K.C** Demonstrate understanding of spoken words, syllables, and sounds (phonemes). • Recognize and produce rhyming words. • Count, pronounce, blend, and segment syllables in spoken words. • Blend and segment onsets and rimes of single-syllable spoken words. • Isolate and pronounce the initial, medial vowel, and final sound (phonemes) in the three-phoneme (CVC) words.	**CC.1.1.1.C** Demonstrate understanding of spoken words, syllables, and sounds (phonemes). • Distinguish long from short vowel sounds in spoken single-syllable words. • Count, pronounce, blend, and segment syllables in spoken and written words. • Orally produce single-syllable words, including consonant blends and digraphs.	Intentionally Blank	Intentionally Blank	Intentionally Blank	Intentionally Blank
			• Isolate and pronounce initial, medial vowel, and final sounds (phonemes) in spoken single-syllable words. • Add or substitute individual sounds (phonemes) in one-syllable words to make new words.				
Phonics and Word	**CC.1.1.PK.D** Develop	**CC.1.1.K.D** Know and apply	**CC.1.1.1.D** Know and apply	**CC.1.1.2.D** Know and apply	**CC.1.1.3.D** Know and apply	**CC.1.1.4.D** Know and apply	**CC.1.1.5.D** Know and apply

Recognition	beginning phonics and word skills. • Associate some letters with their names and sounds. • Identify familiar words and environmental print.	grade-level phonics and word analysis skills in decoding words. • Demonstrate basic knowledge of one-to-one letter-sound correspondence. • Associate the long and short sounds with common spellings for the five major vowels. • Read grade-level high-frequency sight words with automaticity. • Distinguish between similarly spelled words by identifying the sounds of the letters that differ.	grade-level phonics and word analysis skills in decoding words. • Identify common consonant diagraphs, final-e, and common vowel teams. • Decode one- and two-syllable words with common patterns. • Read grade-level words with inflectional endings. • Read grade-appropriate irregularly spelled words.	grade-level phonics and word analysis skills in decoding words. • Distinguish long and short vowels when reading regularly spelled one-syllable words. • Decode two-syllable words with long vowels and words with common prefixes and suffixes. • Read grade-level high-frequency sight words and words with inconsistent but common spelling-sound correspondences.	grade-level phonics and word analysis skills in decoding words. • Identify and know the meaning of the most common prefixes and derivational suffixes. • Decode words with common Latin suffixes. • Decode multisyllable words. • Read grade-appropriate irregularly spelled words.	grade-level phonics and word analysis skills in decoding words. • Use combined knowledge of all letter-sound correspondences, syllabication patterns, and morphology to read accurately unfamiliar multisyllabic words.	grade-level phonics and word analysis skills in decoding words. • Use combined knowledge of all letter-sound correspondences, syllabication patterns, and morphology to read accurately unfamiliar multisyllabic words.
				• Read grade-appropriate irregularly spelled words.			
Fluency	Intentionally Blank	**CC.1.1.K.E** Read emergent-reader text with purpose and understanding.	**CC.1.1.1.E** Read with accuracy and fluency to support comprehension. • Read on-level text with purpose and understanding. • Read on-level text orally with accuracy, appropriate rate, and expression on successive readings. • Use context to confirm or self-correct word recognition and understanding, rereading as necessary.	**CC.1.1.2.E** Read with accuracy and fluency to support comprehension. • Read on-level text with purpose and understanding. • Read on-level text orally with accuracy, appropriate rate, and expression on successive readings. • Use context to confirm or self-correct word recognition and understanding, rereading as necessary.	**CC.1.1.3.E** Read with accuracy and fluency to support comprehension. • Read on-level text with purpose and understanding. • Read on-level text orally with accuracy, appropriate rate, and expression on successive readings. • Use context to confirm or self-correct word recognition and understanding, rereading as necessary.	**CC.1.1.4.E** Read with accuracy and fluency to support comprehension. • Read on-level text with purpose and understanding. • Read on-level text orally with accuracy, appropriate rate, and expression on successive readings. • Use context to confirm or self-correct word recognition and understanding, rereading as necessary.	**CC.1.1.5.E** Read with accuracy and fluency to support comprehension. • Read on-level text with purpose and understanding. • Read on-level text orally with accuracy, appropriate rate, and expression on successive readings. • Use context to confirm or self-correct word recognition and understanding, rereading as necessary.

1.2	**Reading Informational Text** **Students read, understand, and respond to informational text—with an emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with a focus on textual evidence.**						
	Grade Pre K	**Grade K**	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Grade 5**
Key Ideas and Details Main Idea	**CC.1.2.PK.A** With prompting and support, retell key details of text that support a provided main idea.	**CC.1.2.K.A** With prompting and support, identify the main idea and retell key details of text.	**CC.1.2.1.A** Identify the main idea and retell key details of text.	**CC.1.2.2.A** Identify the main idea of a multiparagraph text as well as the focus of specific paragraphs within the text.	**CC.1.2.3.A** Determine the main idea of a text; recount the key details and explain how they support the main idea. E03.B-K.1.1.2	**CC.1.2.4.A** Determine the main idea of a text and explain how it is supported by key details; summarize the text. E04.B-K.1.1.2	**CC.1.2.5.A** Determine two or more main ideas in a text and explain how they are supported by key details; summarize the text. E05.B-K.1.1.2
Key Ideas and Details Text Analysis	**CC.1.2.PK.B** Answer questions about a text.	**CC.1.2.K.B** With prompting and support, answer questions about key details in a text.	**CC.1.2.1.B** Ask and answer questions about key details in a text.	**CC.1.2.2.B** Ask and answer questions such as who, what, where, when, why, and how to demonstrate understanding of key details in a text.	**CC.1.2.3.B** Ask and answer questions about the text and make inferences from text; refer to text to support responses. E03.B-K.1.1.1	**CC.1.2.4.B** Refer to details and examples in text to support what the text says explicitly and make inferences. E04.B-K.1.1.1	**CC.1.2.5.B** Cite textual evidence by quoting accurately from the text to explain what the text says explicitly and make inferences. E05.B-K.1.1.1

Key Ideas and Details Text Analysis	**CC.1.2.PK.C** With prompting and support, make connections between information in a text and personal experiences.	**CC.1.2.K.C** With prompting and support, make a connection between two individuals, events, ideas, or pieces of information in a text.	**CC.1.2.1.C** Describe the connection between two individuals, events, ideas, or pieces of information in a text.	**CC.1.2.2.C** Describe the connection between a series of events, concepts, or steps in a procedure within a text.	**CC.1.2.3.C** Explain how a series of events, concepts, or steps in a procedure is connected within a text, using language that pertains to time, sequence, and cause/effect. E03.B-K.1.1.3	**CC.1.2.4.C** Explain events, procedures, ideas, or concepts in a text, including what happened and why, based on specific information in the text. E04.B-K.1.1.3	**CC.1.2.5.C** Explain the relationships or interactions between two or more individuals, events, ideas, or concepts in a text based on specific information in the text. E05.B-K.1.1.3
Craft and Structure Point of View	Intentionally Blank	Intentionally Blank	Intentionally Blank	Intentionally Blank	**CC.1.2.3.D** Explain the point of view of the author. E03.B-C.2.1.1	**CC.1.2.4.D** Compare and contrast an event or topic told from two different points of view. E04.B-C.2.1.1	**CC.1.2.5.D** Analyze multiple accounts of the same event or topic, noting important similarities and differences in the point of view they represent. E05.B-C.2.1.1
Craft and Structure Text Structure	**CC.1.2.PK.E** Identify the front cover, back cover, and title page of a book.	**CC.1.2.K.E** Identify parts of a book (title, author) and parts of a text (beginning, end, details).	**CC.1.2.1.E** Use various text features and search tools to locate key facts or information in a text.	**CC.1.2.2.E** Use various text features and search tools to locate key facts or information in a text efficiently.	**CC.1.2.3.E** Use text features and search tools to locate and interpret information. E03.B-C.2.1.2	**CC.1.2.4.E** Use text structure to interpret information (e.g., chronology, comparison, cause/effect, problem/solution). E04.B-C.2.1.2	**CC.1.2.5.E** Use text structure, in and among texts, to interpret information (e.g., chronology, comparison, cause/effect, problem/solution). E05.B-C.2.1.2
Craft and Structure Vocabulary	**CC.1.2.PK.F** With prompting and support, answer questions about unfamiliar words read aloud from a text.	**CC.1.2.K.F** With prompting and support, ask and answer questions about unknown words in a text.	**CC.1.2.1.F** Ask and answer questions to help determine or clarify the meaning of words and phrases in a text.	**CC.1.2.2.F** Determine the meaning of words and phrases as they are used in grade-level text including multiple-meaning words.	**CC.1.2.3.F** Determine the meaning of words and phrases as they are used in grade-level text, distinguishing literal from nonliteral meaning as well as shades of meaning among related words. E03.B-V.4.1.1 E03.B-V.4.1.2	**CC.1.2.4.F** Determine the meaning of words and phrases as they are used in grade-level text, including figurative language. E04.B-V.4.1.1 E04.B-V.4.1.2	**CC.1.2.5.F** Determine the meaning of words and phrases as they are used in grade-level text, including interpretation of figurative language. E05.B-V.4.1.1 E05.B-V.4.1.2
Integration of Knowledge and Ideas Diverse Media	**CC.1.2.PK.G** With prompting and support, answer questions to connect illustrations to the written word.	**CC.1.2.K.G** Answer questions to describe the relationship between illustrations and the text in which they appear.	**CC.1.2.1.G** Use the illustrations and details in a text to describe its key ideas.	**CC.1.2.2.G** Explain how graphic representations contribute to and clarify a text.	**CC.1.2.3.G** Use information gained from text features to demonstrate understanding of a text. E03.B-C.3.1.3	**CC.1.2.4.G** Interpret various presentations of information within a text or digital source and explain how the information contributes to an understanding of text in which it appears. E04.B-C.3.1.3	**CC.1.2.5.G** Draw on information from multiple print or digital sources, demonstrating the ability to locate an answer to a question quickly or to solve a problem efficiently. E05.B-C.3.1.3
Integration of Knowledge and Ideas Evaluating Arguments	Intentionally Blank	**CC.1.2.K.H** With prompting and support, identify the reasons an author gives to support points in a text.	**CC.1.2.1.H** Identify the reasons an author gives to support points in a text.	**CC.1.2.2.H** Describe how reasons support specific points the author makes in a text.	**CC.1.2.3.H** Describe how an author connects sentences and paragraphs in a text to support particular points. E03.B-C.3.1.1	**CC.1.2.4.H** Explain how an author uses reasons and evidence to support particular points in a text. E04.B-C.3.1.1	**CC.1.2.5.H** Determine how an author supports particular points in a text through reasons and evidence. E05.B-C.3.1.1
Integration of Knowledge and Ideas Analysis Across Texts	**CC.1.2.PK.I** With prompting and support, identify basic similarities and differences between two texts read aloud on the same topic.	**CC.1.2.K.I** With prompting and support, identify basic similarities and differences between two texts (read or read aloud) on the same topic.	**CC.1.2.1.I** Identify basic similarities in and differences between two texts on the same topic.	**CC.1.2.2.I** Compare and contrast the most important points presented by two texts on the same topic.	**CC.1.2.3.I** Compare and contrast the most important points and key details presented in two texts on the same topic. E03.B-C.3.1.2	**CC.1.2.4.I** Integrate information from two texts on the same topic to demonstrate understanding of that topic. E04.B-C.3.1.2	**CC.1.2.5.I** Integrate information from several texts on the same topic to demonstrate understanding of that topic. E05.B-C.3.1.2

Vocabulary Acquisition and Use	**CC.1.2.PK.J** Use new vocabulary and phrases acquired in conversations and being read to.	**CC.1.2.K.J** Use words and phrases acquired through conversations, reading, and being read to, and responding to texts.	**CC.1.2.1.J** Use words and phrases acquired through conversations, reading, and being read to, and responding to texts, including words that signal connections and relationships between the words and phrases.	**CC.1.2.2.J** Acquire and use grade-appropriate conversational, general academic, and domain-specific words and phrases.	**CC.1.2.3.J** Acquire and use accurately grade-appropriate conversational, general academic, and domain-specific words and phrases, including those that signal spatial and temporal relationships. E03.B-V.4.1.1 E03.B-V.4.1.2	**CC.1.2.4.J** Acquire and use accurately grade-appropriate conversational, general academic, and domain-specific words and phrases, including those that signal precise actions, emotions, or states of being and that are basic to a particular topic. E04.B-V.4.1.1 E04.B-V.4.1.2.	**CC.1.2.5.J** Acquire and use accurately grade-appropriate conversational, general academic, and domain-specific words and phrases, including those that signal contrast, addition, and other logical relationships. E05.B-V.4.1.1 E05.B-V.4.1.2.
Vocabulary Acquisition and Use	**CC.1.2.PK.K** With prompting and support, clarify unknown words or phrases read aloud.	**CC.1.2.K.K** Determine or clarify the meaning of unknown or multiple-meaning words and phrases based upon grade-level reading and content.	**CC.1.2.1.K** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content.	**CC.1.2.2.K** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing from a range of strategies and tools.	**CC.1.2.3.K** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E03.B-V.4.1.1	**CC.1.2.4.K** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E04.B-V.4.1.1	**CC.1.2.5.K** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E05.B-V.4.1.1
Range of Reading	**CC.1.2.PK.L** With prompting and support, actively engage in group reading activities with purpose and understanding.	**CC.1.2.K.L** Actively engage in group reading activities with purpose and understanding.	**CC.1.2.1.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.2.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.3.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.4.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.5.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.

1.3	**Reading Literature** **Students read and respond to works of literature—with emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with focus on textual evidence.**						
	Grade Pre K	**Grade K**	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Grade 5**
Key Ideas and Details Theme	**CC.1.3.PK.A** With prompting and support, retell a familiar story in sequence with picture support.	**CC.1.3.K.A** With prompting and support, retell familiar stories including key details.	**CC.1.3.1.A** Retell stories, including key details, and demonstrate understanding of their central message or lesson.	**CC.1.3.2.A** Recount stories and determine their central message, lesson, or moral.	**CC.1.3.3.A** Determine the central message, lesson, or moral in literary text; explain how it is conveyed in text. E03.A-K.1.1.2	**CC.1.3.4.A** Determine a theme of a text from details in the text; summarize the text. E04.A-K.1.1.2	**CC.1.3.5.A** Determine a theme of a text from details in the text, including how characters in a story or drama respond to challenges or how the speaker in a poem reflects upon a topic; summarize the text. E05.A-K.1.1.2
Key Ideas and Details Text Analysis	**CC.1.3.PK.B** Answer questions about a particular story (who, what, how, when, and where).	**CC.1.3.K.B** Answer questions about key details in a text.	**CC.1.3.1.B** Ask and answer questions about key details in a text.	**CC.1.3.2.B** Ask and answer questions such as who, what, where, when, why, and how to demonstrate understanding of key details in a text.	**CC.1.3.3.B** Ask and answer questions about the text and make inferences from text, referring to text to support responses. E03.A-K.1.1.1	**CC.1.3.4.B** Cite relevant details from text to support what the text says explicitly and make inferences. E04.A-K.1.1.1	**CC.1.3.5.B** Cite textual evidence by quoting accurately from the text to explain what the text says explicitly and make inferences. E05.A-K.1.1.1

1.3	**Reading Literature** **Students read and respond to works of literature—with emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with focus on textual evidence.**						
	Grade Pre K	**Grade K**	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Grade 5**
Key Ideas and Details Literary Elements	**CC.1.3.PK.C** With prompting and support, answer questions to identify characters, settings, and major events in a story.	**CC.1.3.K.C** With prompting and support, identify characters, settings, and major events in a story.	**CC.1.3.1.C** Describe characters, settings, and major events in a story, using key details.	**CC.1.3.2.C** Describe how characters in a story respond to major events and challenges.	**CC.1.3.3.C** Describe characters in a story and explain how their actions contribute to the sequence of events. E03.A-K.1.1.3	**CC.1.3.4.C** Describe in depth a character, setting, or event in a story or drama, drawing on specific details in the text. E04.A-K.1.1.3	**CC.1.3.5.C** Compare and contrast two or more characters, settings, or events in a story or drama, drawing on specific details in the text. E05.A-K.1.1.3
Craft and Structure Point of View	**CC.1.3.PK.D** With prompting and support, name the author and illustrator of a story.	**CC.1.3.K.D** Name the author and illustrator of a story and define the role of each in telling the story.	**CC.1.3.1.D** Identify who is telling the story at various points in a text.	**CC.1.3.2.D** Acknowledge differences in the points of views of characters, including by speaking in a different voice for each character when reading dialogue aloud.	**CC.1.3.3.D** Explain the point of view of the author. E03.A-C.2.1.1	**CC.1.3.4.D** Compare and contrast an event or topic told from two different points of view. E04.A-C.2.1.1	**CC.1.3.5.D** Analyze multiple accounts of the same event or topic, noting important similarities and differences in the point of view they represent. E05.A-C.2.1.1
Craft and Structure Text Structure	**CC.1.3.PK.E** With prompting and support, recognize common types of text.	**CC.1.3.K.E** Recognize common types of text.	**CC.1.3.1.E** Explain major differences between books that tell stories and books that give information, drawing on a wide reading or range of text types.	**CC.1.3.2.E** Describe the overall structure of a story, including describing how the beginning introduces the story and the ending concludes the action.	**CC.1.3.3.E** Refer to parts of texts when writing or speaking about a text using such terms as chapter, scene, and stanza and describe how each successive part builds upon earlier sections.	**CC.1.3.4.E** Explain major differences between poems, drama, and prose and refer to the structural elements of each when writing or speaking about a text.	**CC.1.3.5.E** Explain how a series of chapters, scenes, or stanzas fits together to provide the overall structure of a particular story, drama, or poem.
Craft and Structure Vocabulary	**CC.1.3.PK.F** Answer questions about unfamiliar words read aloud from a story.	**CC.1.3.K.F** Ask and answer questions about unknown words in a text.	**CC.1.3.1.F** Identify words and phrases in stories or poems that suggest feelings or appeal to the senses.	**CC.1.3.2.F** Describe how words and phrases supply rhythm and meaning in a story, poem, or song.	**CC.1.3.3.F** Determine the meaning of words and phrases as they are used in grade-level text, distinguishing literal from nonliteral meaning as well as shades of meaning among related words. E03.A-V.4.1.1 E03.A-V.4.1.2	**CC.1.3.4.F** Determine the meaning of words and phrases as they are used in grade-level text, including figurative language. E04.A-V.4.1.1 E04.A-V.4.1.2	**CC.1.3.5.F** Determine the meaning of words and phrases as they are used in grade-level text, including interpretation of figurative language. E05.A-V.4.1.1 E05.A-V.4.1.2
Integration of Knowledge and Ideas Sources of Information	**CC.1.3.PK.G** Describe pictures in books using details.	**CC.1.3.K.G** Make connections between the illustrations and the text in a story (read or read aloud).	**CC.1.3.1.G** Use illustrations and details in a story to describe characters, setting, or events.	**CC.1.3.2.G** Use information from illustrations and words, in print or digital text, to demonstrate understanding of characters, setting, or plot.	**CC.1.3.3.G** Explain how specific aspects of a text's illustrations contribute to what is conveyed by the words in a story (e.g., create mood, emphasize aspects of a character or setting).	**CC.1.3.4.G** Make connections between the text of a story or drama and a visual or oral presentation of the text, identifying where each version reflects specific descriptions and directions in the text.	**CC.1.3.5.G** Analyze how visual and multimedia elements contribute to the meaning, tone, or beauty of a text (e.g., graphic novel, multimedia presentation of fiction, folktale, myth, poem).
Integration of Knowledge and Ideas Text Analysis	**CC.1.3.PK.H** Answer questions to compare and contrast the adventures and experiences of characters in familiar stories.	**CC.1.3.K.H** Compare and contrast the adventures and experiences of characters in familiar stories.	**CC.1.3.1.H** Compare and contrast the adventures and experiences of characters in stories.	**CC.1.3.2.H** Compare and contrast two or more versions of the same story by different authors or from different cultures.	**CC.1.3.3.H** Compare and contrast the themes, settings, and plots of stories written by the same author about the same or similar characters. E03.A-C.3.1.1	**CC.1.3.4.H** Compare and contrast similar themes, topics, and patterns of events in literature, including texts from different cultures. E04.A-C.3.1.1	**CC.1.3.5.H** Compare and contrast texts in the same genre on their approaches to similar themes and topics as well as additional literary elements. E05.A-C.3.1.1
Vocabulary Acquisition and Use Strategies	**CC.1.3.PK.I** With prompting and support, clarify unknown words or phrases read aloud.	**CC.1.3.K.I** Determine or clarify the meaning of unknown or multiple-meaning	**CC.1.3.1.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases	**CC.1.3.2.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases	**CC.1.3.3.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases	**CC.1.3.4.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases	**CC.1.3.5.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases

		words and phrases based upon grade-level reading and content.	based on grade-level reading and content.	based on grade-level reading and content, choosing from a range of strategies and tools.	based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E03.A-V.4.1.1	based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E04.A-V.4.1.1	based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E05.A-V.4.1.1
Vocabulary Acquisition and Use	**CC.1.3.PK.J** Use new vocabulary and phrases acquired in conversations and being read to.	**CC.1.3.K.J** Use words and phrases acquired through conversations, reading, and being read to, and responding to texts.	**CC.1.3.1.J** Use words and phrases acquired through conversations, reading, and being read to, and responding to texts, including words that signal connections and relationships between the words and phrases.	**CC.1.3.2.J** Acquire and use grade-appropriate conversational, general academic, and domain-specific words and phrases.	**CC.1.3.3.J** Acquire and use accurately grade-appropriate conversational, general academic, and domain-specific words and phrases, including those that signal spatial and temporal relationships. E03.A-V.4.1.1 E03.A-V.4.1.2	**CC.1.3.4.J** Acquire and use accurately grade-appropriate conversational, general academic, and domain-specific words and phrases, including those that signal precise actions, emotions, or states of being and that are basic to a particular topic. E04.A-V.4.1.1 E04.A-V.4.1.2.	**CC.1.3.5.J** Acquire and use accurately grade-appropriate conversational, general academic, and domain-specific words and phrases, including those that signal contrast, addition, and other logical relationships. E05.A-V.4.1.1 E05.A-V.4.1.2.
Range of Reading	**CC.1.3.PK.K** With prompting and support, actively engage in group reading activities with purpose and understanding.	**CC.1.3.K.K** Actively engage in group reading activities with purpose and understanding.	**CC.1.3.1.K** Read and comprehend literature on grade level, reading independently and proficiently.	**CC.1.3.2.K** Read and comprehend literature on grade level, reading independently and proficiently.	**CC.1.3.3.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.	**CC.1.3.4.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.	**CC.1.3.5.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.

1.4	**Writing** **Students write for different purposes and audiences. Students write clear and focused text to convey a well-defined perspective and appropriate content.**						
	Grade Pre K	**Grade K**	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Grade 5**
Informative/Explanatory	**CC.1.4.PK.A Draw/dictate to compose informative/ explanatory texts examining a topic.**	**CC.1.4.K.A Use a combination of drawing, dictating, and writing to compose informative/ explanatory texts.**	**CC.1.4.1.A Write informative/ explanatory texts to examine a topic and convey ideas and information.**	**CC.1.4.2.A Write informative/ explanatory texts to examine a topic and convey ideas and information clearly.**	**CC.1.4.3.A Write informative/ explanatory texts to examine a topic and convey ideas and information clearly.**	**CC.1.4.4.A Write informative/ explanatory texts to examine a topic and convey ideas and information clearly.**	**CC.1.4.5.A Write informative/ explanatory texts to examine a topic and convey ideas and information clearly.**
Informative/ Explanatory Focus	**CC.1.4.PK.B** With prompting and support, draw/dictate about one specific topic.	**CC.1.4.K.B** Use a combination of drawing, dictating, and writing to focus on one specific topic.	**CC.1.4.1.B** Identify and write about one specific topic.	**CC.1.4.2.B** Identify and introduce the topic.	**CC.1.4.3.B** Identify and introduce the topic. E03.C.1.2.1	**CC.1.4.4.B** Identify and introduce the topic clearly. E04.C.1.2.1 E04.E.1.1.1	**CC.1.4.5.B** Identify and introduce the topic clearly. E05.C.1.2.1 E05.E.1.1.1
Informative/Explanatory Content	**CC.1.4.PK.C** With prompting and support, generate ideas to convey information.	**CC.1.4.K.C** With prompting and support, generate ideas and details to convey information that relates to the chosen topic.	**CC.1.4.1.C** Develop the topic with two or more facts.	**CC.1.4.2.C** Develop the topic with facts and/or definitions.	**CC.1.4.3.C** Develop the topic with facts, definitions, details, and illustrations, as appropriate. E03C.1.2.2	**CC.1.4.4.C** Develop the topic with facts, definitions, concrete details, quotations, or other information and examples related to the topic; include illustrations and multimedia when useful to aiding comprehension. E04C.1.2.2 E04E.1.1.2	**CC.1.4.5.C** Develop the topic with facts, definitions, concrete details, quotations, or other information and examples related to the topic; include illustrations and multimedia when useful to aiding comprehension. E05C.1.2.2 E05E.1.1.2
Informative/Explanatory Organization	**CC.1.4.PK.D** With prompting	**CC.1.4.K.D** Make logical	**CC.1.4.1.D** Group	**CC.1.4.2.D** Group	**CC.1.4.3.D** Create an	**CC.1.4.4.D** Group related	**CC.1.4.5.D** Group related

	and support, make logical connections between drawing and dictation.	connections between drawing and dictation/writing.	information and provide some sense of closure.	information and provide a concluding statement or section.	organizational structure that includes information grouped and connected logically with a concluding statement or section. E03.C.1.2.1 E03.C.1.2.3 E03.C.1.2.4	information in paragraphs and sections, linking ideas within categories of information using words and phrases; provide a concluding statement or section; include formatting when useful to aiding comprehension. E04.C.1.2.1 E04.C.1.2.3 E04.C.1.2.5 E04.E.1.1.1 E04.E.1.1.3 E04.E.1.1.5	information logically linking ideas within and across categories of information using words, phrases, and clauses; provide a concluding statement or section; include formatting when useful to aiding comprehension. E05.C.1.2.1 E05.C.1.2.3 E05.C.1.2.6 E05.E.1.1.1 E05.E.1.1.3 E05.E.1.1.6
Informative/Explanatory Style	Intentionally Blank	**CC.1.4.K.E** With prompting and support, illustrate using details and dictate/write using descriptive words.	**CC.1.4.1.E** Choose words and phrases for effect.	**CC.1.4.2.E** Choose words and phrases for effect.	**CC.1.4.3.E** Choose words and phrases for effect. E03.D.2.1.1	**CC.1.4.4.E** Use precise language and domain-specific vocabulary to inform about or explain the topic. E04.C.1.2.4 E04.D.2.1.1 E04.D.2.1.3 E04.E.1.1.4	**CC.1.4.5.E** Write with an awareness of style. • Use precise language and domain-specific vocabulary to inform about or explain the topic. • Use sentences of varying length. E05.C.1.2.4 E05.C.1.2.5 E05.D.2.1.1 E05.D.2.1.2 E05.D.2.1.3 E05.D.2.1.4 E05.E.1.1.3 E05.E.1.1.4 E05.E.1.1.5
Informative/Explanatory Conventions of Language	Intentionally Blank	**CC.1.4.K.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize the first word in a sentence and the pronoun I. • Recognize and use end punctuation. • Spell simple words phonetically.	**CC.1.4.1.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize dates and names of people. • Use end punctuation; use commas in dates and words in series. • Spell words drawing on common spelling patterns, phonemic awareness, and spelling conventions.	**CC.1.4.2.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize proper nouns. • Use commas and apostrophes appropriately. • Spell words drawing on common spelling patterns. • Consult reference material as needed.	**CC.1.4.3.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E03.D.1.1.1 E03.D.1.1.2 E03.D.1.1.3 E03.D.1.1.4 E03.D.1.1.5 E03.D.1.1.6 E03.D.1.1.7 E03.D.1.1.8 E03.D.1.1.9 E03.D.1.2.1 E03.D.1.2.2 E03.D.1.2.3 E03.D.1.2.4 E03.D.1.2.5 E03.D.1.2.6	**CC.1.4.4.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E04.D.1.1.1 E04.D.1.1.2 E04.D.1.1.3 E04.D.1.1.4 E04.D.1.1.5 E04.D.1.1.6 E04.D.1.1.7 E04.D.1.1.8 E04.D.1.2.1 E04.D.1.2.2 E04.D.1.2.3 E04.D.1.2.4	**CC.1.4.5.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E05.D.1.1.1 E05.D.1.1.2 E05.D.1.1.3 E05.D.1.1.4 E05.D.1.1.5 E05.D.1.1.6 E05.D.1.1.7 E05.D.1.1.8 E05.D.1.2.1 E05.D.1.2.2 E05.D.1.2.3 E05.D.1.2.4 E05.D.1.2.5
Opinion/ Argumentative	**Intentionally Blank**	**CC.1.4.K.G** **Use a combination of drawing, dictating, and writing to compose opinion pieces on familiar topics.**	**CC.1.4.1.G** **Write opinion pieces on familiar topics.**	**CC.1.4.2.G** **Write opinion pieces on familiar topics or texts.**	**CC.1.4.3.G** **Write opinion pieces on familiar topics or texts.**	**CC.1.4.4.G** **Write opinion pieces on topics or texts.**	**CC.1.4.5G** **Write opinion pieces on topics or texts.**

Opinion/Argumentative Focus	Intentionally Blank	**CC.1.4.K.H** Form an opinion by choosing between two given topics.	**CC.1.4.1.H** Form an opinion by choosing among given topics.	**CC.1.4.2.H** Identify the topic and state an opinion.	**CC.1.4.3.H** Introduce the topic and state an opinion on the topic. E03.C.1.1.1	**CC.1.4.4.H** Introduce the topic and state an opinion on the topic. E04.C.1.1.1 E04.E.1.1.1	**CC.1.4.5.H** Introduce the topic and state an opinion on the topic. E05.C.1.1.1 E05.E.1.1.1
Opinion/Argumentative Content	Intentionally Blank	**CC.1.4.K.I** Support the opinion with reasons.	**CC.1.4.1.I** Support the opinion with reasons related to the opinion.	**CC.1.4.2.I** Support the opinion with reasons that include details connected to the opinion.	**CC.1.4.3.I** Support an opinion with reasons. E03.C.1.1.2	**CC.1.4.4.I** Provide reasons that are supported by facts and details. E04.C.1.1.2 E04.E.1.1.2	**CC.1.4.5.I** Provide reasons that are supported by facts and details; draw from credible sources. E05.C.1.1.2 E05.E.1.1.2
Opinion/Argumentative Organization	Intentionally Blank	**CC.1.4.K.J** Make logical connections between drawing and writing.	**CC.1.4.1.J** Create an organizational structure that includes reasons and provides some sense of closure.	**CC.1.4.2.J** Create an organizational structure that includes reasons and includes a concluding statement.	**CC.1.4.3.J** Create an organizational structure that includes reasons linked in a logical order with a concluding statement or section. E03.C.1.1.1 E03.C.1.1.3 E03.C.1.1.4	**CC.1.4.4.J** Create an organizational structure that includes related ideas grouped to support the writer's purpose and linked in a logical order with a concluding statement or section related to the opinion. E04.C.1.1.1 E04.C.1.1.3 E04.C.1.1.4 E04.E.1.1.1 E04.E.1.1.3 E04.E.1.1.5	**CC.1.4.5.J** Create an organizational structure that includes related ideas grouped to support the writer's purpose; link opinion and reasons using words, phrases, and clauses; provide a concluding statement or section related to the opinion. E05.C.1.1.1 E05.C.1.1.3 E05.C.1.1.5 E05.E.1.1.1 E05.E.1.1.3 E05.E.1.1.6
Opinion/Argumentative Style	Intentionally Blank	Intentionally Blank	**CC.1.4.1.K** Use a variety of words and phrases.	**CC.1.4.2.K** Use a variety of words and phrases to appeal to the audience.	**CC.1.4.3.K** Use a variety of words and sentence types to appeal to the audience. E03.D.1.1.9 E03.D.2.1.1	**CC.1.4.4.K** Choose words and phrases to convey ideas precisely. E04.D.2.1.1 E04.D.2.1.3 E04.E.1.1.4	**CC.1.4.5.K** Write with an awareness of style. • Use sentences of varying length. • Expand, combine, and reduce sentences for meaning, reader/listener interest, and style. E05.C.1.1.4 E05.D.2.1.1 E05.D.2.1.2 E05.D.2.1.3 E05.D.2.1.4 E05.E.1.1.4 E05.E.1.1.5
Opinion/Argumentative Conventions of Language	Intentionally Blank	**CC.1.4.K.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize the first word in a sentence and the pronoun I. • Recognize and use end punctuation. • Spell simple words phonetically.	**CC.1.4.1.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize dates and names of people. • Use end punctuation; use commas in dates and words in series. • Spell words	**CC.1.4.2.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize proper nouns. • Use commas and apostrophes appropriately. • Spell words drawing on common spelling	**CC.1.4.3.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E03.D.1.1.1 E03.D.1.1.2 E03.D.1.1.3 E03.D.1.1.4 E03.D.1.1.5 E03.D.1.1.6 E03.D.1.1.7 E03.D.1.1.8	**CC.1.4.4.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E04.D.1.1.1 E04.D.1.1.2 E04.D.1.1.3 E04.D.1.1.4 E04.D.1.1.5 E04.D.1.1.6 E04.D.1.1.7 E04.D.1.1.8 E04.D.1.2.1	**CC.1.4.5.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E05.D.1.1.1 E05.D.1.1.2 E05.D.1.1.3 E05.D.1.1.4 E05.D.1.1.5 E05.D.1.1.6 E05.D.1.1.7 E05.D.1.1.8 E05.D.1.2.1

			drawing on common spelling patterns, phonemic awareness, and spelling conventions.	patterns. • Consult reference material as needed.	E03.D.1.1.9 E03.D.1.2.1 E03.D.1.2.2 E03.D.1.2.3 E03.D.1.2.4 E03.D.1.2.5 E03.D.1.2.6	E04.D.1.2.2 E04.D.1.2.3 E04.D.1.2.4	E05.D.1.2.2 E05.D.1.2.3 E05.D.1.2.4 E05.D.1.2.5
Narrative	**CC.1.4.PK.M** **Dictate narratives to describe real or imagined experiences or events.**	**CC.1.4.K.M** **Use a combination of drawing, dictating, and writing to compose narratives that describe real or imagined experiences or events.**	**CC.1.4.1.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.2.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.3.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.4.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.5.M** **Write narratives to develop real or imagined experiences or events.**
Narrative Focus	**CC.1.4.PK.N** Establish who and what the narrative will be about.	**CC.1.4.K.N** Establish who and what the narrative will be about.	**CC.1.4.1.N** Establish who and what the narrative will be about.	**CC.1.4.2.N** Establish a situation and introduce a narrator and/or characters.	**CC.1.4.3.N** Establish a situation and introduce a narrator and/or characters. E03C.1.3.1	**CC.1.4.4.N** Orient the reader by establishing a situation and introducing a narrator and/or characters. E04C.1.3.1	**CC.1.4.5.N** Orient the reader by establishing a situation and introducing a narrator and/or characters. E05C.1.3.1
Narrative Content	**CC.1.4.PK.O** With prompting and support describe experiences and events.	**CC.1.4.K.O** Describe experiences and events.	**CC.1.4.1.O** Include thoughts and feelings to describe experiences and events.	**CC.1.4.2.O** Include thoughts and feelings to describe experiences and events to show the response of characters to situations.	**CC.1.4.3.O** Use dialogue and descriptions of actions, thoughts, and feelings to develop experiences and events or show the response of characters to situations. E03C.1.3.2	**CC.1.4.4.O** Use dialogue and descriptions to develop experiences and events or show the responses of characters to situations; use concrete words and phrases and sensory details to convey experiences and events precisely. E04.C.1.3.2 E04.C.1.3.4	**CC.1.4.5.O** Use narrative techniques such as dialogue, description, and pacing, to develop experiences and events or show the responses of characters to situations; use concrete words and phrases and sensory details to convey experiences and events precisely. E05.C.1.3.2 E05.C.1.3.4
Narrative Organization	**CC.1.4.PK.P** Recount a single event and tell about the events in the order in which they occurred.	**CC.1.4.K.P** Recount a single event or several loosely linked events, tell about the events in the order in which they occurred, and provide a reaction to what happened.	**CC.1.4.1.P** Recount two or more appropriately sequenced events using temporal words to signal event order and provide some sense of closure.	**CC.1.4.2.P** Organize a short sequence of events, using temporal words to signal event order; provide a sense of closure.	**CC.1.4.3.P** Organize an event sequence that unfolds naturally, using temporal words and phrases to signal event order; provide a sense of closure. E03.C.1.3.1 E03.C.1.3.3 E03.C.1.3.4	**CC.1.4.4.P** Organize an event sequence that unfolds naturally, using a variety of transitional words and phrases to manage the sequence of events; provide a conclusion that follows from the narrated experiences and events. E04.C.1.3.1 E04.C.1.3.3 E04.C.1.3.5	**CC.1.4.5.P** Organize an event sequence that unfolds naturally, using a variety of transitional words and phrases to manage the sequence of events; provide a conclusion that follows from the narrated experiences and events. E05.C.1.3.1 E05.C.1.3.3 E05.C.1.3.5
Narrative Style	Intentionally Blank	Intentionally Blank	**CC.1.4.1.Q** Use a variety of words and phrases.	**CC.1.4.2.Q** Choose words and phrases for effect	**CC.1.4.3.Q** Choose words and phrases for effect. E03.D.2.1.1	**CC.1.4.4.Q** Choose words and phrases to convey ideas precisely. E04.C.1.3.4 E04.D.2.1.1 E04.D.2.1.3 E04.E.1.1.4	**CC.1.4.5.Q** Write with an awareness of style. • Use sentences of varying length. • Expand, combine, and reduce sentences for meaning, reader/listener interest, and style.

							E05.C.1.3.4 E05.D.2.1.1 E05.D.2.1.2 E05.D.2.1.3 E05.D.2.1.4
Narrative Conventions of Language	Intentionally Blank	**CC.1.4.K.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize first word in sentence and pronoun I. • Recognize and use end punctuation. • Spell simple words phonetically.	**CC.1.4.1.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize dates and names of people. • Use end punctuation; use commas in dates and words in series. • Spell words drawing on common spelling patterns, phonemic awareness, and spelling conventions.	**CC.1.4.2.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. • Capitalize proper nouns. • Use commas and apostrophes appropriately. • Spell words drawing on common spelling patterns. • Consult reference material as needed.	**CC.1.4.3.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E03.D.1.1.1 E03.D.1.1.2 E03.D.1.1.3 E03.D.1.1.4 E03.D.1.1.5 E03.D.1.1.6 E03.D.1.1.7 E03.D.1.1.8 E03.D.1.1.9 E03.D.1.2.1 E03.D.1.2.2 E03.D.1.2.3 E03.D.1.2.4 E03.D.1.2.5 E03.D.1.2.6	**CC.1.4.4.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E04.D.1.1.1 E04.D.1.1.2 E04.D.1.1.3 E04.D.1.1.4 E04.D.1.1.5 E04.D.1.1.6 E04.D.1.1.7 E04.D.1.1.8 E04.D.1.2.1 E04.D.1.2.2 E04.D.1.2.3 E04.D.1.2.4	**CC.1.4.5.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E05.D.1.1.1 E05.D.1.1.2 E05.D.1.1.3 E05.D.1.1.4 E05.D.1.1.5 E05.D.1.1.6 E05.D.1.1.7 E05.D.1.1.8 E05.D.1.2.1 E05.D.1.2.2 E05.D.1.2.3 E05.D.1.2.4 E05.D.1.2.5
Response to Literature	Intentionally Blank	Intentionally Blank	Intentionally Blank	Intentionally Blank	**CC.1.4.3.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and informational texts.	**CC.1.4.4.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and informational texts. E04.E.1.1.1 E04.E.1.1.2 E04.E.1.1.3 E04.E.1.1.4 E04.E.1.1.5	**CC.1.4.5.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and informational texts. E05.E.1.1.1 E05.E.1.1.2 E05.E.1.1.3 E05.E.1.1.4 E05.E.1.1.5 E05.E.1.1.6
Production and Distribution of Writing Writing Process	**CC.1.4.PK.T** With guidance and support from adults and peers, respond to questions and suggestions, add details as needed.	**CC.1.4.K.T** With guidance and support from adults and peers, respond to questions and suggestions from peers and add details to strengthen writing as needed.	**CC.1.4.1.T** With guidance and support from adults and peers, focus on a topic, respond to questions and suggestions from peers and add details to strengthen writing as needed.	**CC.1.4.2.T** With guidance and support from adults and peers, focus on a topic and strengthen writing as needed by revising and editing.	**CC.1.4.3.T** With guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, and editing.	**CC.1.4.4.T** With guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, and editing.	**CC.1.4.5.T** With guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach.
Technology and Publication	Intentionally Blank	**CC.1.4.K.U** With guidance and support, explore a variety of digital tools to produce and publish writing or in collaboration with peers.	**CC.1.4.1.U** With guidance and support, use a variety of digital tools to produce and publish writing including in collaboration with peers.	**CC.1.4.2.U** With guidance and support, use a variety of digital tools to produce and publish writing including in collaboration with peers.	**CC.1.4.3.U** With guidance and support, use technology to produce and publish writing (using keyboarding skills) as well as to interact and collaborate with others.	**CC.1.4.4.U** With some guidance and support, use technology, including the Internet, to produce and publish writing as well as to interact and collaborate with others; demonstrate sufficient command of	**CC.1.4.5.U** With some guidance and support, use technology, including the Internet, to produce and publish writing as well as to interact and collaborate with others; demonstrate sufficient command of

						keyboarding skills to type a minimum of one page in a single sitting.	keyboarding skills to type a minimum of two pages in a single sitting.
Conducting Research	**CC.1.4.PK.V** Ask questions about topics of personal interest to gain information; with teacher guidance and support, locate information on the chosen topic.	**CC.1.4.K.V** Participate in individual or shared research projects on a topic of interest.	**CC.1.4.1.V** Participate in individual or shared research and writing projects.	**CC.1.4.2.V** Participate in individual or shared research and writing projects.	**CC.1.4.3.V** Conduct short research projects that build knowledge about a topic.	**CC.1.4.4.V** Conduct short research projects that build knowledge through investigation of different aspects of a topic.	**CC.1.4.5.V** Conduct short research projects that use several sources to build knowledge through investigation of different aspects of a topic.
Credibility, Reliability, and Validity of Sources	**CC.1.4.PK.W** With guidance and support, recall information from experiences or books.	**CC.1.4.K.W** With guidance and support, recall information from experiences or gather information from provided sources to answer a question.	**CC.1.4.1.W** With guidance and support, recall information from experiences or gather information from provided sources to answer a question.	**CC.1.4.2.W** Recall information from experiences or gather information from provided sources to answer a question.	**CC.1.4.3.W** Recall information from experiences or gather information from print and digital sources; take brief notes on sources and sort evidence into provided categories.	**CC.1.4.4.W** Recall relevant information from experiences or gather relevant information from print and digital sources; take notes and categorize information, and provide a list of sources.	**CC.1.4.5.W** Recall relevant information from experiences or gather relevant information from print and digital sources; summarize or paraphrase information in notes and finished work, and provide a list of sources.
Range of Writing	Intentionally Blank	**CC.1.4.K.X** Write routinely over short time frames.	**CC.1.4.1.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.2.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.3.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.4.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.5.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.

1.5	**Speaking and Listening** **Students present appropriately in formal speaking situations, listen critically, and respond intelligently as individuals or in group discussions.**						
	Grade Pre K	**Grade K**	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Grade 5**
Comprehension and Collaboration Collaborative Discussion	**CC.1.5.PK.A** Participate in collaborative conversations with peers and adults in small and larger groups.	**CC.1.5.K.A** Participate in collaborative conversations with peers and adults in small and larger groups.	**CC.1.5.1.A** Participate in collaborative conversations with peers and adults in small and larger groups.	**CC.1.5.2.A** Participate in collaborative conversations with peers and adults in small and larger groups.	**CC.1.5.3.A** Engage effectively in a range of collaborative discussions on grade-level topics and texts, building on others' ideas and expressing their own clearly.	**CC.1.5.4.A** Engage effectively in a range of collaborative discussions on grade-level topics and texts, building on others' ideas and expressing their own clearly.	**CC.1.5.5.A** Engage effectively in a range of collaborative discussions on grade-level topics and texts, building on others' ideas and expressing their own clearly.
Comprehension and Collaboration Critical Listening	**CC.1.5.PK.B** Answer questions about key details in a text read aloud or information presented orally or through other media.	**CC.1.5.K.B** Ask and answer questions about key details in a text read aloud or information presented orally or through other media.	**CC.1.5.1.B** Confirm understanding of a text read aloud or information presented orally or through other media by asking and answering questions about key details and requesting clarification if something is not understood.	**CC.1.5.2.B** Recount or describe key ideas or details from a text read aloud or information presented orally or through other media.	**CC.1.5.3.B** Determine the main ideas and supporting details of a text read aloud or information presented in diverse media formats, including visually, quantitatively, and orally.	**CC.1.5.4.B** Paraphrase portions of a text read aloud or information presented in diverse media and formats, including visually, quantitatively, and orally.	**CC.1.5.5.B** Summarize the main points of written text read aloud or information presented in diverse media and formats, including visually, quantitatively, and orally.

Comprehension and Collaboration Evaluating Information	**CC.1.5.PK.C** Respond to what a speaker says in order to follow directions, seek help, or gather information.	**CC.1.5.K.C** Ask and answer questions in order to seek help, get information, or clarify something that is not understood.	**CC.1.5.1.C** Ask and answer questions about what a speaker says in order to gather additional information or clarify something that is not understood.	**CC.1.5.2.C** Ask and answer questions about what a speaker says in order to clarify comprehension, gather additional information, or deepen understanding of a topic or issue.	**CC.1.5.3.C** Ask and answer questions about information from a speaker, offering appropriate detail.	**CC.1.5.4.C** Identify the reasons and evidence a speaker provides to support particular points.	**CC.1.5.5.C** Summarize the points a speaker makes and explain how each claim is supported by reasons and evidence.
Presentation of Knowledge and Ideas Purpose, Audience, and Task	**CC.1.5.PK.D** Using simple sentences, share stories, familiar experiences, and interests, speaking clearly enough to be understood by most audiences.	**CC.1.5.K.D** Share stories, familiar experiences, and interests, speaking clearly enough to be understood by all audiences using appropriate volume.	**CC.1.5.1.D** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly.	**CC.1.5.2.D** Tell a story or recount an experience with appropriate facts and relevant, descriptive details, speaking audibly in coherent sentences.	**CC.1.5.3.D** Report on a topic or text, tell a story, or recount an experience with appropriate facts and relevant, descriptive details; speak clearly with adequate volume, appropriate pacing, and clear pronunciation.	**CC.1.5.4.D** Report on a topic or text, tell a story, or recount an experience in an organized manner, using appropriate facts and relevant, descriptive details to support main ideas or themes; speak clearly with adequate volume, appropriate pacing, and clear pronunciation.	**CC.1.5.5.D** Report on a topic or present an opinion, sequencing ideas logically and using appropriate facts and relevant, descriptive details to support main ideas or themes; speak clearly with adequate volume, appropriate pacing, and clear pronunciation.
Presentation of Knowledge and Ideas Context	**CC.1.5.PK.E** Using simple sentences, express thoughts, feelings, and ideas, speaking clearly enough to be understood by most audiences.	**CC.1.5.K.E** Speak audibly and express thoughts, feelings, and ideas clearly.	**CC.1.5.1.E** Produce complete sentences when appropriate to task and situation.	**CC.1.5.2.E** Produce complete sentences when appropriate to task and situation in order to provide requested detail or clarification.	**CC.1.5.3.E** Speak in complete sentences when appropriate to task and situation in order to provide requested detail or clarification.	**CC.1.5.4.E** Differentiate between contexts that require formal English versus informal situations.	**CC.1.5.5.E** Adapt speech to a variety of contexts and tasks, using formal English when appropriate to task and situation.
Integration of Knowledge and Ideas Multimedia	Intentionally Blank	Intentionally Blank	**CC.1.5.1.F** Add drawings or other visual displays when sharing aloud to clarify ideas, thoughts, and feelings.	**CC.1.5.2.F** Add drawings or other visual displays to presentations when appropriate to clarify ideas, thoughts, and feelings.	**CC.1.5.3.F** Create engaging audio recordings of stories or poems that demonstrate fluid reading at an understandable pace; add visual displays when appropriate to emphasize or enhance certain facts or details.	**CC.1.5.4.F** Add audio recordings and visual displays to presentations when appropriate to enhance the development of main ideas or themes.	**CC.1.5.5.F** Include multimedia components and visual displays in presentations when appropriate to enhance the development of main ideas or themes.
Conventions of Standard English	**CC.1.5.PK.G** Demonstrate command of the conventions of standard English when speaking, based on prekindergarten level and content.	**CC.1.5.K.G** Demonstrate command of the conventions of standard English when speaking, based on kindergarten level and content.	**CC.1.5.1.G** Demonstrate command of the conventions of standard English when speaking, based on Grade 1 level and content.	**CC.1.5.2.G** Demonstrate command of the conventions of standard English when speaking, based on Grade 2 level and content.	**CC.1.5.3.G** Demonstrate command of the conventions of standard English when speaking, based on Grade 3 level and content.	**CC.1.5.4.G** Demonstrate command of the conventions of standard English when speaking, based on Grade 4 level and content.	**CC.1.5.5.G** Demonstrate command of the conventions of standard English when speaking, based on Grade 5 level and content.

Pennsylvania Core Standards for English Language Arts
Grades 6-12

INTRODUCTION

These standards describe what students should know and be able to do with the English language, prekindergarten through Grade 12. The standards provide the targets for instruction and student learning essential for success in all academic areas, not just language arts classrooms. Although the standards are not a curriculum or a prescribed series of activities, school entities will use them to develop a local school curriculum that will meet local students' needs.

Five standard categories are designed to provide a Pre K-12 continuum to reflect the demands of a college- and career-ready graduate:

Standard 1:	*Foundational Skills* begin at prekindergarten and focus on early childhood, with some standards reflected through Grade 5. These foundational skills are a necessary and important component of an effective, comprehensive reading program designed to develop proficient readers with the capacity to comprehend text, both literary and informational, across disciplines.
Standard 2:	*Reading Informational Text* enables students to read, understand, and respond to informational text.
Standard 3:	*Reading Literature* enables students to read, understand, and respond to works of literature.
Standard 4:	*Writing* develops the skills of informational, argumentative, and narrative writing, as well as the ability to engage in evidence-based analysis of text and research.
Standard 5:	*Speaking and Listening* focuses students on communication skills that enable critical listening and effective presentation of ideas.

With a focus on college and career readiness, the instructional shifts as reflected in Common Core are evident throughout the PA Core Standards:

• Balancing the reading of informational and literary texts so that students can access nonfiction and authentic texts, as well as literature

• Focusing on close and careful reading of text so that students are learning from the text

• Building a staircase of complexity (i.e., each grade level requires a ‘‘step’’ of growth on the ‘‘staircase’’) so that students graduate college or career ready

• Supporting writing from sources (i.e., using evidence from text to inform or make an argument) so that students use evidence and respond to the ideas, events, facts, and arguments presented in the texts they read

• Stressing an academically focused vocabulary so that students can access more complex texts

The English Language Arts Standards also provide parents and community members with information about what students should know and be able to do as they progress through the educational program and at graduation. With a clearly defined target provided by the standards, parents, students, educators, and community members become partners in learning. Each standard implies an end-of-year goal—with the understanding that exceeding the standard is an even more desirable end goal.

Note: The Aligned Eligible Content is displayed with the standard statement. On Standard Aligned System portal, it is a live link.

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Foundational Skills (Pre K-5)… **1.1**

Students gain a working knowledge of concepts of print, alphabetic principle, and other basic conventions. These foundational skills are not an end in and of themselves; rather, students apply them as effective readers.

• Book Handling

• Print Concepts

• Phonological Awareness

• Phonics and Word Recognition

• Fluency

Reading Informational Text… **1.2**

Students read, understand, and respond to informational text-with an emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with a focus on textual evidence.

• Key Ideas and Details

• Craft and Structure

• Integration of Knowledge and Ideas

• Vocabulary Acquisition and Use

• Range of Reading

Reading Literature… **1.3**

Students read and respond to works of literature-with an emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with a focus on textual evidence.

• Key Ideas and Details

• Craft and Structure

• Integration of Knowledge and Ideas

• Vocabulary Acquisition and Use

• Range of Reading

Writing… **1.4**

Students write for different purposes and audiences. Students write clear and focused text to convey a well-defined perspective and appropriate content.

• Informative/Explanatory

• Opinion/Argumentative

• Narrative

• Response to Literature

• Production and Distribution of Writing

• Technology and Publication

• Conducting Research

• Credibility, Reliability, and Validity of Sources

• Range of Writing

***Speaking and Listening…* 1.5**

Students present appropriately in formal speaking situations, listen critically, and respond intelligently as individuals or in group discussions.

• Comprehension and Collaboration

• Presentation of Knowledge and Ideas

• Integration of Knowledge and Ideas

• Conventions of Standard English

1.2	**Reading Informational Text** **Students read, understand, and respond to informational text—with an emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with focus on textual evidence.**				
	Grade 6	**Grade 7**	**Grade 8**	**Grades 9-10**	**Grades 11-12**
Key Ideas and Details Main Idea	**CC.1.2.6.A** Determine the central idea of a text and how it is conveyed through particular details; provide a summary of the text distinct from personal opinions or judgments. E06.B-K.1.1.2	**CC.1.2.7.A** Determine two or more central ideas in a text and analyze their development over the course of the text; provide an objective summary of the text. E07.B-K.1.1.2	**CC.1.2.8.A** Determine a central idea of a text and analyze its development over the course of the text, including its relationship to supporting ideas; provide an objective summary of the text. E08.B-K.1.1.2	**CC.1.2.9-10.A** Determine a central idea of a text and analyze its development over the course of the text, including how it emerges and is shaped and refined by specific details; provide an objective summary of the text. L.N.1.3.1 L.N.1.3.2 L.N.2.3.3	**CC.1.2.11-12.A** Determine and analyze the relationship between two or more central ideas of a text, including the development and interaction of the central ideas; provide an objective summary of the text.
Key Ideas and Details Text Analysis	**CC.1.2.6.B** Cite textual evidence to support analysis of what the text says explicitly, as well as inferences and/or generalizations drawn from the text. E06.B-K.1.1.1	**CC.1.2.7.B** Cite several pieces of textual evidence to support analysis of what the text says explicitly, as well as inferences, conclusions, and/or generalizations drawn from the text. E07.B-K.1.1.1	**CC.1.2.8.B** Cite the textual evidence that most strongly supports an analysis of what the text says explicitly, as well as inferences, conclusions, and/or generalizations drawn from the text. E08.B-K.1.1.1	**CC.1.2.9-10.B** Cite strong and thorough textual evidence to support analysis of what the text says explicitly, as well as inferences and conclusions based on an author's explicit assumptions and beliefs about a subject. L.N.1.3.1 L.N.2.1.1 L.N.2.1.2	**CC.1.2.11-12.B** Cite strong and thorough textual evidence to support analysis of what the text says explicitly, as well as inferences and conclusions based on and related to an author's implicit and explicit assumptions and beliefs.
Key Ideas and Details Text Analysis	**CC.1.2.6.C** Analyze in detail how a key individual, event, or idea is introduced, illustrated, and elaborated in a text. E06.B-K.1.1.3	**CC.1.2.7.C** Analyze the interactions between individuals, events, and ideas in a text. E07.B-K.1.1.3	**CC.1.2.8.C** Analyze how a text makes connections among and distinctions between individuals, ideas, or events. E08.B-K.1.1.3	**CC.1.2.9-10.C** Apply appropriate strategies to analyze, interpret, and evaluate how an author unfolds an analysis or series of ideas or events, including the order in which the points are made, how they are introduced and developed, and the connections that are drawn between them. L.N.1.1.3 L.N.1.3.3 L.N.2.3.3 L.N.2.3.5 L.N.2.4.1 L.N.2.4.3	**CC.1.2.11-12.C** Analyze the interaction and development of a complex set of ideas, sequence of events, or specific individuals over the course of the text.
Craft and Structure Point of View	**CC.1.2.6.D** Determine an author's point of view or purpose in a text and explain how it is conveyed in the text.	**CC.1.2.7.D** Determine an author's point of view or purpose in a text and analyze how the author distinguishes	**CC.1.2.8.D** Determine an author's point of view or purpose in a text and analyze how the author acknowledges	**CC.1.2.9-10.D** Determine an author's particular point of view and analyze how rhetoric advances the point of view.	**CC.1.2.11-12.D** Evaluate how an author's point of view or purpose shapes the content and style of a text.

	E06.B-C.2.1.1	his or her position from that of others. E07.B-C.2.1.1	and responds to conflicting evidence or viewpoints. E08.B-C.2.1.1	L.N.2.3.6	
Craft and Structure Text Structure	**CC.1.2.6.E** Analyze the author's structure through the use of paragraphs, chapters, or sections. E06.B-C.2.1.2	**CC.1.2.7.E** Analyze the structure of the text through evaluation of the author's use of graphics, charts, and the major sections of the text. E07.B-C.2.1.2	**CC.1.2.8.E** Analyze the structure of the text through evaluation of the author's use of specific sentences and paragraphs to develop and refine a concept. E08.B-C.2.1.2	**CC.1.2.9-10.E** Analyze in detail how an author's ideas or claims are developed and refined by particular sentences, paragraphs, or larger portions of a text. L.N.1.1.3 L.N.2.4.1 L.N.2.4.3	**CC.1.2.11-12.E** Analyze and evaluate the effectiveness of the structure an author uses in his or her exposition or argument, including whether the structure makes points clear, convincing, and engaging.
Craft and Structure Vocabulary	**CC.1.2.6.F** Determine the meaning of words and phrases as they are used in grade-level reading and content, including interpretation of figurative language in context. E06.B-V.4.1.1 E06.B-V.4.1.2	**CC.1.2.7.F** Determine the meaning of words and phrases as they are used in grade-level reading and content, including interpretation of figurative, connotative, and technical meanings. E07.B-V.4.1.1 E07.B-V.4.1.2 E07.B-C.2.1.3	**CC.1.2.8.F** Analyze the influence of the words and phrases in a text including figurative, connotative, and technical meanings, and how they shape meaning and tone. E08.B-V.4.1.1 E08.B-V.4.1.2 E08.B-C.2.1.3	**CC.1.2.9-10.F** Analyze how words and phrases shape meaning and tone in texts. L.N.1.1.4	**CC.1.2.11-12.F** Evaluate how words and phrases shape meaning and tone in texts.
Integration of Knowledge and Ideas Diverse Media	**CC.1.2.6.G** Integrate information presented in different media or formats (e.g., visually, quantitatively) as well as in words to develop a coherent understanding of a topic or issue.	**CC.1.2.7.G** Compare and contrast a text to an audio, video, or multimedia version of the text, analyzing each medium's portrayal of the subject (e.g., how the delivery of a speech affects the impact of the words).	**CC.1.2.8.G** Evaluate the advantages and disadvantages of using different mediums (e.g., print or digital text, video, multimedia) to present a particular topic or idea.	**CC.1.2.9-10.G** Analyze various accounts of a subject told in different mediums (e.g., a person's life story in both print and multimedia), determining which details are emphasized in each account. L.N.2.2.3	**CC.1.2.11-12.G** Integrate and evaluate multiple sources of information presented in different media or formats (e.g., visually, quantitatively) as well as in words in order to address a question or solve a problem.
Integration of Knowledge and Ideas Evaluating Arguments	**CC.1.2.6.H** Evaluate an author's argument by examining claims and determining if they are supported by evidence. E06.B-C.3.1.1	**CC.1.2.7.H** Evaluate an author's argument, reasoning, and specific claims for the soundness of the argument and the relevance of the evidence. E07.B-C.3.1.1	**CC.1.2.8.H** Evaluate an author's argument, reasoning, and specific claims for the soundness of the arguments and the relevance of the evidence. E08.B-C.3.1.1	**CC.1.2.9-10.H** Delineate and evaluate the argument and specific claims in a text, assessing the validity of reasoning and relevance of evidence. L.N.2.5.4 L.N.2.5.5 L.N.2.5.6	**CC.1.2.11-12.H** Analyze seminal texts based upon reasoning, premises, purposes, and arguments.
Integration of Knowledge and Ideas Analysis Across Texts	**CC.1.2.6.I** Examine how two authors present similar information in different types of text. E06.B-C.3.1.2	**CC.1.2.7.I** Analyze how two or more authors present and interpret facts on the same topic. E07.B-C.3.1.2	**CC.1.2.8.I** Analyze two or more texts that provide conflicting information on the same topic and identify where the texts disagree on matters of fact or interpretation. E08.B-C.3.1.2	**CC.1.2.9-10.I** Analyze seminal U.S. documents of historical and literary significance, including how they address related themes and concepts.	**CC.1.2.11-12.I** Analyze foundational U.S. and world documents of historical, political, and literary significance for their themes, purposes, and rhetorical features.
Vocabulary Acquisition and Use	**CC.1.2.6.J** Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression. E06.B-V.4.1.1 E06.B-V.4.1.2	**CC.1.2.7.J** Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression. E07.B-V.4.1.1 E07.B-V.4.1.2	**CC.1.2.8.J** Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression. E08.B-V.4.1.1 E08.B-V.4.1.2	**CC.1.2.9-10.J** Acquire and use accurately general academic and domain-specific words and phrases, sufficient for reading, writing, speaking, and listening at the college- and career-readiness level; demonstrate independence in gathering vocabulary knowledge when considering a word or phrase important to comprehension or expression. L.N.1.2.4 L.N.1.2.1 L.N.1.2.2 L.N.1.2.3	**CC.1.2.11-12.J** Acquire and use accurately general academic and domain-specific words and phrases, sufficient for reading, writing, speaking, and listening at the college- and career-readiness level; demonstrate independence in gathering vocabulary knowledge when considering a word or phrase important to comprehension or expression.
Vocabulary Acquisition	**CC.1.2.6.K** Determine or clarify the	**CC.1.2.7.K** Determine or clarify the	**CC.1.2.8.K** Determine or clarify the	**CC.1.2.9-10.K** Determine or clarify the	**CC.1.2.11-12.K** Determine or clarify the

and Use	meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E06.B-V.4.1.1	meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E07.B-V.4.1.1	meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E08.B-V.4.1.1	meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. L.N.1.2.1 L.N.1.2.2 L.N.1.2.3 L.N.1.2.4	meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools.
Range of Reading	**CC.1.2.6.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.7.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.8.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.9-10.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.	**CC.1.2.11-12.L** Read and comprehend literary nonfiction and informational text on grade level, reading independently and proficiently.

1.3	**Reading Literature** **Students read and respond to works of literature—with an emphasis on comprehension, vocabulary acquisition, and making connections among ideas and between texts with a focus on textual evidence.**				
	Grade 6	**Grade 7**	**Grade 8**	**Grades 9-10**	**Grades 11-12**
Key Ideas and Details Theme	**CC.1.3.6.A** Determine a theme or central idea of a text and how it is conveyed through particular details; provide a summary of the text distinct from personal opinions or judgments. E06.A-K.1.1.2	**CC.1.3.7.A** Determine a theme or central idea of a text and analyze its development over the course of the text; provide an objective summary of the text. E07.A-K.1.1.2	**CC.1.3.8.A** Determine a theme or central idea of a text and analyze its development over the course of the text, including its relationship to the characters, setting, and plot; provide an objective summary of the text. E08.A-K.1.1.2	**CC.1.3.9-10.A** Determine a theme or central idea of a text and analyze in detail its development over the course of the text, including how it emerges and is shaped and refined by specific details; provide an objective summary of the text. L.F.1.1.2 L.F.1.3.1 L.F.1.3.2 L.F.2.3.4	**CC.1.3.11-12.A** Determine and analyze the relationship between two or more themes or central ideas of a text, including the development and interaction of the themes; provide an objective summary of the text.
Key Ideas and Details Text Analysis	**CC.1.3.6.B** Cite textual evidence to support analysis of what the text says explicitly, as well as inferences and/or generalizations drawn from the text. E06.A-K.1.1.1	**CC.1.3.7.B** Cite several pieces of textual evidence to support analysis of what the text says explicitly, as well as inferences, conclusions, and/or generalizations drawn from the text. E07.A-K.1.1.1	**CC.1.3.8.B** Cite the textual evidence that most strongly supports an analysis of what the text says explicitly, as well as inferences, conclusions, and/or generalizations drawn from the text. E08.A-K.1.1.1	**CC.1.3.9-10.B** Cite strong and thorough textual evidence to support analysis of what the text says explicitly, as well as inferences and conclusions based on an author's explicit assumptions and beliefs about a subject. L.F.1.1.1 L.F.1.3.1 L.F.2.1.2	**CC.1.3.11-12.B** Cite strong and thorough textual evidence to support analysis of what the text says explicitly, as well as inferences and conclusions based on and related to an author's implicit and explicit assumptions and beliefs.
Key Ideas and Details Literary Elements	**CC.1.3.6.C** Describe how a particular story or drama's plot unfolds in a series of episodes, as well as how the characters respond or change as the plot moves toward a resolution. E06.A-K.1.1.3	**CC.1.3.7.C** Analyze how particular elements of a story or drama interact and how setting shapes the characters or plot. E07.A-K.1.1.3	**CC.1.3.8.C** Analyze how particular lines of dialogue or incidents in a story or drama propel the action, reveal aspects of a character, or provoke a decision. E08.A-K.1.1.3	**CC.1.3.9-10.C** Analyze how complex characters develop over the course of a text, interact with other characters, and advance the plot or develop the theme. L.F.1.1.3 L.F.2.3.1 L.F.2.3.4	**CC.1.3.11-12.C** Analyze the impact of the author's choices regarding how to develop and relate elements of a story or drama.
Craft and Structure Point of View	**CC.1.3.6.D** Determine an author's purpose in a text and explain how it is conveyed in a text. E06.A-C.2.1.1	**CC.1.3.7.D** Analyze how an author develops and contrasts the points of view of different characters or narrators in a text. E07.A-C.2.1.1	**CC.1.3.8.D** Analyze how differences in the points of view of the characters and the audience or reader (e.g., created through the use of dramatic irony) create such effects as suspense or humor. E08.A-C.2.1.1	**CC.1.3.9-10.D** Determine the point of view of the text and analyze the impact the point of view has on the meaning of the text. L.F.2.3.6	**CC.1.3.11-12.D** Evaluate how an author's point of view or purpose shapes the content and style of a text.

Craft and Structure Text Structure	**CC.1.3.6.E** Analyze how the structure of a text contributes to the development of theme, setting, and plot. E06.A-C.2.1.2	**CC.1.3.7.E** Analyze how the structure or form of a text contributes to its meaning. E07.A-C.2.1.2	**CC.1.3.8.E** Compare and contrast the structure of two or more texts and analyze how the differing structure of each text contributes to its meaning and style. E08.A-C.2.1.2	**CC.1.3.9-10.E** Analyze how an author's choices concerning how to structure a text, order events within it, and manipulate time create an effect. L.F.1.1.3 L.F.2.3.2 L.F.2.3.3 L.F.2.5.3	**CC.1.3.11-12.E** Evaluate the structure of texts including how specific sentences, paragraphs, and larger portions of the texts relate to each other and the whole.
Craft and Structure Vocabulary	**CC.1.3.6.F** Determine the meaning of words and phrases as they are used in grade-level reading and content, including interpretation of figurative language in context. E06.A-C.2.1.3 E06.A-V.4.1.1 E06.A-V.4.1.2	**CC.1.3.7.F** Determine the meaning of words and phrases as they are used in grade-level reading and content, including interpretation of figurative, connotative meanings. E07.A-C.2.1.3 E07.A-V.4.1.1 E07.A-V.4.1.2	**CC.1.3.8.F** Analyze the influence of the words and phrases in a text including figurative and connotative meanings and how they shape meaning and tone. E08.A-C.2.1.3 E08.A-V.4.1.1 E08.A-V.4.1.2	**CC.1.3.9-10.F** Analyze how words and phrases shape meaning and tone in texts. L.F.2.3.5 L.F.2.5.1	**CC.1.3.11-12.F** Evaluate how words and phrases shape meaning and tone in texts.
Integration of Knowledge and Ideas Sources of Information	**CC.1.3.6.G** Compare and contrast the experiences of reading a story, drama, or poem to listening to or viewing an audio, video, or live version of the text, including contrasting what is ''seen'' and ''heard'' when reading the text to what is perceived when listening or watching.	**CC.1.3.7.G** Compare and contrast a written story, drama, or poem to its audio, filmed, staged, or multimedia version, analyzing the effects of techniques unique to each medium (e.g., lighting, sound, color, or camera focus and angles in a film).	**CC.1.3.8.G** Analyze the extent to which a filmed or live production of a story or drama stays faithful to or departs from the text or script, evaluating the choices made by directors or actors.	**CC.1.3.9-10.G** Analyze the representation of a subject or a key scene in two different artistic mediums, including what is emphasized or absent in each treatment. L.F.2.2.1 L.F.2.2.3 L.F.2.2.4	**CC.1.3.11-12.G** Analyze multiple interpretations of a story, drama, or poem (e.g., recorded or live production of a play or recorded novel or poetry), evaluating how each version interprets the source text. (Include at least one play by Shakespeare and one play by an American dramatist.)
Integration of Knowledge and Ideas	**CC.1.3.6.H** Compare and contrast texts in different forms or genres in terms of their approaches to similar themes and topics as well as their use of additional literary elements. E06.A-C.3.1.1	**CC.1.3.7.H** Compare and contrast a fictional portrayal of a time, place, or character and a historical account of the same period as a means of understanding how authors of fiction use or alter history. E07.A-C.3.1.1	**CC.1.3.8.H** Analyze how a modern work of fiction draws on themes, patterns of events, or character types from traditional works, including describing how the material is rendered new. E08.A-C.3.1.1	**CC.1.3.9-10.H** Analyze how an author draws on and transforms themes, topics, character types, and/or other text elements from source material in a specific work. L.F.2.2.2 L.F.2.4.1	**CC.1.3.11-12.H** Demonstrate knowledge of foundational works of literature that reflect a variety of genres in the respective major periods of literature, including how two or more texts from the same period treat similar themes or topics.
Vocabulary Acquisition and Use	**CC.1.3.6.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E06.A-V.4.1.1	**CC.1.3.7.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E07.A-V.4.1.1	**CC.1.3.8.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. E08.A-V.4.1.1	**CC.1.3.9-10.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools. L.F.1.2.1 L.F.1.2.2 L.F.1.2.3 L.F.1.2.4	**CC.1.3.11-12.I** Determine or clarify the meaning of unknown and multiple-meaning words and phrases based on grade-level reading and content, choosing flexibly from a range of strategies and tools.
Vocabulary Acquisition and Use	**CC.1.3.6.J** Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression. E06.A-V.4.1.1 E06.A-V.4.1.2	**CC.1.3.7.J** Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression. E07.A-V.4.1.1 E07.A-V.4.1.2	**CC.1.3.8.J** Acquire and use accurately grade-appropriate general academic and domain-specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression. E08.A-V.4.1.1 E08.A-V.4.1.2	**CC.1.3.9-10.J** Acquire and use accurately grade appropriate general academic and domain specific words and phrases; gather vocabulary knowledge when considering a word or phrase important to comprehension or expression. L.F.1.2.1 L.F.1.2.2 L.F.1.2.3 L.F.1.2.4	**CC.1.3.11-12.J** Acquire and use accurately general academic and domain-specific words and phrases, sufficient for reading, writing, speaking, and listening at the college- and career-readiness level; demonstrate independence in gathering vocabulary knowledge when considering a word or phrase important to comprehension or expression.

Range of Reading	**CC.1.3.6.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.	**CC.1.3.7.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.	**CC.1.3.8.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.	**CC.1.3.9-10.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.	**CC.1.3.11-12.K** Read and comprehend literary fiction on grade level, reading independently and proficiently.

1.4	**Writing** **Students write for different purposes and audiences. Students write clear and focused text to convey a well-defined perspective and appropriate content.**				
	Grade 6	**Grade 7**	**Grade 8**	**Grades 9-10**	**Grades 11-12**
Informative/ Explanatory	**CC.1.4.6.A** **Write informative/ explanatory texts to examine a topic and convey ideas, concepts, and information clearly.**	**CC.1.4.7.A** **Write informative/ explanatory texts to examine a topic and convey ideas, concepts, and information clearly.**	**CC.1.4.8.A** **Write informative/ explanatory texts to examine a topic and convey ideas, concepts, and information clearly.**	**CC.1.4.9-10.A** **Write informative/ explanatory texts to examine and convey complex ideas, concepts, and information clearly and accurately.**	**CC.1.4.11-12.A** **Write informative/ explanatory texts to examine and convey complex ideas, concepts, and information clearly and accurately.**
Informative/Explanatory Focus	**CC.1.4.6.B** Identify and introduce the topic for the intended audience. E06.C.1.2.1 E06.E.1.1.1	**CC.1.4.7.B** Identify and introduce the topic clearly, including a preview of what is to follow. E07.C.1.2.1 E07.E.1.1.1	**CC.1.4.8.B** Identify and introduce the topic clearly, including a preview of what is to follow. E08.C.1.2.1 E08.E.1.1.1	**CC.1.4.9-10.B** Write with a sharp, distinct focus identifying topic, task, and audience. C.E.1.1.1	**CC.1.4.11-12.B** Write with a sharp, distinct focus identifying topic, task, and audience.
Informative/Explanatory Content	**CC.1.4.6.C** Develop and analyze the topic with relevant facts, definitions, concrete details, quotations, or other information and examples; include graphics and multimedia when useful to aiding comprehension. E06.C.1.2.2 E06.E.1.1.2	**CC.1.4.7.C** Develop and analyze the topic with relevant facts, definitions, concrete details, quotations, or other information and examples; include graphics and multimedia when useful to aiding comprehension. E07.C.1.2.2 E07.E.1.1.2	**CC.1.4.8.C** Develop and analyze the topic with relevant, well-chosen facts, definitions, concrete details, quotations, or other information and examples; include graphics and multimedia when useful to aiding comprehension. E08.C.1.2.2 E08.E.1.1.2	**CC.1.4.9-10.C** Develop and analyze the topic with relevant, well-chosen, and sufficient facts, extended definitions, concrete details, quotations, or other information and examples appropriate to the audience's knowledge of the topic; include graphics and multimedia when useful to aiding comprehension. C.E.1.1.2	**CC.1.4.11-12.C** Develop and analyze the topic thoroughly by selecting the most significant and relevant facts, extended definitions, concrete details, quotations, or other information and examples appropriate to the audience's knowledge of the topic; include graphics and multimedia when useful to aiding comprehension.
Informative/Explanatory Organization	**CC.1.4.6.D** Organize ideas, concepts, and information using strategies such as definition, classification, comparison/contrast, and cause/effect; use appropriate transitions to clarify the relationships among ideas and concepts; provide a concluding statement or section; include formatting when useful to aiding comprehension. E06.C.1.2.1 E06.C.1.2.3 E06.C.1.2.6 E06.E.1.1.1 E06.E.1.1.3 E06.E.1.1.6	**CC.1.4.7.D** Organize ideas, concepts, and information using strategies such as definition, classification, comparison/contrast, and cause/effect; use appropriate transitions to create cohesion and clarify the relationships among ideas and concepts; provide a concluding statement or section; include formatting when useful to aiding comprehension. E07.C.1.2.1 E07.C.1.2.3 E07.C.1.2.6 E07.E.1.1.1 E07.E.1.1.3 E07.E.1.1.6	**CC.1.4.8.D** Organize ideas, concepts, and information into broader categories; use appropriate and varied transitions to create cohesion and clarify the relationships among ideas and concepts; provide a concluding statement or section; include formatting when useful to aiding comprehension. E08.C.1.2.1 E08.C.1.2.3 E08.C.1.2.6 E08.E.1.1.1 E08.E.1.1.3 E08.E.1.1.6	**CC.1.4.9-10.D** Organize ideas, concepts, and information to make important connections and distinctions; use appropriate and varied transitions to link the major sections of the text; include formatting when useful to aiding comprehension; provide a concluding statement or section. C.E.1.1.3 C.E.1.1.5	**CC.1.4.11-12.D** Organize complex ideas, concepts, and information so that each new element builds on that which precedes it to create a whole; use appropriate and varied transitions and syntax to link the major sections of the text; provide a concluding statement or section that supports the information presented; include formatting when useful to aiding comprehension.
Informative/Explanatory Style	**CC.1.4.6.E** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to inform about or explain the topic.	**CC.1.4.7.E** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to inform about or explain the topic.	**CC.1.4.8.E** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to inform	**CC.1.4.9-10.E** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to manage the complexity of the	**CC.1.4.11-12.E** • Write with an awareness of the stylistic aspects of composition. • Use precise language, domain-specific vocabulary, and

	• Use sentences of varying lengths and complexities. • Develop and maintain a consistent voice. • Establish and maintain a formal style. E06.C.1.2.4 E06.C.1.2.5 E06.D.2.1.1 E06.D.2.1.2 E06.D.2.1.3 E06.D.2.1.4 E06.D.2.1.5 E06.E.1.1.4 E06.E.1.1.5	• Use sentences of varying lengths and complexities. • Develop and maintain a consistent voice. • Establish and maintain a formal style. E07.C.1.2.4 E07.C.1.2.5 E07.D.2.1.1 E07.D.2.1.2 E07.D.2.1.3 E07.D.2.1.4 E07.D.2.1.5 E07.E.1.1.4 E07.E.1.1.5	about or explain the topic. • Use sentences of varying lengths and complexities. • Create tone and voice through precise language. • Establish and maintain a formal style. E08.C.1.2.4 E08.C.1.2.5 E08.D.2.1.1 E08.D.2.1.2 E08.D.2.1.3 E08.D.2.1.4 E08.D.2.1.5 E08.D.2.1.6 E08.E.1.1.4 E08.E.1.1.5	topic. • Establish and maintain a formal style and objective tone while attending to the norms of the discipline in which they are writing. C.E.1.1.4 C.E.2.1.1 C.E.2.1.2 C.E.2.1.3 C.E.2.1.4 C.E.2.1.6 C.E.2.1.7	techniques such as metaphor, simile, and analogy to manage the complexity of the topic. • Establish and maintain a formal style and objective tone while attending to the norms of the discipline in which they are writing.
Informative/Explanatory Conventions of Language	**CC.1.4.6.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E06.D.1.1.1 E06.D.1.1.2 E06.D.1.1.3 E06.D.1.1.4 E06.D.1.1.5 E06.D.1.1.6 E06.D.1.1.7 E06.D.1.1.8 E06.D.1.2.1 E06.D.1.2.2 E06.D.1.2.3	**CC.1.4.7.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E07.D.1.1.1 E07.D.1.1.2 E07.D.1.1.3 E07.D.1.1.4 E07.D.1.1.5 E07.D.1.1.6 E07.D.1.1.7 E07.D.1.1.8 E07.D.1.1.9 E07.D.1.2.1 E07.D.1.2.2 E07.D.1.2.3 E07.D.1.2.4	**CC.1.4.8.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E08.D.1.1.1 E08.D.1.1.2 E08.D.1.1.3 E08.D.1.1.4 E08.D.1.1.5 E08.D.1.1.6 E08.D.1.1.7 E08.D.1.1.8 E08.D.1.1.9 E08.D.1.1.10 E08.D.1.1.11 E08.D.1.2.1 E08.D.1.2.2 E08.D.1.2.3 E08.D.1.2.4 E08.D.1.2.5	**CC.1.4.9-10.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. C.E.1.1.5 C.E.3.1.1 C.E.3.1.2 C.E.3.1.3 C.E.3.1.4 C.E.3.1.5	**CC.1.4.11-12.F** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling.
Opinion/Argumentative	**CC.1.4.6.G** **Write arguments to support claims.**	**CC.1.4.7.G** **Write arguments to support claims.**	**CC.1.4.8.G** **Write arguments to support claims.**	**CC.1.4.9-10.G** **Write arguments to support claims in an analysis of substantive topics.**	**CC.1.4.11-12.G** **Write arguments to support claims in an analysis of substantive topics.**
Opinion/Argumentative Focus	**CC.1.4.6.H** Introduce and state an opinion on a topic. E06.C.1.1.1 E06.E.1.1.1	**CC.1.4.7.H** Introduce and state an opinion on a topic. E07.C.1.1.1 E07.E.1.1.1	**CC.1.4.8.H** Introduce and state an opinion on a topic. E08.C.1.1.1 E08.E.1.1.1	**CC.1.4.9-10.H** Write with a sharp, distinct focus identifying topic, task, and audience. • Introduce the precise claim. C.P.1.1.1	**CC.1.4.11-12.H** Write with a sharp, distinct focus identifying topic, task, and audience. • Introduce the precise, knowledgeable claim.
Opinion/Argumentative Content	**CC.1.4.6.I** Use clear reasons and relevant evidence to support claims, using credible sources and demonstrating an understanding of the topic. E06.C.1.1.2 E06.E.1.1.2	**CC.1.4.7.I** Acknowledge alternate or opposing claims and support claim with logical reasoning and relevant evidence, using accurate, credible sources and demonstrating an understanding of the topic. E07.C.1.1.2 E07.E.1.1.2	**CC.1.4.8.I** Acknowledge and distinguish the claim(s) from alternate or opposing claims and support claim with logical reasoning and relevant evidence, using accurate, credible sources and demonstrating an understanding of the topic. E08.C.1.1.2 E08.E.1.1.2	**CC.1.4.9-10.I** Distinguish the claim(s) from alternate or opposing claims; develop claim(s) fairly, supplying evidence for each while pointing out the strengths and limitations of both in a manner that anticipates the audience's knowledge level and concerns. C.P.1.1.2 C.P.1.1.3	**CC.1.4.11-12.I** Distinguish the claim(s) from alternate or opposing claims; develop claim(s) and counterclaims fairly and thoroughly, supplying the most relevant evidence for each while pointing out the strengths and limitations of both in a manner that anticipates the audience's knowledge level, concerns, values, and possible biases.
Opinion/Argumentative Organization	**CC.1.4.6.J** Organize the claim(s) with	**CC.1.4.7.J** Organize the claim(s) with	**CC.1.4.8.J** Organize the claim(s)	**CC.1.4.9-10.J** Create organization that	**CC.1.4.11-12.J** Create organization that

	clear reasons and evidence clearly; clarify relationships among claim(s) and reasons by using words, phrases, and clauses; provide a concluding statement or section that follows from the argument presented. E06.C.1.1.1 E06.C.1.1.3 E06.C.1.1.5 E06.E.1.1.1 E06.E.1.1.3 E06.E.1.1.6	clear reasons and evidence clearly; clarify relationships among claim(s) and reasons by using words, phrases, and clauses to create cohesion; provide a concluding statement or section that follows from and supports the argument presented. E07.C.1.1.1 E07.C.1.1.3 E07.C.1.1.5 E07.E.1.1.1 E07.E.1.1.3 E07.E.1.1.6	with clear reasons and evidence clearly; clarify relationships among claim(s), counterclaims, reasons, and evidence by using words, phrases, and clauses to create cohesion; provide a concluding statement or section that follows from and supports the argument presented. E08.C.1.1.1 E08.C.1.1.3 E08.C.1.1.5 E08.E.1.1.1 E08.E.1.1.3 E08.E.1.1.6	establishes clear relationships among claim(s), counterclaims, reasons, and evidence; use words, phrases, and clauses to link the major sections of the text, create cohesion, and clarify the relationships between claim(s) and reasons, between reasons and evidence, and between claim(s) and counterclaims; provide a concluding statement or section that follows from and supports the argument presented. C.P.1.1.2 C.P.1.1.3 C.P.2.1.5 C.P.2.1.6	logically sequences claim(s), counterclaims, reasons, and evidence; use words, phrases, and clauses as well as varied syntax to link the major sections of the text to create cohesion and clarify the relationships between claim(s) and reasons, between reasons and evidence, and between claim(s) and counterclaims; provide a concluding statement or section that follows from and supports the argument presented.
Opinion/Argumentative Style	**CC.1.4.6.K** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to inform about or explain the topic. • Use sentences of varying lengths and complexities. • Develop and maintain a consistent voice. • Establish and maintain a formal style. E06.C.1.1.4 E06.D.2.1.1 E06.D.2.1.2 E06.D.2.1.3 E06.D.2.1.4 E06.D.2.1.5 E06.E.1.1.4 E06.E.1.1.5	**CC.1.4.7.K** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to inform about or explain the topic. • Use sentences of varying lengths and complexities. • Develop and maintain a consistent voice. • Establish and maintain a formal style. E07.C.1.1.4 E07.D.2.1.1 E07.D.2.1.2 E07.D.2.1.3 E07.D.2.1.4 E07.D.2.1.5 E07.E.1.1.4 E07.E.1.1.5	**CC.1.4.8.K** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to inform about or explain the topic. • Use sentences of varying lengths and complexities. • Create tone and voice through precise language. • Establish and maintain a formal style. E08.C.1.1.4 E08.D.2.1.1 E08.D.2.1.2 E08.D.2.1.3 E08.D.2.1.4 E08.D.2.1.5 E08.D.2.1.6 E08.E.1.1.4 E08.E.1.1.5	**CC.1.4.9-10.K** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to manage the complexity of the topic. • Establish and maintain a formal style and objective tone while attending to the norms of the discipline in which they are writing. C.P.1.1.4 C.P.2.1.1 C.P.2.1.2 C.P.2.1.3 C.P.2.1.4 C.P.2.1.6 C.P.2.1.7	**CC.1.4.11-12.K** Write with an awareness of the stylistic aspects of composition. • Use precise language and domain-specific vocabulary to manage the complexity of the topic. • Establish and maintain a formal style and objective tone while attending to the norms of the discipline in which they are writing.
Opinion/Argumentative Conventions of Language	**CC.1.4.6.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E06.D.1.1.1 E06.D.1.1.2 E06.D.1.1.3 E06.D.1.1.4 E06.D.1.1.5 E06.D.1.1.6 E06.D.1.1.7 E06.D.1.1.8 E06.D.1.2.1 E06.D.1.2.2 E06.D.1.2.3	**CC.1.4.7.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E07.D.1.1.1 E07.D.1.1.2 E07.D.1.1.3 E07.D.1.1.4 E07.D.1.1.5 E07.D.1.1.6 E07.D.1.1.7 E07.D.1.1.8 E07.D.1.1.9 E07.D.1.2.1 E07.D.1.2.2 E07.D.1.2.3 E07.D.1.2.4	**CC.1.4.8.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E08.D.1.1.1 E08.D.1.1.2 E08.D.1.1.3 E08.D.1.1.4 E08.D.1.1.5 E08.D.1.1.6 E08.D.1.1.7 E08.D.1.1.8 E08.D.1.1.9 E08.D.1.1.10 E08.D.1.1.11 E08.D.1.2.1 E08.D.1.2.2 E08.D.1.2.3 E08.D.1.2.4 E08.D.1.2.5	**CC.1.4.9-10.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. C.P.1.1.5 C.P.3.1.1 C.P.3.1.2 C.P.3.1.3 C.P.3.1.4 C.P.3.1.5	**CC.1.4.11-12.L** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling.

Narrative	**CC.1.4.6.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.7.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.8.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.9-10.M** **Write narratives to develop real or imagined experiences or events.**	**CC.1.4.11-12.M** **Write narratives to develop real or imagined experiences or events.**
Narrative Focus	**CC.1.4.6.N** Engage and orient the reader by establishing a context and introducing a narrator and/or characters. E06.C.1.3.1	**CC.1.4.7.N** Engage and orient the reader by establishing a context and point of view and introducing a narrator and/or characters. E07.C.1.3.1	**CC.1.4.8.N** Engage and orient the reader by establishing a context and point of view and introducing a narrator and/or characters. E08.C.1.3.1	**CC.1.4.9-10.N** Engage and orient the reader by setting out a problem, situation, or observation, establishing one or multiple points of view, and introducing a narrator and/or characters.	**CC.1.4.11-12.N** Engage and orient the reader by setting out a problem, situation, or observation and its significance, establishing one or multiple points of view, and introducing a narrator and/or characters.
Narrative Content	**CC.1.4.6.O** Use narrative techniques such as dialogue, description, and pacing to develop experiences, events, and/or characters; use precise words and phrases, relevant descriptive details, and sensory language to convey experiences and events. E06.C.1.3.2 E06.C.1.3.4	**CC.1.4.7.O** Use narrative techniques such as dialogue, description, and pacing to develop experiences, events, and/or characters; use precise words and phrases, relevant descriptive details, and sensory language to capture the action and convey experiences and events. E07.C.1.3.2 E07.C.1.3.4	**CC.1.4.8.O** Use narrative techniques such as dialogue, description, reflection, and pacing to develop experiences, events, and/or characters; use precise words and phrases, relevant descriptive details, and sensory language to capture the action and convey experiences and events. E08.C.1.3.2 E08.C.1.3.4	**CC.1.4.9-10.O** Use narrative techniques such as dialogue, description, reflection, multiple plotlines, and pacing to develop experiences, events, and/or characters; use precise words and phrases, telling details, and sensory language to convey a vivid picture of the experiences, events, settings, and/or characters.	**CC.1.4.11-12.O** Use narrative techniques such as dialogue, description, reflection, multiple plotlines, and pacing to develop experiences, events, and/or characters; use precise words and phrases, telling details, and sensory language to convey a vivid picture of the experiences, events, settings, and/or characters.
Narrative Organization	**CC.1.4.6.P** Organize an event sequence that unfolds naturally and logically, using a variety of transition words, phrases, and clauses to convey sequence and signal shifts from one time frame or setting to another; provide a conclusion that follows from the narrated experiences and events. E06.C.1.3.1 E06.C.1.3.3 E06.C.1.3.5	**CC.1.4.7.P** Organize an event sequence that unfolds naturally and logically, using a variety of transition words, phrases, and clauses to convey sequence and signal shifts from one time frame or setting to another; provide a conclusion that follows from and reflects on the narrated experiences and events. E07.C.1.3.1 E07.C.1.3.3 E07.C.1.3.5	**CC.1.4.8.P** Organize an event sequence that unfolds naturally and logically using a variety of transition words, phrases, and clauses to convey sequence, signal shifts from one time frame or setting to another and show the relationships among experiences and events; provide a conclusion that follows from and reflects on the narrated experiences or events. E08.C.1.3.1 E08.C.1.3.3 E08.C.1.3.5	**CC.1.4.9-10.P** Create a smooth progression of experiences or events using a variety of techniques to sequence events so that they build on one another to create a coherent whole; provide a conclusion that follows from and reflects on what is experienced, observed, or resolved over the course of the narrative.	**CC.1.4.11-12.P** Create a smooth progression of experiences or events using a variety of techniques to sequence events so that they build on one another to create a coherent whole and build toward a particular tone and outcome; provide a conclusion that follows from and reflects on what is experienced, observed, or resolved over the course of the narrative.
Narrative Style	**CC.1.4.6.Q** Write with an awareness of the stylistic aspects of writing. • Vary sentence patterns for meaning, reader/listener interest, and style. • Use precise language. • Develop and maintain a consistent voice. E06.C.1.3.4 E06.D.2.1.1 E06.D.2.1.2 E06.D.2.1.3 E06.D.2.1.4 E06.D.2.1.5 E06.E.1.1.4	**CC.1.4.7.Q** Write with an awareness of the stylistic aspects of writing. • Choose language that expresses ideas precisely and concisely, recognizing and eliminating wordiness and redundancy. • Use sentences of varying lengths and complexities. • Use precise language. • Develop and maintain a consistent voice. E07.C.1.3.4 E07.D.2.1.1 E07.D.2.1.2 E07.D.2.1.3 E07.D.2.1.4 E07.D.2.1.5	**CC.1.4.8.Q** Write with an awareness of the stylistic aspects of writing. • Use verbs in the active and passive voice and in the conditional and subjunctive mood to achieve particular effects. • Use sentences of varying lengths and complexities. • Create tone and voice through precise language. E08.C.1.3.4 E08.D.2.1.1 E08.D.2.1.2 E08.D.2.1.3	**CC.1.4.9-10.Q** Write with an awareness of the stylistic aspects of writing. • Use parallel structure. • Use various types of phrases and clauses to convey meaning and add variety and interest.	**CC.1.4.11-12.Q** Write with an awareness of the stylistic aspects of writing. • Use parallel structure. • Use various types of phrases and clauses to convey specific meanings and add variety and interest. • Use precise language, domain-specific vocabulary, and techniques such as metaphor, simile, and analogy to manage the complexity of the topic.

			E08.D.2.1.4 E08.D.2.1.5 E08.D.2.1.6		
Narrative Conventions of Language	**CC.1.4.6.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E06.D.1.1.1 E06.D.1.1.2 E06.D.1.1.3 E06.D.1.1.4 E06.D.1.1.5 E06.D.1.1.6 E06.D.1.1.7 E06.D.1.1.8 E06.D.1.2.1 E06.D.1.2.2 E06.D.1.2.3	**CC.1.4.7.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E07.D.1.1.1 E07.D.1.1.2 E07.D.1.1.3 E07.D.1.1.4 E07.D.1.1.5 E07.D.1.1.6 E07.D.1.1.7 E07.D.1.1.8 E07.D.1.1.9 E07.D.1.2.1 E07.D.1.2.2 E07.D.1.2.3 E07.D.1.2.4	**CC.1.4.8.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling. E08.D.1.1.1 E08.D.1.1.2 E08.D.1.1.3 E08.D.1.1.4 E08.D.1.1.5 E08.D.1.1.6 E08.D.1.1.7 E08.D.1.1.8 E08.D.1.1.9 E08.D.1.1.10 E08.D.1.1.11 E08.D.1.2.1 E08.D.1.2.2 E08.D.1.2.3 E08.D.1.2.4 E08.D.1.2.5	**CC.1.4.9-10.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling.	**CC.1.4.11-12.R** Demonstrate a grade-appropriate command of the conventions of standard English grammar, usage, capitalization, punctuation, and spelling.
Response to Literature	**CC.1.4.6.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and literary nonfiction. E06.E.1.1.1 E06.E.1.1.2 E06.E.1.1.3 E06.E.1.1.4 E06.E.1.1.5 E06.E.1.1.6	**CC.1.4.7.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and literary nonfiction. E07.E.1.1.1 E07.E.1.1.2 E07.E.1.1.3 E07.E.1.1.4 E07.E.1.1.5 E07.E.1.1.6	**CC.1.4.8.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and literary nonfiction. E08.E.1.1.1 E08.E.1.1.2 E08.E.1.1.3 E08.E.1.1.4 E08.E.1.1.5 E08.E.1.1.6	**CC.1.4.9-10.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and literary nonfiction.	**CC.1.4.11-12.S** Draw evidence from literary or informational texts to support analysis, reflection, and research, applying grade-level reading standards for literature and literary nonfiction.
Production and Distribution of Writing Writing Process	**CC.1.4.6.T** With guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach.	**CC.1.4.7.T** With some guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on how well purpose and audience have been addressed.	**CC.1.4.8.T** With some guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on how well purpose and audience have been addressed.	**CC.1.4.9-10.T** Develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on addressing what is most significant for a specific purpose and audience.	**CC.1.4.11-12.T** Develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on addressing what is most significant for a specific purpose and audience.
Technology and Publication	**CC.1.4.6.U** Use technology, including the Internet, to produce and publish writing as well as to interact and collaborate with others; demonstrate sufficient command of keyboarding skills to type a minimum of three pages in a single sitting.	**CC.1.4.7.U** Use technology, including the Internet, to produce and publish writing and link to and cite sources as well as to interact and collaborate with others, including linking to and citing sources.	**CC.1.4.8.U** Use technology, including the Internet, to produce and publish writing and present the relationships between information and ideas efficiently as well as to interact and collaborate with others.	**CC.1.4.9-10.U** Use technology, including the Internet, to produce, publish, and update individual or shared writing products, taking advantage of technology's capacity to link to other information and to display information flexibly and dynamically.	**CC.1.4.11-12.U** Use technology, including the Internet, to produce, publish, and update individual or shared writing products in response to ongoing feedback, including new arguments and information.
Conducting Research	**CC.1.4.6.V** Conduct short research projects to answer a question, drawing on several sources and refocusing the inquiry when appropriate.	**CC.1.4.7.V** Conduct short research projects to answer a question, drawing on several sources and generating additional related, focused questions for further research and investigation.	**CC.1.4.8.V** Conduct short research projects to answer a question (including a self-generated question), drawing on several sources and generating additional related,	**CC.1.4.9-10.V** Conduct short as well as more sustained research projects to answer a question (including a self-generated question) or solve a problem; narrow or broaden the inquiry when	**CC.1.4.11-12.V** Conduct short as well as more sustained research projects to answer a question (including a self-generated question) or solve a problem; narrow or broaden the inquiry when

			focused questions that allow for multiple avenues of exploration.	appropriate; synthesize multiple sources on the subject, demonstrating understanding of the subject under investigation.	appropriate; synthesize multiple sources on the subject, demonstrating understanding of the subject under investigation.

1.4	**Writing** **Students write for different purposes and audiences. Students write clear and focused text to convey a well-defined perspective and appropriate content.**				
	Grade 6	**Grade 7**	**Grade 8**	**Grades 9-10**	**Grades 11-12**
Credibility, Reliability, and Validity of Sources	**CC.1.4.6.W** Gather relevant information from multiple print and digital sources; assess the credibility of each source; and quote or paraphrase the data and conclusions of others while avoiding plagiarism and providing basic bibliographic information for sources.	**CC.1.4.7.W** Gather relevant information from multiple print and digital sources, using search terms effectively; assess the credibility and accuracy of each source; and quote or paraphrase the data and conclusions of others while avoiding plagiarism and following a standard format for citation.	**CC.1.4.8.W** Gather relevant information from multiple print and digital sources, using search terms effectively; assess the credibility and accuracy of each source; and quote or paraphrase the data and conclusions of others while avoiding plagiarism and following a standard format for citation.	**CC.1.4.9-10.W** Gather relevant information from multiple authoritative print and digital sources, using advanced searches effectively; assess the usefulness of each source in answering the research question; integrate information into the text selectively to maintain the flow of ideas, avoiding plagiarism and following a standard format for citation.	**CC.1.4.11-12.W** Gather relevant information from multiple authoritative print and digital sources, using advanced searches effectively; assess the strengths and limitations of each source in terms of the task, purpose, and audience; integrate information into the text selectively to maintain the flow of ideas, avoiding plagiarism and overreliance on any one source and following a standard format for citation.
Range of Writing	**CC.1.4.6.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.7.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.8.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.9-10.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	**CC.1.4.11-12.X** Write routinely over extended time frames (time for research, reflection, and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.

1.5	**Speaking and Listening** **Students present appropriately in formal speaking situations, listen critically, and respond intelligently as individuals or in group discussions.**				
	Grade 6	**Grade 7**	**Grade 8**	**Grades 9-10**	**Grades 11-12**
Comprehension and Collaboration Collaborative Discussion	**CC.1.5.6.A** Engage effectively in a range of collaborative discussions, on grade-level topics, texts, and issues, building on others' ideas and expressing their own clearly.	**CC.1.5.7.A** Engage effectively in a range of collaborative discussions, on grade-level topics, texts, and issues, building on others' ideas and expressing their own clearly.	**CC.1.5.8.A** Engage effectively in a range of collaborative discussions, on grade-level topics, texts, and issues, building on others' ideas and expressing their own clearly.	**CC.1.5.9-10.A** Initiate and participate effectively in a range of collaborative discussions on grade-level topics, texts, and issues, building on others' ideas and expressing their own clearly and persuasively.	**CC.1.5.11-12.A** Initiate and participate effectively in a range of collaborative discussions on grade-level topics, texts, and issues, building on others' ideas and expressing their own clearly and persuasively.
Comprehension and Collaboration Critical Listening	**CC.1.5.6.B** Delineate a speaker's argument and specific claims by identifying specific reasons and evidence and recognize arguments or claims not supported by factual evidence.	**CC.1.5.7.B** Delineate a speaker's argument and specific claims, evaluating the soundness of the reasoning and the relevance and sufficiency of the evidence.	**CC.1.5.8.B** Delineate a speaker's argument and specific claims, evaluating the soundness of the reasoning and the relevance and sufficiency of the evidence.	**CC.1.5.9-10.B** Evaluate a speaker's perspective, reasoning, and use of evidence and rhetoric, identifying any fallacious reasoning or exaggerated or distorted evidence.	**CC.1.5.11-12.B** Evaluate how the speaker's perspective, reasoning, and use of evidence and rhetoric affect the credibility of an argument through the author's stance, premises, links among ideas, word choice, points of emphasis, and tone.
Comprehension and Collaboration Evaluating Information	**CC.1.5.6.C** Interpret information presented in diverse media and formats (e.g., visually, quantitatively, orally) and explain how it contributes to a topic, text, or issue under study.	**CC.1.5.7.C** Analyze the main ideas and supporting details presented in diverse media formats (e.g., visually, quantitatively, orally) and explain how the ideas clarify a topic, text, or issue under study.	**CC.1.5.8.C** Analyze the purpose of information presented in diverse media formats (e.g., visually, quantitatively, orally) and evaluate the motives (e.g., social, commercial, political) behind its presentation.	**CC.1.5.9-10.C** Integrate multiple sources of information presented in diverse media or formats (e.g., visually, quantitatively, orally) evaluating the credibility and accuracy of each source.	**CC.1.5.11-12.C** Integrate multiple sources of information presented in diverse formats and media (e.g., visually, quantitatively, orally) in order to make informed decisions and solve problems, evaluating the credibility and accuracy of each source and noting any discrepancies among the data.
Presentation of Knowledge and Ideas Purpose,	**CC.1.5.6.D** Present claims and findings, sequencing ideas logically and using	**CC.1.5.7.D** Present claims and findings, emphasizing salient points in a	**CC.1.5.8.D** Present claims and findings, emphasizing salient points in a focused,	**CC.1.5.9-10.D** Present information, findings, and supporting evidence clearly,	**CC.1.5.11-12.D** Present information, findings, and supporting evidence, conveying a clear

Audience, and Task	pertinent descriptions, facts, and details to accentuate main ideas or themes; use appropriate eye contact, adequate volume, and clear pronunciation.	focused, coherent manner with pertinent descriptions, facts, details, and examples; use appropriate eye contact, adequate volume, and clear pronunciation.	coherent manner with relevant evidence, sound, valid reasoning, and well-chosen details; use appropriate eye contact, adequate volume and clear pronunciation.	concisely, and logically such that listeners can follow the line of reasoning; ensure that the presentation is appropriate to purpose, audience, and task.	and distinct perspective; organization, development, substance, and style are appropriate to purpose, audience, and task.
Presentation of Knowledge and Ideas Context	**CC.1.5.6.E** Adapt speech to a variety of contexts and tasks.	**CC.1.5.7.E** Adapt speech to a variety of contexts and tasks.	**CC.1.5.8.E** Adapt speech to a variety of contexts and tasks.	**CC.1.5.9-10.E** Adapt speech to a variety of contexts and tasks.	**CC.1.5.11-12E** Adapt speech to a variety of contexts and tasks.
Integration of Knowledge and Ideas Multimedia	**CC.1.5.6.F** Include multimedia components and visual displays in presentations to clarify information.	**CC.1.5.7.F** Include multimedia components and visual displays in presentations to clarify claims and findings and emphasize salient points.	**CC.1.5.8.F** Integrate multimedia and visual displays into presentations to add interest, clarify information, and strengthen claims and evidence.	**CC.1.5.9-10.F** Make strategic use of digital media in presentations to add interest and enhance understanding of findings, reasoning, and evidence.	**CC.1.5.11-12.F** Make strategic use of digital media in presentations to add interest and enhance understanding of findings, reasoning, and evidence.
Conventions of Standard English	**CC.1.5.6.G** Demonstrate command of the conventions of standard English when speaking based on Grade 6 level and content.	**CC.1.5.7.G** Demonstrate command of the conventions of standard English when speaking based on Grade 7 level and content.	**CC.1.5.8.G** Demonstrate command of the conventions of standard English when speaking based on Grade 8 level and content.	**CC.1.5.9-10.G** Demonstrate command of the conventions of standard English when speaking based on Grades 9-10 level and content.	**CC.1.5.11-12.G** Demonstrate command of the conventions of standard English when speaking based on Grades 11-12 level and content.

Pennsylvania Core Standards for Mathematics
Grades PreK—High School

INTRODUCTION

The Pennsylvania Core Standards in Mathematics in grades PreK-5 lay a solid foundation in whole numbers, addition, subtraction, multiplication, division, fractions, and decimals. Taken together, these elements support a student's ability to learn and apply more demanding math concepts and procedures. The middle school and high school standards call on students to practice applying mathematical ways of thinking to real world issues and challenges; they prepare students to think and reason mathematically. Additionally, they set a rigorous definition of college and career readiness by demanding that students develop a depth of understanding and ability to apply mathematics to novel situations, as college students and employees regularly do. Although the standards are not a curriculum or a prescribed series of activities, school entities will use them to develop a local school curriculum that will meet local students' needs.

This document includes PA Core Standards for Mathematical Content and Mathematical Practice. The mathematics standards define what students should understand and be able to do. Mathematical Practice Standards describes the habits of mind required to reach a level of mathematical proficiency.

PA Core Standards *Mathematical Content and Mathematical Practice*		
Standards for Mathematical Content	**Standards for Mathematical Practice**	
2.1 Numbers and Operations A) Counting and Cardinality B) Numbers and Operations in Base Ten C) Numbers and Operations—Fractions D) Ratios and Proportional Relationships E) The Number System F) Number and Quantity		• Make sense of problems and persevere in solving them. • Reason abstractly and quantitatively. • Construct viable arguments and critique the reasoning of others. • Model with mathematics. • Use appropriate tools strategically. • Attend to precision. • Look for and make use of structure. • Look for and make sense of regularity in repeated reasoning.
2.2 Algebraic Concepts A) Operations and Algebraic Thinking B) Expressions & Equations C) Functions D) Algebra		
2.3 Geometry A) Geometry		
2.4 Measurement, Data, and Probability A) Measurement and Data B) Statistics and Probability		

Standards cannot be viewed or addressed in isolation, as each standard depends upon or may lead into multiple standards across grades; thus, it is imperative that educators are familiar with both the standards that come before and those that follow a particular grade level. These revised standards reflect instructional shifts that cannot occur without the integrated emphasis on content and practice.

Standards are overarching statements of what a proficient math student should know and be able to do. The Pennsylvania Assessment Anchors and Eligible Content closely align with the revised standards and are an invaluable source for greater detail.

Key Points in Mathematics

• The standards stress both procedural skills and conceptual understanding to ensure students are learning and applying the critical information they need to succeed at higher levels.

• K-5 standards, which provide students with a *solid foundation in whole numbers, addition, subtraction, multiplication, division, fractions, and decimals*, help young students build the foundation to successfully apply more demanding math concepts and procedures, and move into application. They also provide detailed guidance to teachers on how to navigate their way through topics such as *fractions, negative numbers, and geometry*, and do so by maintaining a continuous progression from grade to grade.

• Having built a strong foundation at K-5, students can do hands-on learning in geometry, algebra, and probability and statistics. Students who have mastered the content and skills through the seventh grade will be *well-prepared for algebra* in grade 8.

• High school standards emphasize *practicing applying mathematical ways of thinking to real world issues and challenges*.

The PA Core Standards for Mathematics detail four standard areas: *Numbers and Operations, Algebraic Concepts, Geometry,* and *Measurement, Data, and Probability*. These standard areas are reflective of the reporting categories in the PA Core Assessment Anchors and Eligible Content. The intent of this document is to provide a useful tool for designing curriculum, instruction, and assessment. The grade level curriculum and instructional shifts in mathematics cannot occur without the integrated emphasis on content and practice. The chart below illustrates the four standard areas and the development and progression of the strands, with an understanding that all is framed around the Standards for Mathematical Practice.

Mathematical Standards: Development and Progression											
Standards for Mathematical Practice											
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.							Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.				
	PreK	K	1	2	3	4	5	6	7	8	HS
2.1 Numbers and Operations			(A) Counting & Cardinality								
			(B) Numbers and Operations in Base Ten				(D) Ratios and Proportional Relationships			(F) Number and Quantity	
				(C) Numbers and Operations—Fractions			(E) The Number System				
2.2 Algebraic Concepts			(A) Operations and Algebraic Thinking						(B) Expressions and Equations		(D) Algebra
									(C) Functions		
2.3 Geometry			(A) Geometry								
2.4 Measurement, Data, and Probability			(A) Measurement and Data					(B) Statistics and Probability			

2.1 Numbers and Operations							
The Standards of Mathematical Practices							
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.		
	Grade PreK 2.1.PreK	Grade K 2.1.K	Grade 1 2.1.1	Grade 2 2.1.2	Grade 3 2.1.3	Grade 4 2.1.4	Grade 5 2.1.5
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:							
(A) Counting & Cardinality	**CC.2.1.PreK.A.1** Know number names and the count sequence.		**CC.2.1.K.A.1** Know number names and write and recite the count sequence.	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*
CC.2.1.PreK.A.2 Count to tell the number of objects.	**CC.2.1.K.A.2** Apply one-to-one correspondence to count the						

	number of objects.
CC.2.1.PreK.A.3 Compare numbers.	**CC.2.1.K.A.3** Apply the concept of magnitude to compare numbers and quantities.

2.1 Numbers and Operations							
The Standards of Mathematical Practices							
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.		
	Grade PreK 2.1.PreK	Grade K 2.1.K	Grade 1 2.1.1	Grade 2 2.1.2	Grade 3 2.1.3	Grade 4 2.1.4	Grade 5 2.1.5
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:							
(B) Numbers & Operations in Base Ten			*Intentionally Blank*		**CC.2.1.K.B.1** Use place value to compose and decompose numbers within 19.	**CC.2.1.1.B.1** Extend the counting sequence to read and write numerals to represent objects.	**CC.2.1.2.B.1** Use place-value concepts to represent amounts of tens and ones and to compare three digit numbers.
Intentionally Blank		**CC.2.1.1.B.2** Use place-value concepts to represent amounts of tens and ones and to compare two digit numbers.	**CC.2.1.2.B.2** Use place-value concepts to read, write, and skip count to 1000.	*Intentionally Blank*	**CC.2.1.4.B.2** Use place-value understanding and properties of operations to perform multi-digit arithmetic. M04.A-T.2.1.1 M04.A-T.2.1.2 M04.A-T.2.1.3 M04.A-T.2.1.4	**CC.2.1.5.B.2** Extend an understanding of operations with whole numbers to perform operations including decimals. M05.A-T.2.1.1 M05.A-T.2.1.2 M05.A-T.2.1.3	
CC.2.1.1.B.3 Use place-value concepts and properties of opera-tions to add and subtract within 100.	**CC.2.1.2.B.3** Use place-value understanding and properties of opera-tions to add and subtract within 1000.	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*			

2.1 Numbers and Operations	
The Standards of Mathematical Practices	

Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.		
	Grade PreK 2.1.PreK	Grade K 2.1.K	Grade 1 2.1.1	Grade 2 2.1.2	Grade 3 2.1.3	Grade 4 2.1.4	Grade 5 2.1.5
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:							
(C) Numbers & Operations—Fractions			*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*	**CC.2.1.3.C.1** Explore and develop an understanding of fractions as numbers. M03.A-F.1.1.1 M03.A-F.1.1.2 M03.A-F.1.1.3 M03.A-F.1.1.4 M03.A-F.1.1.5
Intentionally Blank	**CC.2.1.4.C.2** Build fractions from unit fractions by applying and extending previous understandings of operations on whole numbers. M04.A-F.2.1.1 M04.A-F.2.1.2 M04.A-F.2.1.3 M04.A-F.2.1.4 M04.A-F.2.1.5 M04.A-F.2.1.6 M04.A-F.2.1.7	**CC.2.1.5.C.2** Apply and extend previous understandings of multiplication and division to multiply and divide fractions. M05.A-F.2.1.1 M05.A-F.2.1.2 M05.A-F.2.1.3 M05.A-F.2.1.4					
Intentionally Blank	**CC.2.1.4.C.3** Connect decimal notation to fractions, and compare decimal fractions (base 10 denominator, e.g., 19/100). M04.A-F.3.1.1 M04.A-F.3.1.2 M04.A-F.3.1.3	*Intentionally Blank*					

2.2 Algebraic Concepts							
The Standards of Mathematical Practices							
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.		
	Grade PreK 2.2.PreK	Grade K 2.2.K	Grade 1 2.2.1	Grade 2 2.2.2	Grade 3 2.2.3	Grade 4 2.2.4	Grade 5 2.2.5
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:							

(A) Operations and Algebraic Thinking				**CC.2.2.PreK.A.1** Understand addition as putting together and adding to, and understand subtraction as taking apart and taking from.	**CC.2.2.K.A.1** Extend the concepts of putting together and taking apart to add and subtract within 10.	**CC.2.2.1.A.1** Represent and solve problems involving addition and subtraction within 20.	**CC.2.2.2.A.1** Represent and solve problems involving addition and subtraction within 100.	**CC.2.2.3.A.1** Represent and solve problems involving multiplication and division. M03.B-O.1.1.1 M03.B-O.1.1.2 M03.B-O.1.2.1 M03.B-O.1.2.2	**CC.2.2.4.A.1** Represent and solve problems involving the four operations. M04.B-O.1.1.1 M04.B-O.1.1.2 M04.B-O.1.1.3 M04.B-O.1.1.4	**CC.2.2.5.A.1** Interpret and evaluate numerical expressions using order of operations. M05.B-O.1.1.1 M05.B-O.1.1.2
Intentionally Blank	*Intentionally Blank*	**CC.2.2.1.A.2** Understand and apply properties of operations and the relationship between addition and subtraction.	**CC.2.2.2.A.2** Use mental strategies to add and subtract within 20.	**CC.2.2.3.A.2** Understand properties of multiplication and the relationship between multiplication and division. M03.B-O.2.1.1 M03.B-O.2.1.2 M03.B-O.2.2.1	**CC.2.2.4.A.2** Develop and/or apply number theory concepts to find factors and multiples. M04.B-O.2.1.1	*Intentionally Blank*				
Intentionally Blank	*Intentionally Blank*	*Intentionally Blank*	**CC.2.2.2.A.3** Work with equal groups of objects to gain foundations for multiplication.	**CC.2.2.3.A.3** Demonstrate multiplication and division fluency.	*Intentionally Blank*	*Intentionally Blank*				
Intentionally Blank	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*	**CC.2.2.3.A.4** Solve problems involving the four operations, and identify and explain patterns in arithmetic. M03.B-O.3.1.1 M03.B-O.3.1.2 M03.B-O.3.1.3 M03.B-O.3.1.4 M03.B-O.3.1.5 M03.B-O.3.1.6 M03.B-O.3.1.7	**CC.2.2.4.A.4** Generate and analyze patterns using one rule. M04.B-O.3.1.1 M04.B-O.3.1.2 M04.B-O.3.1.3	**CC.2.2.5.A.4** Analyze patterns and relationships using two rules. M05.B-O.2.1.1 M05.B-O.2.1.2				

2.3 Geometry							
The Standards of Mathematical Practices							
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.		
	Grade PreK 2.3.PreK	Grade K 2.3.K	Grade 1 2.3.1	Grade 2 2.3.2	Grade 3 2.3.3	Grade 4 2.3.4	Grade 5 2.3.5
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:							

(A) Geometry	**CC.2.3.PreK.A.1** Identify and describe shapes.	**CC.2.3.K.A.1** Identify and describe two- and three-dimensional shapes.	**CC.2.3.1.A.1** Compose and distinguish between two- and three-dimensional shapes based on their attributes.	**CC.2.3.2.A.1** Analyze and draw two- and three-dimensional shapes having specified attributes.	**CC.2.3.3.A.1** Identify, compare, and classify shapes and their attributes. M03.C-G.1.1.1 M03.C-G.1.1.2	**CC.2.3.4.A.1** Draw lines and angles and identify these in two-dimensional figures. M04.C-G.1.1.1	**CC.2.3.5.A.1** Graph points in the first quadrant on the coordinate plane and interpret these points when solving real world and mathematical problems.

									M05.C-G.1.1.1 M05.C-G.1.1.2
CC.2.3.PreK.A.2 Analyze, compare, create, and compose shapes.	**CC.2.3.K.A.2** Analyze, compare, create, and compose two- and three-dimensional shapes.	**CC.2.3.1.A.2** Use the understanding of fractions to partition shapes into halves and quarters.	**CC.2.3.2.A.2** Use the understanding of fractions to partition shapes into halves, quarters, and thirds.	**CC.2.3.3.A.2** Use the understanding of fractions to partition shapes into parts with equal areas and express the area of each part as a unit fraction of the whole. M03.C-G.1.1.3	**C.2.3.4.A.2** Classify two-dimensional figures by properties of their lines and angles. M04.C-G.1.1.2	**CC.2.3.5.A.2** Classify two-dimensional figures into categories based on an understanding of their properties. M05.C-G.2.1.1			
Intentionally Blank	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*	*Intentionally Blank*	**CC.2.3.4.A.3** Recognize symmetric shapes and draw lines of symmetry. M04.C-G.1.1.3	*Intentionally Blank*			

2.4 Measurement, Data, and Probability									
The Standards of Mathematical Practices									
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.				
	Grade PreK 2.4.PreK	Grade K 2.4.K	Grade 1 2.4.1	Grade 2 2.4.2	Grade 3 2.4.3	Grade 4 2.4.4	Grade 5 2.4.5		
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:									
(A) Measurement and Data			**CC.2.4.PreK.A.1** Describe and compare measurable attributes of length and weight of everyday objects.	**CC.2.4.K.A.1** Describe and compare attributes of length, area, weight, and capacity of everyday objects.	**CC.2.4.1.A.1** Order lengths and measure them both indirectly and by repeating length units.	**CC.2.4.2.A.1** Measure and estimate lengths in standard units using appropriate tools.	**CC.2.4.3.A.1** Solve problems involving measurement and estimation of temperature, liquid volume, mass, and length. M03.D-M.1.2.1 M03.D-M.1.2.2 M03.D-M.1.2.3	**CC.2.4.4.A.1** Solve problems involving measurement and conversions from a larger unit to a smaller unit. M04.D-M.1.1.1 M04.D-M.1.1.2 M04.D-M.1.1.3 M04.D-M.1.1.4	**CC.2.4.5.A.1** Solve problems using conversions within a given measurement system. M05.D-M.1.1.1
Intentionally Blank	*Intentionally Blank*	**CC.2.4.1.A.2** Tell and write time to the nearest half hour using both analog and digital clocks.	**CC.2.4.2.A.2** Tell and write time to the nearest five minutes using both analog and digital clocks.	**CC.2.4.3.A.2** Tell and write time to the nearest minute and solve problems by calculating time intervals. M03.D-M.1.1.1 M03.D-M.1.1.2	**CC.2.4.4.A.2** Translate information from one type of data display to another. M04.D-M.2.1.3	**CC.2.4.5.A.2** Represent and interpret data using appropriate scale. M05.D-M.2.1.2			
Intentionally Blank	*Intentionally Blank*	*Intentionally Blank*	**CC.2.4.2.A.3** Solve problems and make change using coins and	**CC.2.4.3.A.3** Solve problems and make change	*Intentionally Blank*	*Intentionally Blank*			

			paper currency with appropriate symbols.	involving money using a combination of coins and bills. M03.D-M.1.3.1 M03.D-M.1.3.2 M03.D-M.1.3.3					
(A) Measurement and Data			**CC.2.4.PreK.A.4** Classify objects and count the number of objects in each category.	**CC.2.4.K.A.4** Classify objects and count the number of objects in each category.	**CC.2.4.1.A.4** Represent and interpret data using tables/charts.	**CC.2.4.2.A.4** Represent and interpret data using line plots, picture graphs, and bar graphs.	**CC.2.4.3.A.4** Represent and interpret data using tally charts, tables, pictographs, line plots, and bar graphs. M03.D-M.2.1.1 M03.D-M.2.1.2 M03.D-M.2.1.3 M03.D-M.2.1.4	**CC.2.4.4.A.4** Represent and interpret data involving fractions using information provided in a line plot. M04.D-M.2.1.1 M04.D-M.2.1.2	**CC.2.4.5.A.4** Solve problems involving computation of fractions using information provided in a line plot. M05.D-M.2.1.1
Intentionally Blank		*Intentionally Blank*		*Intentionally Blank*		*Intentionally Blank*	**CC.2.4.3.A.5** Determine the area of a rectangle and apply the concept to multiplication and to addition. M03.D-M.3.1.1 M03.D-M.3.1.2	*Intentionally Blank*	**CC.2.4.5.A.5** Apply concepts of volume to solve problems and relate volume to multiplication and to addition. M05.D-M.3.1.1 M05.D-M.3.1.2
CC.2.4.2.A.6 Extend the concepts of addition and subtraction to problems involving length.	**CC.2.4.3.A.6** Solve problems involving perimeters of polygons and distinguish between linear and area measures. M03.D-M.4.1.1	**CC.2.4.4.A.6** Measure angles and use properties of adjacent angles to solve problems. M04.D-M.3.1.1 M04.D-M.3.1.2	*Intentionally Blank*						

2.1 Numbers and Operations					
The Standards of Mathematical Practices					
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.
	2.1.6 Grade 6	2.1.7 Grade 7	2.1.8 Grade 8		2.1.HS High School
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:					
(D) Ratios & Proportional Relationships	**CC.2.1.6.D.1** Understand ratio concepts and use ratio reasoning to	**CC.2.1.7.D.1** Analyze proportional relationships and	*Intentionally Blank*	**(F) Number and Quantity**	**CC.2.1.HS.F.1** Apply and extend the properties of exponents to solve

	solve problems. M06.A-R.1.1.1 M06.A-R.1.1.2 M06.A-R.1.1.3 M06.A-R.1.1.4 M06.A-R.1.1.5	use them to model and solve real-world and mathematical problems. M07.A-R.1.1.1 M07.A-R.1.1.2 M07.A-R.1.1.3 M07.A-R.1.1.4 M07.A-R.1.1.5 M07.A-R.1.1.6				problems with rational exponents. A1.1.1.1.1, A1.1.1.1.2, A1.1.1.3.1, A2.1.2.1.1, A2.1.2.1.2, A2.1.2.1.3, A2.1.2.1.4 **CC.2.1.HS.F.2** Apply properties of rational and irrational numbers to solve real world or mathematical problems. A1.1.1.1.1, A1.1.1.1.2, A1.1.1.3.1, A1.1.1.2.1 **CC.2.1.HS.F.3** Apply quantitative reasoning to choose and interpret units and scales in formulas, graphs, and data displays. A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A1.2.1.2.1, A1.2.1.2.2, A2.2.2.1.1, A2.2.2.1.2, A2.2.3.1.1, A2.2.3.1.2 **CC.2.1.HS.F.4** Use units as a way to understand problems and to guide the solution of multi-step problems. A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A1.2.1.2.1, A1.2.1.2.2, A2.2.2.1.1, A2.2.2.1.2
(E) The Number System			**CC.2.1.6.E.1** Apply and extend previous understandings of multiplication and division to divide fractions by fractions. M06.A-N.1.1.1	**CC.2.1.7.E.1** Apply and extend previous understandings of operations with fractions to operations with rational numbers. M07.A-N.1.1.1 M07.A-N.1.1.2 M07.A-N.1.1.3	**CC.2.1.8.E.1** Distinguish between rational and irrational numbers using their properties. M08.A-N.1.1.1 M08.A-N.1.1.2 A1.1.1.1.1 A1.1.1.1.2	
CC.2.1.6.E.2 Identify and choose appropriate processes to compute fluently with multi-digit numbers. M06.A-N.2.1.1	*Intentionally Blank*		*Intentionally Blank*		**CC.2.1.HS.F.5** Choose a level of accuracy appropriate to limitations on measurement when reporting quantities. A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A1.1.2.2.1, A1.1.2.2.2, A1.1.3.1.1, A1.1.3.1.2, A1.1.3.1.3, A1.1.3.2.1, A1.1.3.2.2, A2.2.3.1.1, A2.2.3.1.2 **CC.2.1.HS.F.6** Extend the knowledge of arithmetic operations and apply to complex numbers. A2.1.1.1.1, A2.1.1.1.2, A2.1.1.2.1, A2.1.1.2.2 **CC.2.1.HS.F.7** Apply concepts of complex numbers in polynomial identities and quadratic equations to solve problems. A2.2.1.1.1, A2.2.1.1.2, A2.2.1.1.3, A2.2.1.1.4	
CC.2.1.6.E.3 Develop and/or apply number theory concepts to						

find common factors and multiples. M06.A-N.2.2.1 M06.A-N.2.2.2 A1.1.1.2.1				

2.1 Numbers and Operations			
The Standards of Mathematical Practices			
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.			Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.
2.1.6 Grade 6	2.1.7 Grade 7	2.1.8 Grade 8	2.1.HS High School
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
CC.2.1.6.E.4 Apply and extend previous understandings of numbers to the system of rational numbers. M06.A-N.3.1.1 M06.A-N.3.1.2 M06.A-N.3.1.3 M06.A-N.3.2.1 M06.A-N.3.2.2 M06.A-N.3.2.3	*Intentionally Blank*	**CC.2.1.8.E.4** Estimate irrational numbers by comparing them to rational numbers. M08.A-N.1.1.3 M08.A-N.1.1.4 M08.A-N.1.1.5 A1.1.1.1.1	

2.2 Algebraic Concepts					
The Standards of Mathematical Practices					
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.
	2.2.6 Grade 6	2.2.7 Grade 7	2.2.8 Grade 8		2.2.HS High School
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:					
(B) Expressions and Equations	**CC.2.2.6.B.1** Apply and extend previous understandings of arithmetic to algebraic expressions. M06.B-E.1.1.1 M06.B-E.1.1.2 M06.B-E.1.1.3 M06.B-E.1.1.4 M06.B-E.1.1.5	**CC.2.2.7.B.1** Apply properties of operations to generate equivalent expressions. M07.B-E.1.1.1	**CC.2.2.8.B.1** Apply concepts of radicals and integer exponents to generate equivalent expressions. M08.B-E.1.1.1 M08.B-E.1.1.2 M08.B-E.1.1.3 M08.B-E.1.1.4 A1.1.1.3.1	**(D) Algebra**	**CC.2.2.HS.D.1** Interpret the structure of expressions to represent a quantity in terms of its context. A1.1.1.5.1, A1.1.1.5.2, A1.1.1.5.3, A2.1.2.2.1, A2.1.2.2.2 **CC.2.2.HS.D.2** Write expressions in equivalent forms to solve problems. A1.1.1.5.1, A1.1.1.5.2, A1.1.1.5.3, A2.1.2.1.1, A2.1.2.1.2, A2.1.2.1.3, A2.1.2.1.4, A2.1.2.2.1, A2.1.2.2.2 **CC.2.2.HS.D.3** Extend the knowledge of arithmetic operations and apply to

						polynomials. A1.1.1.5.1, A1.1.1.5.2, A1.1.1.5.3, A2.1.2.2.1, A2.1.2.2.2
CC.2.2.6.B.2 Understand the process of solving a one-variable equation or inequality and apply it to real-world and mathematical problems. M06.B-E.2.1.1 M06.B-E.2.1.2 M06.B-E.2.1.3 M06.B-E.2.1.4	*Intentionally Blank*	**CC.2.2.8.B.2** Understand the connections between proportional relationships, lines, and linear equations. M08.B-E.2.1.1 M08.B-E.2.1.2 M08.B-E.2.1.3 A1.2.1.2.2		**CC.2.2.HS.D.4** Understand the relationship between zeros and factors of polynomials to make generalizations about functions and their graphs. A2.1.2.2.1, A2.1.2.2.2 **CC.2.2.HS.D.5** Use polynomial identities to solve problems. A1.1.1.5.1, A1.1.1.5.2, A1.1.1.5.3, A2.1.2.2.1, A2.1.2.2.2, A2.1.3.1.1, A2.1.3.1.2, A2.1.3.1.3, A2.1.3.1.4 **CC.2.2.HS.D.6** Extend the knowledge of rational functions to rewrite in equivalent forms. A1.1.1.5.1, A1.1.1.5.2, A1.1.1.5.3, A2.1.3.1.1, A2.1.3.1.2, A2.1.3.1.3, A2.1.3.1.4 **CC.2.2.HS.D.7** Create and graph equations or inequalities to describe numbers or relationships. A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A1.1.2.2.1, A1.1.2.2.2, A1.1.3.1.1, A1.1.3.1.2, A1.1.3.1.3, A1.1.3.2.1, A1.1.3.2.2, A2.1.3.1.1, A2.1.3.1.2, A2.1.3.1.3, A2.1.3.1.4, A2.1.3.2.1, A2.1.3.2.2, A2.2.2.1.1, A2.2.2.1.2, A2.2.2.1.3, A2.2.2.1.4 **CC.2.2.HS.D.8** Apply inverse operations to solve equations or formulas for a given variable. A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A2.1.3.1.1, A2.1.3.1.2,		

				A2.1.3.1.3, A2.1.3.1.4, A2.1.3.2.1, A2.1.3.2.2			
(B) Expressions and Equations	**CC.2.2.6.B.3** Represent and analyze quantitative relationships between dependent and independent variables. M06.B-E.3.1.1 M06.B-E.3.1.2	**CC.2.2.7.B.3** Model and solve real-world and mathematical problems by using and connecting numerical, algebraic, and/or graphical representations. M07.B-E.2.1.1 M07.B-E.2.2.1 M07.B-E.2.2.2 M07.B-E.2.3.1 A1.1.1.4.1	**CC.2.2.8.B.3** Analyze and solve linear equations and pairs of simultaneous linear equations. M08.B-E.3.1.1 M08.B-E.3.1.2 M08.B-E.3.1.3 M08.B-E.3.1.4 M08.B-E.3.1.5 A1.1.2.1.1 A1.1.2.2.1 A1.1.2.2.2	**(D) Algebra**	**CC.2.2.HS.D.9** Use reasoning to solve equations and justify the solution method. A1.1.1.4.1, A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A1.1.2.2.1, A1.1.2.2.2, A1.1.3.1.1, A1.1.3.1.2, A1.1.3.1.3, A2.1.3.1.1, A2.1.3.1.2, A2.1.3.1.3, A2.1.3.1.4, A2.1.3.2.1, A2.1.3.2.2 **CC.2.2.HS.D.10** Represent, solve, and interpret equations/inequalities and systems of equations/inequalities algebraically and graphically. A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A1.1.2.2.1, A1.1.2.2.2, A1.1.3.1.1, A1.1.3.1.2, A1.1.3.1.3, A1.1.3.2.1, A1.1.3.2.2, A2.1.3.1.1, A2.1.3.1.2, A2.1.3.1.3, A2.1.3.1.4		
(C) Functions	*Intentionally Blank*		*Intentionally Blank*		**CC.2.2.8.C.1** Define, evaluate, and compare functions. M08.B-F.1.1.1 M08.B-F.1.1.2 M08.B-F.1.1.3 A1.1.2.1.1 A1.2.1.1.2 A1.2.1.2.1 A1.2.1.2.2 **CC.2.2.8.C.2** Use concepts of functions to model relationships between quantities. M08.B-F.2.1.1 M08.B-F.2.1.2 A1.1.2.1.3 A1.2.1.1.1 A1.2.1.2.2 A1.2.2.1.3 A1.2.2.1.4	**(C) Functions**	**CC.2.2.HS.C.1** Use the concept and notation of functions to interpret and apply them in terms of their context. A1.2.1.1.1, A1.2.1.1.2, A1.2.1.1.3, A1.2.2.1.1, A1.2.2.1.2, A1.2.2.1.3, A1.2.2.1.4, A2.2.1.1.1, A2.2.1.1.2, A2.2.1.1.3, A2.2.1.1.4, G.2.2.2.1, G.2.2.2.2, G.2.2.2.3, G.2.2.2.4, G.2.2.2.5 **CC.2.2.HS.C.2** Graph and analyze functions and use their properties to make connections between the different representations. A1.2.1.1.1, A1.2.1.1.2, A1.2.1.1.3, A1.2.1.2.1, A1.2.1.2.2, A1.2.2.1.1, A2.1.3.1.4, A2.1.3.2.1, A2.1.3.2.2, A2.2.1.1.1, A2.2.1.1.2, A2.2.1.1.3, A2.2.1.1.4 **CC.2.2.HS.C.3** Write functions or sequences that model relationships between two quantities. A1.1.2.1.1, A1.1.2.1.2, A1.1.2.1.3, A1.2.1.1.1, A1.2.1.1.2, A1.2.1.1.3,

						A1.2.1.2.1, A1.2.1.2.2, A1.2.2.1.3, A1.2.2.1.4, A2.1.3.1.1, A2.1.3.1.2, A2.1.3.1.3, A2.1.3.1.4, A2.1.3.2.1, A2.1.3.2.2, A2.2.1.1.1, A2.2.1.1.2, A2.2.1.1.3, A2.2.1.1.4, A2.2.2.1.1, A2.2.2.1.2, A2.2.2.1.3, A2.2.2.1.4 **CC.2.2.HS.C.4** Interpret the effects transformations have on functions and find the inverses of functions. A1.2.1.2.1, A1.2.1.2.2, A2.1.3.1.3, A2.1.3.1.4, A2.1.3.2.1, A2.2.2.1.1, A2.2.2.1.2, A2.2.2.1.3, A2.2.2.1.4, A2.2.2.2.1 **CC.2.2.HS.C.5** Construct and compare linear, quadratic, and exponential models to solve problems. A1.2.2.1.1, A1.2.2.1.2, A1.2.2.1.3, A1.2.2.1.4, A2.1.3.1.1, A2.1.3.1.2, A2.1.3.1.3, A2.1.3.1.4, A2.2.1.1.1, A2.2.1.1.2, A2.2.1.1.3, A2.2.1.1.4, A2.2.2.1.1, A2.2.2.1.2, A2.2.2.1.3, A2.2.2.1.4, A2.2.2.2.1 **CC.2.2.HS.C.6** Interpret functions in terms of the situations they model. A1.2.1.2.1, A1.2.2.1.2, A1.2.2.1.3, A1.2.2.2.1, A2.1.3.1.3, A2.2.1.1.1, A2.2.1.1.2, A2.2.1.1.3, A2.2.1.1.4, A2.2.2.1.3, A2.2.2.1.4, A2.2.2.2.1 **CC.2.2.HS.C.7** Apply radian measure of an angle and the unit circle to analyze the trigonometric functions. **CC.2.2.HS.C.8**

					Choose trigonometric functions to model periodic phenomena and describe the properties of the graphs. **CC.2.2.HS.C.9** Prove the Pythagorean identity and use it to calculate trigonometric ratios. G.1.3.2.1, G.2.1.1.1, G.2.1.1.2

2.3 Geometry						
The Standards of Mathematical Practices						
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.	
	2.3.6 Grade 6	2.3.7 Grade 7	2.3.8 Grade 8		2.3.HS High School	
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:						
(A) Geometry	**CC.2.3.6.A.1** Apply appropriate tools to solve real-world and mathematical problems involving area, surface area, and volume. M06.C-G.1.1.1 M06.C-G.1.1.2 M06.C-G.1.1.3 M06.C-G.1.1.4 M06.C-G.1.1.5 M06.C-G.1.1.6	**CC.2.3.7.A.1** Solve real-world and mathematical problems involving angle measure, area, surface area, circumference, and volume. M07.C-G.2.1.1 M07.C-G.2.1.2 M07.C-G.2.2.1 M07.C-G.2.2.2	**CC.2.3.8.A.1** Apply the concepts of volume of cylinders, cones, and spheres to solve real-world and mathematical problems. M08.C-G.3.1.1 G.2.3.1.2	**(A) Geometry**	**CC.2.3.HS.A.1** Use geometric figures and their properties to represent transformations in the plane. G.1.3.1.1, G.1.3.1.2 **CC.2.3.HS.A.2** Apply rigid transformations to determine and explain congruence. G.1.3.1.1, G.1.3.1.2 **CC.2.3.HS.A.3** Verify and apply geometric theorems as they relate to geometric figures. G.1.2.1.1, G.1.2.1.2, G.1.2.1.3, G.1.2.1.4, G.1.2.1.5, G.1.3.2.1, G.2.2.1.1, G.2.2.1.2, G.2.2.2.1, G.2.2.2.2, G.2.2.2.3, G.2.2.2.4, G.2.2.2.5 **CC.2.3.HS.A.4** Apply the concept of congruence to create geometric constructions. **CC.2.3.HS.A.5** Create justifications based on transformations to establish similarity of plane figures. G.1.3.1.1, G.1.3.1.2 **CC.2.3.HS.A.6** Verify and apply theorems involving similarity as they relate to plane figures. G.1.3.1.1, G.1.3.1.2, G.1.3.2.1 **CC.2.3.HS.A.7** Apply trigonometric ratios to solve problems involving right triangles. G.2.1.1.1, G.2.1.1.2 **CC.2.3.HS.A.8** Apply geometric theorems to verify properties of circles. G.1.1.1.1, G.1.1.1.2,

						G.1.1.1.3, G.1.1.1.4, G.1.3.2.1, G.2.2.3.1 **CC.2.3.HS.A.9** Extend the concept of similarity to determine arc lengths and areas of sectors of circles. G.1.1.1.1, G.1.1.1.2, G.1.1.1.3, G.1.1.1.4, G.2.2.2.1, G.2.2.2.2, G.2.2.2.3, G.2.2.2.4, G.2.2.2.5, G.2.2.3.1
Intentionally Blank	**CC.2.3.7.A.2** Visualize and represent geometric figures and describe the relationships between them. M07.C-G.1.1.1 M07.C-G.1.1.2 M07.C-G.1.1.3 M07.C-G.1.1.4	**CC.2.3.8.A.2** Understand and apply congruence, similarity, and geometric transformations using various tools. M08.C-G.1.1.1 M08.C-G.1.1.2 M08.C-G.1.1.3 M08.C-G.1.1.4 G.1.2.1.1 G.1.2.1.4 G.2.2.1.1				
(A) Geometry	*Intentionally Blank*	*Intentionally Blank*	**CC.2.3.8.A.3** Understand and apply the Pythagorean Theorem to solve problems. M08.C-G.2.1.1 M08.C-G.2.1.2 M08.C-G.2.1.3 G.2.1.1.1 G.2.1.2.1	**(A) Geometry**	**CC.2.3.HS.A.10** Translate between the geometric description and the equation for a conic section. A2.2.1.1.4, A2.2.2.1.1 **CC.2.3.HS.A.11** Apply coordinate geometry to prove simple geometric theorems algebraically. G.2.1.2.1, G.2.1.2.2, G.2.1.2.3 **CC.2.3.HS.A.12** Explain volume formulas and use them to solve problems. G.2.3.1.1, G.2.3.1.2, G.2.3.1.3 **CC.2.3.HS.A.13** Analyze relationships between two-dimensional and three-dimensional objects. G.1.1.1.1, G.1.1.1.2, G.1.1.1.3, G.1.1.1.4, G.1.2.1.1, G.1.2.1.2, G.1.2.1.3, G.1.2.1.4, G.1.2.1.5, G.2.3.2.1 **C.2.3.HS.A.14** Apply geometric concepts to model and solve real world problems. G.2.2.4.1, G.2.3.1.1, G.2.3.1.2, G.2.3.1.3	

2.4 Measurement, Data, and Probability					
The Standards of Mathematical Practices					
Make sense of problems and persevere in solving them. Construct viable arguments and critique the reasoning of others. Use appropriate tools strategically. Look for and make use of structure.					Reason abstractly and quantitatively. Model with mathematics. Attend to precision. Look for and express regularity in repeated reasoning.
	2.4.6 Grade 6	2.4.7 Grade 7	2.4.8 Grade 8		2.4.HS High School
Pennsylvania's public schools shall teach, challenge, and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:					

(B) Statistics and Probability			**CC.2.4.6.B.1** Demonstrate an understanding of statistical variability by displaying, analyzing, and summarizing distributions. M06.D-S.1.1.1 M06.D-S.1.1.2 M06.D-S.1.1.3 M06.D-S.1.1.4	**CC.2.4.7.B.1** Draw inferences about populations based on random sampling concepts. M07.D-S.1.1.1 M07.D-S.1.1.2	**CC.2.4.8.B.1** Analyze and/or interpret bivariate data displayed in multiple representations. M08.D-S.1.1.1 M08.D-S.1.1.2 M08.D-S.1.1.3 A1.2.2.2.1	**(B) Statistics and Probability**	**CC.2.4.HS.B.1** Summarize, represent, and interpret data on a single count or measurement variable. A1.2.2.1.2, A1.2.3.1.1, A1.2.3.2.1, A1.2.3.2.2, A1.2.3.2.3, **CC.2.4.HS.B.2** Summarize, represent, and interpret data on two categorical and quantitative variables. A1.2.1.1.1, A1.2.1.1.2, A1.2.1.1.3, A1.2.1.2.1, A1.2.1.2.2, A1.2.2.2.1, A2.2.1.1.1, A2.2.3.1.1, A2.2.3.1.2 **CC.2.4.HS.B.3** Analyze linear models to make interpretations based on the data. A1.2.2.2.1, A1.2.3.1.1, A1.2.3.2.1, A1.2.3.2.2, A1.2.3.2.3, A2.2.3.1.1, A2.2.3.1.2 **CC.2.4.HS.B.4** Recognize and evaluate random processes underlying statistical experiments. A1.2.3.3.1, A2.2.3.2.1, A2.2.3.2.2, A2.2.3.2.3 **CC.2.4.HS.B.5** Make inferences and justify conclusions based on sample surveys, experiments, and observational studies. A1.2.3.2.1, A1.2.3.2.2, A1.2.3.2.3, A2.2.3.2.1, A2.2.3.2.2, A2.2.3.2.3 **CC.2.4.HS.B.6** Use the concepts of independence and conditional probability to interpret data. A2.2.3.2.1, A2.2.3.2.2, A2.2.3.2.3 **CC.2.4.HS.B.7** Apply the rules of probability to compute probabilities of compound events in a uniform probability model. A1.2.3.3.1, A2.2.3.2.1, A2.2.3.2.2, A2.2.3.2.3
Intentionally Blank	**CC.2.4.7.B.2** Draw informal comparative inferences about two populations. M07.D-S.2.1.1	**CC.2.4.8.B.2** Understand that patterns of association can be seen in bivariate data utilizing frequencies. M08.D-S.1.2.1					
Intentionally Blank	**CC.2.4.7.B.3** Investigate chance processes and develop, use, and evaluate probability models. M07.D-S.3.1.1 M07.D-S.3.2.1 M07.D-S.3.2.2	*Intentionally Blank*					

	M07.D-S.3.2.3 A1.2.3.3.1	

Key Terms for this Document

Standards for Mathematical Contents—These standards define what students should know and be able to do in their study of mathematics.

Standards for Mathematical Practice—These standards describe the processes and proficiencies in which all students grades K-12 should engage. Educators must instill these standards of practice in their students so that they become habitual. The standards for mathematical practice should be used as the vehicle to deliver the standards of mathematical content.

Standard Algorithm—A locally agreed upon method of computation which is conventionally taught for solving mathematical problems.

Decimal Fraction—A fraction whose denominator is a power of ten (examples: 2/100, 8/10). These fractions are commonly expressed as decimals.

Unit Fraction—A rational number written as a fraction where the numerator is one and the denominator is a positive integer (example: 1/20).

Bivariate Data—The data involves two variables and is usually represented as a scatter plot.

Rule—A single operation (examples: add 5, multiply by 2).

APPENDIX B

{Expired}

Authority

The provisions of this Appendix B amended under sections 121, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix B adopted January 4, 2002, effective January 5, 2002, 32 Pa.B. 17; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; corrected March 21, 2014, effective March 1, 2014, 44 Pa.B. 1754, expired July 1, 2025, 55 Pa.B. 1234, unless otherwise noted. Immediately preceding text appears at serial pages (422310) to (422312), (286565) to (286596), (371219) to (371224), (286599) to (286602), (371225) to (371232), (286605) to (286652) and (422313).

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards); 22 Pa. Code § 4.24 (relating to high school graduation requirements); 22 Pa. Code § 4.51 (relating to State assessment system); 22 Pa. Code § 4.51a (relating to Pennsylvania System of School Assessment); and 22 Pa. Code § 4.51b (relating to Keystone Exams).

APPENDIX B-1

Pennsylvania Integrated Standards for Science, Environment, Ecology, Technology and Engineering (Grades K—5)[1, 2], Pennsylvania Integrated Standards for Science, Environment and Ecology (Grades 6—12), and Pennsylvania Technology and Engineering Standards (Grades 6—12)

Authority

The provisions of this Appendix B-1 added under sections 2603-B and 2304-B of the Public School Code of 1949 (24 P.S. § § 26-2603-B and 26-2604-B); amended under sections 290.1, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 2-290.1, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix B-1 added July 15, 2022, effective July 16, 2022, 52 Pa.B. 3946; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234. Immediately preceding text appears at serial pages (410373) to (410396).

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards); 22 Pa. Code § 4.24 (relating to high school graduation requirements); 22 Pa. Code § 4.51 (relating to State assessment system); 22 Pa. Code § 4.51a (relating to Pennsylvania System of School Assessment); and 22 Pa. Code § 4.51b (relating to Keystone Exams).

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[1] Across grades K—5, all of the core ideas in Table 1 are covered, but not every discipline or core idea is reflected at every grade.

[2] The language of the standards is adapted, informed by or taken from the: National Research Council. (2012). A framework for K—12 science education: Practices, crosscutting concepts, and core ideas. North American Association for Environmental Education (2019) K—12 environmental education: Guidelines for excellence; International Society for Technology in Education Standards. (2019). ISTE standards for students; International Technology and Engineering Educators Association (ITEEA) (2020); NGSS Lead States. (2013). Next generation science standards: For states, by states; Standards for technological and engineering literacy: The role of technology and engineering in STEM education. National Council for Agricultural Education. (2015); International Society for Technology in Education. (2019). ISTE Standards for students. Agriculture, food and natural resources (AFNR) career cluster content standards; Pennsylvania State Board of Education. (2002). Academic standards for science and technology; Pennsylvania Department of Education. (2002). Safety guidelines for elementary and technology education teachers; Pennsylvania Department of Education. (n.d.). Pennsylvania career ready skills continuum; Standards for Technological and Engineering Literacy. (2020); Pennsylvania Association for Environmental Educators. (September 2015). Pennsylvania environmental literacy plan Pennsylvania State Board of Education. (2002). Academic standards for environment and ecology. North American Association for Environmental Education. (2014). State environmental literacy plans: 2014 status report.

Kindergarten

Earth and Space Sciences

Earth and Human Activity

1. Use a model to represent the relationship between the needs of different plants or animals (including humans) and the places they live.

2. Ask questions to obtain information about the purpose of weather forecasting to prepare for, and respond to, severe weather.

3. Communicate solutions that will reduce the impact of humans on the land, water, air, and/or other living things in the local environment.

Earth's Systems

1. Use and share observations of local weather conditions to describe patterns over time.

2. Construct an argument supported by evidence for how plants and animals (including humans) can change the environment to meet their needs.

Life Science

From Molecules to Organisms: Structures and Processes

1. Use observations to describe patterns of what plants and animals (including humans) need to survive.

Physical Science

Motion and Stability: Forces and Interactions

1. Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object.

2. Analyze data to determine if a design solution works as intended to change the speed or direction of an object with a push or a pull.

Energy

1. Make observations to determine the effect of sunlight on Earth's surface.

2. Use tools and materials to design and build a structure that will reduce the warming effect of sunlight on an area.

Grade 1

Earth and Space Sciences

Earth's Place in the Universe

1. Use observations of the sun, moon, and stars to describe patterns that can be predicted.

2. Make observations at different times of year to relate the amount of daylight to the time of year.

Life Science

From Molecules to Organisms: Structures and Processes

1. Use materials to design a solution to a human problem by mimicking how plants and/or animals use their external parts to help them survive, grow, and meet their needs.

2. Read texts and use media to determine patterns in behavior of parents and offspring that help offspring survive.

Heredity: Inheritance and Variation of Traits

1. Make observations to construct an evidence-based account that young plants and animals are like, but not exactly like, their parents.

Physical Science

Waves and Their Applications in Technologies for Information Transfer

1. Plan and conduct investigations to provide evidence that vibrating materials can make sound and that sound can make materials vibrate.

2. Make observations to construct an evidence-based account that objects can be seen only when illuminated.

3. Plan and conduct an investigation to determine the effect of placing objects made with different materials in the path of a beam of light.

4. Use tools and materials to design and build a device that uses light or sound to solve the problem of communicating over a distance.

Grade 2

Earth and Space Sciences

Earth's Place in the Universe

1. Use information from several sources to provide evidence that Earth events can occur quickly or slowly.

Earth's Systems

1. Compare multiple solutions designed to slow or prevent wind or water from changing the shape of the land.

2. Develop a model to represent the shapes and kinds of land and bodies of water in an area.

3. Obtain information to identify where water is found on Earth and that it can be solid or liquid.

Life Science

Ecosystems: Interactions, Energy, and Dynamics

1. Plan and conduct an investigation to determine if plants need sunlight and water to grow.

2. Develop a simple model that mimics the function of an animal in dispersing seeds or pollinating plants.

Biological Evolution: Unity and Diversity

1. Make observations of plants and animals to compare the diversity of life in different habitats.

Matter and its Interactions

1. Plan and conduct an investigation to describe and classify different kinds of materials by their observable properties.

2. Analyze data obtained from testing different materials to determine which materials have the properties that are best suited for an intended purpose.

3. Make observations to construct an evidence-based account of how an object made of a small set of pieces can be disassembled and made into a new object.

4. Construct an argument with evidence that some changes caused by heating or cooling can be reversed and some cannot.

Grade 3

Earth and Space Sciences

Earth's Systems

1. Represent data in tables and graphical displays to describe typical weather conditions expected during a particular season.

2. Obtain and combine information to describe climates in different regions of the world.

Earth and Human Activity

1. Make a claim supported by evidence about the merit of a design solution that reduces the impacts of a weather-related hazard.

Life Science

From Molecules to Organisms: Structures and Processes

1. Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

Ecosystems: Interactions, Energy, and Dynamics

1. Construct an argument that some animals form groups that help members survive.

Heredity: Inheritance and Variation of Traits

1. Analyze and interpret data to provide evidence that plants and animals have traits inherited from parents and that variation of these traits exists in a group of similar organisms.

2. Use evidence to support the explanation that traits can be influenced by the environment.

Biological Evolution: Unity and Diversity

1. Analyze and interpret data from fossils to provide evidence of the organisms and the environments in which they lived long ago.

2. Use evidence to construct an explanation for how the variations in characteristics among individuals of the same species may provide advantages in surviving, finding mates, and reproducing.

3. Construct an argument with evidence that in a particular habitat some organisms can survive well, some survive less well, and some cannot survive at all.

4. Make a claim supported by evidence about the merit of a solution to a problem caused when the environment changes and the types of plants and animals that live there may change.

Physical Science

Motion and Stability: Forces and Interactions

1. Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.

2. Make and communicate observations and/or measurements of an object's motion to provide evidence that a pattern can be used to predict future motion.

3. Ask questions to determine cause and effect relationships of electric or magnetic interactions between two objects not in contact with each other.

4. Define a simple design problem that can be solved by applying scientific ideas about magnets.

Grade 4

Earth and Space Sciences

Earth's Place in the Universe

1. Identify evidence from patterns in rock formations and fossils in rock layers to support an explanation for changes in a landscape over time.

Earth's Systems

1. Make observations and/or measurements to provide evidence of the effects of weathering or the rate of erosion by water, ice, wind, or vegetation.

2. Analyze and interpret data from maps to describe patterns of Earth's features.

Earth and Human Activity

1. Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the environment.

2. Generate and compare multiple solutions to reduce the impacts of natural Earth processes on humans.

Life Science

From Molecules to Organisms: Structures and Processes

1. Construct an argument that plants and animals have internal and external structures that function to support survival, growth, behavior, and reproduction.

2. Use a model to describe that animals receive different types of information through their senses, process the information in their brain, and respond to the information in different ways.

Physical Science

Waves and Their Applications in Technologies for Information Transfer

1. Develop a model of waves to describe patterns in terms of amplitude and wavelength and that waves can cause objects to move.

2. Develop a model to describe that light reflecting from objects and entering the eye allows objects to be seen.

3. Generate and compare multiple solutions that use patterns to transfer information.

Energy

1. Use evidence to construct an explanation relating the speed of an object to the energy of that object.

2. Make and communicate observations to provide evidence that energy can be transferred from place to place by sound, light, heat, and electric currents.

3. Ask questions and predict outcomes about the changes in energy that occur when objects collide.

4. Apply scientific ideas to design, test, and refine a device that converts energy from one form to another.

Grade 5

Earth and Space Sciences

Earth's Place in the Universe

1. Support an argument that differences in the apparent brightness of the sun compared to other stars is due to their relative distances from Earth.

2. Represent data in graphical displays to reveal patterns of daily changes in length and direction of shadows, day and night, and the seasonal appearance of some stars in the night sky.

Earth's Systems

1. Develop a model using an example to describe ways the geosphere, biosphere, hydrosphere, and/or atmosphere interact.

2. Describe and graph the amounts and percentages of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth.

Earth and Human Activity

1. Obtain and combine information about ways individual communities use science ideas to protect the Earth's resources and environment.

2. Generate and design possible solutions to a current environmental issue, threat, or concern.

Life Science

From Molecules to Organisms: Structures and Processes

1. Support an argument that plants get the materials they need for growth chiefly from air and water.

Ecosystems: Interactions, Energy, and Dynamics

1. Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment.

Physical Science

Matter and Its Interactions

1. Develop a model to describe that matter is made of particles too small to be seen.

2. Measure and graph quantities to provide evidence that regardless of the type of change that occurs when heating, cooling, or mixing substances, the total weight of matter is conserved.

3. Make and communicate observations and measurements to identify materials based on their properties.

4. Conduct an investigation to determine whether the mixing of two or more substances results in new substances.

5. Interpret and analyze data to make decisions about how to utilize materials based on their properties.

Motion and Stability: Forces and Interactions

1. Support an argument that the gravitational force exerted by Earth on objects is directed down.

Energy

1. Use models to describe that energy in animals' food (used for body repair, growth, motion, and to maintain body warmth) was once energy from the sun.

Standards by Grade Band

Grades K—2: Environmental Literacy and Sustainability

Agricultural and Environmental Systems and Resources

1. Examine how people from different cultures and communities, including one's own, interact and express their beliefs about nature.

2. Categorize ways people harvest, re-distribute, and use natural resources.

Environmental Literacy Skills

1. Explain ways that places differ in their physical characteristics, their meaning, and their value and/or importance.

2. Plan and carry out an investigation to address an issue in their local environment and community.

Grades K—2: Technology and Engineering

Applying, Maintaining, and Assessing Technological Products and Systems

1. Analyze how things work.

2. Identify and use everyday symbols.

3. Describe qualities of everyday products.

Core Concepts of Technology and Engineering

1. Illustrate how systems have parts or components that work together to accomplish a goal.

2. Safely use tools to complete tasks.

3. Explain that materials are selected for use because they possess desirable properties and characteristics.

4. Develop a plan in order to complete a task.

5. Collaborate effectively as a member of a team.

Design in Technology and Engineering Education

1. Apply design concepts, principles, and processes through play and exploration.

2. Demonstrate that designs have requirements.

3. Explain that design is a response to wants and needs.

4. Discuss that all designs have different characteristics that can be described.

5. Illustrate that there are different solutions to a design and that none are perfect.

6. Demonstrate essential skills of the engineering design process.

7. Apply skills necessary for making a design.

History of Technology

1. Discuss how the way people live and work has changed throughout history because of technology.

Impacts of Technology

1. Explain ways that technology helps with everyday tasks.

2. Illustrate helpful and harmful effects of technology.

3. Compare simple technologies to evaluate their impacts.

4. Select ways to reduce, reuse, and recycle resources in daily life.

5. Design new technologies that could improve their daily lives.

Influence of Society on Technological Development

1. Explain the needs and wants of individuals and societies.

2. Explore how technologies are developed to meet individual and societal needs and wants.

3. Investigate the use of technologies in the home and community.

Integration of Knowledge, Technologies, and Practices

1. Apply concepts and skills from technology and engineering activities that reinforce concepts and skills across multiple content areas.

2. Draw connections between technology and human experiences.

Nature and Characteristics of Technology and Engineering

1. Compare the natural world and human-made world.

2. Explain the tools and techniques that people use to help them do things.

3. Demonstrate that creating can be done by anyone.

4. Discuss the roles of scientists, engineers, technologists and others who work with technology.

Grades 3—5: Environmental Literacy and Sustainability

Agricultural and Environmental Systems and Resources

1. Analyze how living organisms, including humans, affect the environment in which they live, and how their environment affects them.

2. Make a claim about the environmental and social impacts of design solutions and civic actions, including their own actions.

Environmental Literacy Skills

1. Investigate how perspectives over the use of resources and the development of technology have changed over time and resulted in conflict over the development of societies and nations.

2. Develop a model to demonstrate how local environmental issues are connected to larger local environment and human systems.

Sustainability and Stewardship

1. Critique ways that people depend on and change the environment.

2. Examine ways you influence your local environment and community by collecting and displaying data.

3. Construct an argument to support whether action is needed on a selected environmental issue and propose possible solutions.

Grades 3—5: Technology and Engineering

Applying, Maintaining, and Assessing Technological Products and Systems

1. Follow directions to complete a technological task.

2. Use appropriate symbols, numbers and words to communicate key ideas about technological products and systems.

3. Identify why a product or system is not working properly.

4. Examine information to assess the trade-offs of using a product or system.

Core Concepts of Technology and Engineering

1. Describe how a subsystem is a system that operates as a part of another larger system.

2. Illustrate how, when parts of a system are missing, it may not work as planned.

3. Identify the resources needed to get a technical job done, such as people, materials, capital, tools, machines, knowledge, energy, and time.

4. Describe the properties of different materials.

5. Demonstrate how tools and machines extend human capabilities, such as holding, lifting, carrying, fastening, separating, and computing.

6. Describe requirements of designing or making a product or system.

7. Create a new product that improves someone's life.

Design in Technology and Engineering Education

1. Illustrate that there are multiple approaches to design.

2. Demonstrate essential skills of the engineering design process.

3. Evaluate designs based on criteria, constraints, and standards.

4. Interpret how good design improves the human condition.

5. Apply universal principles and elements of design.

6. Evaluate the strengths and weaknesses of existing design solutions, including their own solutions.

7. Practice successful design skills.

8. Apply tools, techniques, and materials in a safe manner as part of the design process.

History of Technology

1. Create representations of the tools people made, how they cultivated to provide food, made clothing, and built shelters to protect themselves.

Impacts of Technology

1. Describe the helpful and harmful effects of technology.

2. Judge technologies to determine the best one to use to complete a given task or meet a need.

3. Classify resources used to create technologies as either renewable or nonrenewable.

4. Explain why responsible use of technology requires sustainable management of resources.

5. Predict how certain aspects of their daily lives would be different without given technologies.

Influence of Society on Technological Development

1. Determine factors that influence changes in a society's technological systems or infrastructure.

2. Explain how technologies are developed or adapted when individual or societal needs and wants change.

Integration of Knowledge, Technologies, and Practices

1. Demonstrate how simple technologies are often combined to form more complex systems.

2. Explain how various relationships can exist between technology and engineering and other content areas.

Nature and Characteristics of Technology and Engineering

1. Compare how things found in nature differ from things that are human-made, noting differences and similarities in how they are produced and used.

2. Describe the unique relationship between science and technology, and how the natural world can contribute to the human-made world to foster innovation.

3. Differentiate between the role of scientists, engineers, technologists, and others in creating and maintaining technological systems.

4. Design solutions by safely using tools, materials, and skills.

5. Explain how solutions to problems are shaped by economic, political, and cultural forces.

Pennsylvania Integrated Standards for Science, Environment and Ecology (Grades 6—12)

Standards for Grades 6—8[3, 4]

Physical Science

Structure and Properties of Matter

1. Develop models to describe the atomic composition of simple molecules and extended structures.

2. Gather and make sense of information to describe how synthetic materials come from natural resources and impact society.

3. Develop a model that predicts and describes changes in the particle motion, temperature and state of a pure substance when thermal energy is added or removed.

Chemical Reactions

1. Analyze and interpret data on the properties of substances before and after the substances interact to determine if a chemical reaction has occurred.

2. Develop and use a model to describe how the total number of atoms does not change in a chemical reaction and thus mass is conserved.

3. Undertake a design project to construct, test, and modify a device that either releases or absorbs thermal energy by chemical processes.*

Forces and Interactions

1. Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects.*

2. Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object.

3. Ask questions about data to determine the factors that affect the strength of electric and magnetic forces.

4. Construct and present arguments using evidence to support the claim that gravitational interactions are attractive and depend on the masses of interacting objects.

5. Conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects exerting forces on each other even though the objects are not in contact.

Energy

1. Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass and speed of an object.

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[3] The asterisk (*) indicates that the Performance Expectation is integrating Engineering Design.

[4] The language of the standards is adapted, informed by or taken from the: National Research Council. (2012). A framework for K—12 science education: Practices, crosscutting concepts, and core ideas. North American Association for Environmental Education (2019) K—12 environmental education: Guidelines for excellence; International Society for Technology in Education Standards. (2019). ISTE standards for students; International Technology and Engineering Educators Association (ITEEA) (2020); NGSS Lead States. (2013). Next generation science standards: For states, by states; Standards for technological and engineering literacy: The role of technology and engineering in STEM education. National Council for Agricultural Education. (2015); International Society for Technology in Education. (2019). ISTE Standards for students. Agriculture, food and natural resources (AFNR) career cluster content standards; Pennsylvania State Board of Education. (2002). Academic standards for science and technology; Pennsylvania Department of Education. (2002). Safety guidelines for elementary and technology education teachers; Pennsylvania Department of Education. (n.d.). Pennsylvania career ready skills continuum; Standards for Technological and Engineering Literacy. (2020); Pennsylvania Association for Environmental Educators. (September 2015). Pennsylvania environmental literacy plan Pennsylvania State Board of Education. (2002). Academic standards for environment and ecology. North American Association for Environmental Education. (2014). State environmental literacy plans: 2014 status report.

2. Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system.

3. Apply scientific principles to design, construct, and test a device that either minimizes or maximizes thermal energy transfer.*

4. Plan an investigation to determine the relationships among the energy transferred, the type of matter, the mass, and the change in the average kinetic energy of the particles as measured by the temperature of the sample.

5. Construct, use, and present arguments to support the claim that when the kinetic energy of an object changes, energy is transferred to or from the object.

Waves and Electromagnetic Radiation

1. Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave.

2. Develop and use a model to describe how waves are reflected, absorbed, or transmitted through various materials.

3. Integrate qualitative scientific and technical information to support the claim that digitized signals are a more reliable way to encode and transmit information than analog signals.

Life Science

Structure, Function, and Information Processing

1. Conduct an investigation to provide evidence that living things are made of cells, either one cell or many different numbers and types of cells.

2. Develop and use a model to describe the function of a cell as a whole and the ways that parts of cells contribute to the function.

3. Use arguments supported by evidence for how the body is a system of interacting subsystems composed of groups of cells.

4. Gather and synthesize information about how sensory receptors respond to stimuli by sending messages to the brain for immediate behavior or storage as memories.

Matter and Energy in Organisms and Ecosystems

1. Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms.

2. Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism.

3. Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem.

4. Develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem.

5. Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.

Interdependent Relationships in Ecosystems

1. Construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems.

2. Evaluate competing design solutions for maintaining biodiversity and ecosystem services.*

Growth, Development, and Reproduction of Organisms

1. Use arguments based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants, respectively.

2. Construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms.

3. Develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral effects to the structure and function of the organism.

4. Develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation.

5. Gather and synthesize information about the technologies that have changed the way humans influence the inheritance of desired traits in organisms.

Natural Selection and Adaptations

1. Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth under the assumption that natural laws operate today as in the past.

2. Apply scientific ideas to construct an explanation for anatomical similarities and differences among modern organisms and between modern and fossil organisms to infer evolutionary relationships.

3. Analyze displays of pictorial data to compare patterns of similarities in anatomical structures across multiple species to identify relationships not evident in the fully formed anatomy.

4. Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment.

5. Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

Earth and Space Science

Space Systems

1. Develop and use a model of the Earth-sun-moon system to describe the cyclic patterns of lunar phases, eclipses of the sun and moon, and seasons.

2. Develop and use a model to describe the role of gravity in the motion within galaxies and the solar system.

3. Analyze and interpret data to determine scale properties of objects in the solar system.

History of Earth

1. Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's 4.6-billion-year-old history.

2. Construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales.

3. Analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of past plate motions.

Earth's Systems

1. Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process.

2. Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity.

3. Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes.

Weather and Climate

1. Collect data to provide evidence for how the motion and complex interactions of air masses result in changes in weather conditions.

2. Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates.

3. Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.

Human Impacts

1. Analyze and interpret data on natural hazards to forecast future catastrophic events and inform the development of technologies to mitigate their effects.

2. Apply scientific principles to design a method for monitoring and minimizing human impact on the environment.*

3. Construct an argument supported by evidence for how increases in human population and per capita consumption of natural resources impact Earth's systems.

Engineering, Technology, and Applications of Science

Engineering Design (Define Problems, Develop Solutions and Improve Designs)

1. Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

2. Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.

3. Analyze data from tests to determine similarities and differences among several design solutions to identify the best characteristics of each that can be combined into a new solution to better meet the criteria for success.

4. Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process such that an optimal design can be achieved.

Environmental Literacy and Sustainability

Agricultural and Environmental Systems and Resources

Agricultural Systems

1. Develop a model to describe how agricultural and food systems function, including the sustainable use of natural resources and the production, processing, and management of food, fiber, and energy.

Environment and Society

1. Analyze and interpret data about how different societies (economic and social systems) and cultures use and manage natural resources differently.

Watersheds and Wetlands

1. Develop a model to describe how watersheds and wetlands function as systems, including the roles and functions they serve.

Environmental Literacy Skills

Investigating Environmental Issues

1. Gather, read, and synthesize information from multiple sources to investigate how Pennsylvania environmental issues affect Pennsylvania's human and natural systems.

Environmental Experiences

1. Collect, analyze, and interpret environmental data to describe a local environment.

Evaluating Solutions

1. Obtain and communicate information on how integrated pest management could improve indoor and outdoor environments.

Sustainability and Stewardship

Environmental Sustainability

1. Obtain and communicate information to describe how best management practices and environmental laws are designed to achieve environmental sustainability.

Environmental Stewardship

1. Design a solution to an environmental issue in which individuals and societies can engage as stewards of the environment.

Environmental Justice

1. Construct an explanation that describes regional environmental conditions and their implications on environmental justice and social equity.

Standards for Grades 9—12[5, 6]

Physical Science

Structure and Properties of Matter

1. Use the periodic table as a model to predict the relative properties of elements based on the patterns of electrons in the outermost energy level of atoms.

2. Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles.

3. Develop models to illustrate the changes in the composition of the nucleus of the atom and the energy released during the processes of fission, fusion, and radioactive decay.

4. Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.*

Chemical Reactions

1. Construct and revise an explanation for the outcome of a simple chemical reaction based on the outermost electron states of atoms, trends in the periodic table, and knowledge of the patterns of chemical properties.

2. Develop a model to illustrate that the release or absorption of energy from a chemical reaction system depends upon the changes in total bond energy.

3. Apply scientific principles and evidence to provide an explanation about the effects of changing the temperature or concentration of the reacting particles on the rate at which a reaction occurs.

4. Refine the design of a chemical system by specifying a change in conditions that would produce increased amounts of products at equilibrium.*

5. Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction.

Forces and Interactions

1. Analyze data to support the claim that Newton's second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration.

2. Use mathematical representations to support the claim that the total momentum of a system of objects is conserved when there is no net force on the system.

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[5] The asterisk (*) indicates that the Performance Expectation is integrating Engineering Design.

[6] The language of the standards is adapted, informed by or taken from the: National Research Council. (2012). A framework for K—12 science education: Practices, crosscutting concepts, and core ideas. North American Association for Environmental Education (2019) K—12 environmental education: Guidelines for excellence; International Society for Technology in Education Standards. (2019). ISTE standards for students; International Technology and Engineering Educators Association (ITEEA) (2020); NGSS Lead States. (2013). Next generation science standards: For states, by states; Standards for technological and engineering literacy: The role of technology and engineering in STEM education. National Council for Agricultural Education. (2015); International Society for Technology in Education. (2019). ISTE Standards for students. Agriculture, food and natural resources (AFNR) career cluster content standards; Pennsylvania State Board of Education. (2002). Academic standards for science and technology; Pennsylvania Department of Education. (2002). Safety guidelines for elementary and technology education teachers; Pennsylvania Department of Education. (n.d.). Pennsylvania career ready skills continuum; Standards for Technological and Engineering Literacy. (2020); Pennsylvania Association for Environmental Educators. (September 2015). Pennsylvania environmental literacy plan Pennsylvania State Board of Education. (2002). Academic standards for environment and ecology. North American Association for Environmental Education. (2014). State environmental literacy plans: 2014 status report.

3. Apply scientific and engineering ideas to design, evaluate and refine a device that minimizes the force on a macroscopic object during a collision.*

4. Use mathematical representations of Newton's Law of Gravitation and Coulomb's Law to describe and predict the gravitational and electrostatic forces between objects.

5. Plan and conduct an investigation to provide evidence that an electric current can produce a magnetic field and that a changing magnetic field can produce an electric current.

Energy

1. Create a computational model to calculate the change in the energy of one component in a system when the change in energy of the other component(s) and energy flows in and out of the system are known.

2. Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative positions of particles (objects).

3. Design, build and refine a device that works within given constraints to convert one form of energy into another form of energy.*

4. Plan and conduct an investigation to provide evidence that the transfer of thermal energy when two components of different temperature are combined within a closed system results in a more uniform energy distribution among the components in the system (second law of thermodynamics).

5. Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction.

Waves and Electromagnetic Radiation

1. Use mathematical representations to support a claim regarding relationships among the frequency, wavelength, and speed of waves traveling in various media.

2. Evaluate questions about the advantages of using digital transmission and storage of information.

3. Evaluate the claims, evidence, and reasoning behind the idea that electromagnetic radiation can be described either by a wave model or a particle model and that for some situations one model is more useful than the other.

4. Evaluate the validity and reliability of claims in published materials of the effects that different frequencies of electromagnetic radiation have when absorbed by matter.

5. Communicate technical information about how some technological devices use the principles of wave behavior and wave interactions with matter to transmit and capture information and energy.*

Life Science

Structure and Function

1. Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells.

2. Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.

3. Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis.

Matter and Energy in Organisms and Ecosystems

1. Use a model to illustrate how photosynthesis transforms light energy into stored chemical energy.

2. Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and/or other large carbon-based molecules.

3. Use a model to illustrate that cellular respiration is a chemical process whereby the bonds of food molecules and oxygen molecules are broken and the bonds in new compounds are formed resulting in a net transfer of energy.

4. Construct and revise an explanation based on evidence for the cycling of matter and flow of energy in aerobic and anaerobic conditions.

5. Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem.

6. Develop a model to illustrate the role of photosynthesis and cellular respiration in the cycling of carbon among the biosphere, atmosphere, hydrosphere, and geosphere.

Interdependent Relationships in Ecosystems

1. Use mathematical and/or computational representations to support explanations of factors that affect carrying capacity of ecosystems at different scales.

2. Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems of different scales.

3. Evaluate the claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

4. Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity.*

5. Evaluate the evidence for the role of group behavior on individual and species' chances to survive and reproduce.

6. Create or revise a simulation to test a solution to mitigate the adverse impacts of human activity on biodiversity.*

Inheritance and Variation of Traits

1. Use a model to illustrate the role of cellular division (mitosis) and differentiation in producing and maintaining complex organisms.

2. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring.

3. Make and defend a claim based on evidence that inheritable genetic variations may result from (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors.

4. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population.

Natural Selection and Evolution

1. Communicate scientific information that common ancestry and biological evolution are supported by multiple lines of empirical evidence.

2. Construct an explanation based on evidence that the process of evolution primarily results from four factors: (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment.

3. Apply concepts of statistics and probability to support explanations that organisms with an advantageous heritable trait tend to increase in proportion to organisms lacking this trait.

4. Construct an explanation based on evidence for how natural selection leads to adaptation of populations.

5. Evaluate the evidence supporting claims that changes in environmental conditions may result in (1) increases in the number of individuals of some species, (2) the emergence of new species over time, and (3) the extinction of other species.

Earth and Space Science

Space Systems

1. Develop a model based on evidence to illustrate the life span of the sun and the role of nuclear fusion in the sun's core to release energy in the form of radiation.

2. Construct an explanation of the Big Bang theory based on astronomical evidence of light spectra, the motion of distant galaxies, and the composition of matter in the universe.

3. Communicate scientific ideas about the way stars, over their life cycle, produce elements.

4. Use mathematical or computational representations to predict the motion of orbiting objects in the solar system.

History of Earth

1. Evaluate evidence of the past and current movements of continental and oceanic crust and the theory of plate tectonics to explain the ages of crustal rocks.

2. Apply scientific reasoning and evidence from ancient Earth materials, meteorites, and other planetary surfaces to construct an account of Earth's formation and early history.

3. Develop a model to illustrate how Earth's internal and surface processes operate at different spatial and temporal scales to form continental and ocean-floor features.

Earth's Systems

1. Analyze geoscience data to make the claim that one change to Earth's surface can create feedback that causes changes to other Earth systems.

2. Develop a model based on evidence of Earth's interior to describe the cycling of matter by thermal convection.

3. Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes.

4. Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere.

5. Construct an argument based on evidence about the simultaneous coevolution of Earth's systems and life on Earth.

Weather and Climate

1. Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate.

2. Analyze geoscience data and the results from global climate models to make an evidence-based forecast of the current rate of global or regional climate change and associated future impacts to Earth systems.

Human Sustainability

1. Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

2. Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.*

3. Create a computational simulation to illustrate the relationships among management of natural resources, the sustainability of human populations, and biodiversity.

4. Evaluate or refine a technological solution that reduces the impact of human activities on natural systems.*

5. Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity.

Engineering, Technology, and Applications of Science

Engineering Design (Define Problems, Develop Solutions and Improve Designs)

1. Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants.

2. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

3. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts.

4. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem with numerous criteria and constraints on interactions within and between systems relevant to the problem.

Environmental Literacy and Sustainability

Agricultural and Environmental Systems and Resources

Agricultural Systems

1. Analyze and interpret how issues, trends, technologies, and policies impact agricultural, food, and environmental systems and resources.

Environment and Society

1. Apply research and analytical skills to evaluate the conditions and motivations that lead to conflict, cooperation, and change among individuals, groups, and nations.

Watersheds and Wetlands

1. Analyze and interpret how issues, trends, technologies, and policies impact watersheds and water resources.

Environmental Literacy Skills

Investigating Environmental Issues

1. Apply research and analytical skills to systematically investigate environmental issues ranging from local issues to those that are regional or global in scope.

Environmental Experiences

1. Plan and conduct an investigation utilizing environmental data about a local environmental issue.

Evaluating Solutions

1. Evaluate and communicate the effect of integrated pest management practices on indoor and outdoor environments.

Sustainability and Stewardship

Environmental Sustainability

1. Analyze and evaluate how best management practices and environmental laws achieve sustainability of natural resources.

Environmental Stewardship

1. Design and evaluate solutions in which individuals and societies can promote stewardship in environmental quality and community well-being.

Environmental Justice

1. Analyze and interpret data on a regional environmental condition and its implications on environmental justice and social equity.

Technology and Engineering Academic Standards[7]

Grades 6—8

Nature and Characteristics of Technology and Engineering

1. Consider historical factors that have contributed to the development of technologies and human progress.

2. Engage in a research and development process to simulate how inventions and innovations have evolved through systematic tests and refinements.

3. Differentiate between inputs, processes, outputs, and feedback in technological systems.

4. Demonstrate how systems thinking involves considering relationships between every part, as well as how the systems interact with the environment in which it is used.

5. Create an open-loop system that has no feedback path and requires human intervention.

6. Create a closed-loop system that has a feedback path and requires no human intervention.

7. Predict outcomes of a future product or system at the beginning of the design process.

8. Apply informed problem-solving strategies to the improvement of existing devices or processes or the development of new approaches.

9. Explain how technology and engineering are closely linked to creativity, which can result in both intended and unintended innovations.

10. Compare how different technologies involve different sets of processes.

Integration of Knowledge, Technologies, and Practices

1. Compare, contrast, and identify overlap between the contributions of science, technology, engineering, and mathematics in the development of technological systems.

2. Analyze how different technological systems often interact with economic, environmental, and social systems.

3. Adapt and apply an existing product, system, or process to solve a problem in a different setting.

4. Demonstrate how knowledge gained from other content areas affects the development of technological products and systems.

Applying, Maintaining, Assessing and Evaluating Technological Products and Systems

1. Examine the ways that technology can have both positive and negative effects at the same time.

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[7] The language of the standards is adapted, informed or from the: International Technology and Engineering Educators Association (ITEEA). (2020). Standards for technological and engineering literacy: The role of technology and engineering in STEM education. Pennsylvania State Board of Education. (2002). Academic standards for science and technology; Pennsylvania Department of Education. (2002). Safety guidelines for elementary and technology education teachers; Pennsylvania Department of Education. (n.d.). Pennsylvania career ready skills continuum.

2. Analyze how the creation and use of technologies consumes renewable, non-renewable, and inexhaustible resources; creates waste; and may contribute to environmental challenges.

3. Consider the impacts of a proposed or existing technology and devise strategies for reducing, reusing, and recycling waste caused by its creation.

4. Analyze examples of technologies that have changed the way people think, interact, live, and communicate.

5. Hypothesize what alternative outcomes (individual, cultural, and/or environmental) might have resulted had a different technological solution been selected.

6. Analyze how an invention or innovation was influenced by the context and circumstances in which it is developed.

7. Evaluate trade-offs based on various perspectives as part of a decision process that recognizes the need for careful compromises among competing factors.

8. Research information from various sources to use and maintain technological products or systems.

9. Use tools, materials, and machines to safely diagnose, adjust, and repair systems.

10. Use devices to control technological systems.

11. Design methods to gather data about technological systems.

12. Interpret the accuracy of information collected.

13. Use instruments to gather data on the performance of everyday products.

Design Thinking in Technology and Engineering Education

1. Apply a technology and engineering design thinking process.

2. Develop innovative products and systems that solve problems and extend capabilities based on individual or collective needs and wants.

3. Illustrate the benefits and opportunities associated with different approaches to design.

4. Create solutions to problems by identifying and applying human factors in design.

5. Evaluate and assess the strengths and weaknesses of various design solutions given established principles and elements of design.

6. Refine design solutions to address criteria and constraints.

7. Defend decisions related to a design problem.

Grades 9—12

Nature and Characteristics of Technology & Engineering

1. Evaluate how technology and engineering have been powerful forces in reshaping the social, cultural, political, and economic landscapes throughout history.

2. Relate how technological and engineering developments have been evolutionary, often the result of a series of refinements to basic inventions or technological knowledge.

3. Identify and explain how the evolution of civilization has been directly affected by, and has in turn affected, the development and use of tools, materials, and processes.

4. Analyze how the Industrial Revolution resulted in the development of mass production, sophisticated transportation and communication systems, advanced construction practices, and improved education and leisure time.

5. Investigate the widespread changes that have resulted from the Information Age, which has placed emphasis on the processing and exchange of information.

6. Analyze the rate of technological and engineering development and predict future diffusion and adoption of new innovations and technologies.

7. Demonstrate the use of conceptual, graphical, virtual, mathematical, and physical modeling to identify conflicting considerations before the entire system is developed and to aid in design decision making.

8. Analyze the stability of a technological system and how it is influenced by all of the components in the system, especially those in the feedback loop.

9. Troubleshoot and improve a flawed system embedded within a larger technological, social, or environmental system.

10. Use project management tools, strategies, and processes in planning, organizing, and controlling work.

11. Implement quality control as a planned process to ensure that a product, service, or system meets established criteria.

Integration of Knowledge, Technologies, and Practices

1. Assess how similarities and differences among scientific, technological, engineering, and mathematical knowledge and skills contributed to the design of a product or system.

2. Develop a plan that incorporates knowledge from science, mathematics, and other disciplines to design or improve a technological product or system.

3. Analyze how technology transfer occurs when a user applies an existing innovation developed for one function for a different purpose.

4. Evaluate how technology enhances opportunities for new products and services through globalization.

5. Connect technological and engineering progress to the advancement of other areas of knowledge and vice versa.

Applying, Maintaining, Assessing, and Evaluating Technological Products and Systems

1. Develop a solution to a technological problem that has the least negative environmental and social impact.

2. Develop a device or system for the marketplace.

3. Evaluate ways that technology and engineering can impact individuals, society, and the environment.

4. Critique whether existing or proposed technologies use resources sustainably.

5. Critically assess and evaluate a technology that minimizes resource use and resulting waste to achieve a goal.

6. Evaluate a technological innovation that arose from a specific society's unique need or want.

7. Evaluate how technology and engineering advancements alter human health and capabilities.

8. Evaluate a technological innovation that was met with societal resistance impacting its development.

9. Use various approaches to communicate processes and procedures for using, maintaining, and assessing technological products and systems.

10. Synthesize data and analyze trends to make decisions about technological products, systems, or processes.

11. Interpret laws, regulations, policies, and other factors that impact the development and use of technology.

Design Thinking in Technology and Engineering Education

1. Apply a broad range of design skills to a design thinking process.

2. Implement and critique principles, elements, and factors of design.

3. Evaluate and define the purpose of a design.

4. Conduct research to inform intentional inventions and innovations that address specific needs and wants.

5. Analyze and use relevant and appropriate design thinking processes to solve technological and engineering problems.

6. Implement the best possible solution to a design using an explicit process.

7. Apply principles of human-centered design.

8. Optimize a design by addressing desired qualities within criteria and constraints while considering trade-offs.

9. Use a design thinking process to design an appropriate technology for use in a different culture.

10. Apply appropriate design thinking processes to diagnose, adjust, and repair systems to ensure precise, safe, and proper functionality.

11. Recognize and explain how their community and the world around them informs technological development and engineering design.

12. Safely apply an appropriate range of making skills to a design thinking process.

APPENDIX C

Academic Standards for Civics and Government and Geography and History

Authority

The provisions of this Appendix C amended under sections 121, 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 1-121, 2-290.1, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix C adopted January 10, 2003, effective January 12, 2003, 33 Pa.B. 283; amended February 28, 2014, effective March 1, 2014, 44 Pa.B. 1131; corrected March 21, 2014, effective March 1, 2014, 44 Pa.B. 1754; amended February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234, corrected March 21, 2025, 55 Pa.B. 2255. Immediately preceding text appears at serial pages (422621) to (422622), (294915) to (294986), (422623) to (422626), (294991) to (295012), (371235) to (371244) and (422339) to (422340).

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards).

Academic Standards for Civics and Government

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XIV. INTRODUCTION

This document includes Academic Standards for Civics and Government that describe what students should know and be able to do in four areas:

• 5.1. Principles and Documents of Government

• 5.2. Rights and Responsibilities of Citizenship

• 5.3. How Government Works

• 5.4. How International Relationships Function

The Civics and Government Academic Standards describe what students should know and be able to do at four grade levels (third, sixth, ninth and twelfth). Throughout the standard statements, concepts found in lower grades must be developed more fully throughout higher grade levels.

The Pennsylvania Constitution of 1790 was the basis for the Free Public School Act of 1834 that is the underpinning of today's system of schools operating throughout the Commonwealth. These schools were created to educate children to be useful citizens, loyal to the principles upon which our Republic was founded, and aware of their duties as citizens to maintain those ideals.

The Academic Standards for Civics and Government are based on the Public School Code of 1949 which directs ''. . . teaching and presentation of the principles and ideals of the American republican representative form of government as portrayed and experienced by the acts and policies of the framers of the Declaration of Independence and framers of the Constitution of the United States andBill of Rights. . .''. The intent of the Code is that such instruction ''shall have for its purpose also instilling into every boy and girl who comes out of public, private and parochial schools their solemn duty and obligation to exercise intelligently their voting privilege and to understand the advantages of the American republican form of government as compared with various other forms of governments.''

The Academic Standards for Civics and Government consist of four standard categories (designated as 5.1., 5.2., 5.3., and 5.4.). Each category has a number of standards statements designated by a capital letter. Some standard statements have bulleted items known as standard descriptors. The standard descriptors are items within the document to illustrate and enhance the standard statement. The categories, statements and descriptors are regulations. The descriptors may be followed by an "e.g.". The "e.g.'s" are examples to clarify what type of information could be taught. These are suggestions and the choice of specific content is a local decision as is the method of instruction.

Civics and Government along with Economics, Geography and History are identified as Social Studies in Chapter 4. This identification is consistent with citizenship education in Chapter 49 and Chapter 354. Based on these regulations, Social Studies/Citizenship Programs should include the four sets of standards as an entity in developing a scope and sequence for curriculum and planned instruction.

A glossary is included to assist the reader in clarifying terminology contained in the standards.

5.1. Principles and Documents of Government			
5.1.3. GRADE 3	**5.1.6. GRADE 6**	**5.1.9. GRADE 9**	**5.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
A. Describe what government is.	A. Explain the purpose of government.	A. Identify and explain the major arguments advanced for the necessity of government.	A. Evaluate the major arguments advanced for the necessity of government.
B. Explain the purposes of rules and laws and why they are important in the classroom, school, community, state and nation.	B. Explain the importance of the rule of law for the protection of individual rights and the common good in the community, state, nation and world.	B. Describe historical examples of the importance of the rule of law. • Sources • Purposes • Functions	B. Analyze the sources, purposes and functions of law.
C. Define the principles and ideals shaping government. • Justice • Truth • Diversity of people and ideas • Patriotism • Common good • Liberty • Rule of law • Leadership • Citizenship	C. Describe the principles and ideals shaping government. • Equality • Majority rule/Minority rights • Popular sovereignty • Privacy • Checks and balances • Separation of powers	C. Analyze the principles and ideals that shape government. • Constitutional government • Liberal democracy • Classical republicanism • Federalism	C. Evaluate the importance of the principles and ideals of civic life.

5.1. Principles and Documents of Government			
5.1.3. GRADE 3	**5.1.6. GRADE 6**	**5.1.9. GRADE 9**	**5.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
D. Identify the document which created Pennsylvania.	D. Explain the basic principles and ideals within documents of Pennsylvania government. • Charter of 1681 • Charter of Privileges • Pennsylvania Constitution • Pennsylvania Declaration of Rights	D. Interpret significant changes in the basic documents shaping the government of Pennsylvania. • The Great Law of 1682 • Constitution of 1776 • Constitution of 1790 • Constitution of 1838 • Constitution of 1874 • Constitution of 1968	D. Analyze the principles and ideals that shape the government of Pennsylvania and apply them to the government. • The Charter of 1681 • Charter of Privileges • PA Constitution, its revisions and Amendments

5.1. Principles and Documents of Government			
5.1.3. GRADE 3	**5.1.6. GRADE 6**	**5.1.9. GRADE 9**	**5.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
E. Identify documents of United States government. • Declaration of Independence • Constitution of the United States • Bill of Rights	E. Explain the basic principles and ideals within documents of United States government.	E. Analyze the basic documents shaping the government of the United States. • Magna Carta • English Bill of Rights • Mayflower Compact • Articles of Confederation • Declaration of Independence • Federalist papers • Anti-federalist writings • United States Constitution	E. Evaluate the principles and ideals that shape the United States and compare them to documents of government.

5.1. Principles and Documents of Government			
5.1.3. GRADE 3	**5.1.6. GRADE 6**	**5.1.9. GRADE 9**	**5.1.12. GRADE 12**

Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
F. Explain the meaning of a preamble. • Constitution of the United States • Pennsylvania Constitution	F. Explain the meaning of the Preamble to the Constitution of the Commonwealth of Pennsylvania and compare it to the Preamble of the Constitution of the United States.	F. Contrast the individual rights created by the Pennsylvania Constitution and those created by the Constitution of the United States.	F. Analyze and assess the rights of the people as listed in the Pennsylvania Constitution and the Constitution of the United States.
G. Describe the purpose of the United States Flag, The Pledge of Allegiance and The National Anthem.	G. Describe the proper use, display and respect for the United States Flag and explain the significance of patriotic activities. • Reciting The Pledge of Allegiance • Standing for The National Anthem	G. Describe the procedures for proper uses, display and respect for the United States Flag as per the National Flag Code.	G. Analyze and interpret the role of the United States Flag in civil disobedience and in patriotic activities.

5.1. Principles and Documents of Government			
5.1.3. GRADE 3	**5.1.6. GRADE 6**	**5.1.9. GRADE 9**	**5.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
H. Identify framers of documents of governments. • Pennsylvania • United States	H. Describe the roles played by the framers of the basic documents of governments of Pennsylvania and the United States.	H. Explain and interpret the roles of framers of basic documents of government from a national and Pennsylvania perspective.	H. Analyze the competing positions held by the framers of the basic documents of government of Pennsylvania and United States.
I. Explain why government is necessary in the classroom, school, community, state and nation and the basic purposes of government in Pennsylvania and the United States.	I. Describe and compare the making of rules by direct democracy and by republican form of government.	I. Explain the essential characteristics of limited and unlimited governments and explain the advantages and disadvantages of systems of government. • Confederal • Federal • Unitary	I. Analyze historical examples of the importance of the rule of law explaining the sources, purposes and functions of law.
J. Explain the importance of respect for the property and the opinions of others.	J. Describe how the government protects individual and property rights and promotes the common good.	J. Explain how law protects individual rights and the common good.	J. Analyze how the law promotes the common good and protects individual rights.

5.1. Principles and Documents of Government			
5.1.3. GRADE 3	**5.1.6. GRADE 6**	**5.1.9. GRADE 9**	**5.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
K. Identify symbols and political holidays. • Pennsylvania (e.g., Charter Day, Liberty Bell, Keystone State) • United States (e.g., Presidents' Day, Statue of Liberty, White House)	K. Describe the purpose of symbols and holidays.	K. Explain why symbols and holidays were created and the ideals they commemorate.	K. Analyze the roles of symbols and holidays in society.
L. Identify ways courts resolve conflicts involving principles and ideals of government.	L. Explain the role of courts in resolving conflicts involving the principles and ideals of government. • Local • State • Federal	L. Interpret Pennsylvania and United States court decisions that have impacted the principles and ideals of government.	L. Analyze Pennsylvania and United States court decisions that have affected principles and ideals of government in civic life. • Civil rights • Commerce • Judicial review • Federal supremacy

5.1. Principles and Documents of Government			
5.1.3. GRADE 3	**5.1.6. GRADE 6**	**5.1.9. GRADE 9**	**5.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
M. Identify portions of famous speeches and writings that reflect the basic principles and ideals of government (e.g., "I have a dream," Reverend Martin Luther King; "One small step for man," Neil Armstrong).	M. Explain the basic principles and ideals found in famous speeches and writings (e.g., "Governments, like clocks, go from the motion people give them," William Penn; "A date that will live in infamy," Franklin D. Roosevelt).	M. Interpret the impact of famous speeches and writings on civic life (e.g., *The Gospel of Wealth, Declaration of Sentiments)*.	M. Evaluate and analyze the importance of significant political speeches and writings in civic life (e.g., *Diary of Anne Frank, Silent Spring*).
Basic concepts found in lower grades for standard statements and their descriptors must be developed more fully throughout higher grade levels.			

5.2. Rights and Responsibilities of Citizenship

5.2.3. GRADE 3	5.2.6. GRADE 6	5.2.9. GRADE 9	5.2.12. GRADE 12
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
A. Identify examples of the rights and responsibilities of citizenship. • Personal rights • Political rights • Economic rights • Personal responsibili-ties • Civic responsibilities	A. Compare rights and responsibilities of citizenship. • Political rights • Economic rights • Personal responsibilities of the individual and to society • Civic responsibilities of the individual and to society • Traits of character of individuals and to a republican form of government	A. Contrast the essential rights and responsibilities of citizens in systems of government. • Autocracy • Democracy • Oligarchy • Republic	A. Evaluate an individual's civic rights, responsibilities and duties in various governments.
B. Identify personal rights and responsibilities.	B. Explain the relationship between rights and responsibilities.	B. Analyze citizens' rights and responsibilities in local, state and national government.	B. Evaluate citizen's participation in government and civic life.

5.2. Rights and Responsibilities of Citizenship			
5.2.3. GRADE 3	**5.2.6. GRADE 6**	**5.2.9. GRADE 9**	**5.2.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
C. Identify sources of conflict and disagreement and different ways conflicts can be resolved.	C. Explain ways citizens resolve conflicts in society and government.	C. Analyze skills used to resolve conflicts in society and government.	C. Interpret the causes of conflict in society and analyze techniques to resolve those conflicts.
D. Identify the importance of political leadership and public service in the school, community, state and nation.	D. Describe the importance of political leadership and public service.	D. Analyze political leadership and public service in a republican form of government.	D. Evaluate political leadership and public service in a republican form of government.
E. Describe ways citizens can influence the decisions and actions of government.	E. Identify examples of the rights and responsibilities of citizenship.	E. Explain the importance of the political process to competent and responsible participation in civic life.	E. Analyze how participation in civic and political life leads to the attainment of individual and public goals.
F. Explain the benefits of following rules and laws and the consequences of violating them.	F. Describe the impact of the consequences of violating rules and laws in a civil society.	F. Analyze the consequences of violating laws of Pennsylvania compared to those of the United States.	F. Evaluate how individual rights may conflict with or support the common good.
G. Identify ways to participate in government and civic life.	G. Explain the importance of participating in government and civic life.	G. Analyze political and civic participation in government and society.	G. Evaluate what makes a competent and responsible citizen.

5.2. Rights and Responsibilities of Citizenship			
5.2.3. GRADE 3	**5.2.6. GRADE 6**	**5.2.9. GRADE 9**	**5.2.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
Basic concepts found in lower grades for standard statements and their descriptors must be developed more fully throughout higher grade levels.			

5.3. How Government Works			
5.3.3. GRADE 3	**5.3.6. GRADE 6**	**5.3.9. GRADE 9**	**5.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. . .			
A. Identify the elected representative bodies responsible for making local, Pennsylvania and United States laws.	A. Compare the structure, organization and operation of local, state and national governments.	A. Explain the structure, organization and operation of the local, state and national governments including domestic and national policy-making.	A. Analyze and evaluate the structure, organization and operation of the local, state and national governments including domestic and national policy-making.
B. Identify the role of the three branches of government. • Executive • Legislative • Judicial	B. Describe the responsibilities and powers of the three branches of government.	B. Compare the responsibilities and powers of the three branches within the national government.	B. Analyze the responsibilities and powers of the national government.
C. Identify reasons for rules and laws in the school and community.	C. Explain how government actions affect citizens' daily lives.	C. Explain how a bill becomes a law on a federal, state, and local level.	C. Evaluate the process of how a bill becomes the law on a federal, state, and local levels.

5.3. How Government Works			
5.3.3. GRADE 3	**5.3.6. GRADE 6**	**5.3.9. GRADE 9**	**5.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. . .			
D. Identify services performed by the local, state and national	D. Describe how local, state and national governments implement their services.	D. Explain how independent government agencies create, amend and enforce	D. Evaluate how independent government agencies create, amend

governments.		regulatory policies. • Local (e.g., Zoning Board) • State (e.g., Pennsylvania Public Utility Commission) • National (e.g., Federal Communications Commission)	and enforce regulations.
E. Identify positions of authority at school and in local, state and national governments.	E. Identify major leaders of local, state and national governments, their primary duties and their political party affiliation.	E. Explain how citizens participate in choosing their leaders through political parties, campaigns and elections.	E. Evaluate the roles of political parties in election campaigns.

5.3. How Government Works			
5.3.3. GRADE 3	**5.3.6. GRADE 6**	**5.3.9. GRADE 9**	**5.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. . .			
F. Explain what an election is.	F. Describe the voting process. • Pennsylvania • United States	F. Explain the election process. • Voter registration • Primary Elections • Caucuses • Political party conventions • General Elections • Electoral College	F. Evaluate the elements of the election process.
G. Explain why being treated fairly is important.	G. Describe how the government protects individual rights. • Presumption of Innocence • Right to Counsel • Trial by Jury • Bill of Rights	G. Explain how the government protects individual rights. • Equal protection • Habeas Corpus • Right Against Self Incrimination • Double Jeopardy • Right of Appeal • Due Process	G. Evaluate how the government protects or curtails individual rights and analyze the impact of supporting or opposing those rights.

5.3. How Government Works			
5.3.3. GRADE 3	**5.3.6. GRADE 6**	**5.3.9. GRADE 9**	**5.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. . .			
H. Identify individual interests and explain ways to influence others.	H. Identify individual interests and how they impact government.	H. Analyze how interest groups provide opportunities for citizens to participate in the political process.	H. Evaluate the impact of interest groups on the political process.
I. Explain why taxes are necessary and identify who pays them.	I. Describe why and how government raises money to pay for its operations and services.	I. Analyze how and why government raises money to pay for its operation and services.	I. Evaluate how and why government raises money to pay for its operations and services.
J. Identify the role of the media in society.	J. Describe the influence of media in reporting issues.	J. Analyze the importance of freedom of the press.	J. Evaluate the role of media in political life in the United States and explain the role of the media in setting the public agenda.
K. Identify different ways people govern themselves.	K. Describe forms of government. • Limited • Unlimited	K. Identify and explain systems of government. • Autocracy • Democracy • Oligarchy • Republic	K. Evaluate the strengths and weaknesses of various systems of government. • Autocracy • Democracy • Oligarchy • Republic
Basic concepts found in lower grades for standard statements and their descriptors must be developed more fully throughout higher grade levels.			

5.4. How International Relationships Function			
5.4.3. GRADE 3	**5.4.6. GRADE 6**	**5.4.9. GRADE 9**	**5.4.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. . .			
A. Identify how customs and traditions influence governments.	A. Explain the concept of nation-states.	A. Explain how the United States is affected by policies of nation-states, governmental and non-governmental organizations.	A. Analyze the impact of international economic, technological and cultural developments on the government of the United States.
B. Recognize that the world is divided into various political units.	B. Describe how nation-states coexist in the world community.	B. Explain the role of the United States in world affairs.	B. Analyze the United States' interaction with other nations and governmental groups in world events.
C. Identify ways in which countries interact with the United States.	C. Describe the governments of the countries bordering the United States and their relationships with the United States.	C. Explain the effects United States political ideas have had on other nations.	C. Compare how past and present United States' policy interests have changed over time and analyze the impact on future international relationships.

5.4. How International Relationships Function			
5.4.3. GRADE 3	**5.4.6. GRADE 6**	**5.4.9. GRADE 9**	**5.4.12. GRADE 12**

Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. . .			
D. Identify treaties and other agreements between or among nations.	D. Describe the processes that resulted in a treaty or agreement between the United States and another nation-state.	D. Contrast how the three branches of federal government function in foreign policy.	D. Explain how foreign policy is developed and implemented.
E. Identify how nations work together to solve problems.	E. Explain how nations work together on common environmental problems, natural disasters and trade.	E. Explain the development and the role of the United Nations and other international organizations, both governmental and non-governmental.	E. Compare the purposes and functions of international organizations. • Governmental (e.g., NATO, World Court, OAS) • Nongovernmental (e.g., International Red Cross, Amnesty International, World Council of Churches)

XV. GLOSSARY

Amendment (Constitutional):	**Changes in, or additions to, a constitution. Proposed by a two-thirds vote of both houses of Congress or by a convention called by Congress at the request of two-thirds of the state legislatures. Ratified by approval of three-fourths of the state.**
Articles of Confederation:	**First framework of government of the United States, 1781. Created a weak national government, replaced in 1789 by the Constitution of the United States.**
Authority:	Right to control or direct the actions of others, legitimized by law, morality, custom or consent.
Autocracy:	A government in which one person possesses unlimited power.
Bill of Rights:	First Ten Amendments to the Constitution. Ratified in 1791, these amendments limit government power and protect basic rights and liberties of individuals.
Caucuses:	A private meeting of members of a political party to plan action or to select delegates for a nominating convention. The term also refers to distinct groups, either official or unofficial, in Congress, as in the black caucus in the House of Representatives.
Checks and balances:	Constitutional mechanisms that authorize each branch of government to share powers with the other branches and thereby check their activities. For example, the president may veto legislation passed by Congress, the Senate must confirm major executive appointments and the courts may declare acts of Congress unconstitutional.
Citizen:	Member of a political society who therefore owes allegiance to and is entitled to protection by and from the government.
Citizenship:	Status of being a member of a state; one who owes allegiance to the government and is entitled to protection by and from the government.
Civic life:	A manner of existence of an individual concerned with the affairs of communities and the common good rather than solely in pursuit of private and personal interests.
Civic responsibilities:	Obligation of citizens to take part in the governance of the school, community, tribe, state or nation.
Civil disobedience:	Refusal to obey laws. This tactic is usually passive and nonviolent, aimed at bringing injustices to the attention of lawmakers and the public at large. An example of civil disobedience was the American Civil Rights Movement in the 1950s and 1960s.
Civil rights:	Protections and privileges given to all United States citizens by the Constitution and Bill of Rights.
Civil society:	The spheres of voluntary individual, social and economic relationships and organizations that although limited by law are not part of governmental institutions.
Classical republicanism:	Refers to government that seeks the public or common good rather than the good of a particular group or class of society.
Common or public good:	Benefit or interest of a politically organized society as a whole.
Confederal:	Relating to a league of independent states.
Constitutional government:	A form of authority in which a legal structure details the powers available to each branch of government and the rights of the individual in relation to the government. Any action by government that is not in accord with the Constitution is considered illegitimate.
Democracy:	Form of government in which political control is exercised by the people, either directly or through their elected representatives.
Diplomacy:	The art and practice of conducting negotiations between nations.
Direct democracy:	Form of government in which the people completely exercise political decisions.
Diversity:	State of being different; variety.
Documents of government:	Papers necessary for the organization and powers of government.
Double jeopardy:	A concept established by law that says a person cannot be tried twice for the same offense. It is part of the Fifth Amendment, which states that ''no person shall . . . be subject for the same offense to be twice put in jeopardy of life or limb.''
Due process of law:	Right of every citizen to be protected against arbitrary action by government.
Economic rights:	Financial choices and privileges that individuals may select without government prohibition. Economic rights would include: right to own property, change employment, operate a business and join a labor union.
Electoral College:	The group of presidential electors that casts the official votes for president after the presidential election. Each state has a number of electors equal to the total of its members in the Senate and House of Representatives.
Enumerated powers:	Powers that are specifically granted to Congress by Article I, Section 8 of the Constitution.
Equal protection:	An idea that no individual or group may receive special privileges from nor be unjustly discriminated against by the political authority of the legal system.
Equality:	The condition of possessing substantially the same rights, privileges and immunities, and being substantially responsible for the same duties as other members of society.
Federal Supremacy Clause:	Article VI of the Constitution provides that the Constitution and all federal laws and treaties shall be the ''Supreme Law of the Land.'' Therefore, all federal laws take precedence over state and local laws.
Federal system (or Federalism):	Form of political organization in which governmental power is divided between a central government and territorial subdivisions (e.g., in the United States—the national, state and local governments).
Federalism:	The distribution of power in a government between a central authority and states and the distribution of power among states with most powers retained by central government.

Foreign Policy:	Actions of the federal government directed to matters beyond United States' borders, especially relations with other countries.
Government:	Institutions and procedures through which a territory and its people are ruled.
Habeas Corpus:	Court order demanding that the individual in custody be brought into court and shown the cause for detention. Habeas corpus is guaranteed by the Constitution and can be suspended only in the case of rebellion or invasion.
Individual responsibility:	Fulfilling the moral and legal obligations of membership in society.
Individual rights:	Just claims due a person by law, morality or tradition as opposed to those due to groups.
Interest group:	Organized body of individuals who share same goals and try to influence public policy to meet those goals.
International organizations:	Groups formed by nation-states to achieve common political, social or economic goals.
Judicial Review:	Doctrine that permits the federal courts to declare unconstitutional, and thus null and void, acts of the Congress, the executive branch and the states. The precedent for judicial review was established in the 1803 case of *Marbury v. Madison*.
Justice:	That which may be obtained through fair distribution of benefits and burdens, fair correction of wrongs and injuries, or use of fair procedures in gathering information and making decisions.
Leadership:	State or condition of one who guides or governs.
Liberal Democracy:	Government that recognizes that the individual has rights that exist independently of government and which ought to be protected by and against government.
Liberty:	Freedom from restraint under conditions essential to the equal enjoyment of the same right by others.
Limited government:	A legal structure where officials in authority do not have enormous power. The Constitution of the United States limits government through methods of checks and balances.
Majority rule:	Decision by more than half of those participating in the decision-making process.
Minority rights:	Opportunities that a member is entitled to have, or to receive from others within the limits of the law, even though he/she may not be part of the controlling group.
Nation-state:	Divisions of the world in which each state claims sovereignty over defined territory and jurisdiction over everyone within it. These states interact using diplomacy, formal agreements and sanctions that may be peaceful or may involve the use of force.
NATO:	North Atlantic Treaty Organization, an international transatlantic partnership consisting of various European states, the United States and Canada, which was designed through cooperation, consultation and collective defense to maintain peace and promote stability throughout Europe.
Non-governmental organization:	A group in a free society that is not a part of any government institution and does not derive its power from government.
OAS:	Organization of American States, an international governmental organization formed by the states of North and South America for security and the protection of mutual interests.
Oligarchy:	A government in which a small group exercises control. These systems are usually based on wealth, military power or social position.
Patriotism:	A feeling of pride in and respect for one's country.
Personal rights:	Private legal privileges and decisions that individuals are free to participate in without intervention from government. Personal rights would include the right to vote, petition, assemble, and seek public office.
Political party:	Any group, however loosely organized, that seeks to elect government officials under a given label.
Political rights:	Legal claims by citizens to participate in government and be treated fairly. Political rights would include the right to vote, petition, assemble, and seek public office.
Popular sovereignty:	The concept that ultimate political authority rests with the people to create, alter or abolish governments.
Presumption of innocence:	The legal concept that a criminal defendant is not guilty until the prosecution proves every element of the crime, beyond a reasonable doubt.
Privacy:	The right to be left alone; the right of an individual to withhold one's self and one's property from public scrutiny if one so chooses.
Public service:	Action of benefit to local, state or national communities through appointed or elected office.
Representative Democracy:	Form of government in which power is held by the voters and is exercised indirectly through elected representatives who make decisions.
Republic:	Form of government in which political control is exercised through elected representatives.
Republican form of government:	System of government in which power is held by the voters and is exercised by elected representatives responsible for promoting the common welfare.
Right against self-incrimination:	Individual right found in the Fifth Amendment to the United States Constitution that prevents an individual from being forced to testify against himself or herself.
Right of appeal:	The right to seek review by a superior court of an injustice done or error committed by an inferior court, whose judgment or decision the court above is called upon to correct or reverse.
Right to counsel:	Individual right found in the Sixth Amendment to the Constitution that requires criminal defendants to have access to legal representation.
Rule of Law:	Principle that every member of a society, even a ruler, must follow the law.
Separation of powers:	Distribution among the branches of government to ensure that the same person or group will not make the law, enforce the law and interpret the law.
State:	A commonwealth; a nation; a civil power.
Treaty:	Formal agreement between or among sovereign nations to create or restrict rights and responsibilities. In the United States all treaties must be approved by a two-thirds vote in the Senate.
Trial by jury:	Individual right found in the Sixth and Seventh Amendment of the Constitution that guarantees a person an impartial jury.
Truth:	Agreement of thought and reality that can eventually be verified.
Unitary government:	An authoritative system in which all regulatory power is vested in a central government from which regional and local governments derive their powers (e.g., Great Britain and France as well as the American states within their spheres of authority).
United Nations:	International organization comprising most of the nation-states of the world. It was formed in 1945 to promote peace, security and economic development.
Unlimited government:	A legal structure where officials in authority have unrestricted power. Examples of unlimited governments would be authoritarian or totalitarian systems without restraints on their power.
World Court:	Court in the Hague, the Netherlands, set up by the United Nations treaty to which nations may voluntarily submit disputes.

(*Editor's Note* Under § 4.12(a)(3)(iv), the academic standards for economics which appeared at 22 Pa. Code, Appendix C, pages 4-227—4-248, serial pages (294939)—(294960) expired July 1, 2026. The academic standards for economics in Appendix C-1 apply beginning July 1, 2026.)

Academic Standards for Geography

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XX. INTRODUCTION

This document includes Academic Standards for Geography that describe what students should know and be able to do in four areas:

• 7.1. Basic Geographic Literacy

• 7.2. The Physical Characteristics of Places and Regions

• 7.3. The Human Characteristics of Places and Regions

• 7.4. The Interactions Between People and Places

The Geography Standards describe what students should know and be able to do at four grade levels (third, sixth, ninth and twelfth). They reflect the increasingly complex and sophisticated understanding of geography that students are expected to achieve as they progress through school. Throughout the standards, all grade levels must address the local-to-global progression (scales). Basic concepts found in lower grade levels must be developed more fully at higher grade levels.

Geography is the science of space and place on Earth's surface. Its subject matter is the physical and human phenomena that make up the world's environments and places. These standards build on using geographic tools as a means for asking and answering geographic questions; setting information into a range of spatial contexts; recognizing places and regions as human concepts; understanding the physical processes that have shaped Earth's surface and the patterns resulting from those processes; identifying the relationships between people and environments; recognizing the characteristics and distribution of people and cultures on Earth's surface; focusing on the spatial patterns of settlements and their resulting political structures; and exploring the networks of economic interdependence and the importance of resources.

At each grade level, instructional content should be selected to support the development of geographic understanding. In the primary grade levels (1-3), the emphasis should be on identifying the basic characteristics of the world (answering the what question); at the intermediate grade levels (4-6), the emphasis should be on describing spatial patterns of phenomena (answering the where and when questions); at the middle grade levels (7-9), the emphasis should be on explaining spatial patterns of phenomena (answering the how question); and at high school grade levels (10-12), the emphasis should be on analyzing spatial patterns of phenomena (answering the why question). Although the emphasis may focus on specific questions, these questions may be encountered at any grade level.

Geography is an integrative discipline that enables students to apply geography skills and knowledge to life situations at home, at work and in the community. Therefore, these standards should be cross-walked with those in Civics and Government, Economics and History to create an interdisciplinary

view of the world. Topics and concepts in geography directly relate to standard statements in Environment and Ecology, Economics, Mathematics, Science and Technology and Civics and Government.

Teachers should employ the Five Fundamental Themes of Geography while proceeding through the Academic Standards for Geography. The relationship between the themes and the standards is clear. The standards describe what students should know and be able to do while the themes provide a clear conceptual basis for teachers and students to use in organizing their knowledge.

These are the Five Fundamental Themes of Geography:

Theme	*Description*
Location	**The absolute and relative position of a place on Earth's surface**
Place	How physical and human characteristics define and distinguish a place
Human-Environment Interactions	How humans modify and adapt to natural settings
Movement	How people, ideas and materials move between and among locations
Regions	How an area displays unity in terms of physical and human characteristics

The academic standards for Geography consist of four standard categories (designated as 7.1., 7.2., 7.3., and 7.4.). Each category has two to five standard statements (designated by a capital letter). Most standard statements have bulleted items known as standard descriptors. The standard descriptors are items within the document to illustrate and enhance the standard statement. The categories, statements and descriptors are regulations. The descriptors may be followed by an "e.g." The "e.g.'s" are examples to clarify what type of information could be taught. These are suggestions and the choice of specific content is a local decision as is the method of instruction.

Geography along with Civics and Government, Economics, and History are identified as Social Studies in Chapter 4. This identification is consistent with citizenship education in Chapter 49 and Chapter 354. Based on these regulations, Social Studies/Citizenship programs should include the four sets of standards as an entity in developing a scope and sequence for curriculum and planned instruction.

A glossary is included to assist the reader in clarifying terminology contained in the standards.

7.1. Basic Geographic Literacy			
7.1.3. GRADE 3	**7.1.6. GRADE 6**	**7.1.9. GRADE 9**	**7.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
A. Identify geographic tools and their uses. • Characteristics and purposes of different geographic representations • Maps and basic map elements • Globes • Graphs • Diagrams • Photographs • Geographic representations to display spatial information • Sketch maps • Thematic maps • Mental maps to describe the human and physical features of the local area	A. Describe geographic tools and their uses. • Basis on which maps, graphs and diagrams are created • Aerial and other photographs • Reference works • Field observations • Surveys • Geographic representations to display spatial information • Absolute location • Relative location • Flows (e.g., goods, people, traffic) • Topography • Historic events • Mental maps to organize an understanding of the human and physical features of Pennsylvania and the home county • Basic spatial elements for depicting the patterns of physical and human features	A. Explain geographic tools and their uses. • Development and use of geographic tools • Geographic information systems [GIS] • Population pyramids • Cartograms • Satellite-produced images • Climate graphs • Access to computer-based geographic data (e.g., Internet, CD-ROMs) • Construction of maps • Projections • Scale • Symbol systems • Level of generalization • Types and sources of data • Geographic representations to track spatial patterns • Weather • Migration • Environmental change (e.g., tropical forest reduction, sea-level changes)	A. Analyze data and issues from a spatial perspective using the appropriate geographic tools. • Spatial patterns of human features that change over time (e.g., intervening opportunity, distance decay, central place theory, locational preference) • Physical patterns of physical features that change over time (e.g., climate change, erosion, ecological invasion and succession) • Human and physical features of the world through mental maps

7.1. Basic Geographic Literacy			
7.1.3. GRADE 3	**7.1.6. GRADE 6**	**7.1.9. GRADE 9**	**7.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
	• Point, line, area, location, distance, scale • Map grids • Alpha-numeric system • Cardinal and intermediate directions	• Mental maps to organize and understand the human and physical features of the United States	

B. Identify and locate places and regions. • Physical features • Continents and oceans • Major landforms, rivers and lakes in North America • Local community • Human features • Countries (i.e., United States, Mexico, Canada) • States (i.e., Pennsylvania, Delaware, Maryland, New Jersey, New York, Ohio, West Virginia) • Cities (i.e., Philadelphia, Erie, Altoona, Pittsburgh, Scranton, Harrisburg, Johnstown, Allentown, Washington D.C., Baltimore, New York, Toronto, Cleveland • Local community • Regions as areas with unifying geographic characteristics • Physical regions (e.g., landform regions, climate regions, river basins)	B. Describe and locate places and regions. • Coordinate systems (e.g., latitude and longitude, time zones) • Physical features • In the United States (e.g., Great Lakes, Rocky Mountains, Great Plains) • In Pennsylvania (e.g., Coastal Plain, Piedmont, Appalachians) • Human features • Countries (e.g., United Kingdom, Argentina, Egypt) • Provinces (e.g., Ontario, Quebec, Nova Scotia) • Major human regions (e.g., Mid Atlantic, New England, Southwest) • States (e.g., California, Massachusetts, Florida) • Major cities (e.g., London, Los Angeles, Tokyo) • Counties (e.g., Lancaster, Lackawanna, Jefferson)	B. Explain and locate places and regions. • How regions are created to interpret Earth's complexity (i.e., the differences among formal regions, functional regions, perceptual regions) • How characteristics contribute to regional changes (e.g., economic development, accessibility, demographic change) • How culture and experience influence perceptions of places and regions • How structures and alliances impact regions • Development (e.g., First vs. Third World, North vs. South) • Trade (e.g., NAFTA, the European Union) • International treaties (e.g., NATO, OAS)	B. Analyze the location of places and regions. • Changing regional characteristics (e.g., short- and long-term climate shifts; population growth or decline; political instability) • Criteria to define a region (e.g., the reshaping of south Florida resulting from changing migration patterns; the US-Mexico border changes as a function of NAFTA; metropolitan growth in the Philadelphia region) • Cultural change (e.g., influence on people's perceptions of places and regions)

7.1. Basic Geographic Literacy			
7.1.3. GRADE 3	**7.1.6. GRADE 6**	**7.1.9. GRADE 9**	**7.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
• Human regions (e.g., neighborhoods, cities, states, countries)	• Townships (e.g., Dickinson, Lower Mifflin, Southampton) • Ways in which different people view places and regions (e.g., places to visit or to avoid) • Community connections to other places • Dependence and interdependence • Access and movement	• How regions are connected (e.g., watersheds and river systems, patterns of world trade, cultural ties, migration)	
Basic Geography Literacy must include local-to-global progression (scales) for all students at all grade levels for the standard statements and their descriptors. Basic concepts introduced in lower grade levels must be developed more fully throughout higher grade levels. Portions of Basic Geography Literacy relate directly to the Mathematics Standards.			

7.2 The Physical Characteristics of Places and Regions			
7.2.3. GRADE 3	**7.2.6. GRADE 6**	**7.2.9. GRADE 9**	**7.2.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. . .			
A. Identify the physical characteristics of places and regions. • Physical properties • Landforms (e.g., plains, hills, plateaus and mountains) • Bodies of water (e.g., rivers, lakes, seas and oceans) • Weather and climate • Vegetation and animals • Earth's basic physical systems • Lithosphere • Hydrosphere • Atmosphere • Biosphere	A. Describe the physical characteristics of places and regions. • Components of Earth's physical systems (e.g., clouds, storms, relief and elevation [topography], tides, biomes, tectonic plates) • Comparison of the physical characteristics of different places and regions (e.g., soil, vegetation, climate, topography) • Climate types (e.g., marine west coast, humid continental, tropical wet and dry)	A. Explain the physical characteristics of places and regions including spatial patterns of Earth's physical systems. • Climate regions • Landform regions	A. Analyze the physical characteristics of places and regions including the interrelationships among the components of Earth's physical systems. • Biomes and ecosystem regions • Watersheds and river basins • World patterns of biodiversity
B. Identify the basic physical processes that affect the physical characteristics of	B. Describe the physical processes that shape patterns on Earth's surface. • Earth-sun relationships (i.e.,	B. Explain the dynamics of the fundamental processes that underlie the operation of Earth's physical	B. Analyze the significance of physical processes in shaping the character of places and regions.

places and regions. • Earth-sun relationships (i.e., seasons and length of daylight, weather and climate) • Extreme physical events (e.g., earthquakes, floods, hurricanes, tornadoes)	differences between equinoxes and solstices, reasons they occur and their relationship to latitude) • Climate influences (e.g., elevation, latitude, nearby ocean currents) • Climate change, (e.g., global warming/cooling, decertification, glaciations) • Plate tectonics • Hydrologic cycle	systems. • Wind systems • Water cycle • Erosion/deposition cycle • Plate tectonics • Ocean currents • Natural hazards	• Circulation of the oceans • Ecosystem processes • Atmospheric systems • Extreme natural events

The Physical Characteristics of Places and Regions must include local-to-global progression (scales) for all students at all grade levels for the standard statements and their descriptors. Basic concepts must be developed more fully throughout higher grade levels. Portions of Physical Characteristics of Places and Regions relate directly to Science and Technology and Environment and Ecology standards.

7.3 The Human Characteristics of Places and Regions			
7.3.3. GRADE 3	**7.3.6. GRADE 6**	**7.3.9. GRADE 9**	**7.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to. .			
A. Identify the human characteristics of places and regions by their population characteristics. • The number and distribution of people in the local community • Human movement in the local community (e.g., mobility in daily life, migration)	A. Describe the human characteristics of places and regions by their population characteristics. • Spatial distribution, size, density and demographic characteristics of population at the county and state level. • Causes of human movement • Mobility (e.g., shopping, commuting, recreation) • Migration models (e.g., push/pull factors, barriers to migration)	A. Explain the human characteristics of places and regions by their population characteristics. • Spatial distribution, size, density and demographic characteristics of population at the state and National level • Demographic structure of a population (e.g., life expectancy, fertility rate, mortality rate, infant mortality rate, population growth rate, the demographic transition model) • Effects of different types and patterns of human movement • Mobility (e.g., travel for business) • Migration (e.g., rural to urban, short term vs. long term, critical distance)	A. Analyze the significance of human activity in shaping places and regions by their population characteristics: • Spatial distribution, size, density and demographic characteristics of population at the international level • Demographic trends and their impacts on patterns of population distribution (e.g., carrying capacity, changes in fertility, changes in immigration policy, the mobility transition model) • Impact of movement on human systems (e.g., refugees, guest workers, illegal aliens)
B. Identify the human characteristics of places and regions by their cultural characteristics. • Components of culture (e.g., language, belief systems and customs, social organizations, foods, ethnicity) • Ethnicity of people in the local community (e.g., customs, celebrations, languages, religions)	B. Describe the human characteristics of places and regions by their cultural characteristics. • Ethnicity of people at the county and state levels (e.g., customs, celebrations, languages, religions) • Spatial arrangement of cultures creates distinctive landscapes (e.g., cultural regions based on languages, customs, religion, building styles as in the Pennsylvania German region)	B. Explain the human characteristics of places and regions by their cultural characteristics. • Ethnicity of people at national levels (e.g., customs, celebrations, languages, religions) • Culture distribution (e.g., ethnic enclaves and neighborhoods) • Cultural diffusion (e.g., acculturation and assimilation, cultural revivals of language)	B. Analyze the significance of human activity in shaping places and regions by their cultural characteristics. • Cultural conflicts (e.g., over language (Canada), over political power (Spain), over economic opportunities (Mexico)) • Forces for cultural convergence (e.g., the diffusion of foods, fashions, religions, language)
C. Identify the human characteristics of places and regions by their settlement characteristics. • Types of settlements (e.g., villages, towns, suburbs, cities, metropolitan areas) • Factors that affect where people settle (e.g., water, resources, transportation)	C. Describe the human characteristics of places and regions by their settlement characteristics. • Current and past settlement patterns in the local area • Factors that affect the growth and decline of settlements (e.g., immigration, transportation development, depletion of natural resources, site and situation)	C. Explain the human characteristics of places and regions by their settlement characteristics. • Current and past settlement patterns in Pennsylvania and the United States • Forces that have re-shaped modern settlement patterns (e.g., central city decline, suburbanization, the development of transport systems) • Internal structure of cities (e.g., manufacturing zones, inner and outer suburbs, the location of infrastructure)	C. Analyze the significance of human activity in shaping places and regions by their settlement characteristics. • Description of current and past settlement patterns at the international scale (e.g., global cities) • Use of models of the internal structure of cities (e.g., concentric zone, sector, multiple nuclei) • Forces that have reshaped settlement patterns (e.g., commuter railroads, urban freeways, the development of megalopolis and edge cities)
D. Identify the human characteristics of places and regions by their economic activities. • Location factors in the spatial distribution of economic activities (e.g., market, transportation, workers, materials) • Producers of consumer products and services (e.g., bread, pizza, television, shopping malls) • Products of farms and factories at the local and regional level (e.g.,	D. Describe the human characteristics of places and regions by their economic activities. • Spatial distribution of economic activities in the local area (e.g., patterns of agriculture, forestry, mining, retailing, manufacturing, services) • Factors that influence the location and spatial distribution of economic activities (e.g., market size for different types of business, accessibility, modes of transportation used to move people, goods and materials) • Spatial distribution of resources and their relationship to population distribution	D. Explain the human characteristics of places and regions by their economic activities. • Spatial distribution of economic activities in Pennsylvania and the United States (e.g., patterns of agriculture, forestry, mining, retailing, manufacturing, services) • Factors that shape spatial patterns of economic activity both Nationally and internationally (e.g., comparative advantage in location of economic activities; changes in resource trade; disruption of trade flows) • Technological changes that affect the definitions of, access to, and use of natural	D. Analyze the significance of human activity in shaping places and regions by their economic characteristics. • Changes in spatial distribution of economic activities at the global scale (e.g., patterns of agriculture, forestry, mining, retailing, manufacturing, services) • Forces that are reshaping business (e.g., the information economy, business globalization, the development of off-shore activities) • Effects of changes and movements in factors of production (e.g., resources, labor, capital)

mushrooms, milk, snack foods, furniture) • Spatial distribution of resources • Non-renewable resources • Renewable resources • Flow resources (e.g., water power, wind power)	• Historical settlement patterns and natural resource use (e.g.,	resources (e.g., the role of exploration, extraction, use and depletion of resources)	

7.3 The Human Characteristics of Places and Regions			
7.3.3. GRADE 3	**7.3.6. GRADE 6**	**7.3.9. GRADE 9**	**7.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to.			
	waterpower sites along the FallLine) • Natural resource-based industries (e.g., agriculture, mining, fishing, forestry)		
E. Identify the human characteristics of places and regions by their political activities. • Type of political units (e.g., townships, boroughs, towns, cities, counties, states, countries (nation-state)) • Political units in the local area	E. Describe the human characteristics of places and regions by their political activities. • Spatial pattern of political units in Pennsylvania • Functions of political units (e.g., counties, municipalities, townships, school districts, PA General Assembly districts (House and Senate), U.S. Congressional districts, states)	E. Explain the human characteristics of places and regions by their political activities. • Spatial pattern of political units in the United States • Geographic factors that affect decisions made in the United States (e.g., territorial expansion, boundary delineation, allocation of natural resources) • Political and public policies that affect geography (e.g., open space, urban development)	E. Analyze the significance of human activity in shaping places and regions by their political characteristics: • Spatial pattern of political units in the global system • Role of new political alliances on the international level (e.g., multinational organizations, worker's unions, United Nations' organizations) • Impact of political conflicts (e.g., secession, fragmentation, insurgencies, invasions)
The Human Characteristics of Places and Regions must include local-to-global progression (scales) for all students at all grade levels for the standard statements and their descriptors. Basic concepts found in lower grade levels must be developed more fully throughout higher grade levels. Portions of Human Characteristics of Places and Regions relate directly to the Civics and Government and Economics Standards.			

7.4 The Interactions Between People and Places			
7.4.3. GRADE 3	**7.4.6. GRADE 6**	**7.4.9. GRADE 9**	**7.4.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
A. Identify the impacts of physical systems on people. • How people depend on, adjust to and modify physical systems on a local scale (e.g., soil quality and agriculture, snowfall and daily activities, drought and water use) • Ways in which natural hazards affect human activities (e.g., storms, lightning, flooding)	A. Describe the impacts of physical systems on people. • How people depend on, adjust to and modify physical systems on regional scale (e.g., coastal industries, development of coastal communities, flood control) • Ways in which people adjust to life in hazard-prone areas (e.g., California and earthquakes, Florida and hurricanes, Oklahoma and tornadoes)	A. Explain the impacts of physical systems on people. • How people depend on, adjust to and modify physical systems on National scale (e.g., soil conservation programs, projects of The Corps of Engineers) • Ways in which people in hazard-prone areas adjust their ways of life (e.g., building design in earthquake areas, dry-farming techniques in drought-prone areas)	A. Analyze the impacts of physical systems on people. • How people depend on, adjust to and modify physical systems on international scales (e.g., resource development of oil, coal, timber) • Ways in which people modify ways of life to accommodate different environmental contexts (e.g., building in permafrost areas; the role of air-conditioning in the United States South and Southwest; the development of enclosed spaces for movement in cold climates)
B. Identify the impacts of people on physical systems. • Effects of energy use (e.g., water quality, air quality, change in natural vegetation) • Ways humans change local ecosystems (e.g., land use, dams and canals on waterways, reduction and extinction of species)	B. Describe the impacts of people on physical systems. • Changing spatial patterns on Earth's surface that result from human activities (e.g., lake desiccation as in the Aral Sea, construction of dikes, dams and storm surge barriers in the Netherlands, designation of State parks and forests throughout Pennsylvania) • Ways humans adjust their impact on the habitat (e.g., Endangered Species Act, replacement of wetlands, logging and replanting trees)	B. Explain the impacts of people on physical systems. • Forces by which people modify the physical environment (e.g., increasing population; new agricultural techniques; industrial processes and pollution) • Spatial effects of activities in one region on another region (e.g., scrubbers on power plants to clean air, transportation systems such as Trans-Siberian Railroad, potential effects of fallout from nuclear power plant accidents)	B. Analyze the impacts of people on physical systems. • How people develop international agreements to manage environmental issues (e.g., Rio de Janeiro Agreement, the Law of the Sea, the Antarctica Treaty) • How local and regional processes can have global effects (e.g., wind and hydroelectric power transmitted across regions, water use and irrigation for crop production) • Sustainability of resources (e.g., reforestation, conservation) • World patterns of resource distribution and utilization (e.g., oil trade, regional electrical grids)
The Interactions Between People and Places must include local to global scales for all students at all grade levels for the standard statements and their descriptors. Basic concepts found in lower grade levels must be developed more fully throughout higher grade levels.			

XXI. GLOSSARY

Absolute location:	**The position of a point on Earth's surface that can usually be described by latitude and longitude. Another example of absolute location would be the use of a nine digit zip code and street address.**
Acculturation:	**The process of adopting the traits of a cultural group.**
Assimilation:	The acceptance, by one culture group or community, of cultural traits associated with another.
Atmosphere:	The body of gases, aerosols and other materials that surrounds Earth and is held close by gravity. It extends about twelve miles from Earth's surface.
Barriers to migration:	Factors that keep people from moving (e.g., lack of information about potential destination, lack of funds to cover the costs of moving, regulations that control migration).
Basic map elements:	Materials included on geographic representations. These include title, directions, date of map, mapmaker's name, a legend and scale. Often a geographic grid, the source of information and sometimes an index of places on the map are also included.
Biomes:	A community of living organisms of a single major ecological region.
Biosphere:	The domain of Earth that includes all plant and animal life forms.
Boundary:	The limit or extent within which a system exists or functions, including a social group, a state or physical features.
Capital:	One of the factors of production of goods and services. Capital can be goods (e.g., factories and equipment, highways, information, communications systems) and/or funds (investment and working capital) used to increase production and wealth. Other factors are land, water and labor.
Cardinal directions:	The four main points of the compass; north, east, south and west.
Carrying capacity:	Maximum population that an area can support over time depending upon environmental conditions, human interventions and interdependence.
Central Place Theory:	The conceptual framework that explains the size, spacing and distribution of settlements and their economic relationships with their market areas.
Climate:	Long-term patterns and trends in weather elements and atmospheric conditions.
Climate graph (climagraph):	A diagram that combines average monthly temperature and precipitation data for a particular place.
Comparative advantage:	The specialization by a given area in the production of one or a few commodities for which it has a particular edge (e.g., labor quality, resources availability, production costs).
Concentric Zone Model:	A framework that proposes that urban functions and the associated land uses are arranged in rings that grow outward from a central area. One of three models developed to explain how cities and metropolitan areas are arranged internally. The other models are the Sector and the Multiple Nuclei.
Country:	Unit of political space often referred to as a state or nation-state.
Culture:	Learned behavior of people, which includes their belief systems and languages, their social relationships, their institutions and organizations and their material goods—food, clothing, buildings, tools and machines.
Cultural diffusion:	The spread of cultural elements from one culture to another.
Cultural landscape:	The human imprint on the physical environment; the humanized image as created or modified by people.
Demographic change:	Variation in population size, composition, rates of growth, density, fertility and mortality rates and patterns of migration.
Density:	The population or number of objects per unit area (e.g., per square kilometer or mile).
Decertification:	The spread of desert conditions in arid and semiarid regions resulting from a combination of climatic changes and increasing human pressures (e.g., overgrazing, removal of vegetation, cultivation of marginal land).
Desiccation:	See lake desiccation.
Developed country:	An area of the world that is technologically advanced, highly urbanized and wealthy and has generally evolved through both economic and demographic transitions.
Diffusion:	The spread of people, ideas, technology and products among places.
Distance decay:	The tendency for the acceptance of new ideas and technologies to decrease with distance from their source.
Earthquake:	Vibrations and shock waves caused by the sudden movement of tectonic plates along fracture zones, called faults, in Earth's crust.
Ecosystem (ecological system):	A network formed by the interaction of all living organisms (plants, animals, humans) with each other and with the physical and chemical factors of the environment in which they live.
Elevation:	Height of a point or place above sea level (e.g., Mount Everest has an elevation of 29,028 feet above sea level).
Enclaves:	A country, territorial or culturally distinct unit enclosed within a larger country or community.
Environment:	Everything in and on Earth's surface and its atmosphere within which organisms, communities or objects exist.
Equilibrium:	The point in the operation of a system when driving forces and resisting forces are in balance.
Equinoxes:	The two days during the calendar year (usually September 23 and March 21) when all latitudes have twelve hours of both daylight and darkness and the sun is directly overhead at the Equator.
Erosional processes:	The removal and transportation of weathered (loose) rock material by water, wind, waves and glaciers. Deposition is the end result of erosion and occurs when transported material is dropped.
Fall line:	A linear connection joining the waterfalls on numerous rivers and streams that marks the point where each river and stream descends from the upland and the limit of the navigability of each river (e.g., the narrow boundary zone between the coastal plain and the Piedmont in the Eastern United States where there are falls and rapids on streams and rivers as they drop from the more resistant rocks of the Piedmont onto the softer rocks of the coastal plain).
Fertility rate:	A measure of the number of children a woman will have during her child-bearing years (15 to 49 years of age) in comparison to the adult female population in a particular place.
Formal region:	An area defined by the uniformity or homogeneity of certain characteristics (e.g., precipitation, landforms, subculture).
Functional region:	An area united by a strong core (node) or center of human population and activity (e.g., banking linkages between large cities and smaller cities and towns).
Geographic Information System:	A geographic database that contains information about the distribution of physical and human characteristics of places. In order to test hypotheses, maps of one characteristic or a combination can be produced from the database to analyze the data relationships.
Geographic scale:	The size of Earth's surface being studied. Study areas vary from local to regional to global. Scale also refers to the relationship between the size of space on a map and the size of that space on Earth's surface. Maps are referred to as large scale if they are of smaller (local) areas and small scale if they represent much or all of the Earth's surface. Map scale is expressed as a bar graph or representative fraction.
Global warming:	The theory that Earth's atmosphere is gradually warming due to the buildup of certain gases, including carbon dioxide and methane, which are released by human activities. The increased levels of these gases cause added heat energy from Earth to be absorbed by the

	atmosphere instead of being lost in space.
Globe:	A scale model of Earth that correctly represents area, relative size and shape of physical features, distance between points and true compass direction.
Grid:	A pattern of lines on a chart or map, such as those representing latitude and longitude, which helps determine absolute location and assists in the analysis of distribution patterns.
Human features:	Tangible and intangible ideas associated with the culture, society and economy of places or areas. These include the spatial arrangement of land uses including transportation, the design of buildings and the nature and timing of activities that people conduct in these spaces.
Hydroelectric power:	Electrical energy generated by the force of falling water which rotates turbines housed in power plants in dams on rivers.
Hydrosphere:	The water realm of Earth which includes water contained in the oceans, lakes, rivers, ground, glaciers and water vapor in the atmosphere.
Infant mortality rate:	The annual number of deaths among infants under 1 year of age for every 1,000 live births. It usually provides an indication of health care levels. The United States, for example, has a 1994 rate of 8.3 infant deaths per 1,000 live births while Angola has a rate of 137 infant deaths per 1,000 births.
Interdependence:	Ideas, goods and services in one area affect decisions and events in other areas reducing self-sufficiency.
Intermediate directions:	The points of the compass that fall between north and east, north and west, south and east, south and west (e.g., NE, NW, SE, SW).
Intervening opportunity:	An alternate area that is a source of a product or service or a destination in the case of migration.
Lake desiccation:	The reduction in water level (drying out) of an inland water body.
Landform:	The shape, form or nature of a specific physical feature of Earth's surface (e.g., plain, hill, plateau, mountain).
Land use:	The range of uses of Earth's surface made by humans. Uses are classified as urban, rural, agricultural, forested, etc. with more specific sub-classifications useful for specific purposes (e.g., low-density residential, light industrial, nursery crops).
Life expectancy:	The average number of remaining years a person can expect to live under current mortality levels in a society. Life expectancy at birth is the most common use of this measure.
Lithosphere:	The uppermost portion of the solid Earth including soil, land and geologic formations.
Location:	The position of a point on Earth's surface expressed by means of a grid (absolute) or in relation (relative) to the position of other places.
Map:	A graphic representation of a portion of Earth that is usually drawn to scale on a flat surface.
Materials:	Raw or processed substances that are used in manufacturing (secondary economic activities). Most substances used in factories are already manufactured to some degree and come from other factories rather than from sources of raw materials.
Megalopolis:	The intermingling of two or more large metropolitan areas into a continuous or almost continuous built-up urban complex; sometimes referred to as a conurbation.
Mental map:	A geographic representation which conveys the cognitive image a person has of an area, including knowledge of features and spatial relationships as well as the individual's perceptions and attitudes regarding the place; also known as a cognitive map.
Metropolitan area:	The Federal Office of Management and Budget's designation for the functional area surrounding and including a central city; has a minimum population of 50,000; is contained in the same county as the central city; and includes adjacent counties having at least 15% of their residents working in the central city's county.
Migration:	The act or process of people moving from one place to another with the intent of staying at the destination permanently or for a relatively long period of time.
Multinational organizations:	An association of nations aligned around a common economic or political cause (e.g., the Organization of Petroleum Exporting Countries, the Organization of American States).
Multiple Nuclei Model:	A representation of urban structure based on the idea that the functional areas (land use) of cities develop around various points rather than just one in the Central Business District.
Municipality:	A political unit incorporated for local self-government (e.g., Pennsylvania's boroughs, townships).
NAFTA:	North American Free Trade Agreement. NAFTA is an accord to establish clear and mutually advantageous rules governing commerce among Canada, Mexico and the United States.
NATO:	North Atlantic Treaty Organization. An international transatlantic partnership consisting of various European states, the United States and Canada, which was designed through cooperation, consultation and collective defense to maintain peace and promote stability throughout Europe.
Nation:	A cultural concept for a group of people bound together by a strong sense of shared values and cultural characteristics including language, religion and common history.
Natural hazard:	An event in the physical environment, such as a hurricane or earthquake, that is destructive to human life and property.
Natural resource:	An element of the physical environment that people value and use to meet a need for fuel, food, industrial product or something else of value.
Nonrenewable resource:	A finite element that cannot be replaced once it is used (e.g., petroleum, minerals).
Ocean currents:	The regular and consistent horizontal flow of water in the oceans, usually in response to persistent patterns of circulation in the atmosphere.
OAS:	Organization of American States. An international governmental organization formed by the nation-states of North America and South America for security and the protection of mutual interests.
OPEC:	The Organization of Petroleum Exporting Countries; international cartel of thirteen nations designed to promote collective pricing of petroleum, unified marketing policies and regulation of petroleum extraction.
Perceptual region:	Ideas that people have about the character of areas based on impressions from a variety of sources of information including other individuals and media. Mental maps can be used to access these ideas to find out what people think about particular areas.
Physical feature:	An aspect of a place or area that derives from the physical environment.
Physical process:	A course or method of operation that produces, maintains or alters Earth's physical system (e.g., glacial eroding, depositing landforms).
Place:	An area with distinctive human and physical characteristics; these characteristics give it meaning and character and distinguish it from other areas.
Plate tectonics:	The theory that Earth's surface is composed of rigid slabs or plates (see tectonic plates). The divergence, convergence and slipping side-by-side of the different plates is responsible for present-day configurations of continents, ocean basins and major mountain ranges and valley systems.

Pollution:	The direct or indirect process resulting from human action by which any part of the environment is made potentially or actually unhealthy, unsafe or hazardous to the welfare of the organisms which live in it.
Population density:	The number of individuals occupying an area derived from dividing the number of people by the area they occupy (e.g., 2,000 people divided by ten square miles = 200 people per square mile).
Population pyramid:	A bar graph showing the distribution by gender and age of a country's population.
Primary economic activity:	The production of naturally existing or culturally improved resources (i.e., agriculture, ranching, forestry, fishing, extraction of minerals and ores).
Pull factors:	In migration theory, the social, political, economic and environmental attractions of new areas that draw people away from their previous location.
Push factors:	In migration theory, the social, political, economic and environmental forces that drive people from their previous location.
Region:	An area with one or more common characteristics or features that give it a measure of consistency and make it different from surrounding areas.
Relative location:	The site of a place or region in relation to other places or regions (e.g., northwest, downstream).
Renewable resource:	A substance that can be regenerated if used carefully (e.g., fish, timber).
Resource:	An aspect of the physical environment that people value and use to meet a need for fuel, food, industrial product or something else of value.
Satellite image:	A representation produced by a variety of sensors (e.g., radar, microwave detectors, scanners) that measure and record electromagnetic radiation. The collected data are turned into digital form for transmission to ground receiving stations. The data can be reconverted into imagery in a form resembling a photograph.
Scale:	On maps the relationship or ratio between a linear measurement on a map and the corresponding distance on Earth's surface. For example, the scale 1:1,000,000 means one unit (inch or centimeter) on the map represents 1,000,000 of the same units on Earth's surface. Also refers to the size of places or regions being studied.
Sector Model:	A theory of urban structure that recognizes the impact of transportation on land prices within the city and the resulting tendency for functional areas to be organized into sectors.
Secondary economic activity:	Processing of raw and manufactured materials into products with added value.
Settlement pattern:	The spatial distribution and arrangement of human habitations (e.g., rural, urban).
Site:	The specific location where something may be found including its physical setting (e.g., on a floodplain).
Situation:	The general location of something in relation to other places or features of a larger region (e.g., in the center of a group of cities).
Soil:	Unconsolidated material found at the surface of Earth, which is divided into layers (or horizons) characterized by the accumulation or loss of organic and inorganic compounds. Loam types and depths vary greatly over Earth's surface and are very much influenced by climate, organisms, rock type, local relief, time and human activity.
Spatial:	Pertains to space on Earth's surface.
Spatial distribution:	The distribution of physical and human elements on Earth's surface.
Spatial organization:	The arrangement on Earth's surface of physical and human elements.
Suburbanization:	The shift in population from living in higher density urban areas to lower density developments on the edge of cities.
System:	A collection of entities that are linked and interrelated (e.g., the hydrologic cycle, cities, transportation modes).
Technology:	Application of knowledge to meet the goals, goods and services needed and desired by people.
Tectonic plates:	Sections of Earth's rigid crust that move as distinct units on a plastic-like ledge (mantle) on which they rest. As many as twenty different plates have been identified, but only seven are considered to be major (e.g., Eurasian Plate, South American Plate).
Thematic map:	A geographic representation of a specific spatial distribution, theme or topic (e.g., population density, cattle production, climates of the world).
Time zone:	A division of Earth, usually 15 degrees longitude, within which the time at the central meridian of the division represents the whole division.
Topography:	The shape of Earth's surface.
Water cycle:	The continuous circulation of water from the oceans, through the air, to the land and back to the sea. Water evaporates from oceans, lakes, rivers and the land surfaces and transpires from vegetation. It condenses into clouds in the atmosphere that may result in precipitation returning water to the land. Water then seeps into the soil or flows out to sea completing the circulation. Also known as Hydrologic Cycle.

Academic Standards for History

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XXIII. INTRODUCTION

This document includes Academic Standards for History that describe what students should know and be able to do in four areas:

• 8.1. Historical Analysis and Skills Development

• 8.2. Pennsylvania History

• 8.3. United States History

• 8.4. World History

The History Standards describe what students should know and be able to do at four grade levels (third, sixth, ninth and twelfth). They reflect an understanding of chronological events and the application of historical thinking skills in viewing the human record. These academic standards provide an organizing content for schools.

The Academic Standards for History are grounded in the Public School Code of 1949 which directs ''... study in the history and government of that portion of America which has become the United States of America, and of the Commonwealth of Pennsylvania . . .''. Chapter 4—Academic Standards and Assessment in § 4.21 (relating to elementary education; primary and intermediate levels) reinforces the School Code by indicating that the history of the United States and the history of the Commonwealth must be taught once by the end of elementary school. In addition, § 4.22 (relating to middle level education) indicates

that planned instruction in the history and cultures of the United States, the Commonwealth and world shall be provided. Chapter 4 also states that planned instruction shall be provided in the history and cultures of the United States, the Commonwealth and world in § 4.23 (relating to high school education).

To support the intent of the Public School Code and Chapter 4, this document creates four standard categories. The four standard categories were designed to meld historical thinking (8.1. Historical Analysis and Skills Development) with historical understanding (8.2. Pennsylvania History, 8.3. United States History, and 8.4. World History) to describe what students should know and be able to do.

Standard category 8.1. Historical Analysis and Skill Development provides the basis for learning the content within the other three standard categories. The intent of the history standards is to instill in each student an ability to comprehend chronology, develop historical comprehension, evaluate historical interpretation and to understand historical research. One should not view these standards as a list of facts to recall, rather as stated in the opening phrase to the Pennsylvania, United States and World standard categories, ''Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze the interaction of cultural, economic, geographic, political and social relations.''

These standards provide a history framework to permit every school and teacher to create planned instruction. The content within this document is general and does not represent a course or even a portion thereof. Every school is encouraged to move beyond these standards. These standards are merely a starting point for the study of history. Planned instruction to meet these standards is required; however, the methodology, resources and time are not recommended nor implied.

History is a discipline that interprets and analyzes the past. It is a narrative—a story. In order to tell the story it is not sufficient to simply recall facts; it is also necessary to understand the context of the time and place and to apply historical thinking skills. It is with this concept established, that the content delineated in Pennsylvania, United States and World histories should be approached. Having established the need to move beyond recall, it is the intent of these standards to give students throughout Pennsylvania a common cultural literacy.

Pennsylvania, United States, and World History standard categories use the same four standard statements to guide teachers in developing planned instruction. The four standard statements are: (A) Political and Cultural Contributions of Individuals and Groups; (B) Primary Documents, Material Artifacts and Historical Places; (C) How Continuity and Change Has Influenced History; (D) Conflict and Cooperation Among Social Groups and Organizations. The chart, Four Standard Statements within the Academic Standards for History: An Overview outlines standard statements and descriptors.

Although the standard statements are similar across grade levels and standard categories, the degree of comprehension, changes in content and shifts in chronology differ. Although different grade levels outline different chronological periods within the standards, it is intended that the specified chronological eras be linked to past learnings and that all eras be linked to the present. Linking to past learnings and the present is important, but so is addressing the standard statements in more depth. Therefore the following chronological time periods for the standard categories are established for the standard categories.

Pennsylvania and United States History		**World History**	
Grades 1-3	**Beginnings to Present**	**Grades 1-3**	**Beginnings to Present**
Grades 4-6	Beginnings to 1824	Grades 4-6	Beginnings to Present
Grades 7-9	1787 to 1914	Grades 7-9	Beginnings to 1500
Grades 10-12	1890 to Present	Grades 10-12	1450 to Present

Districts are encouraged to delineate each chronological period into less expansive historical eras within their planned instruction. The content listed in grade levels 1-3, 4-6, 7-9 and 10-12 should be age appropriate for the students in those grade levels and the reader should interpret each standard descriptor in that manner.

The Academic Standards for History consist of four standard categories (designated as 8.1., 8.2., 8.3., and 8.4.). Each category has four standard statements (designated A, B, C, and D). Most standard statements have bulleted items known as standard descriptors. The standard descriptors are items within the document to illustrate and enhance the standard statement. The categories, statements and descriptors are the regulations. The descriptors many times are followed by an ''e.g..'' The ''e.g.'s'' are examples to clarify what type of information could be taught. These are suggestions and the choice of specific content is a local decision as is the method of instruction.

History along with civics and government, economics and geography are identified as social studies in Chapter 4. This identification is consistent with citizenship education in Chapters 49 and 354 (relating to certification of professional personnel; and preparation of professional educators). Based on these regulations, social studies/citizenship programs should include the four sets of standards as an entity in developing a scope and sequence for curriculum and planned instruction.

A glossary is included to assist the reader in understanding terminology contained in the standards.

Four Standard Statements within the Academic Standards for History: An Overview	
Political and Cultural Contributions of Individuals and Groups • Inhabitants (cultures, subcultures, groups) • Political Leaders (monarchs, governors, elected officials) • Military Leaders (generals, noted military figures) • Cultural and Commercial Leaders (entrepreneurs, corporate executives, artists, entertainers, writers) • Innovators and Reformers (inventors, philosophers, religious leaders, social change agents, improvers of technology)	**How Continuity and Change Have Influenced History** • Belief Systems and Religions (ideas, beliefs, values) • Commerce and Industry (jobs, trade, environmental change, labor systems, entertainment) • Innovations (ideas, technology, methods and processes) • Politics (political party systems, administration of government, rules, regulations and laws, political and judicial interpretation) • Transportation (methods of moving people and goods over time, transportation routes, circulation systems) • Settlement Patterns and Expansion (population density and diversity, settlement types, land use, colonization) • Social Organization (social structure, identification of social groups, families, groups and communities, education, school population, suffrage, civil rights) • Women's Movement (changing roles of women, social and political movements, breaking barriers, role models)
Primary Documents, Material Artifacts and Historical Places	**Conflict and Cooperation Among Social Groups and Organizations** • Domestic Instability (political unrest, natural and man-made disasters,

• Documents, Writings and Oral Traditions (government documents, letters and diaries, fiction and non-fiction works, newspapers and other media, folklore) • Artifacts, Architecture and Historic Places (historic sites and places, museums and museum collections, official and popular cultural symbols, material culture)	genocide) • Ethnic and Racial Relations (racism and xenophobia, ethnic and religious prejudices, collective and individual actions) • Immigration and Migration (causes of population shifts, xenophobia, intercultural activity) • Labor Relations (strikes and collective bargaining, working conditions over time, labor/management identity) • Military Conflicts (causes, conduct and impact of military conflicts, wars and rebellions)

Each standard statement outlines its respective standard descriptors. Each standard descriptor suggests content that may be addressed. These are not all encompassing and local planned instruction is *not* limited to these examples.

8.1. Historical Analysis and Skills Development			
8.1.3. GRADE 3	**8.1.6. GRADE 6**	**8.1.9. GRADE 9**	**8.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
A. Understand chronological thinking and distinguish between past, present and future time. • Calendar time • Time lines • Continuity and change • Events (time and place)	A. Understand chronological thinking and distinguish between past, present and future time. • Calendar time • Time lines • People and events in time • Patterns of continuity and change • Sequential order • Context for events	A. Analyze chronological thinking. • Difference between past, present and future • Sequential order of historical narrative • Data presented in time lines • Continuity and change • Context for events	A. Evaluate chronological thinking. • Sequential order of historical narrative • Continuity and change • Context for events knowledge and skills needed to . . .
B. Develop an understanding of historical sources. • Data in historical maps • Visual data from maps and tables • Mathematical data from graphs and tables • Author or historical source	B. Explain and analyze historical sources. • Literal meaning of a historical passage • Data in historical and contemporary maps, graphs and tables • Author or historical source • Multiple historical perspectives • Visual evidence • Mathematical data from graphs and tables	B. Analyze and interpret historical sources. • Literal meaning of historical passages • Data in historical and contemporary maps, graphs, and tables • Different historical perspectives • Data from maps, graphs and tables • Visual data presented in historical evidence	B. Synthesize and evaluate historical sources. • Literal meaning of historical passages • Data in historical and contemporary maps, graphs and tables • Different historical perspectives • Data presented in maps, graphs and tables • Visual data presented in historical evidence

8.1. Historical Analysis and Skills Development			
8.1.3. GRADE 3	**8.1.6. GRADE 6**	**8.1.9. GRADE 9**	**8.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to . . .			
C. Understand fundamentals of historical interpretation. • Difference between fact and opinion • The existence of multiple points of view • Illustrations in historical stories • Causes and results	C. Explain the fundamentals of historical interpretation. • Difference between fact and opinion • Multiple points of view • Illustrations in historical stories • Causes and results • Author or source of historical narratives	C. Analyze the fundamentals of historical interpretation. • Fact versus opinion • Reasons/causes for multiple points of view • Illustrations in historical documents and stories • Causes and results • Author or source used to develop historical narratives • Central issue	C. Evaluate historical interpretation of events. • Impact of opinions on the perception of facts • Issues and problems in the past • Multiple points of view • Illustrations in historical stories and sources • Connections between causes and results • Author or source of historical narratives' points of view • Central issue
D. Understand historical research. • Event (time and place) • Facts, folklore and fiction • Formation of historical question • Primary sources • Secondary sources • Conclusions (e.g., storytelling, role playing, diorama)	D. Describe and explain historical research. • Historical events (time and place) • Facts, folklore and fiction • Historical questions • Primary sources • Secondary sources • Conclusions (e.g., simulations, group projects, skits and plays)	D. Analyze and interpret historical research. • Historical event (time and place) • Facts, folklore and fiction • Historical questions • Primary sources • Secondary sources • Conclusions (e.g., History Day projects, mock trials, speeches) • Credibility of evidence	D. Synthesize historical research. • Historical event (time and place) • Facts, folklore and fiction • Historical questions • Primary sources • Secondary sources • Conclusions (e.g., Senior Projects, research papers, debates) • Credibility of evidence
			Pennsylvania History, 8.3. United States History and 8.4. World History.

Historical Analysis and Skill Development are learned through and applied to the standards statements and their descriptors for 8.2 Pennsylvania History, 8.3 United States History and 8.4 World History

8.2. Pennsylvania History			
8.2.3. GRADE 3	**8.2.6. GRADE 6**	**8.2.9. GRADE 9**	**8.2.12. GRADE 12**

Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			
A. Understand the political and cultural contributions of individuals and groups to Pennsylvania history. • William Penn • Benjamin Franklin • Pennsylvanians impacting American Culture (e.g., John Chapman, Richard Allen, Betsy Ross, Mary Ludwig Hayes, Rachel Carson, Elizabeth Jane Cochran, Marian Anderson) • Local historical figures in municipalities and counties.	A. Identify and explain the political and cultural contributions of individuals and groups to Pennsylvania history from Beginnings to 1824. • Inhabitants (e.g., Native Americans, Europeans, Africans) • Military Leaders (e.g., Anthony Wayne, Oliver H. Perry, John Muhlenberg) • Political Leaders (e.g., William Penn, Hannah Penn, Benjamin Franklin) • Cultural and Commercial Leaders (e.g., Robert Morris, John Bartram, Albert Gallatin) • Innovators and Reformers (e.g., Society of Friends, Richard Allen, Sybilla Masters)	A. Analyze the political and cultural contributions of individuals and groups to Pennsylvania history from 1787 to 1914. • Political Leaders (e.g., James Buchanan, Thaddeus Stevens, Andrew Curtin) • Military Leaders (e.g., George Meade, George McClellan, John Hartranft) • Cultural and Commercial Leaders (e.g., John J. Audubon, Rebecca Webb Lukens, Stephen Foster) • Innovators and Reformers (e.g., George Westinghouse, Edwin Drake, Lucretia Mott)	A. Evaluate the political and cultural contributions of individuals and groups to Pennsylvania history from 1890 to Present. • Political Leaders (e.g., Gifford Pinchot, Genevieve Blatt, K. Leroy Irvis) • Military Leaders (e.g., Tasker H. Bliss, Henry ''Hap'' Arnold, George C. Marshall) • Cultural and Commercial Leaders (e.g., Milton Hershey, Marian Anderson, Fred Rogers) • Innovators and Reformers (e.g., Frank Conrad, Rachel Carson, Joseph Rothrock)
B. Identify and describe primary documents, material artifacts and historic sites important in Pennsylvania history. • Documents, Writings and Oral Traditions (e.g., Penn's Charter, Pennsylvania ''Declaration of Rights') • Artifacts, Architecture and Historic Places (e.g., Local historical sites, museum collections, Independence Hall) • Liberty Bell • Official Commonwealth symbols (e.g., tree, bird, dog, insect)	B. Identify and explain primary documents, material artifacts and historic sites important in Pennsylvania history from Beginnings to 1824. • Documents, Writings and Oral Traditions (e.g., Charter of Privileges, The Gradual Abolition of Slavery Act of 1780, *Letters from a Pennsylvania Farmer*) • Artifacts, Architecture and Historic Places (e.g., Conestoga Wagon, Pennsylvania rifle, Brig Niagara)	B. Identify and analyze primary documents, material artifacts and historic sites important in Pennsylvania history from 1787 to 1914. • Documents, Writings and Oral Traditions (e.g., Pennsylvania Constitutions of 1838 and 1874, The ''Gettysburg Address,'' *The Pittsburgh Survey)* • Artifacts, Architecture and Historic Places (e.g., Gettysburg, Eckley Miners' Village, Drake's Well)	B. Identify and evaluate primary documents, material artifacts and historic sites important in Pennsylvania history from 1890 to Present. • Documents, Writings and Oral Traditions (e.g., Constitution of 1968, *Silent Spring* by Rachel Carson, Pennsylvania historical markers) • Artifacts, Architecture and Historic Places (e.g., 28th Division Shrine, Fallingwater, Levittown, Allegheny Ridge heritage corridor)
C. Identify and describe how continuity and change have influenced Pennsylvania history. • Belief Systems and Religions (e.g., Native Americans, early settlers, contemporary religions) • Commerce and Industry (e.g., jobs, trade, environmental change) • Innovations (e.g., technology, ideas, processes) • Politics (e.g., rules, regulations, laws) • Settlement Patterns (e.g., farms, towns, rural communities, cities) • Social Organization (e.g., relationships of individuals, families, groups, communities; ability to be educated) • Transportation (e.g., methods of moving people and goods over time) • Women's Movement (e.g., changes in roles and rights over time)	C. Identify and explain how continuity and change have influenced Pennsylvania history from the Beginnings to 1824. • Belief Systems and Religions (e.g., Native Americans, Quakers) • Commerce and Industry (e.g., iron production, sailing, fur trade) • Innovations (e.g., steam boat, Conestoga Wagon) • Politics (e.g., The Mason-Dixon Line, Pennsylvania's acquisition and detachment of the ''lower three counties,'' movements of State capital) • Settlement Patterns (e.g., native settlements, Westward expansion, development of towns) • Social Organization (e.g., trade and development of cash economy, African Methodist Episcopal Church founded, schools in the colony)	C. Identify and analyze how continuity and change have influenced Pennsylvania history from 1787 to 1914. • Belief Systems and Religions (e.g., Ephrata Cloister, Harmonists, Amish, immigrant influences) • Commerce and Industry (e.g., mining coal, producing iron, harvesting timber) • Innovations (e.g., John Roebling's steel cable, steel-tipped plow, improved techniques for making iron, steel and glass) • Politics (e.g., Fugitive Slave Act reaction, canal system legislation, The Free School Act of 1834) • Settlement Patterns (e.g., farms and growth of urban centers)	C. Identify and evaluate how continuity and change have influenced Pennsylvania history from the 1890s to Present. • Belief Systems and Religions (e.g., Buddhism, Christianity, Hinduism, Islam, Judaism) • Commerce and Industry (e.g., work of defense industries, rise and decline of the steel industry, increase of service industries) • Innovations (e.g., polio vaccine, air pollution examined, nuclear power plants) • Politics (e.g., Great Depression special legislative session, creation of the state income tax) • Settlement Patterns (e.g., growth and decline of cities, coal towns, Pittsburgh Renaissance)

8.2. Pennsylvania History			
8.2.3. GRADE 3	**8.2.6. GRADE 6**	**8.2.9. GRADE 9**	**8.2.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			
	• Transportation (e.g., trade routes, turnpikes, post roads) • Women's Movement (e.g., voting qualifications, role models)	• Social Organization (e.g., the Philadelphia Centennial Exposition of 1876, prohibition of racial discrimination in schools) • Transportation (e.g., canals, National Road, Thompson's Horseshoe Curve) • Women's Movement (e.g., work of the Equal Rights League of Pennsylvania)	• Social Organization (e.g., creation of the State Soil Conservation Commission, First Amendment challenges to education, social services) • Transportation (e.g., Pennsylvania Turnpike, Interstate highways, international airports) • Women's Movement (e.g., League of Women Voters, Commission for Women)

8.2. Pennsylvania History			
8.2.3. GRADE 3	**8.2.6. GRADE 6**	**8.2.9. GRADE 9**	**8.2.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			

D. Identify and describe conflict and cooperation among social groups and organizations in Pennsylvania history. • Domestic Instability (e.g., political, economic and geographic impact on daily activities) • Ethnic and Racial Relations (e.g., treatment of various ethnic and racial groups in history) • Labor Relations (e.g., working conditions, over time) • Immigration (e.g., diverse groups inhabiting the state) • Military Conflicts (e.g., struggle for control)	D. Identify and explain conflict and cooperation among social groups and organizations in Pennsylvania history from Beginnings to 1824. • Domestic Instability (e.g., religious diversity, toleration and conflicts, incursion of the Iroquois) • Ethnic and Racial Relations (e.g., Penn's Treaties with Indians, the Underground Railroad, the abolition of slavery) • Labor Relations (e.g., indentured servants, working conditions) • Immigration (e.g., Germans, Irish) • Military Conflicts (e.g., Dutch, Swedish and English struggle for control of land, Wyoming Massacre, The Whiskey Rebellion)	D. Identify and analyze conflict and cooperation among social groups and organizations in Pennsylvania history from 1787 to 1914. • Domestic Instability (e.g., impact of war, 1889 Johnstown Flood) • Ethnic and Racial Relations (e.g., Christiana riots, disenfranchisement and restoration of suffrage for African-Americans, Carlisle Indian School) • Labor Relations (e.g., National Trade Union, The "Molly Maguires," Homestead steel strike) • Immigration (e.g., Anti-Irish Riot of 1844, new waves of immigrants) • Military Conflicts (e.g., Battle of Lake Erie, the Mexican War, the Civil War)	D. Identify and evaluate conflict and cooperation among social groups and organizations in Pennsylvania history from 1890 to Present. • Domestic Instability (e.g., The Great Depression, Three-Mile Island nuclear accident, floods of 1936, 1972 and 1977) • Ethnic and Racial Relations (e.g., segregation, desegregation, racial profiling) • Labor Relations (e.g., strikes, work stoppages, collective bargaining) • Immigration (e.g., increased immigration from Europe, migration of African-Americans from the South, influx of Hispanic and Asian peoples) • Military Conflicts (e.g., World War I, World War II, Persian Gulf War)
Standard Category 8.1. Historical Analysis and Skills Development should be applied to the above standard statements and descriptors. Suggested chronology for grade levels 4-6, 7-9 and 10-12 focus on a particular century; however, instruction is encouraged that draws on prior and later events in history so that students may develop a seamless view of the world.			

8.3. United States History			
8.3.3. GRADE 3	**8.3.6. GRADE 6**	**8.3.9. GRADE 9**	**8.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			
A. Identify contributions of individuals and groups to United States history. • George Washington • Thomas Jefferson • Abraham Lincoln • Theodore Roosevelt • Franklin D. Roosevelt • Individuals who are role models (e.g., Abigail Adams, Sacajawea, Frederick Douglass, Clara Barton, Jackie Robinson, Rosa Parks, Archbishop Patrick Flores, Jamie Escalante, Sally Ride, Tiger Woods, Cal Ripken, Jr., Sammy Sosa)	A. Identify and explain the political and cultural contributions of individuals and groups to United States history from Beginnings to 1824. • Native Americans, Africans and Europeans • Political Leaders (e.g., John Adams, Thomas Jefferson, John Marshall) • Military Leaders (e.g. George Washington, Meriwether Lewis, Henry Knox) • Cultural and Commercial Leaders (e.g., Paul Revere, Phyllis Wheatley, John Rolfe) • Innovators and Reformers (e.g., Ann Hutchinson, Roger Williams, Junipero Serra)	A. Identify and analyze the political and cultural contributions of individuals and groups to United States history from 1787 to 1914. • Political Leaders (e.g., Daniel Webster, Abraham Lincoln, Andrew Johnson) • Military Leaders (e.g., Andrew Jackson, Robert E. Lee, Ulysses S. Grant) • Cultural and Commercial Leaders (e.g., Jane Addams, Jacob Riis, Booker T. Washington) • Innovators and Reformers (e.g., Alexander G. Bell, Frances E. Willard, Frederick Douglass)	A. Identify and evaluate the political and cultural contributions of individuals and groups to United States history from 1890 to Present. • Political Leaders (e.g., Theodore Roosevelt, Woodrow Wilson, Franklin D. Roosevelt) • Military Leaders (e.g., John Pershing, Douglas MacArthur, Dwight D. Eisenhower) • Cultural and Commerical Leaders (e.g., Abby Aldrich Rockefeller, Langston Hughes, Alan Greenspan) • Innovators and Reformers (e.g., Wilbur and Orville Wright, John L. Lewis, Reverend Dr. Martin Luther King)
B. Identify and describe primary documents, material artifacts and historic sites important in United States history. • Documents (e.g., Declaration of Independence, U.S. Constitution, Bill of Rights) • Writings and Communications (e.g., Pledge of Allegiance, famous quotations and sayings) • Historic Places (e.g., The White House, Mount Rushmore, Statue of Liberty) • The Flag of the United States	B. Identify and explain primary documents, material artifacts and historic sites important in United States history from Beginnings to 1824. • Documents (e.g., Mayflower Compact, Northwest Ordinance, Washington's Farewell Address) • 18th Century Writings and Communications (e.g., Paine's *Common Sense*; Franklin's "Join, or Die," Henry's "Give me liberty or give me death") • Historic Places (e.g., Cahokia Mounds, Spanish Missions, Jamestown)	B. Identify and analyze primary documents, material artifacts and historic sites important in United States history from 1787 to 1914. • Documents (e.g., Fugitive Slave Law, Treaty of Guadalupe Hidalgo, Emancipation Proclamation) • 19th Century Writings and Communications (e.g., Stowe's *Uncle Tom's Cabin*, Brown's "Washed by Blood," Key's Star Spangled Banner) • Historic Places (e.g., The Alamo, Underground Railroad sites, Erie Canal)	B. Identify and evaluate primary documents, material artifacts and historic sites important in United States history from 1890 to Present. • Documents (e.g., Treaty of Versailles, North Atlantic Treaty, Neutrality Acts) • 20th Century Writings and Communication (e.g., Coolidge's "The Business of America is Business," King's "I Have A Dream," Armstrong's "One Small Step for Man") • Historic Places (e.g., Ellis Island, Pearl Harbor, Los Alamos)

8.3. United States History			
8.3.3. GRADE 3	**8.3.6. GRADE 6**	**8.3.9. GRADE 9**	**8.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			
C. Identify important changes in United States history (e.g., Belief Systems and Religions, Commerce and Industry, Innovations, Politics, Settlement Patterns and Expansion, Social Organization, Transportation, Women's Movement).	C. Explain how continuity and change has influenced United States history from Beginnings to 1824. • Belief Systems and Religions (e.g., impact on daily life, colonial government established religions, communal sects)	C. Analyze how continuity and change has influenced United States history from 1787 to 1914. • Belief Systems and Religions (e.g., 19th century trends and movements) • Commerce and Industry (e.g., growth of manufacturing	C. Evaluate how continuity and change has influenced United States history from 1890 to Present. • Belief Systems and Religions (e.g., 20th century movements, religions of recent immigrants) • (Commerce and Industry (e.g.,

	• Commerce and Industry (e.g., fur trade, development of cash crops) • Innovations (e.g., cotton gin, Whitney; wooden clock, Banneker; stove, Franklin) • Politics (e.g., Hamilton's defense of John Peter Zenger, The Great Compromise, *Marbury v. Madison*) • Settlement Patterns (e.g., frontier settlements, slave plantation society, growth of cities) • Social Organization (e.g., community structure on the frontier, cultural and language barriers)	industries, economic nationalism) • Innovations (e.g., Brooklyn Bridge, refrigerated shipping, telephone) • Politics (e.g., election of 1860, impeachment of Andrew Johnson, Jim Crow Laws) • Settlement Patterns and Expansion (e.g., Manifest Destiny, successive waves of immigrants, purchase of Alaska and Hawaii) • Social Organization (e.g., social class differences, women's rights and antislavery movement, education reforms)	corporations, conglomerates, multinational corporations) • Innovations (e.g., The Tin Lizzie, radio, World Wide Web) • Politics (e.g., New Deal legislation, *Brown v. Topeka*, isolationist/non-isolationist debate) • Settlement Patterns (e.g., suburbs, large urban centers, decline of city population) • Social Organization (e.g., compulsory school laws, court decisions expanding individual rights, technological impact)

8.3. United States History			
8.3.3. GRADE 3	**8.3.6. GRADE 6**	**8.3.9. GRADE 9**	**8.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			
	• Transportation and Trade (e.g., methods of overland travel, water transportation, National Road) • Women's Movement (e.g., roles and changing status of women, Margaret Brent's vote, soldier Deborah Sampson)	• Transportation and Trade (e.g., Pony Express, telegraph, Transcontinental Railroad) • Women's Movement (e.g., roles in the Civil War, medical college for women, Seneca Falls Conference)	• Transportation and Trade (e.g., expansion and decline of railroads, increased mobility, Internet) • Women's Movement (e.g., right to vote, women in the war effort, Women's Peace Party)
D. Identify conflict and cooperation among social groups and organizations in United States history. • Domestic Instability (e.g., impact on daily activities) • Ethnic and Racial Relations (e.g., treatment of minority groups in history) • Labor Relations (e.g., working conditions over time) • Immigration (e.g., diverse groups inhabiting the state) • Military Conflicts (e.g., struggle for control)	D. Identify and explain conflict and cooperation among social groups and organizations in United States history from Beginnings to 1824. • Domestic Instability (e.g., Salem Witch Trials, Shays Rebellion, religious persecution) • Ethnic and Racial Relations (e.g., cooperation between and among Native Americans and European settlers, slave uprisings, "Colored" troops in the Revolution) • Labor Relations (e.g., early union efforts, 10-hour day, women's role) • Immigration and Migration (e.g., western settlements, Louisiana Purchase, European immigration)	D. Identify and analyze conflict and cooperation among social groups and organizations in United States history from 1787 to 1914. • Domestic Instability (e.g., wartime confiscation of private property, abolitionist movement, Reconstruction) • Ethnic and Racial Relations (e.g., Cherokee Trail of Tears, slavery and the Underground Railroad, draft riots) • Labor Relations (e.g., female and child labor, trade unionism, strike breakers) • Immigration and Migration (e.g., Manifest Destiny, eastern and southern European immigration, Chinese Exclusion Act)	D. Identify and evaluate conflict and cooperation among social groups and organizations in United States history from 1890 to the Present. • Domestic Instability (e.g., Great Depression, assassination of political and social leaders, terrorist threats) • Ethnic and Racial Relations (e.g., internment camps for Japanese Americans, Montgomery Alabama Bus Boycott, land tensions with Native Americans) • Labor Relations (e.g., rise and decline of industrial unions, free trade agreements, imports impact on domestic employment) • Immigration and Migration (e.g., anti-immigrant attitudes, quota laws, westward and southward migration)

8.3. United States History			
8.3.3. GRADE 3	**8.3.6. GRADE 6**	**8.3.9. GRADE 9**	**8.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			
	• Military Conflicts (e.g., French and Indian War, American Revolutionary War, War of 1812)	• Military Conflicts (e.g., Native American opposition to expansion and settlement, Civil War, Spanish-American War)	• Military Conflicts (e.g., World War I, World War II, War on Terrorism)

Standard Category 8.1. Historical Analysis and Skills Development should be applied to the above standard statements and descriptors. Suggested chronology for grade levels 4-6, 7-9 and 10-12 focus on a particular century; however, instruction is encouraged that draws on prior or later events in history so that students may develop a seamless view of the world.

8.4. World History			
8.4.3. GRADE 3	**8.4.6. GRADE 6**	**8.4.9. GRADE 9**	**8.4.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to analyze cultural, economic, geographic, political and social relations to . . .			
A. Identify individuals and groups who have made significant political and cultural contributions to world history. • Africa (e.g., Nefertiti, Mansa Musa, Nelson Mandela) • Americas (e.g., Montezuma, Simon Bolivar, Fidel Castro)	A. Identify and explain how individuals and groups made significant political and cultural contributions to world history. • Africa (e.g., Nelson Mandela, Desmond Tutu, F. W. de Klerk, Pieter Botha, African National Congress) • Americas (e.g., Pizarro, Atahualpa,	A. Analyze the significance of individuals and groups who made major political and cultural contributions to world history before 1500. • Political and Military Leaders (e.g., King Ashoka, Montezuma I, Ghenghis Khan, William the Conqueror) • Cultural and Commercial Leaders	A. Evaluate the significance of individuals and groups who made major political and cultural contributions to world history since 1450. • Political and Military Leaders (e.g., Askia Daud, Simon Bolivar, Napoleon Bonaparte, Mao Zedong) • Cultural and Commercial Leaders (e.g.,

• Asia (e.g., Hammurabi, Mohandas Gandhi, Benazir Bhutto) • Europe (e.g., Julius Ceasar, Joan of Arc, Pope John Paul)	Aztecs, Incas, Montezuma, Cortez) • Asia (e.g., Tokugawa Ieyasu, Toyotomi clan, shogun Iemitsu, Commodore Perry, daimyo) • Europe (e.g., Pope Leo X, John Calvin, John Wesley, Martin Luther, Ignatius of Loyola)	(e.g., Mansa Musa, Yak Pac, Cheng Ho, Marco Polo) • Innovators and Reformers (e.g., Erastostenes, Tupac Inka Yupenqui, Johannes Gutenberg)	Chinua Achebe, Gabriel Garcia Marquiez, Akira Kurosa, Christopher Columbus) • Innovators and Reformers (e.g., Nelson Mandela, Louis-Joseph Papineau, Mohandas Gandhi, Alexander Fleming)
B. Identify historic sites and material artifacts important to world history. • Africa (e.g., Pyramids, treasures of Tutankhamen, Nefertiti's sculpture) • Americas (e.g., Olmec ritualistic centers, Mayan pyramids, arrowheads) • Asia (e.g., Code of Hammurabi, Ziggurat at Ur, canals) • Europe (e.g., ancient megaliths, Arc de Triomphe, Acropolis)	B. Identify and explain important documents, material artifacts and historic sites in world history. • Africa (e.g., Prohibition of Marriages Act, prison on Robben Island) • Americas (e.g., Tenochtitlan, Aztec masks) • Asia (e.g., samurai sword, Commodore Perry's Black Ships) • Europe (e.g., Luther's Ninety-Five Theses, Wittenberg Castle Church)	B. Analyze historical documents, material artifacts and historic sites important to world history before 1500. • Documents, Writings and Oral Traditions (e.g., Rosetta Stone, Aztec glyph writing, Dead Sea Scrolls, Magna Carta) • Artifacts, Architecture and Historic Places (e.g., Ethiopian rock churches, Mayan pyramids, Nok terra cotta figures, megaliths at Stonehenge) • Historic districts (e.g., Memphis and its Necropolis, Sanctuary of Machu Picchu, Old City of Jerusalem and its Walls, Centre of Rome and the Holy See)	B. Evaluate historical documents, material artifacts and historic sites important to world history since 1450. • Documents, Writings and Oral Traditions (e.g., Declaration of the International Conference on Sanctions Against South Africa; Monroe Doctrine, Communist Manifesto, Luther's Ninety-five Theses) • Artifacts, Architecture and Historic Places (e.g., Robben Island, New York Trade Center, Hiroshima Ground Zero Memorial, Nazi concentration camps) • Historic districts (e.g., Timbuktu, Centre of Mexico City and Xochimilco, Taj Mahal and Gardens, Kremlin and Red Square)
C. Compare similarities and differences between earliest civilizations and life today. (e.g., Africa, Egypt; Asia, Babylonia; Americas, Olmec; Europe, Neolithic settlements).	C. Identify and explain how continuity and change has affected belief systems, commerce and industry, innovations, settlement patterns, social organizations, transportation and women's roles in world history. • Africa (e.g., Apartheid) • Americas (e.g., European conquest) • Asia (e.g., Japanese society prior to the Meiji Restoration) • Europe (e.g., Impact of the Great Schism and Reformation)	C. Analyze how continuity and change throughout history has impacted belief systems and religions, commerce and industry, innovations, settlement patterns, social organization, transportation and roles of women before 1500. • Africa • Americas • Asia • Europe	C. Evaluate how continuity and change throughout history has impacted belief systems and religions, commerce and industry, innovations, settlement patterns, social organization, transportation and roles of women since 1450. • Africa • Americas • Asia • Europe
D. Identify how conflict and cooperation among social groups and organizations affected world history. • Domestic Instability (e.g., political, economic and geographic impact on normal activities) • Labor Relations (e.g., working conditions over time) • Racial and Ethnic Relations (e.g., treatment of various ethnic and racial groups in history) • Immigration and migration (e.g., diverse groups inhabiting a territory) • Military Conflicts (e.g., struggle for control)	D. Explain how conflict and cooperation among social groups and organizations affected world history • Africa (e.g., imperialism) • Americas (e.g., European diseases) • Asia (e.g., trade routes) • Europe (e.g., Counter reformation)	D. Analyze how conflict and cooperation among social groups and organizations impacted world history through 1500 in Africa, Americas, Asia and Europe • Domestic Instability • Ethnic and Racial Relations • Labor Relations • Immigration and Migration • Military Conflicts	D. Evaluate how conflict and cooperation among social groups and organizations impacted world history from 1450 to Present in Africa, Americas, Asia and Europe. • Domestic Instability • Ethnic and Racial Relations • Labor Relations • Immigration and Migration • Military Conflicts
Standard Category 8.1. Historical Analysis and Skills Development should be applied to the above standard statements and descriptors. Suggested chronology in organizing the content for grade levels 7-9 and 10-12 use the 15th century as the dividing point; however, instruction is encouraged that draws on prior and later events in history so that students may develop a seamless view of the world.			

Pennsylvania Core Standards for Reading in History and Social Studies
Grades 6-12

INTRODUCTION

These standards describe what students in the social studies classroom should know and be able to do with the English language in reading, grade 6 through 12. The standards provide the targets for instruction and student learning essential for success in all academic areas, not just language arts classrooms. Although the standards are not a curriculum or a prescribed series of activities, school entities will use them to develop a local school curriculum that will meet local students' needs.

The standards below begin at grade 6; standards for K-5 reading in history/social studies, science, and technical subjects are integrated into the K-5 Reading standards.

The English Language Arts Standards for History and Social Studies also provide parents and community members with information about what students should know and be able to do as they progress through the educational program and at graduation. With a clearly defined target provided by the standards, parents, students, educators and community members become partners in learning. Each standard implies an end of year goal—with the understanding that exceeding the standard is an even more desirable end goal.

8.5	**Reading Informational Text** **Students read, understand, and respond to informational text—with emphasis on comprehension, making connections among ideas and between texts with focus on textual evidence.**				
	GRADE 6-8	**GRADE 9-10**	**GRADE 11-12**		
Key Ideas and Details			CC.8.5.6-8.A. Cite specific textual evidence to support analysis of primary and secondary sources.	CC.8.5.9-10.A. Cite specific textual evidence to support analysis of primary and secondary sources, attending to such features as the date and origin of the information.	CC.8.5.11-12.A. Cite specific textual evidence to support analysis of primary and secondary sources, connecting insights gained from specific details to an understanding of the text as a whole.
CC.8.5.6-8.B. Determine the central ideas or information of a primary or secondary source; provide an accurate summary of the source distinct from prior knowledge or opinions.	CC.8.5.9-10.B. Determine the central ideas or information of a primary or secondary source; provide an accurate summary of how key events or ideas develop over the course of the text.	CC.8.5.11-12.B. Determine the central ideas or information of a primary or secondary source; provide an accurate summary that makes clear the relationships among the key details and ideas.			
CC.8.5.6-8.C. Identify key steps in a text's description of a process related to history/social studies (e.g., how a bill becomes law, how interest rates are raised or lowered).	CC.8.5.9-10.C. Analyze in detail a series of events described in a text; determine whether earlier events caused later ones or simply preceded them.	CC.8.5.11-12.C. Evaluate various explanations for actions or events and determine which explanation best accords with textual evidence, acknowledging where the text leaves matters uncertain.			
Craft and Structure			CC.8.5.6-8.D. Determine the meaning of words and phrases as they are used in a text, including vocabulary specific to domains related to history/social studies.	CC.8.5.9-10.D. Determine the meaning of words and phrases as they are used in a text, including vocabulary describing political, social, or economic aspects of history/social science.	CC.8.5.11-12.D. Determine the meaning of words and phrases as they are used in a text, including analyzing how an author uses and refines the meaning of a key term over the course of a text (e.g., how Madison defines faction in Federalist No. 10).
CC.8.5.6-8.E. Describe how a text presents information (e.g., sequentially, comparatively, causally).	CC.8.5.9-10.E. Analyze how a text uses structure to emphasize key points or advance an explanation or analysis.	CC.8.5.11-12.E. Analyze in detail how a complex primary source is structured, including how key sentences, paragraphs, and larger portions of the text contribute to the whole.			
CC.8.5.6-8.F. Identify aspects of a text that reveal an author's point of view or purpose (e.g., loaded language, inclusion or avoidance of particular facts).	CC.8.5.9-10.F. Compare the point of view of two or more authors for how they treat the same or similar topics, including which details they include and emphasize in their respective accounts.	CC.8.5.11-12.F. Evaluate authors' differing points of view on the same historical event or issue by assessing the authors' claims, reasoning, and evidence.			
Integration of Knowledge and Ideas			CC.8.5.6-8.G. Integrate visual information (e.g., in charts, graphs, photographs, videos, or maps) with other information in print and digital texts.	CC.8.5.9-10.G. Integrate quantitative or technical analysis (e.g., charts, research data) with qualitative analysis in print or digital text.	CC.8.5.11-12.G. Integrate and evaluate multiple sources of information presented in diverse formats and media (e.g., visually, quantitatively, as well as in words) in order to address a question or solve a problem.

CC.8.5.6-8.H. Distinguish among fact, opinion, and reasoned judgment in a text.	CC.8.5.9-10.H. Assess the extent to which the reasoning and evidence in a text support the author's claims.	CC.8.5.11-12.H. Evaluate an author's premises, claims, and evidence by corroborating or challenging them with other information.	
CC.8.5.6-8.I. Analyze the relationship between a primary and secondary source on the same topic.	CC.8.5.9-10.I. Compare and contrast treatments of the same topic in several primary and secondary sources.	CC.8.5.11-12.I. Integrate information from diverse sources, both primary and secondary, into a coherent understanding of an idea or event, noting discrepancies among sources.	
Range and Level of Complex Texts	CC.8.5.6-8.J. By the end of grade 8, read and comprehend history/social studies texts in the grades 6-8 text complexity band independently and proficiently.	CC.8.5.9-10.J. By the end of grade 10, read and comprehend history/social studies texts in the grades 9-10 text complexity band independently and proficiently.	CC.8.5.11-12.J. By the end of grade 12, read and comprehend history/social studies texts in the grades 11-CCR text complexity band independently and proficiently.

Pennsylvania Core Standards for Writing in History and Social Studies *Grades 6-12*

INTRODUCTION

These standards describe what students in the social studies classroom should know and be able to do with the English language in writing, grade 6 through 12. The standards provide the targets for instruction and student learning essential for success in all academic areas, not just language arts classrooms. Although the standards are not a curriculum or a prescribed series of activities, school entities will use them to develop a local school curriculum that will meet local students' needs.

The standards below begin at grade 6; standards for K-5 reading in history/social studies, science, and technical subjects are integrated into the K-5 Writing standards.

The English Language Arts Standards for History and Social Studies also provide parents and community members with information about what students should know and be able to do as they progress through the educational program and at graduation. With a clearly defined target provided by the standards, parents, students, educators and community members become partners in learning. Each standard implies an end of year goal—with the understanding that exceeding the standard is an even more desirable end goal.

8.6	**Writing** **Students write for different purposes and audiences. Students write clear and focused text to convey a well-defined perspective and appropriate content.**		
	GRADES 6-8	**GRADES 9-10**	**GRADES 11-12**
Text Types and Purposes	CC.8.6.6-8.A. Write arguments focused on *discipline-specific content*. • Introduce claim(s) about a topic or issue, acknowledge and distinguish the claim(s) from alternate or opposing claims, and organize the reasons and evidence logically. • Support claim(s) with logical reasoning and relevant, accurate data and evidence that demonstrate an understanding of the topic or text, using credible sources. • Use words, phrases, and clauses to create cohesion and clarify the relationships among claim(s), counterclaims, reasons, and evidence. • Establish and maintain a	CC.8.6.9-10.A. Write arguments focused on *discipline-specific content*. • Introduce precise claim(s), distinguish the claim(s) from alternate or opposing claims, and create an organization that establishes clear relationships among the claim(s), counterclaims, reasons, and evidence. • Develop claim(s) and counterclaims fairly, supplying data and evidence for each while pointing out the strengths and limitations of both claim(s) and counterclaims in a discipline-appropriate form and in a manner that anticipates the audience's knowledge level and concerns. • Use words, phrases, and clauses to link the major	CC.8.6.11-12.A. Write arguments focused on *discipline-specific content*. • Introduce precise, knowledgeable claim(s), establish the significance of the claim(s), distinguish the claim(s) from alternate or opposing claims, and create an organization that logically sequences the claim(s), counterclaims, reasons, and evidence. • Develop claim(s) and counterclaims fairly and thoroughly, supplying the most relevant data and evidence for each while pointing out the strengths and limitations of both claim(s) and counterclaims in a discipline-appropriate form that anticipates the audience's knowledge level,

	formal style. • Provide a concluding statement or section that follows from and supports the argument presented.	sections of the text, create cohesion, and clarify the relationships between claim(s) and reasons, between reasons and evidence, and between claim(s) and counterclaims. • Establish and maintain a formal style and objective tone while attending to the norms and conventions of the discipline in which they are writing. • Provide a concluding statement or section that follows from or supports the argument presented.	concerns, values, and possible biases. • Use words, phrases, and clauses as well as varied syntax to link the major sections of the text, create cohesion, and clarify the relationships between claim(s) and reasons, between reasons and evidence, and between claim(s) and counterclaims. • Establish and maintain a formal style and objective tone while attending to the norms and conventions of the discipline in which they are writing. • Provide a concluding statement or section that follows from or supports the argument presented.		
	CC.8.6.6-8.B.* Write informative/explanatory texts, including the narration of historical events, scientific procedures/experiments, or technical processes. • Introduce a topic clearly, previewing what is to follow; organize ideas, concepts, and information into broader categories as appropriate to achieving purpose; include formatting (e.g., headings), graphics (e.g., charts, tables), and multimedia when useful to aiding comprehension. • Develop the topic with relevant, well-chosen facts, definitions, concrete details, quotations, or other information and examples. • Use appropriate and varied transitions to create cohesion and clarify the relationships among ideas and concepts. • Use precise language and domain-specific vocabulary to inform about or explain the topic. • Establish and maintain a formal style and objective tone. • Provide a concluding statement or section that follows from and supports the information or explanation presented.	CC.8.6.9-10.B.* Write informative/explanatory texts, including the narration of historical events, scientific procedures/experiments, or technical processes. • Introduce a topic and organize ideas, concepts, and information to make important connections and distinctions; include formatting (e.g., headings), graphics (e.g., figures, tables), and multimedia when useful to aiding comprehension. • Develop the topic with well-chosen, relevant, and sufficient facts, extended definitions, concrete details, quotations, or other information and examples appropriate to the audience's knowledge of the topic. • Use varied transitions and sentence structures to link the major sections of the text, create cohesion, and clarify the relationships among ideas and concepts. • Use precise language and domain-specific vocabulary to manage the complexity of the topic and convey a style appropriate to the discipline and context as well as to the expertise of likely readers. • Establish and maintain a formal style and objective tone while attending to the norms and conventions of the discipline in which they are writing. • Provide a concluding statement or section that follows from and supports the information or explanation presented (e.g., articulating implications or the significance of the topic).	CC.8.6.11-12.B.* Write informative/explanatory texts, including the narration of historical events, scientific procedures/experiments, or technical processes. • Introduce a topic and organize complex ideas, concepts, and information so that each new element builds on that which precedes it to create a unified whole; include formatting (e.g., headings), graphics (e.g., figures, tables), and multimedia when useful to aiding comprehension. • Develop the topic thoroughly by selecting the most significant and relevant facts, extended definitions, concrete details, quotations, or other information and examples appropriate to the audience's knowledge of the topic. • Use varied transitions and sentence structures to link the major sections of the text, create cohesion, and clarify the relationships among complex ideas and concepts. • Use precise language, domain-specific vocabulary and techniques such as metaphor, simile, and analogy to manage the complexity of the topic; convey a knowledgeable stance in a style that responds to the discipline and context as well as to the expertise of likely readers. • Provide a concluding statement or section that follows from and supports the information or explanation provided (e.g., articulating implications or the significance of the topic).		
Production and Distribution of Writing			CC.8.6.6-8.C. Produce clear and coherent writing in which the development, organization, and style are appropriate to task, purpose, and audience.	CC.8.6.9-10.C. Produce clear and coherent writing in which the development, organization, and style are appropriate to task, purpose, and audience.	CC.8.6.11-12.C. Produce clear and coherent writing in which the development, organization, and style are appropriate to task, purpose, and audience.
CC.8.6.6-8.D. With some guidance and support from peers and adults, develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on how well purpose and	CC.8.6.9-10.D. Develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on addressing what is most significant for a specific purpose and audience.	CC.8.6.11-12.D. Develop and strengthen writing as needed by planning, revising, editing, rewriting, or trying a new approach, focusing on addressing what is most significant for a specific purpose and audience.			

audience have been addressed.					
CC.8.6.6-8.E. Use technology, including the Internet, to produce and publish writing and present the relationships between information and ideas clearly and efficiently.	CC.8.6.9-10.E. Use technology, including the Internet, to produce, publish, and update individual or shared writing products, taking advantage of technology's capacity to link to other information and to display information flexibly and dynamically.	CC.8.6.11-12.E. Use technology, including the Internet, to produce, publish, and update individual or shared writing products in response to ongoing feedback, including new arguments or information.			
Research to Build and Present Knowledge			CC.8.6.6-8.F. Conduct short research projects to answer a question (including a self-generated question), drawing on several sources and generating additional related, focused questions that allow for multiple avenues of exploration.	CC.8.6.9-10.F. Conduct short as well as more sustained research projects to answer a question (including a self-generated question) or solve a problem; narrow or broaden the inquiry when appropriate; synthesize multiple sources on the subject, demonstrating understanding of the subject under investigation.	CC.8.6.11-12.F. Conduct short as well as more sustained research projects to answer a question (including a self-generated question) or solve a problem; narrow or broaden the inquiry when appropriate; synthesize multiple sources on the subject, demonstrating understanding of the subject under investigation.
CC.8.6.6-8.G. Gather relevant information from multiple print and digital sources, using search terms effectively; assess the credibility and accuracy of each source; and quote or paraphrase the data and conclusions of others while avoiding plagiarism and following a standard format for citation.	CC.8.6.9-10.G. Gather relevant information from multiple authoritative print and digital sources, using advanced searches effectively; assess the usefulness of each source in answering the research question; integrate information into the text selectively to maintain the flow of ideas, avoiding plagiarism and following a standard format for citation.	CC.8.6.11-12.G. Gather relevant information from multiple authoritative print and digital sources, using advanced searches effectively; assess the strengths and limitations of each source in terms of the specific task, purpose, and audience; integrate information into the text selectively to maintain the flow of ideas, avoiding plagiarism and overreliance on any one source and following a standard format for citation.			
CC.8.6.6-8.H. Draw evidence from informational texts to support analysis, reflection, and research.	CC.8.6.9-10.H. Draw evidence from informational texts to support analysis, reflection, and research.	CC.8.6.11-12.H. Draw evidence from informational texts to support analysis, reflection, and research.			
Range of Writing	CC.8.6.6-8.I. Write routinely over extended time frames (time for reflection and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	CC.8.6.9-10.I. Write routinely over extended time frames (time for reflection and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.	CC.8.6.11-12.I. Write routinely over extended time frames (time for reflection and revision) and shorter time frames (a single sitting or a day or two) for a range of discipline-specific tasks, purposes, and audiences.		

* Students' narrative skills continue to grow in these grades. The Standards require that students be able to incorporate narrative elements effectively into arguments and informative/explanatory texts. In history/social studies, students must be able to incorporate narrative accounts into their analyses of individuals or events of historical import. In science and technical subjects, students must be able to write precise enough descriptions of the step-by-step procedures they use in their investigations or technical work that others can replicate them and (possibly) reach the same results.

XXIV. GLOSSARY

Artifact:	**Any object made by human work or skill.**
Beginnings:	**A demarcation of time designating studies to commence with the written historical record.**
Central issue:	The primary concern from which other problems or matters are derived. For example, today's world migration flows are a central issue from which other concerns such as terrorist threats may arise.
Chronology:	The science of measuring time and of dating events. Examples include BCE (before the common era) and CE (common era). Another reference to chronology is CA, around the time, circa.
Conflict:	The opposition of persons or groups that gives rise to dramatic action. Such actions could include the use of force as in combat.

Culture:	The skills and arts of a given people in a given period of time or a civilization.
Document:	Anything written or printed used to record or prove something.
Historical evidence:	Something that makes something else noticeable, obvious or evident.
Historical passage:	An article or section of a longer work that has importance to the past.
Innovation:	The introduction of something new; an idea, method or devise.
Interpretation:	Explanation or to reply to a situation in order to make sense of it (e.g., a time period, an individual's actions).
Memorial:	An object or ceremony serving as a remembrance for a person, group, day, site or event.
Museum:	A historical display in a building, room, etc. for exhibiting artistic, historical or scientific objects.
Present:	A demarcation of time designating studies to the current year.
Opinion:	A belief based not on certainty but on what seems to be true or probable.
Strike:	A work stoppage by employees organized against the management of a business entity.
Time lines:	A measure of a period during which something exists or happens; usually displayed in chronological order on a graph or linear lines.
War:	A conflict in which two or more nations or two or more entities inside a nation are at odds.
Xenophobia:	An intense fear or dislike of groups unknown or not within one's experience including the group's customs and culture.

(*Editor's Note*: Under § 4.12(a)(3)(iv), the academic standards for economics in Appendix C-1 apply beginning July 1, 2026.)

APPENDIX C-1

Academic Standards for Economics

Grades K—12

Authority

The provisions of this Appendix C-1 added under sections 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix C-1 added February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234.

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards).

Overview

Economics is concerned with the behavior of individuals and institutions engaged in the production, exchange, and consumption of goods and services. For students to succeed in the workforce, knowledge of how the world works is critical. Every student should know the characteristics of economic systems, how markets establish prices, how scarcity and choice affect the allocation of resources, the global nature of economic interdependence, and how work and earnings impact productivity. These standards are intended to provide direction in learning how economic activity impacts everyday life through the areas of fundamentals of economics, microeconomics, macroeconomics, and money and financial institutions. The standards are grade banded and reflect the increasing complexity and sophistication that students are expected to achieve as they progress through school.

Number	Areas
6.1	**Fundamentals of Economics**
6.2	Microeconomics
6.3	Macroeconomics
6.4	Money and Economic Institutions

Appendix C-1. Academic Standards for Economics

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
1. Fundamentals of Economics				
Economic systems	Intentionally blank	Intentionally blank	**6.1.6-8.A** Describe and identify the characteristics of traditional, command, and market systems.	**6.1.9-12.A** Evaluate the strengths and weaknesses of traditional, command, and market economies.
Fundamental economic questions	**6.1.K-2.B** Explain how limited resources require us to make choices and share when we cannot have everything we want.	**6.1.3-5.B** Identify the three basic questions that all economic systems attempt to answer: • What goods and services should be produced? • How will goods and services be produced? • Who will consume goods and services?	**6.1.6-8.B** Explain how traditional, command, and market economies answer the basic economic questions: • What goods and services should be produced? • How will goods and services be produced? • Who will consume goods and services?	**6.1.9-12.B** Analyze the impact of traditional, command, and market economies on the United States economy.
Scarcity	Intentionally blank	**6.1.3-5.C** Define scarcity and how it impacts personal, household, and community decision-making.	**6.1.6-8.C** Describe ways to deal with scarcity within various communities.	**6.1.9-12.C** Analyze actions taken as a result of scarcity in decision-making.
Economic choice	Intentionally blank	Intentionally blank	**6.1.6-8.D** Explain that scarcity is the condition of not having all the goods and services wanted because resources are limited.	**6.1.9-12.D** Analyze how choices made by individuals, firms, or government officials are constrained by available resources.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Economic resources	Intentionally blank	Intentionally blank	**6.1.6-8.E** Explain how resources can be used in different ways to produce different goods and services.	**6.1.9-12.E** Evaluate the allocation of resources used to produce goods and services.
Marginal analysis	Intentionally blank	**6.1.3-5.F** Explain the costs and benefits of an economic decision.	**6.1.6-8.F** Explain how the additional cost and additional benefit of a choice affects decision-making (marginal analysis).	**6.1.9-12.F** Evaluate regional, national, or international economic decisions using marginal analysis.
Opportunity cost	Intentionally blank	**6.1.3-5.G** Define opportunity cost and describe the opportunity cost of a personal choice.	**6.1.6-8.G** Explain the opportunity cost of a public choice from different perspectives.	**6.1.9-12.G** Analyze the opportunity cost of decisions by individuals, businesses, communities, and nations.
Incentives	Intentionally blank	**6.1.3-5.H** Explain how negative and positive incentives affect choices.	**6.1.6-8.H** Explain how incentives affect the behaviors of workers, savers, consumers, and producers.	**6.1.9-12.H** Evaluate in terms of marginal analysis how incentives influence decisions of consumers, producers, and policymakers.
Specialization	**6.1.K-2.I** Describe why people divide tasks to make work easier.	**6.1.3-5.I** Define specialization and identify examples of interdependence.	**6.1.6-8.I** Describe how specialization may lead to increased production and consumption.	**6.1.9-12.I** Analyze the relationship between specialization and the standard of living.
Trade	**6.1.K-2.J** Explain why people trade.	**6.1.3-5.J** Explain why countries trade.	**6.1.6-8.J** Explain how trade may improve a society's standard of living.	**6.1.9-12.J** Analyze the relationships between trade, competition, and productivity.
Foreign trade	Intentionally blank	**6.1.3-5.K** Identify and define imports, exports, and international trade.	**6.1.6-8.K** Explain why governments sometimes subsidize or restrict trade (i.e., tariffs or quotas).	**6.1.9-12.K** Evaluate how a country might benefit by lowering or removing trade barriers.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Global interdependence	Intentionally blank	Intentionally blank	**6.1.6-8.L** Explain how specialization and trade lead to interdependence.	**6.1.9-12.L** Analyze how consumers and producers participate in the global production and consumption of goods or services.
Comparative advantage	Intentionally blank	**6.1.3-5.M** Explain why some products are produced locally while others are not.	**6.1.6-8.M** Explain how opportunity costs influence where goods and services are produced locally and regionally.	**6.1.9-12.M** Explain that individuals and nations have a comparative advantage in the production of goods or services if they can produce a product at a lower opportunity cost than other individuals or nations.
		2. Microeconomics		
Circular flow	**6.2.K-2.A** Define and identify goods and services.	**6.2.3-5.A** Describe how consumers purchase goods and services from producers.	**6.2.6-8.A** Explain the flow of goods, services, and resources in an economy.	**6.2.9-12.A** Analyze the circular flow of products, resources, and money in a mixed economy.
Competition	Intentionally blank	**6.2.3-5.B** Identify ways local businesses compete for customers.	**6.2.6-8.B** Describe the advantages and disadvantages of competition for consumers in markets.	**6.2.9-12.B** Define noncompetitive markets and explain how consumers and producers operate in those markets.
Monopolies	Intentionally blank	**6.2.3-5.C** Define and provide an example of a monopoly.	**6.2.6-8.C** Identify historical examples of monopolies in the United States.	**6.2.9-12.C** Evaluate the economic roles of governments in free markets.
Supply and demand	Intentionally blank	**6.2.3-5.D** Explain how markets bring together buyers and sellers.	**6.2.6-8.D** Explain how supply and demand affect the prices of goods and services.	**6.2.9-12.D** Predict how changes in supply and demand affect equilibrium price and quantity sold.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Economic patterns	Intentionally blank	**6.2.3-5.E** Explain how the location of resources and transportation have affected Pennsylvania economic patterns.	**6.2.6-8.E** Explain how the location of resources, transportation, and technology have affected United States economic patterns.	**6.2.9-12.E** Explain how the location of resources, transportation, and technology have affected international economic patterns.
Wages	Intentionally blank	Intentionally blank	**6.2.6-8.F** Define wages and explain how wages are determined by the supply of and demand for workers.	**6.2.9-12.F** Analyze the factors influencing wages, including demand for goods and services produced, productivity of labor, labor unions, and education and skills.
Productivity	Intentionally blank	Intentionally blank	**6.2.6-8.G** Describe how productivity is measured and identify ways in which a person can improve his or her productivity.	**6.2.9-12.G** Evaluate how changes in education, incentives, technology, and capital investment alter productivity.
Types of businesses	**6.2.K-2.H** Identify businesses in your community.	**6.2.3-5.H** Describe businesses that provide goods and businesses that provide services.	**6.2.6-8.H** Identify and explain the characteristics of the three types of businesses: sole proprietorship, partnership, and corporation.	**6.2.9-12.H** Analyze the costs and benefits of organizing a business as a sole proprietorship, partnership, or corporation.
Profit and loss	Intentionally blank	**6.2.3-5.I** Define profit and loss.	**6.2.6-8.I** Explain how profits and losses serve as incentives.	**6.2.9-12.I** Analyze the role of profits and losses in the allocation of resources in a market economy.
Entrepreneurship	**6.2.K-2.J** Define entrepreneurship and identify entrepreneurs in the local community.	**6.2.3-5.J** Identify historical and contemporary entrepreneurs in Pennsylvania.	**6.2.6-8.J** Investigate leading entrepreneurs in Pennsylvania and the United States and describe the characteristics of successful entrepreneurship (including risks and rewards).	**6.2.9-12.J** Evaluate the impact of entrepreneurs on the economy.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
3. Macroeconomics				
Economic indicators	Intentionally blank	**6.3.3-5.A** Define unemployment and inflation.	**6.3.6-8.A** Define macroeconomic indicators such as the Consumer Price Index (CPI), gross domestic product (GDP), and unemployment rate.	**6.3.9-12.A** Evaluate the relationship between economic growth and changes in macroeconomic indicators such as the Consumer Price Index (CPI), gross domestic product (GDP), and unemployment rate.
Business cycles	Intentionally blank	Intentionally blank	**6.3.6-8.B** Describe historical examples of expansion, recession, and depression in the United States.	**6.3.9-12.B** Compare current and historical examples of expansion, recession, and depression.
Fiscal policy	Intentionally blank	Intentionally blank	**6.3.6-8.C** Define fiscal policies available to the federal government.	**6.3.9-12.C** Identify fiscal and monetary policies that address inflation, unemployment, economic growth, and the national debt and deficit.
Federal budget	Intentionally blank	Intentionally blank	Intentionally blank	**6.3.9-12.D** Explain how deficit spending by the federal government increases the national debt and the impact it has on the economy.
Taxation	**6.3.K-2.E** Define a tax and identify examples.	**6.3.3-5.E** Explain how taxes affect the price of goods and services.	**6.3.6-8.E** Define the three different types of tax structures (progressive, proportional, and regressive) and different tax bases, such as income, wealth, property, etc.	**6.3.9-12.E** Evaluate the impact of different tax structures (progressive, proportional, and regressive) and different tax bases (income, wealth, property, etc.) on households and firms.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Tax policy	Intentionally blank	**6.3.3-5.F** Explain how the government pays for goods and services through taxes.	**6.3.6-8.F** Compare the taxation policies of local, state, and national government levels in the economy.	**6.3.9-12.F** Analyze the social, political, and economic consequences of changes in tax policy.
Economic role of government	Intentionally blank	Intentionally blank	**6.3.6-8.G** Analyze the economic roles of governments in market economies in promoting growth and stability and providing legal frameworks (laws and regulations).	**6.3.9-12.G** Evaluate the economic roles of governments in a globalized economy (e.g., trade policy, income inequality, distribution of wealth).
Externalities	Intentionally blank	**6.3.3-5.H** Identify unintended consequences of market activities (e.g., pollution, noise, traffic).	**6.3.6-8.H** Identify policies governments can use to address unintended consequences (i.e., externalities) of market activities.	**6.3.9-12.H** Evaluate policies governments can use to address unintended consequences (i.e., externalities) of market activities.
Public goods and services	**6.3.K-2.I** Define public goods and services provided by the government.	**6.3.3-5.I** Distinguish between public and private goods and services.	**6.3.6-8.I** Explain why the government provides public goods and services (e.g., education, infrastructure, national defense, safety and health).	**6.3.9-12.I** Evaluate government decisions in prioritizing public goods and services.
4. Money and Economic Institutions				
Money	Intentionally blank	**6.4.3-5.A** Explain the advantages and disadvantages of using money instead of bartering.	**6.4.6-8.A** Identify and explain various forms of payments used in markets.	**6.4.9-12.A** Explain how the Federal Reserve defines money.
Exchange rates	Intentionally blank	**6.4.3-5.B** Explain why most countries create their own form of money.	**6.4.6-8.B** Explain why you have to exchange currencies to purchase goods when you visit another country.	**6.4.9-12.B** Explain how changes in exchange rates affect the relative prices of foreign goods and services.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Monetary policy	Intentionally blank	Intentionally blank	Intentionally blank	**6.4.9-12.C** Explain how changes in the money supply and monetary policy affect interest rates throughout the economy.
Banking	**6.4.K-2.D** Define saving and explain why people use banks to save.	**6.4.3-5.D** Explain how banks bring savers and borrowers together.	**6.4.6-8.D** Explain how banks and other financial institutions channel funds from savers to borrowers and investors.	**6.4.9-12.D** Evaluate the role of banks in promoting economic growth in the economy.
Federal reserve system	Intentionally blank	Intentionally blank	**6.4.6-8.E** Explain the history, structure, and purposes of the Federal Reserve System.	**6.4.9-12.E** Analyze Federal Reserve policies designed to raise or lower interest rates.
Nongovernmental organizations	Intentionally blank	Intentionally blank	**6.4.6-8.F** Define nongovernmental economic institutions.	**6.4.9-12.F** Evaluate the role of nongovernmental economic institutions.
Interest rate policy	Intentionally blank	Intentionally blank	**6.4.6-8.G** Explain the impact of higher or lower interest rates for savers, borrowers, consumers, and producers.	**6.4.9-12.G** Evaluate benefits and costs of changes in interest rates for individuals and society.

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APPENDIX D
Academic Standards for the Arts and Humanities and Health, Safety and Physical Education

Source

The provisions of this Appendix D adopted January 10, 2003, effective January 11, 2003, 33 Pa.B. 255.

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards).

Academic Standards for the Arts and Humanities

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XXVI. INTRODUCTION

The Academic Standards for the Arts and Humanities describe what students should know and be able to do at the end of grades 3, 5, 8 and 12 in the visual and performing arts and the understanding about humanities context within the arts. The arts include dance, music, theatre and visual arts. The arts and the humanities are interconnected through the inclusion of history, criticism and aesthetics. In addition, the humanities include literature and language, philosophy, social studies and world languages. The areas encompassed in the humanities such as jurisprudence, comparative religions and ethics are included among other standards documents. The interconnected arts and humanities areas are divided into these standards categories:

• 9.1. Production, Performance and Exhibition of Dance, Music, Theatre and Visual Arts

• 9.2. Historical and Cultural Contexts

• 9.3. Critical Response

• 9.4. Aesthetic Response

The Academic Standards for the Arts and Humanities define the content for planned instruction that will result in measurable gains for all students in knowledge and skills and provide a basis of learning for continued study in the arts. The unifying themes of production, history, criticism and aesthetics are common to each area of study within the Academic Standards in the Arts and Humanities.

• Dance Education is a kinesthetic art form that satisfies the human need to respond to life experiences through movement of the physical being.

• Music Education is an aural art form that satisfies the human need to respond to life experiences through singing, listening and/or playing an instrument.

• Theatre Education is an interdisciplinary art form that satisfies the human need to express thoughts and feelings through written text, dramatic interpretation and multimedia production.

• Visual Arts Education is a spatial art form that satisfies the human need to respond to life experiences through images, structures and tactile works.

• Humanities Education is the understanding and integration of human thought and accomplishment.

Knowledge of the Academic Standards for the Arts and Humanities incorporates carefully developed and integrated components:

• Application of problem solving skills

• Extensive practice in the comprehension of basic symbol systems and abstract concepts

• Application of technical skills in practical production and performance

• Comprehension and application of the creative process

• Development and practice of creative thinking skills

• Development of verbal and nonverbal communication skills

These standards provide the targets essential for success in student learning in arts and humanities. They describe the expectations for students' achievement and performance throughout their education in Pennsylvania schools. Utilizing these standards, school entities can develop a local school curriculum that will meet their students' needs.

The arts represent society's capacity to integrate human experience with individual creativity. Comprehensive study of the arts provides an opportunity for all students to observe, reflect and participate both in the arts of their culture and the cultures of others. Sequential study in the arts and humanities provides the knowledge and the analytical skills necessary to evaluate and critique a media-saturated culture. An arts education contributes to the development of productive citizens who have gained creative and technological knowledge necessary for employment in the 21st Century.

A glossary is included to assist the reader in understanding terminology contained in the standards.

9.1. Production, Performance and Exhibition of Dance, Music, Theatre and Visual Arts			
9.1.3. GRADE 3	**9.1.5. GRADE 5**	**9.1.8. GRADE 8**	**9.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Know and use the elements and principles of each art form to create works in the arts and humanities. • Elements • Dance: • energy/force • space • time • Music: • duration • intensity • pitch • timbre • Theatre: • scenario • script/text • set design • Visual Arts: • color • form/shape • line • space • texture • value • Principles • Dance: • choreography • form • genre • improvisation • style • technique • Music: • composition • form • genre • harmony • rhythm • texture • Theatre: • balance • collaboration • discipline • emphasis • focus • intention • movement • rhythm • style voice • Visual Arts: • balance • contrast • emphasis/focal point • movement/rhythm • proportion/scale • repetition unity/harmony			
B. Recognize, know, use and demonstrate a variety of appropriate arts elements and principles to produce, review and revise original works in the arts. • Dance: • move • perform • read and notate dance • create and choreograph • improvise • Music: • sing • play an instrument • read and notate music • compose and arrange • improvise • Theatre: • stage productions • read and write scripts • improvise • interpret a role • design sets • direct • Visual Arts: • paint • draw • craft • sculpt • print • design for environment, communication, multi-media			
C. Recognize and use fundamental vocabulary within each of the arts forms.	C. Know and use fundamental vocabulary within each of the arts forms.	C. Identify and use comprehensive vocabulary within each of the arts forms.	C. Integrate and apply advanced vocabulary to the arts forms.
D. Use knowledge of varied styles within each art form through a performance or exhibition of unique work.	D. Describe and use knowledge of a specific style within each art form through a performance or exhibition of a unique work.	D. Demonstrate knowledge of at least two styles within each art form through performance or exhibition of unique works.	D. Demonstrate specific styles in combination through the production or performance of a unique work of art (e.g., a dance composition that combines jazz dance and African dance).
E. Demonstrate the ability to define objects, express emotions, illustrate an action or relate an experience through creation of works in the arts.	E. Know and demonstrate how arts can communicate experiences, stories or emotions through the production of works in the arts.	E. Communicate a unifying theme or point of view through the production of works in the arts.	E. Delineate a unifying theme through the production of a work of art that reflects skills in media processes and techniques.
F. Identify works of others through a performance or exhibition (e.g., exhibition of student paintings based on the study of Picasso).	F. Describe works of others through performance or exhibition in two art forms.	F. Explain works of others within each art form through performance or exhibition.	F. Analyze works of arts influenced by experiences or historical and cultural events through production, performance or exhibition.
G. Recognize the function of rehearsals and practice sessions.	G. Identify the function and benefits of rehearsal and practice sessions.	G. Explain the function and benefits of rehearsal and practice sessions.	G. Analyze the effect of rehearsal and practice sessions.
H. Handle materials, equipment and tools safely at work and performance spaces. • Identify materials used. • Identify issues of cleanliness related to the arts. • Recognize some mechanical/electrical equipment.	H. Use and maintain materials, equipment and tools safely at work and performance spaces. • Describe some materials used. • Describe issues of cleanliness related to the arts. • Describe types of mechanical/electrical equipment usage.	H. Demonstrate and maintain materials, equipment and tools safely at work and performance spaces. • Analyze the use of materials. • Explain issues of cleanliness related to the arts. • Explain the use of mechanical/electrical equipment.	H. Incorporate the effective and safe use of materials, equipment and tools into the production of works in the arts at work and performance spaces. • Evaluate the use and applications of materials. • Evaluate issues of cleanliness related to the arts.

• Recognize differences in selected physical space/environments. • Recognize the need to select safe props/stage equipment. • Identify methods for storing materials in the arts.	• Know how to work in selected physical space/environments. • Identify the qualities of safe props/stage equipment. • Describe methods for storing materials in the arts.	• Demonstrate how to work in selected physical space/environment. • Demonstrate the selection of safe props/stage equipment. • Demonstrate methods for storing materials in the arts.	• Evaluate the use and applications of mechanical/electrical equipment. • Evaluate differences among selected physical space/environment. • Evaluate the use and applications of safe props/stage equipment. • Evaluate the use and apply safe methods for storing materials in the arts.
I. Identify arts events that take place in schools and in communities.	I. Describe arts events that take place in schools and in communities.	I. Know where arts events, performances and exhibitions occur and how to gain admission.	I. Distinguish among a variety of regional arts events and resources and analyze methods of selection and admission.
J. Know and use traditional and contemporary technologies for producing, performing and exhibiting works in the arts or the works of others. • Know and use traditional technologies (e.g., charcoal, pigments, clay, needle/thread, quill pens, stencils, tools for wood carving, looms, stage equipment). • Know and use contemporary technologies (e.g., CDs/software, audio/sound equipment, polymers, clays, board-mixers, photographs, recorders).	J. Apply traditional and contemporary technologies for producing, performing and exhibiting works in the arts or the works of others. • Experiment with traditional technologies (e.g., ceramic/wooden tools, earthen clays, masks, instruments, folk shoes, etching tools, folk looms). • Experiment with contemporary technologies (e.g., color fills on computers, texture methods on computers, fonts/point systems, animation techniques, video teleconferencing, multimedia techniques, internet access, library computer card catalogues).	J. Incorporate specific uses of traditional and contemporary technologies within the design for producing, performing and exhibiting works in the arts or the works of others. • Explain and demonstrate traditional technologies (e.g., paint, tools, sponges, weaving designs, instruments, natural pigments/glazes). • Explain and demonstrate contemporary technologies (e.g., MIDI keyboards, internet design, computers, interactive technologies, audio/sound equipment, board-mixer, video equipment, computerized lighting design).	J. Analyze and evaluate the use of traditional and contemporary technologies for producing, performing and exhibiting works in the arts or the works of others. • Analyze traditional technologies (e.g., acid printing, etching methods, musical instruments, costume materials, eight track recording, super 8 movies). • Analyze contemporary technologies (e.g., virtual reality design, instrument enhancements, photographic tools, broadcast equipment, film cameras, preservation tools, web graphics, computer generated marching band designs).
K. Know and use traditional and contemporary technologies for furthering knowledge and understanding in the humanities.	K. Apply traditional and contemporary technology in furthering knowledge and understanding in the humanities.	K. Incorporate specific uses of traditional and contemporary technologies in furthering knowledge and understanding in the humanities.	K. Analyze and evaluate the use of traditional and contemporary technologies in furthering knowledge and understanding in the humanities.

9.2. Historical and Cultural Contexts			
9.2.3. GRADE 3	**9.2.5. GRADE 5**	**9.2.8. GRADE 8**	**9.2.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to identify, compare, contrast and analyze works in the arts in their historical and cultural context appropriate for each grade level in concert with districts' social studies, literature and language standards.			
A. Explain the historical, cultural and social context of an individual work in the arts.			
B. Relate works in the arts chronologically to historical events (e.g., 10,000 B.C. to present).			
C. Relate works in the arts to varying styles and genre and to the periods in which they were created (e.g., Bronze Age, Ming Dynasty, Renaissance, Classical, Modern, Post-Modern, Contemporary, Futuristic, others).			
D. Analyze a work of art from its historical and cultural perspective.			
E. Analyze how historical events and culture impact forms, techniques and purposes of works in the arts (e.g., Gilbert and Sullivan operettas).			
F. Know and apply appropriate vocabulary used between social studies and the arts and humanities.			
G. Relate works in the arts to geographic regions: • Africa • Asia • Australia • Central America • Europe • North America • South America			
H. Identify, describe and analyze the work of Pennsylvania Artists in dance, music, theatre and visual arts.			
I. Identify, explain and analyze philosophical beliefs as they relate to works in the arts (e.g., classical architecture, rock music, Native American dance, contemporary American musical theatre).			
J. Identify, explain and analyze historical and cultural differences as they relate to works in the arts (e.g., plays by Shakespeare, works by Michelangelo, ethnic dance and music).			
K. Identify, explain and analyze traditions as they relate to works in the arts (e.g., story telling—plays, oral histories— poetry, work songs—blue grass).			
L. Identify, explain and analyze common themes, forms and techniques from works in the arts (e.g., Copland and Graham's *Appalachian Spring* and Millet's *The Gleaners*).			

9.3. Critical Response			
9.3.3. GRADE 3	**9.3.5. GRADE 5**	**9.3.8. GRADE 8**	**9.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Recognize critical processes used in the examination of works in the arts and humanities. • Compare and contrast	A. Identify critical processes in the examination of works in the arts and humanities. • Compare and contrast	A. Know and use the critical process of the examination of works in the arts and humanities. • Compare and contrast	A. Explain and apply the critical examination processes of works in the arts and humanities. • Compare and contrast

• Analyze • Interpret • Form and test hypotheses • Evaluate/form judgments	• Analyze • Interpret • Form and test hypotheses • Evaluate/form judgments	• Analyze • Interpret • Form and test hypotheses • Evaluate/form judgments	• Analyze • Interpret • Form and test hypotheses • Evaluate/form judgments
B. Know that works in the arts can be described by using the arts elements, principles and concepts (e.g., use of color, shape and pattern in Mondrian's *Broadway Boogie-Woogie*; use of dynamics, tempo, texture in Ravel's *Bolero*).	B. Describe works in the arts comparing similar and contrasting characteristics (e.g., staccato in Grieg's *In the Hall of the Mountain King* and in tap dance).	B. Analyze and interpret specific characteristics of works in the arts within each art form (e.g., pentatonic scales in Korean and Indonesian music).	B. Determine and apply criteria to a person's work and works of others in the arts (e.g., use visual scanning techniques to critique the student's own use of sculptural space in comparison to Julio Gonzales' use of space in *Woman Combing Her Hair)*.
C. Know classification skills with materials and processes used to create works in the arts (e.g., sorting and matching textiles, musical chants, television comedies).	C. Classify works in the arts by forms in which they are found (e.g., farce, architecture, graphic design).	C. Identify and classify styles, forms, types and genre within art forms (e.g., modern dance and the ethnic dance, a ballad and a patriotic song).	C. Apply systems of classification for interpreting works in the arts and forming a critical response.
D. Explain meanings in the arts and humanities through individual works and the works of others using a fundamental vocabulary of critical response.	D. Compare similar and contrasting important aspects of works in the arts and humanities based on a set of guidelines using a comprehensive vocabulary of critical response.	D. Evaluate works in the arts and humanities using a complex vocabulary of critical response.	D. Analyze and interpret works in the arts and humanities from different societies using culturally specific vocabulary of critical response.
E. Recognize and identify types of critical analysis in the arts and humanities. • Contextual criticism • Formal criticism • Intuitive criticism	E. Describe and use types of critical analysis in the arts and humanities. • Contextual criticism • Formal criticism • Intuitive criticism	E. Interpret and use various types of critical analysis in the arts and humanities. • Contextual criticism • Formal criticism • Intuitive criticism	E. Examine and evaluate various types of critical analysis of works in the arts and humanities. • Contextual criticism • Formal criticism • Intuitive criticism
F. Know how to recognize and identify similar and different characteristics among works in the arts (e.g., Amish and Hawaiian quilts, Navaho weavings and Kente cloth from West Africa).	F. Know how to recognize the process of criticism in identifying and analyzing characteristics among works in the arts.	F. Apply the process of criticism to identify characteristics among works in the arts.	F. Analyze the processes of criticism used to compare the meanings of a work in the arts in both its own and present time.
G. Know and demonstrate what a critic's position or opinion is related to works in the arts and humanities (e.g., I like patriotic songs because ...; The movie was enjoyed for its exceptional special effects).	G. Describe a critic's position or opinion about selected works in the arts and humanities (e.g., student's presentation of a critical position on Walt Disney's *Evolution of Mickey and Minnie Mouse*).	G. Compare and contrast critical positions or opinions about selected works in the arts and humanities (e.g., critic's review and comparison of Alvin Ailey's *Revelations to Tchaikovsky's Swan Lake*).	G. Analyze works in the arts by referencing the judgments advanced by arts critics as well as one's own analysis and critique.

9.4. Aesthetic Response			
9.4.3. GRADE 3	**9.4.5. GRADE 5**	**9.4.8. GRADE 8**	**9.4.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Know how to respond to a philosophical statement about works in the arts and humanities (e.g., "Can artworks that depict or are about ugly or unpleasant things ever be beautiful?").	A. Identify uses of expressive symbols that show philosophical meanings in works in the arts and humanities (e.g., American TV ads versus Asian TV ads).	A. Compare and contrast examples of group and individual philosophical meanings of works in the arts and humanities (e.g., group discussions on musical theatre versus the individual's concept of musical theatre).	A. Evaluate an individual's philosophical statement on a work in the arts and its relationship to one's own life based on knowledge and experience.
B. Know how to communicate an informed individual opinion about the meaning of works in the arts (e.g., works of an artist of the month).	B. Investigate and communicate multiple philosophical views about works in the arts.	B. Compare and contrast informed individual opinions about the meaning of works in the arts to others (e.g., debate philosophical opinions within a listserve or at an artist's website).	B. Describe and analyze the effects that works in the arts have on groups, individuals and the culture (e.g., Orson Welles' 1938 radio broadcast, *War of the Worlds)*.
C. Recognize that the environment of the observer influences individual aesthetic responses to works in the arts (e.g., the effect of live music as opposed to listening to the same piece on a car radio).	C. Identify the attributes of various audiences' environments as they influence individual aesthetic response (e.g., Beatles' music played by the Boston Pops versus video taped concerts from the 1970s).	C. Describe how the attributes of the audience's environment influence aesthetic responses (e.g., the ambiance of the theatre in a performance of Andrew Lloyd Weber's *Cats*).	C. Compare and contrast the attributes of various audiences' environments as they influence individual aesthetic response (e.g., viewing traditional *Irish* dance at county fair versus the performance of *River Dance* in a concert hall).
D. Recognize that choices made by artists regarding subject matter and themes communicate ideas through works in the arts and humanities (e.g., artist's interpretation through the use of classical ballet of the American West in Agnes De Mille's *Rodeo*).	D. Explain choices made regarding media, technique, form, subject matter and themes that communicate the artist's philosophy within a work in the arts and humanities (e.g., selection of stage lighting in Leonard Bernstein's *West Side Story* to communicate mood).	D. Describe to what purpose philosophical ideas generated by artists can be conveyed through works in the arts and humanities (e.g., T. Ganson's *Destructive Periods in Russia During Stalin's and Deniken's Leadership* conveys her memories and emotions of a specific incident).	D. Analyze and interpret a philosophical position identified in works in the arts and humanities.

XXVII. GLOSSARY

Aesthetics:	**A branch of philosophy that focuses on the nature of beauty, the nature and value of the arts and the inquiry processes and human responses they produce.**
Aesthetic criteria:	**Standards on which to make judgments about the artistic merit of a work of art, derived from cultural and emotional values and cognitive meaning.**
Aesthetic response:	A philosophical reply to works in the arts.
Artistic choices:	Selections made by artists in order to convey meaning.
Arts resource:	An outside community asset (e.g., performances, exhibitions, performers, artists).
Assess:	To analyze and determine the nature and quality of the process/product through means appropriate to the art form.
Community:	A group of people who share a common social, historical, regional or cultural heritage.
Contemporary technology:	Tools, machines or implements emerging and used today for the practice or production of works in the arts.
Context:	A set of interrelated background conditions (e.g., social, economic, political) that influence and give meaning to the development and reception of thoughts, ideas or concepts and that define specific cultures and eras.
Create:	To produce works in the arts using materials, techniques, processes, elements, principles and analysis.
Critical analysis:	The process of examining and discussing the effective uses of specific aspects of works in the arts.
Contextual criticism:	Discussion and evaluation with consideration of factors surrounding the origin and heritage to works in the arts and humanities.
Formal Criticism:	Discussion and evaluation of the elements and principles essential to works in the arts and humanities.
Intuitive Criticism:	Discussion and evaluation of one's subjective insight to works in the arts and humanities.
Critical process:	The use of sequential examination through comparison, analysis, interpretation, formation and testing of hypothesis and evaluation to form judgments.
Critical response:	The act or process of describing and evaluating the media, processes and meanings of works in the arts and making comparative judgments.
Culture:	The way of life of a group of people, including customs, beliefs, arts, institutions and worldview. Culture is acquired through many means and is always changing.
Elements:	Core components that support the principles of the arts.
Genre:	A type or category (e.g., music—opera, oratorio; theater—tragedy, comedy; dance—modern, ballet; visual arts—pastoral, scenes of everyday life).
Humanities:	The branch of learning that connects the fine arts, literature, languages, philosophy and cultural science. The humanities are concerned with the understanding and integration of human thought and accomplishment.
Improvisation:	Spontaneous creation requiring focus and concentration.
MIDI keyboard:	(Musical Instrument Digital Interface) A piece of equipment that interacts with a computer that uses a MIDI language set-up to notate and play music.
Multimedia:	The combined use of media, such as movies, CD-ROMs, television, radio, print and the Internet for entertainment and publicity.
Original works in the arts:	Dance, music, theatre and visual arts pieces created by performing or visual artists.
Principles:	Essential assumptions, basic or essential qualities determining intrinsic characteristics.
Style:	A distinctive or characteristic manner of expression.
Technique:	Specific skills and details employed by an artist, craftsperson or performer in the production of works in the arts.
Timbre:	A unique quality of sound.
Traditions:	Knowledge, opinions and customs a group feels is so important that members continue to practice it and pass it on to other generations.
Traditional technology:	Tools, machines or implements used for the historical practice or production of works in the arts.
Vocabulary:	Age and content appropriate terms used in the instruction of the arts and humanities that demonstrate levels of proficiency as defined in local curriculum (i.e., fundamental—grade 3, comprehensive—grade 5, discriminating—grade 8 and advanced—grade 12).

Academic Standards for Health, Safety and Physical Education

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XXIX. INTRODUCTION

This document includes Academic Standards for Health, Safety and Physical Education in these categories:

• 10.1 Concepts of Health

• 10.2 Healthful Living

• 10.3 Safety and Injury Prevention

• 10.4 Physical Activity

• 10.5 Concepts, Principles and Strategies of Movement

The Academic Standards for Health, Safety and Physical Education describe what students should know and be able to do by the end of third, sixth, ninth and twelfth grade. The standards are sequential across the grade levels and reflect the increasing complexity and rigor that students are expected to achieve. The Standards define the content for planned instruction that will result in measurable gains for all students in knowledge and skill. School entities will use these standards to develop local school curriculum and assessments that will meet the needs of the students.

The Academic Standards for Health, Safety and Physical Education provide students with the knowledge and skills that will enable them to achieve and maintain a physically active and healthful life. The attainment of these standards will favorably impact their lives and the lives of those around them. By becoming and remaining physically, mentally, socially and emotionally healthy, students will increase their chances of achieving to their highest academic potential.

The Academic Standards for Health, Safety and Physical Education provide parents with specific information about the knowledge and skills students should be developing as they progress through their educational programs. With the standards serving as clearly defined targets, parents, students, teachers and community members will be able to become partners in helping children achieve educational success.

A glossary is included to assist the reader in understanding terminology contained in the standards.

10.1. Concepts of Health			
10.1.3. GRADE 3	**10.1.6. GRADE 6**	**10.1.9. GRADE 9**	**10.1.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Identify and describe the stages of growth and development. • infancy • childhood • adolescence • adulthood • late adulthood	A. Describe growth and development changes that occur between childhood and adolescence and identify factors that can influence these changes. • education • socioeconomic	A. Analyze factors that impact growth and development between adolescence and adulthood. • relationships (e.g., dating, friendships, peer pressure) • interpersonal communication • risk factors (e.g., physical inactivity, substance abuse, intentional/unintentional injuries, dietary patterns) • abstinence • STD and HIV prevention • community	A. Evaluate factors that impact growth and development during adulthood and late adulthood. • acute and chronic illness • communicable and non-communicable disease • health status • relationships (e.g., marriage, divorce, loss) • career choice • aging process • retirement
B. Identify and know the location and function of the major body organs and systems. • circulatory • respiratory • muscular • skeletal • digestive	B. Identify and describe the structure and function of the major body systems. • nervous • muscular • integumentary • urinary • endocrine • reproductive • immune	B. Analyze the interdependence existing among the body systems.	B. Evaluate factors that impact the body systems and apply protective/preventive strategies. • fitness level • environment (e.g., pollutants, available health care) • health status (e.g., physical, mental, social) • nutrition
C. Explain the role of the food guide pyramid in helping people eat a healthy diet. • food groups • number of servings • variety of food • nutrients	C. Analyze nutritional concepts that impact health. • caloric content of foods • relationship of food intake and physical activity (energy output) • nutrient requirements • label reading • healthful food selection	C. Analyze factors that impact nutritional choices of adolescents. • body image • advertising • dietary guidelines • eating disorders • peer influence • athletic goals	C. Analyze factors that impact nutritional choices of adults. • cost • food preparation (e.g., time, skills) • consumer skills (e.g., understanding food labels, evaluating fads) • nutritional knowledge • changes in nutritional requirements (e.g., age, physical activity level)
D. Know age appropriate drug information. • definition of drugs • effects of drugs • proper use of medicine • healthy/unhealthy risk-taking (e.g. inhalant use, smoking) • skills to avoid drugs	D. Explain factors that influence childhood and adolescent drug use. • peer influence • body image (e.g., steroids, enhancers) • social acceptance • stress • media influence • decision-making/refusal skills • rules, regulations and laws • consequences	D. Analyze prevention and intervention strategies in relation to adolescent and adult drug use. • decision-making/refusal skills • situation avoidance • goal setting • professional assistance (e.g., medical, counseling. support groups) • parent involvement	D. Evaluate issues relating to the use/non-use of drugs. • psychology of addiction • social impact (e.g., cost, relationships) • chemical use and fetal development • laws relating to alcohol, tobacco and chemical substances • impact on the individual • impact on the community
E. Identify types and causes of common health problems of children. • infectious diseases (e.g., colds, flu, chickenpox) • noninfectious diseases (e.g., asthma, hay fever, allergies, lyme disease) • germs • pathogens • heredity	E. Identify health problems that can occur throughout life and describe ways to prevent them. • Diseases (e.g., cancer diabetes, STD/HIV/AIDS, cardiovascular disease) • Preventions (i.e. do not smoke, maintain proper weight, eat a balanced diet, practice sexual abstinence, be physically active)	E. Analyze how personal choice, disease and genetics can impact health maintenance and disease prevention.	E. Identify and analyze factors that influence the prevention and control of health problems. • research • medical advances • technology • government policies/regulations

10.2. Healthful Living			
10.2.3. GRADE 3	**10.2.6. GRADE 6**	**10.2.9. GRADE 9**	**10.2.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Identify personal hygiene practices and community helpers that promote health and prevent the spread of disease.	A. Explain the relationship between personal health practices and individual well-being. • immunizations • health examinations	A. Identify and describe health care products and services that impact adolescent health practices.	A. Evaluate health care products and services that impact adult health practices.
B. Identify health-related information. • signs and symbols • terminology • products and services	B. Explain the relationship between health-related information and consumer choices. • dietary guidelines/food selection	B. Analyze the relationship between health-related information and adolescent consumer choices. • tobacco products • weight control products	B. Assess factors that impact adult health consumer choices. • access to health information • access to health care

	• sun exposure guidelines/sunscreen selection		• cost • safety
C. Identify media sources that influence health and safety.	C. Explain the media's effect on health and safety issues.	C. Analyze media health and safety messages and describe their impact on personal health and safety.	C. Compare and contrast the positive and negative effects of the media on adult personal health and safety.
D. Identify the steps in a decision making process.	D. Describe and apply the steps of a decision making process to health and safety issues.	D. Analyze and apply a decision making process to adolescent health and safety issues.	D. Examine and apply a decision making process to the development of short and long-term health goals.
E. Identify environmental factors that affect health. • pollution (e.g., air, water, noise, soil) • waste disposal • temperature extremes • insects/animals	E. Analyze environmental factors that impact health. • indoor air quality (e.g., second-hand smoke, allergens) • chemicals, metals, gases (e.g., lead, radon, carbon monoxide) • radiation • natural disasters	E. Explain the interrelationship between the environment and personal health. • ozone layer/skin cancer • availability of health care/individual health • air pollution/respiratory disease • breeding environments/lyme disease/West Nile virus	E. Analyze the interrelationship between environmental factors and community health. • public health policies and laws/health promotion and disease prevention • individual choices/maintenance of environment • recreational opportunities/health status

10.3. Safety and Injury Prevention			
10.3.3. GRADE 3	**10.3.6 GRADE 6**	**10.3.9. GRADE 9**	**10.3.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Recognize safe/unsafe practices in the home, school and community. • general (e.g., fire, electrical, animals) • modes of transportation (e.g., pedestrian, bicycle, vehicular) • outdoor (e.g., play, weather, water) • safe around people (e.g., safe/unsafe touch, abuse, stranger, bully)	A. Explain and apply safe practices in the home, school and community. • emergencies (e.g., fire, natural disasters) • personal safety (e.g., home alone, latch key, harassment) • communication (e.g., telephone, Internet) • violence prevention (e.g., gangs, weapons)	A. Analyze the role of individual responsibility for safe practices and injury prevention in the home, school and community. • modes of transportation (e.g., pedestrian, bicycle, vehicular, passenger, farm vehicle, all-terrain vehicle) • violence prevention in school • self-protection in the home • self-protection in public places	A. Assess the personal and legal consequences of unsafe practices in the home, school or community. • loss of personal freedom • personal injury • loss of income • impact on others • loss of motor vehicle operator's license
B. Recognize emergency situations and explain appropriate responses. • importance of remaining calm • how to call for help • simple assistance procedures • how to protect self	B. Know and apply appropriate emergency responses. • basic first aid • Heimlich maneuver • universal precautions	B. Describe and apply strategies for emergency and long-term management of injuries. • rescue breathing • water rescue • self-care • sport injuries	B. Analyze and apply strategies for the management of injuries. • CPR • advanced first aid
C. Recognize conflict situations and identify strategies to avoid or resolve. • walk away • I-statements • refusal skills • adult intervention	C. Describe strategies to avoid or manage conflict and violence. • anger management • peer mediation • reflective listening • negotiation	C. Analyze and apply strategies to avoid or manage conflict and violence during adolescence. • effective negotiation • assertive behavior	C. Analyze the impact of violence on the victim and surrounding community.
D. Identify and use safe practices in physical activity settings (e.g., proper equipment, knowledge of rules, sun safety, guidelines of safe play, warm-up, cool-down).	D. Analyze the role of individual responsibility for safety during physical activity.	D. Analyze the role of individual responsibility for safety during organized group activities.	D. Evaluate the benefits, risks and safety factors associated with self-selected life-long physical activities.

10.4. Physical Activity			
10.4.3. GRADE 3	**10.4.6. GRADE 6**	**10.4.9. GRADE 9**	**10.4.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Identify and engage in physical activities that promote physical fitness and health.	A. Identify and engage in moderate to vigorous physical activities that contribute to physical fitness and health.	A. Analyze and engage in physical activities that are developmentally/individually appropriate and support achievement of personal fitness and activity goals.	A. Evaluate and engage in an individualized physical activity plan that supports achievement of personal fitness and activity goals and promotes life-long participation.
B. Know the positive and negative effects of regular participation in moderate to vigorous physical activities.	B. Explain the effects of regular participation in moderate to vigorous physical activities on the body systems.	B. Analyze the effects of regular participation in moderate to vigorous physical activities in relation to adolescent health improvement. • stress management • disease prevention • weight management	B. Analyze the effects of regular participation in a self-selected program of moderate to vigorous physical activities. • social • physiological • psychological

C. Know and recognize changes in body responses during moderate to vigorous physical activity. • heart rate • breathing rate	C. Identify and apply ways to monitor and assess the body's response to moderate to vigorous physical activity. • heart rate monitoring • checking blood pressure • fitness assessment	C. Analyze factors that affect the responses of body systems during moderate to vigorous physical activities. • exercise (e.g., climate, altitude, location, temperature) • healthy fitness zone • individual fitness status (e.g., cardiorespiratory fitness, muscular endurance, muscular strength, flexibility) • drug/substance use/abuse	C. Evaluate how changes in adult health status may affect the responses of the body systems during moderate to vigorous physical activity. • aging • injury • disease
D. Identify likes and dislikes related to participation in physical activities.	D. Describe factors that affect childhood physical activity preferences. • enjoyment • personal interest • social experience • opportunities to learn new activities • parental preference • environment	D. Analyze factors that affect physical activity preferences of adolescents. • skill competence • social benefits • previous experience • activity confidence	D. Evaluate factors that affect physical activity and exercise preferences of adults. • personal challenge • physical benefits • finances • motivation • access to activity • self-improvement
E. Identify reasons why regular participation in physical activities improves motor skills.	E. Identify factors that have an impact on the relationship between regular participation in physical activity and the degree of motor skill improvement. • success-oriented activities • school-community resources • variety of activities • time on task	E. Analyze factors that impact on the relationship between regular participation in physical activity and motor skill improvement. • personal choice • developmental differences • amount of physical activity • authentic practice	E. Analyze the interrelationships among regular participation in physical activity, motor skill improvement and the selection and engagement in lifetime physical activities.
F. Recognize positive and negative interactions of small group activities. • roles (e.g., leader, follower) • cooperation/sharing • on task participation	F. Identify and describe positive and negative interactions of group members in physical activities. • leading • following • teamwork • etiquette • adherence to rules	F. Analyze the effects of positive and negative interactions of adolescent group members in physical activities. • group dynamics • social pressure	F. Assess and use strategies for enhancing adult group interaction in physical activities. • shared responsibility • open communication • goal setting

10.5. Concepts, Principles and Strategies of Movement			
10.5.3. GRADE 3	**10.5.6. GRADE 6**	**10.5.9. GRADE 9**	**10.5.12. GRADE 12**
Pennsylvania's public schools shall teach, challenge and support every student to realize his or her maximum potential and to acquire the knowledge and skills needed to:			
A. Recognize and use basic movement skills and concepts. • locomotor movements (e.g., run, leap, hop) • non-locomotor movements (e.g., bend, stretch, twist) • manipulative movements (e.g., throw, catch, kick) • relationships (e.g., over, under, beside) • combination movements (e.g., locomotor, non-locomotor, manipulative) • space awareness (e.g., self-space, levels, pathways, directions) • effort (e.g., speed, force)	A. Explain and apply the basic movement skills and concepts to create and perform movement sequences and advanced skills.	A. Describe and apply the components of skill-related fitness to movement performance. • agility • balance • coordination • power • reaction time • speed	A. Apply knowledge of movement skills, skill-related fitness and movement concepts to identify and evaluate physical activities that promote personal lifelong participation.
B. Recognize and describe the concepts of motor skill development using appropriate vocabulary. • form • developmental differences • critical elements • feedback	B. Identify and apply the concepts of motor skill development to a variety of basic skills. • transfer between skills • selecting relevant cues • types of feedback • movement efficiency • product (outcome/result)	B. Describe and apply concepts of motor skill development that impact the quality of increasingly complex movement. • response selection • stages of learning a motor skill i.e. verbal cognitive, motor, automatic • types of skill i.e. discrete, serial, continuous	B. Incorporate and synthesize knowledge of motor skill development concepts to improve the quality of motor skills. • open and closed skills • short-term and long-term memory • aspects of good performance
C. Know the function of practice.	C. Describe the relationship between practice and skill development.	C. Identify and apply practice strategies for skill improvement.	C. Evaluate the impact of practice strategies on skill development and improvement.
D. Identify and use principles of exercise to improve movement and fitness activities. • frequency/how often to exercise • intensity/how hard to exercise • time/how long to exercise • type/what kind of exercise	D. Describe and apply the principles of exercise to the components of health-related and skill-related fitness. • cardiorespiratory endurance • muscular strength • muscular endurance	D. Identify and describe the principles of training using appropriate vocabulary. • specificity • overload • progression • aerobic/anaerobic	D. Incorporate and synthesize knowledge of exercise principles, training principles and health and skill-related fitness components to create a fitness program for personal use.

	• flexibility • body composition	• circuit/interval • repetition/set	
E. Know and describe scientific principles that affect movement and skills using appropriate vocabulary. • gravity • force production/absorption • balance • rotation	E. Identify and use scientific principles that affect basic movement and skills using appropriate vocabulary. • Newton's Laws of Motion • application of force • static/dynamic balance • levers • flight	E. Analyze and apply scientific and biomechanical principles to complex movements. • centripetal/centrifugal force • linear motion • rotary motion • friction/resistance • equilibrium • number of moving segments	E. Evaluate movement forms for appropriate application of scientific and biomechanical principles. • efficiency of movement • mechanical advantage • kinetic energy • potential energy • inertia • safety
F. Recognize and describe game strategies using appropriate vocabulary. • faking/dodging • passing/receiving • moving to be open • defending space • following rules of play	F. Identify and apply game strategies to basic games and physical activities. • give and go • one on one • peer communication	F. Describe and apply game strategies to complex games and physical activities. • offensive strategies • defensive strategies • time management	F. Analyze the application of game strategies for different categories of physical activities. • individual • team • lifetime • outdoor

XXX. GLOSSARY

Abstinence:	**Choosing not to do something or completely giving something up in order to gain something.**
Acute illness:	**A health condition of sudden onset, sharp rises and short course.**
Adolescence:	The period of life beginning with puberty and ending with completed growth.
Aerobic:	Physical activity or exercise done at a steady pace for an extended period of time so that the heart can supply as much oxygen as the body needs (e.g., walking, running, swimming, cycling).
Agility:	A component of physical fitness that relates to the ability to rapidly change the position of the entire body in space with speed and accuracy.
AIDS	Acquired Immune Deficiency Syndrome: a condition that results when infection with HIV causes a breakdown of the body's ability to fight other infections.
Allergen:	A substance that stimulates the production of antibodies and subsequently results in allergic reactions (e.g., mold spores, cat/dog dander, dust).
Anaerobic:	Physical activity or exercise done in short, fast bursts so that the heart cannot supply oxygen as fast as the body needs (e.g., sprinting, weightlifting, football).
Assertive:	The expression of thoughts and feelings without experiencing anxiety or threatening others.
Automatic Stage of Learning:	Movement responses flow and the individual can focus on what to do without thinking about it.
Balance:	A skill-related component of physical fitness that relates to the maintenance of equilibrium while stationary or moving.
Biomechanical principles:	The science concerned with the action of forces, internal or external, on the living body.
Body composition:	A health-related component of physical fitness that relates to the percentage of fat tissue and lean tissue in the body.
Body systems:	Anatomically or functionally related parts of the body (e.g., skeletal, nervous, immune, circulatory systems).
Caloric content:	The amount of energy supplied by food. The more calories in the food, the more fattening.
Cardiorespiratory fitness:	A health related component of physical fitness relating to the ability of the circulatory and respiratory systems to supply oxygen during sustained physical activity.
Centrifugal:	The force that seems to pull an object away from the center as it moves in a circle.
Centripetal:	The force that is required to keep an object moving around a circular path.
Chronic illness:	A health condition of long duration or frequent recurrence.
Circuit training:	Exercise program, similar to an obstacle course, in which the person goes from one place to another doing a different exercise at each place.
Closed:	Skills that are performed in an environment that does not change or that changes very little, such as archery or the foul shot in basketball.
Communicable:	Illness caused by pathogens that enter the body through direct or indirect contact and can be transmitted from one host to another.
Community helpers:	Any group or individual who plays a role in health promotion or disease prevention such as doctors, nurses, dentists, teachers, parents, firemen, policemen, trash collectors, animal control officers.
Continuous:	Two or more repetitions of the same skill such as dribbling in basketball or soccer.
Cool-down:	Brief, mild exercise done after vigorous exercise to help the body safely return to a resting state.
Coordination:	A skill-related component of physical fitness that relates to the ability to use the senses together with body parts in performing motor tasks smoothly and accurately.
CPR:	A first aid technique, which involves rescue breathing and chest (heart) compressions, that is used to revive a person whose heart has stopped beating.
Critical elements:	The important parts of a skill.
Decision-making process:	An organized approach to making choices.
Developmental differences:	Learners are at different levels in their motor, cognitive, emotional, social and physical development. The learners' developmental status will affect their ability to learn or improve.
Developmentally appropriate:	Motor skill development and change that occur in an orderly, sequential fashion and are age and experience related.
Directions:	Forward, backward, left, right, up, down.
Discrete:	Single skill performed in isolation from other motor skills such as the soccer penalty kick and golf stroke.
Dynamic balance:	Equilibrium used when in motion, starting and stopping.

Eating disorders:	Food-related dysfunction in which a person changes eating habits in a way that is harmful to the mind or body (e.g., bulimia, anorexia nervosa).
Efficiency of movement:	The state or quality of competence in performance with minimum expenditure of time and effort.
Equilibrium:	State in which there is no change in the motion of a body.
Feedback:	Information given to the learner about how to improve or correct a movement.
Flexibility:	A health-related component of physical fitness that relates to the range of motion available at a joint.
Food guide pyramid:	A visual tool used to help people plan healthy diets according to the Dietary Guidelines for America.
Force:	Any external agent that causes a change in the motion of a body.
Form:	Manner or style of performing a movement according to recognized standards of technique.
Good performance:	The ability to correctly select what to do and the ability to execute the selection appropriately.
Health:	A state of complete physical, mental and social well-being; not merely the absence of disease and infirmity.
Health education:	Planned, sequential K—12 program of curricula and instruction that helps students develop knowledge, attitudes and skills related to the physical, mental, emotional and social dimensions of health.
Health-related fitness:	Components of physical fitness that have a relationship with good health. Components are cardiorespiratory endurance, muscular strength and endurance, flexibility and body composition.
Heimlich maneuver:	A first aid technique that is used to relieve complete airway obstruction.
HIV:	Human immunodeficiency virus that infects cells of the immune system and other tissues and causes acquired immunodeficiency syndrome (AIDS).
I-statement:	A statement describing a specific behavior or event and the effect that behavior or event has on a person and the feelings that result.
Inertia:	A body at rest will remain at rest and a body in motion will remain in motion unless acted upon by a force.
Inhalant:	Chemicals that produce vapors that act on the central nervous system and alter a user's moods, perceptions, feelings, personality and behavior such as airplane glue and aerosols.
Integumentary system:	Body system composed of the skin, hair, nails and glands.
Intensity:	How hard a person should exercise to improve fitness.
Interval training:	An anaerobic exercise program that consists of runs of short distance followed by rest.
Kinetic:	Energy that an object possesses because it is moving, such as a pitched baseball or a person running.
Levels:	Positions of the body (e.g., high, medium, low).
Linear motion:	Movement which occurs in a straight path.
Locomotor movement:	Movements producing physical displacement of the body, usually identified by weight transference via the feet. Basic locomotor steps are the walk, run, hop and jump as well as the irregular rhythmic combinations of the skip, slide and gallop.
Long-term memory:	Ability to recall information that was learned days or even years ago.
Manipulative movements:	Control of objects with body parts and implements. Action causes an object to move from one place to another.
Mechanical advantage:	The ratio between the force put into a machine and the force that comes out of the same machine.
Media sources:	Various forms of mass communication such as television, radio, magazines, newspapers and Internet.
Moderate physical activity:	Sustained, repetitive, large muscle movements (e.g., walking, running, cycling) done at less than 60% of maximum heart rate for age. Maximum heart rate is 220 beats per minute minus participant's age.
Motor skills:	Non-fitness abilities that improve with practice and relate to one's ability to perform specific sports and other motor tasks (e.g., tennis serve, shooting a basketball).
Motor stage of learning:	Individual working to perfect the motor skill and makes conscious adjustments to the environment.
Movement skills:	Proficiency in performing nonlocomotor, locomotor and manipulative movements that are the foundation for participation in physical activities.
Muscular endurance:	A health-related component of physical fitness that relates to the ability of a muscle to continue to perform without fatigue.
Muscular strength:	A health-related component of physical fitness that relates to the ability of the muscle to exert force.
Newton's Laws of Motion:	Three laws by Sir Isaac Newton that explain the relations between force and the motions produced by them: The Law of Inertia, Force and Acceleration, Reacting Forces.
Noncommunicable:	Illness that is not caused by a pathogen that is not transmitted from one host to another.
Nonlocomotor movement:	Movements that do not produce physical displacement of the body.
Nutrient:	A basic component of food that nourishes the body.
Open:	Skill performed in an environment that varies or is unpredictable such as the tennis forehand or the soccer pass.
Overload:	A principle of exercise that states that the only way to improve fitness is to exercise more than the normal.
Pathways:	Patterns of travel while performing locomotor movements (e.g., straight, curved, zigzag).
Physical activity:	Bodily movement that is produced by the contraction of skeletal muscle and which substantially increases energy expenditure.
Physical education:	Planned, sequential, movement-based program of curricula and instruction that helps students develop knowledge, attitudes, motor skills, self-management skills and confidence needed to adapt and maintain a physically active life.
Physical fitness:	A set of attributes that people have or achieve and that relate to their ability to perform physical activity. Generally accepted to consist of health-related fitness and skill-related fitness.
Potential:	Energy stored in a body because of its position such as the crouch position prior to a jump.
Power:	A skill-related component of physical fitness that relates to the rate at which one can perform work.
Principles of exercise:	Guidelines to follow to obtain the maximum benefits from physical activity and exercise.
Principles of training:	Guidelines to follow to obtain the maximum benefits from an exercise plan.
Progression:	A principle of exercise that states that a person should start slowly and increase exercise gradually.
Reaction time:	A skill-related component of physical fitness that relates to the time elapsed between stimulation and the beginning of the response to it.
Reflective listening:	An active listening skill in which the individual lets others know he/she has heard and understands what has been said.
Refusal skills:	Systematic ways to handle situations in which a person wants to say no to an action and/or leave an environment that threatens health or safety, breaks laws, results in lack of respect for self and others or disobeys guidelines set by responsible adults.
Repetitions:	Number of times an exercise is repeated.
Rescue breathing:	Technique used to supply air to an individual who is not breathing.

Rotary motion:	Force that produces movement that occurs around an axis or center point such as a somersault.
Safety education:	Planned, sequential program of curricula and instruction that helps students develop the knowledge, attitudes and confidence needed to protect them from injury.
Self-space:	All the space that the body or its parts can reach without traveling from a starting location.
Serial:	Two or more different skills performed with each other such as fielding a ball and throwing it or dribbling a basketball and shooting it.
Set:	A group of several repetitions.
Short-term memory:	Ability to recall recently learned information, such as within the past few seconds or minutes.
Skill-related fitness:	Consists of components of physical fitness that have a relationship with enhanced performance in sports and motor skills. The components are agility, balance, coordination, power, reaction time and speed.
Specificity:	A principle of exercise that states that specific kinds of exercises must be done to develop specific aspects of the body and specific aspects of fitness.
Speed:	A skill-related component of physical fitness that relates to the ability to perform a movement or cover a distance in a short period of time.
Static balance:	Maintaining equilibrium while holding a pose or remaining motionless.
STD:	Sexually transmitted disease.
Universal precautions:	An approach to infection control. All human blood and body fluids are treated as if known to be infectious.
Warm-up:	Brief, mild exercise that is done to get ready for more vigorous exercise.
Verbal cognitive stage of learning:	The individual is attempting to move from verbal instruction to trying to figure out how to actually do the skill. The first attempts at the skill are generally mechanical and success is inconsistent. The individual thinks through each step of the movement.
Vigorous physical activity:	Sustained, repetitive, large muscle movements (e.g., running, swimming, soccer) done at 60% or more of maximum heart rate for age. Maximum heart rate is 220 beats per minute minus the participant's age. Activity makes person sweat and breathe hard.

(*Editor's Note*: Under § 4.12(a)(7), the academic standards for family and consumer sciences in Appendix D-1 apply beginning July 1, 2026.)

APPENDIX D-1

Academic Standards for Family and Consumer Sciences

Grades K—12

Authority

The provisions of this Appendix D-1 added under sections 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix D-1 added February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234.

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards).

Overview

This document includes Academic Standards for Family and Consumer Sciences at four grade levels (K—2, 3—5, 6—8, 9—12), with the emphasis on what students will know and be able to do in the following areas:

Number	Areas
11.1	**Food Science and Nutrition**
11.2	Resource Management
11.3	Human Development, Family, and Relationships
11.4	Education and Early Childhood Development
11.5	Personal Finance and Consumer Skills
11.6	Career, Community, and Family Connections

The Academic Standards for Family and Consumer Sciences emphasize the crucial role of families in shaping individual and community well-being. This educational approach underscores the shared responsibility of meeting diverse needs, promoting lifelong learning, and contributing to the community. The

standards aim to prepare students not just for personal, family, and work responsibilities but also to empower them for the challenges of living and working in a diverse, global society, fostering holistic individual development and community advancement.

Family and Consumer Sciences (FCS) is a field of study focused on the science and art of living and working well in our complex world, encompassing family and consumer sciences, career exploration, community and family connections, consumer and family resource management, parenting and child development, food, nutrition, wellness, and family and interpersonal relationships.

Appendix D-1. Academic Standards for Family and Consumer Sciences

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
		1. Food Science and Nutrition		
Food supply	**11.1.K-2.A** Identify the production steps that food travels from the farm to the consumer.	**11.1.3-5.A** Demonstrate knowledge of techniques used to evaluate food in various forms.	**11.1.6-8.A** Evaluate factors that affect food safety from production through consumption.	**11.1.9-12.A** Predict factors that affect food safety and security from production through consumption.
Safety and sanitation	**11.1.K-2.B** Describe personal hygiene techniques in food handling.	**11.1.3-5.B** Describe safe food-handling techniques.	**11.1.6-8.B** Demonstrate food safety and sanitation procedures.	**11.1.9-12.B** Evaluate the role of government agencies in safeguarding our food supply
Nutrition analysis	**11.1.K-2.C** Explain the importance of eating a varied diet in maintaining health.	**11.1.3-5.C** Establish factors that affect food choices.	**11.1.6-8.C** Analyze factors that influence nutrition and wellness practices across the life span.	**11.1.9-12.C** Model the ability to acquire, handle, and use foods to meet nutrition and wellness needs of individuals and families across the life span.
Meal management	Intentionally blank	**11.1.3-5.D** Identify components of a well-balanced meal.	**11.1.6-8.D** Hypothesize the effectiveness of the use of meal management principles.	**11.1.9-12.D** Evaluate the application of nutrition and meal-planning principles in the selection, planning, preparation, and serving of meals that meet the specific nutritional needs of individuals across their life span.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Culinary math & measurement	**11.1.K-2.E** Name and classify culinary measurement tools.	**11.1.3-5.E** Demonstrate measurement skills used in food preparation.	**11.1.6-8.E** Apply measurement and math skills in following recipe directions.	**11.1.9-12.E** Apply principles of measurement, portion control, conversions, food cost analysis and control, menu terminology, and menu pricing to menu planning.
Culinary equipment and techniques	**11.1.K-2.F** Name basic kitchen measuring tools and cooking equipment.	**11.1.3-5.F** Describe the use of common cooking tools and equipment.	**11.1.6-8.F** Prepare a recipe using fundamental culinary skills and techniques.	**11.1.9-12.F** Apply the fundamentals of time, temperature, and cooking techniques to preparing, cooking, cooling, reheating, and holding a variety of foods.
Food science	**11.1.K-2.G** Classify foods according to the senses.	**11.1.K-3-5.G** Understand and identify the physical changes that occur during food preparation.	**11.1.6-8.G** Demonstrate food science through principles of food biology and chemistry.	**11.1.9-12.G** Analyze the relevance of scientific principles to food processing, preparation, and packaging.
Nutrition and health	**11.1.K-2.H** Classify foods by food group within the current USDA dietary guidelines.	**11.1.3-5.H** Describe a well-balanced daily menu using the current USDA dietary guidelines.	**11.1.6-8.H** Examine the nutritional needs of individuals and families in relation to health and wellness across the life span.	**11.1.9-12.H** Evaluate nutrition principles, food plans, preparation techniques, and specialized dietary plans.
Calories and energy	Intentionally blank	**11.1.3-5.I** Describe the relationship between food input and energy output.	**11.1.6-8.I** Analyze the energy and nutrient requirements for individuals at various stages of the life cycle.	**11.1.9-12.I** Analyze the breakdown of foods, the absorption of nutrients, and their conversion to energy by the body.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Careers	**11.1.K-3.J** Identify career choices that relate to food.	**11.1.3-5.J** Describe career choices within food services.	**11.1.6-8.J** Investigate career choices within food services, sciences, and nutrition industries.	**11.1.9-12.J** Analyze opportunities for employment, entrepreneurial endeavors, and emerging careers in food services, sciences, services, and nutrition industries.
		2. Resource Management		
Resource sustainability	**11.2.K-2.A** Identify waste and practice ways to reduce, reuse, and recycle.	**11.2.3-5.A** Explain the decision to use or not use resources based on resource availability and sustainability.	**11.2.6-8.A** Analyze sustainability practices and their effect on future renewable and non-renewable resources.	**11.2.9-12.A** Develop a global view to weigh consumer and design decisions with the parameters of sustainability, socioeconomic, and cultural contexts.
Individual and family resources	**11.2.K-2.B** Identify individual and family resources and responsibilities.	**11.2.3-5.B** Develop planning skills to organize individual and family resources and responsibilities.	**11.2.6-8.B** Apply planning skills to manage individual and family resources such as food, clothing, shelter, time, and human capital.	**11.2.9-12.B** Examine strategies for managing individual and family resources such as food, clothing, shelter, health care, recreation, transportation, time, and human capital.
Housing and interior design	Intentionally blank	**11.2.3-5.C** Explain the importance of organizing space for efficiency.	**11.2.6-8.C** Apply elements of design to create products and environments.	**11.2.9-12.C** Evaluate housing and design concepts and theories, including sustainability and universal design, in relation to available resources and options.
Consumer strategies	Intentionally blank	**11.2.3-5.D** Identify information in care instructions, safety precautions, and recommended use of consumable goods.	**11.2.6-8.D** Evaluate different strategies to safely obtain and maintain consumer goods and services.	**11.2.9-12.D** Compare and contrast the selection of goods and services by applying safe and effective consumer strategies.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Textiles, fashion and apparel	**11.2.K-2.E** Identify resources used for clothing.	**11.2.3-5.E** Select clothing for specific tasks and/or environment.	**11.2.6-8.E** Assess effects of design construction, care, use, and maintenance of textiles.	**11.2.9-12.E** Demonstrate skills needed to produce, alter, repair, and care for textiles and fashion apparel.
Laundering	**11.2.K-2.F** Identify how to care for clothing.	**11.2.3-5.F** Select proper laundering methods for clothing and other textiles.	**11.2.6-8.F** Demonstrate laundering processes aligned with industry standards and regulations.	**11.2.9-12.F** Analyze procedures for the selection of textiles, chemicals, and equipment associated with laundry.
Careers	Intentionally blank	**11.2.3-5.G** Identify career choices in the field of Family and Consumer Sciences.	**11.2.6-8.G** Investigate career choices within resource management, including human, environmental, textiles and apparel, and consumer products in the field of Family and Consumer Sciences.	**11.2.9-12.G** Analyze opportunities for employment, entrepreneurial endeavors, and emerging careers within resource management, including human, environmental, and consumer products, in the field of Family and Consumer Sciences.
		3. Human Development, Family, and Relationships		
Role of family	**11.3.K-2.A** Explain daily activities that fulfill family functions in meeting responsibilities.	**11.3.3-5.A** Compare how different cultures meet family responsibilities within various family structures.	**11.3.6-8.A** Analyze the effects of change and family transitions over the life span.	**11.3.9-12.A** Evaluate the impact of family and its effects on the well-being of individuals and society.
Family life cycle	Intentionally blank	**11.3.3-5.B** Identify the characteristics of the stages of the family life cycle.	**11.3.6-8.B** Explain the influences of family life cycle stages on the needs of families and communities.	**11.3.9-12.B** Hypothesize the impact of present family life cycle trends on the global community.
Interpersonal communications	**11.3.K-2.C** Identify how to resolve conflict using interpersonal communication skills.	**11.3.3-5.C** Describe positive and negative interactions within patterns of interpersonal communication.	**11.3.6-8.C** Justify the significance of interpersonal communication skills in the practical reasoning method of decision-making.	**11.3.9-12.C** Evaluate the effectiveness of using interpersonal communication skills to resolve conflict.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Human development and well-being	**11.3.K-2.D** Identify similarities and differences between self and others.	**11.3.3-5.D** Describe how areas of development impact human growth across the life cycle.	**11.3.6-8.D** Discuss factors that influence human growth, development, and well-being.	**11.3.9-12.D** Analyze strategies that promote human growth, development, and well-being.
Careers	Intentionally blank	**11.3.3-5.E** Identify career and work choices within family and human services.	**11.3.6-8.E** Investigate career and work choices within family and human services.	**11.3.9-12.E** Analyze opportunities for employment, entrepreneurial endeavors, and emerging careers within family and human services.
4. Education and Early Childhood Development				
Developmental stages	Intentionally blank	Intentionally blank	**11.4.6-8.A** Identify child guidance practices for each stage of child development.	**11.4.9-12.A** Demonstrate the integration of theories of child development and their impact on parenting and childcare.
Developmental needs	Intentionally blank	Intentionally blank	**11.4.6-8.B** Identify learning activities that meet the developmental needs of learners.	**11.4.9-12.B** Demonstrate integration of academic standards that address language, culture, learning styles, and early childhood experiences.
Learning environments	Intentionally blank	**11.4.3-5.C** Establish the characteristics of a healthy learning environment.	**11.4.6-8.C** Evaluate various environments to determine if they provide the characteristics of a proper learning environment.	**11.4.9-12.C** Analyze developmentally appropriate and culturally responsive practices to evaluate early childhood education providers and services.
Health and safety	Intentionally blank	**11.4.3-5.D** Identify health and safety needs for children at each stage of child development.	**11.4.6-8.D** Evaluate health and safety hazards relating to children at each stage of child development.	**11.4.9-12.D** Analyze current issues in health and safety affecting children at each stage of child development.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Literacy	Intentionally blank	Intentionally blank	**11.4.6-8.E** Explain how storytelling, story reading, and writing enhance literacy development in children.	**11.4.9-12.E** Identify characteristics of quality literature for children that develop the child's imagination, creativity, reading, and writing skills.
Collaborative relationships	**11.4.K-2.F** Identify community resources and services.	**11.4.3-5.F** Establish relationships that are positive and supportive of others.	**11.4.6-8.F** Evaluate support systems that provide services for parents and families.	**11.4.9-12.F** Demonstrate skills for building and maintaining positive collaborative relationships with children, youth, and adults in their family, school, and community environments.
Careers	Intentionally blank	**11.4.K-2.G** Identify career and work choices within education and early childhood.	**11.4.6-8.G** Investigate career choices within early childhood, education, and other careers in the field.	**11.4.9-12.G** Analyze opportunities for employment, entrepreneurial endeavors, and emerging careers in education and early childhood.
5. Personal Finance and Consumer Skills				
Consumer rights and responsibilities	**11.5.K-2.A** Describe self as consumer.	**11.5.3-5.A** Explain consumer rights and responsibilities: • To be safe, • To be informed, • To be heard, • To choose, • To redress.	**11.5.6-8.A** Explain how consumer rights and responsibilities are protected.	**11.5.9-12.A** Analyze factors including socioeconomic and cultural influences that affect consumer advocacy

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Financial planning and budgeting	Intentionally blank	**11.5.3-5.B** Define the difference between a short-term and long-term goal.	**11.5.6-8.B** Develop short-term and long-term financial goals for a family budget.	**11.5.9-12.B** Demonstrate the ability to manage and maintain individual and family budgets.
Financial planning and budgeting	Intentionally blank	Intentionally blank	Intentionally blank	**11.5.9-12.C** Analyze components of a family financial planning process that reflect the distinction between wants, needs, values, goals, and economic resources.
Financial resources	Intentionally blank	**11.5.3-5.D** Explain the relationship between work and income within the family.	**11.5.6-8.D** Compare/evaluate the financial advantages and disadvantages of various incomes.	**11.5.9-12.D** Demonstrate the ability to use knowledge and skills to manage one's income, spending, and savings effectively for a lifetime of financial security.
Careers	Intentionally blank	Intentionally blank	**11.5.6-8.E** Investigate career choices within financial service industries.	**11.5.9-12.E** Analyze opportunities for employment, entrepreneurial endeavors, and emerging careers within financial service industries.
6. Career, Community, and Family Connections				
Action plans	Intentionally blank	**11.6.3-5.A** Identify the importance of routines and schedules while differentiating between short- and long-term goals.	**11.6.6-8.A** Explain the Family, Career and Community Leaders of America (FCCLA) action planning procedure (identify concerns, set a goal, form a plan, act, and follow up) and how to apply it to family, work, and community decisions.	**11.6.9-12.A** Evaluate the effectiveness of action plans that integrate personal, work, family, and community responsibilities.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Teamwork and leadership	Intentionally blank	**11.6.3-5.B** Classify the components of effective teamwork and leadership.	**11.6.6-8.B** Assess the use of teamwork and leadership skills in accomplishing the work of families and communities.	**11.6.9-12.B** Analyze teamwork and leadership skills and their application in various family dynamics and employability skills.
Role of technology	Intentionally blank	**11.6.3-5.C** Describe the roles of technology within the family and community in maintaining a safe and healthy living environment.	**11.6.6-8.** Assess the use of technology and its effect on the quality of life.	**11.6.9-12.C** Analyze and evaluate emerging technologies that impact individuals, families, and communities.
Service providers	Intentionally blank	**11.6.3-5.D** Identify the public and nonpublic services that are available to serve families within the community.	Intentionally blank	**11.6.9-12.D** Analyze how public, nonpublic, and for-profit service providers serve the family.

APPENDIX E

{Expired}

(*Editor's Note*: Under § 4.12(a)(5), the academic standards for career education and work which appeared at 22 Pa. Code, Appendix E, pages 4-356.9— 4-375, serial pages (422361)—(422362), (320293)—(320300), (371245)—(371246), (320303)—(320310) and (422363) expired July 1, 2026. The academic standards for career education and work in Appendix E-1 apply beginning July 1, 2026.)

Source

The provisions of this Appendix E adopted July 7, 2006, effective July 8, 2006, 36 Pa.B. 3528, expired July 1, 2026, 55 Pa.B. 1234.

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards).

(*Editor's Note*: Under § 4.12(a)(5), the academic standards for career education and work in Appendix E-1 apply beginning July 1, 2026.)

APPENDIX E-1

Academic Standards for Career Education and Work

Grades K—12

Authority

The provisions of this Appendix E-1 added under sections 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix E-1 added February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234.

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards).

Overview

Pennsylvania's economic future depends on having a well-educated and skilled workforce. Career Education and Work standards reflect the increasing complexity and sophistication that students experience as they progress through school, focusing on the skills and continuous learning and innovation required for students to succeed in a rapidly changing workplace. The standards are written as grade-banded standards built around the concepts of career awareness and exploration, employability skills, growth and advancement, and personal interests and career planning.

Number	Areas
13.1	**Career Awareness and Exploration**
13.2	Employability Skills
13.3	Growth and Advancement
13.4	Personal Interests and Career Planning

Appendix E-1. Academic Standards for Career Education and Work

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
1. Career Awareness and Exploration				
Educational requirements and opportunities	**13.1.K-2.A** Identify that there are different ways to prepare for careers.	**13.1.3-5.A** Investigate how people prepare for different careers that may include various levels of education, experiences, and training.	**13.1.6-8.A** Analyze the relationship of school subjects, extracurricular activities, and community experiences to career preparation.	**13.1.9-12.A** Analyze the relationship between career choices and career preparation opportunities to choose a path.
Entrepreneurship	**13.1.K-2.B** Identify entrepreneurial character traits of historical and contemporary entrepreneurs and ways to integrate entrepreneurial traits into schoolwide activities and events (e.g., posters to advertise, create ideas).	**13.1.3-5.B** Define entrepreneurship and the types of entrepreneurship (e.g., social, public service, small business) in the community; compare and contrast the risks and rewards of entrepreneurship as a career through self-employment, in for-profit or non-profit organizations; practice entrepreneurial traits in schoolwide activities and events.	**13.1.6-8.B** Compare and contrast entrepreneurship and other self-employment opportunities to other employment types; evaluate how entrepreneurial character traits influence career opportunities in any type of job; identify how the 4 p's (product, price, place, and promotion) play a role in entrepreneurial branding.	**13.1.9-12.B** Analyze entrepreneurship and other self-employment opportunities as they relate to personal career goals and corporate opportunities; evaluate successful entrepreneurs in both non-profit and for-profit organizations and how the 4 p's play a role in branding.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Factors in career decisions	**13.1.K-2.C** Describe the work done by community helpers and business owners.	**13.1.3-5.C** Describe external factors that influence career choices, such as, but not limited to: • Geographic location • Salaries/benefits • Work schedule • Working conditions	**13.1.6-8.C** Compare and contrast multiple careers using a variety of career research resources.	**13.1.9-12.C** Analyze external factors that impact employment opportunities, such as, but not limited to: • Competition • Global influences • Job growth • Job openings • Labor supply • Potential advancement • Potential earnings
Local, national, and global labor market	**13.1.K-2.D** Describe what a job is and the reasons people may have jobs.	**13.1.3-5.D** Identify the range of jobs available in the community.	**13.1.6-8.D** Use multiple resources or tools to research the job availability in a local, national, and global job market.	**13.1.9-12.D** Describe employment outlook, organized by in-demand occupations, clusters, pathways, or other methods.
Resources in school and community	**13.1.K-2.E** Participate in school and community events that share career awareness opportunities.	**13.1.3-5.E** Describe the benefits of participating in hands-on career awareness/preparation activities.	**13.1.6-8.E** Identify, participate in, and evaluate school- and community-based opportunities for career awareness/preparation experiences.	**13.1.9-12.E** Utilize school- and community-based opportunities for career awareness/preparation.
Rights of workers	Intentionally blank	**13.1.3-5.F** Explain why employers pay people for their labor.	**13.1.6-8.F** Understand what labor laws are and how they impact worker wages and safety.	**13.1.9-12.F** Examine basic labor laws to determine the rights and responsibilities of both employers and employees.
2. Employability Skills				

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Critical thinking and problem solving	**13.2.K-2.A** Practice solving problems individually and in groups.	**13.2.3-5.A** Develop the traits needed to solve problems.	**13.2.6-8.A** Apply critical thinking to solve real-life problems.	**13.2.9-12.A** Identify solutions to existing conditions or problems using critical thinking skills.
Oral and written communication	**13.2.K-2.B** Use context clues to identify how communication varies within different social situations in varied environments.	**13.2.3-5.B** Use context clues to determine how communication varies within different social situations in varied environments.	**13.2.6-8.B** Develop formal communication that demonstrates professional etiquette and observes social protocols when communicating verbally, nonverbally, and in writing.	**13.2.9-12.B** Use professional etiquette and observe social protocols when communicating verbally, nonverbally, and in writing.
Teamwork and collaboration	**13.2.K-2.C** Demonstrate cooperation and positive interactions with classmates, recognizing that people have different backgrounds, experiences, beliefs, and ideas.	**13.2.3-5.C** Demonstrate the ability to work well in group or team interactions, recognizing that people have different backgrounds, experiences, beliefs, and ideas.	**13.2.6-8.C** Practice effective group interactions, acknowledging people's different backgrounds, experiences, beliefs, and ideas.	**13.2.9-12.C** Demonstrate shared responsibility for collaborative work and the value of contributions made by team members with various perspectives.
Technology	**13.2.K-2.D** Demonstrate proper and safe Internet and instructional technology use.	**13.2.3-5.D** Demonstrate proper and safe Internet and instructional technology use.	**13.2.6-8.D** Demonstrate proficient use of technologies by using various types of technology.	**13.2.9-12.D** Use job-specific technologies to accomplish work responsibilities in a productive manner.
Time and project management	**13.2.K-2.E** Discuss how time is structured at home and school.	**13.2.3-5.E** Develop a personal schedule based on activities and responsibilities at both home and school.	**13.2.6-8.E** Demonstrate effective time and project management strategies.	**13.2.9-12.E** Evaluate time and project management strategies and their application to both personal and work situations.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Entrepreneurial mindset	Intentionally blank	**13.2.3-5.F** List and describe several characteristics of the Entrepreneurial Mindset characteristics.	**13.2.6-8.F** Explain how Entrepreneurial Mindset characteristics and skills can help one be successful in their career.	**13.2.9-12.F** Apply Entrepreneurial Mindset characteristics and skills to school, work, and extracurricular environments.
		3. Growth and Advancement		
Career growth	**13.3.K-2.A** Communicate an original opinion and describe the justification for that opinion.	**13.3.3-5.A** Identify situations that may require self- advocacy, such as: personal preference, values, etc.	**13.3.6-8.A** Identify factors that may inform decisions for career changes, such as loss of employment, lifestyle changes, workplace conflicts, work value changes, and job market changes.	**13.3.9-12.A** Develop strategies for navigating career and life transitions, such as coping with loss of employment, resigning from a job, negotiating a raise or promotion, and employing transferable skills.
Lifelong learning	**13.3.K-2.B** Recognize the importance of improving skills and learning new ones.	**13.3.3-5.B** Reflect on skill improvement and identify opportunities to build new skills.	**13.3.6-8.B** Explain how decisions regarding education, skill building, and work impact other major life decisions.	**13.3.9-12.B** Identify sources of lifelong learning to strengthen existing skills, develop new skills, and maximize career growth.
Strategic goals	**13.3.K-2.C** Set and achieve a goal.	**13.3.3-5.C** Set and achieve S.M.A.R.T. goals.	**13.3.6-8.C** Practice feedback and personal reflection to evaluate success towards setting and completing goals.	**13.3.9-12.C** Incorporate feedback and personal reflection to evaluate success towards completing goals and determine a next course of action.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Work ethic	**13.3.K-2.D** Build an awareness of the importance of a positive work ethic as a means to learn and grow.	**13.3.3-5.D** Identify attributes of a positive work ethic.	**13.3.6-8.D** Demonstrate a positive work ethic.	**13.3.9-12.D** Analyze and reflect upon the efficacy of your work ethic.
4. Personal Interests and Career Planning				
Develop a personal brand	**13.4.K-2.A** Describe qualities of people who are successful.	**13.4.3-5.A** Identify personal strengths and successes.	**13.4.6-8.A** Identify ways to market yourself as a job candidate.	**13.4.9-12.A** Strategically market yourself as a job candidate.
Identify personal interests	**13.4.K-2.B** Identify current interests and skills.	**13.4.3-5.B** Connect personal interests and abilities to potential career choices.	**13.4.6-8.B** Relate careers to personal interests, abilities, and aptitudes.	**13.4.9-12.B** Evaluate career options based on personal interests, abilities, aptitudes, achievements, and goals.
Research careers	**13.4.K-2.C** Explore career choices and identify the knowledge and skills associated with different types of careers.	**13.4.3-5.C** Discuss resources available in researching career opportunities.	**13.4.6-8.C** Evaluate a variety of methods to research careers that align with personal interests, values, strengths, and career goals.	**13.4.9-12.C** Use a variety of methods to research careers that align with personal interests, values, strengths, and career goals.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Understand a career path	**13.4.K-2.D** Explain how workers in their careers use what is learned in the classroom.	**13.4.3-5.D** Document personal interests, abilities, and academic strengths.	**13.4.6-8.D** Build a career plan to describe planned action steps that may include, but not be limited to: • Electives • Extracurricular activities • Coursework • Career interests • Strengths and abilities • Educational and training opportunities	**13.4.9-12.D** Evolve career plan to enact additional action steps that may include, but not be limited to: • Electives • Extracurricular activities • Coursework • Career interests • Strengths and abilities • Educational and training opportunities • Work-based learning • Industry-recognized credentials

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(*Editor's Note*: Under § 4.12(a)(10), the academic standards for personal finance in Appendix F apply beginning July 1, 2026.)

APPENDIX F

Academic Standards for Personal Finance

Grades K—12

Authority

The provisions of this Appendix F added under sections 290.1, 1551, 2603-B and 2604-B of the Public School Code of 1949 (24 P.S. § § 2-290.1, 15-1551, 26-2603-B and 26-2604-B).

Source

The provisions of this Appendix F added February 7, 2025, effective February 8, 2025, 55 Pa.B. 1234.

Cross References

This appendix cited in 22 Pa. Code § 4.12 (relating to academic standards).

Overview

Personal Finance focuses on the fundamentals of personal finance, income, spending, saving and investing, risk and insurance, and credit, with a goal to develop individuals who can manage their personal finances. Personal finance is also known as personal financial literacy and financial literacy. The standards are written as grade-banded standards built around the areas below and topics that make up the areas of study.

Number	Areas
17.1	**Personal Finance Fundamentals**
17.2	Income
17.3	Spending
17.4	Saving and Investing
17.5	Risk and Insurance
17.6	Credit

Appendix F. Academic Standards for Personal Finance

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
		1. Personal Finance Fundamentals		
Financial goal setting and decision making	**17.1.K-2.A** Identify short-term financial goals and steps people can take to achieve them.	**17.1.3-5.A** Describe reasons people set financial goals.	**17.1.6-8.A** Compare short-, intermediate-, and long-term financial goals.	**17.1.9-12.A** Determine the financial impact of various long-term goals (e.g., lifestyle, family, education).
	17.1.K-2.B Explain how limited personal financial resources affect the choices people make.	**17.1.3-5.B** Identify the opportunity cost of various financial decisions, including ones related to earning, spending, and saving.	**17.1.6-8.B** Explain why the opportunity cost of a decision might differ from one person or situation to another.	**17.1.9-12.B** Apply a systematic decision-making process, including opportunity costs, to setting and achieving financial goals.
Financial mindset and behaviors	**17.1.K-2.C** Describe how friends and family can impact a person's attitudes toward money.	**17.1.3-5.C** Explain how life circumstances and experiences can alter attitudes toward money.	**17.1.6-8.C** Analyze how people differ in their attitudes toward money.	**17.1.9-12.C** Analyze the impact of various factors on a person's financial mindset and decisions.
	Intentionally blank	**17.1.3-5.D** Identify financial decisions people make that may not be in their best interest.	**17.1.6-8.D** Describe behavioral biases (e.g., mental accounting, anchoring, confirmation bias) that can affect financial decisions.	**17.1.9-12.D** Evaluate strategies for dealing with behavioral biases (e.g., loss aversion, experiential bias, mental accounting) and other obstacles to managing personal finances.
	Intentionally blank	**17.1.3-5.E** Describe conversations people have about money.	**17.1.6-8.E** Describe important financial conversations people have and with whom they have them.	**17.1.9-12.E** Assess the value of sharing financial goals and information with others.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Financial services	**17.1.K-2.F** Describe ways people use financial institutions (e.g., depositing money, obtaining cash).	**17.1.3-5.F** Identify products and services provided by financial institutions.	**17.1.6-8.F** Identify personal information needed to establish a financial account.	**17.1.9-12.F** Compare various financial service providers (e.g., banks, credit unions, check cashers, brokerage firms) and the types of accounts and services each provides.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.1.9-12.G** Communicate the process of opening financial accounts and the factors to consider when selecting financial institutions and professionals.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.1.9-12.H** Evaluate the use of financial technology to access financial services and make financial decisions.
Financial record keeping	Intentionally blank	Intentionally blank	**17.1.6-8.I** Describe information people document and track for their personal finances.	**17.1.9-12.I** Develop a system for documenting and organizing personal financial records, both paper and electronic.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.1.9-12.J** Explain the financial implications of wills, powers of attorney, and naming beneficiaries for various accounts.
Consumer protection	Intentionally blank	Intentionally blank	Intentionally blank	**17.1.9-12.K** Explain the role of various state and federal financial regulators

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
				and consumer protection agencies.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.1.9-12.L** Describe the issues addressed by various laws and regulations that impact or safeguard a person's finances.
		2. Income		
Sources of income	**17.2.K-2.A** Identify reasons people and households require income.	**17.2.3-5.A** Describe how people receive income (e.g., work, gifts, renting property to others).	**17.2.6-8.A** Compare compensation methods, including hourly wage, salary, commission, tips, and piecework.	**17.2.9-12.A** Explain various types of income (e.g., earned, unearned, passive, active) and their sources (e.g., work, rentals, investments, government programs).
	Intentionally blank	Intentionally blank	Intentionally blank	**17.2.9-12.B** Describe sources of retirement income and how they relate to individual investment choices, employer-sponsored retirement plans, and government programs.
Factors influencing income	**17.2.K-2.C** Predict the knowledge and skills needed for various jobs.	**17.2.3-5.C** Explain ways people improve their ability to earn income through education, training, and experience.	**17.2.6-8.C** Analyze connections between the amount individuals might earn in various careers and the education or training that is required.	**17.2.9-12.C** Use data to support an individual's decision to obtain or forgo post-secondary education based on the associated costs and anticipated future income.

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Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
	Intentionally blank	Intentionally blank	**17.2.6-8.D** Compare ways people pay for education and training, including scholarships, grants, savings, military service, apprenticeships, and loans.	**17.2.9-12.D** Research options to pay for education and training, ways to reduce the total cost, and steps needed to obtain financial aid.
	Intentionally blank	Intentionally blank	**17.2.6-8.E** Explain factors that may impact a person's future income and employment status, including changes in technology, demand for workers, and the cost of post-secondary education.	**17.2.9-12.E** Evaluate the impacts of technology, labor markets, and economic conditions and trends on a person's employment potential.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.2.9-12.F** Explain the impact of employee benefits (e.g., health insurance, retirement savings plans, education reimbursement programs) on an individual's finances.
Self-employment and supplemental income	**17.2.K-2.G** Identify ways people earn income through entrepreneurship.	**17.2.3-5.G** Describe how people earn income through entrepreneurship and supplemental employment.	**17.2.6-8.G** Compare various forms of self-employment.	**17.2.9-12.G** Analyze the financial impact of a person's decision to own a business, work as an independent contractor, or be employed.
Income and payroll taxes	Intentionally blank	Intentionally blank	**17.2.6-8.H** Interpret a pay statement and explain the relationship between gross pay, net pay, and payroll deductions.	**17.2.9-12.H** Calculate the impact of taxes and payroll deductions on income.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.2.9-12.I** Complete various federal, state, and local tax forms.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
			3. Spending	
Spending decisions	**17.3.K-2.A** Describe spending choices people make in everyday life and factors that influence them.	**17.3.3-5.A** Explain factors that influence a person's spending decisions and the impact these has on how they prioritize their wants.	**17.3.6-8.A** Demonstrate making an informed purchase decision by evaluating price, product claims, and information from a variety of sources.	**17.3.9-12.A** Develop a process for making informed spending decisions, including factors to consider (e.g., product features, price, durability, environmental or societal impact, reliability of information).
	Intentionally blank	Intentionally blank	**17.3.6-8.B** Apply mathematical skills to assess the impact of various ways retailers express prices (e.g., coupons, discounts, unit price).	**17.3.9-12.B** Compare ways people can lower the price they pay for goods and services (e.g., online tools, discount retailers, negotiating, secondhand items).
Developing a budget	**17.3.K-2.C** Differentiate between money that is received and money that is spent.	**17.3.3-5.C** Provide examples of household spending and sources of income.	**17.3.6-8.C** Compare ways people organize and track their spending and income.	**17.3.9-12.C** Develop a personal approach to keeping track of income and spending.
	17.3.K-2.D Describe how people use money for different purposes, including spending, saving, and sharing with others.	**17.3.3-5.D** Construct a simple budget (e.g., for a family, individual, or school event).	**17.3.6-8.D** Explain the components of a personal budget (e.g., income, expenses, planned savings) and how budgeting can help people achieve their financial goals.	**17.3.9-12.D** Evaluate various budgeting approaches (e.g., 50-30-20, zero-based) and methods (e.g., envelope system, spreadsheets, online tools).
	Intentionally blank	Intentionally blank	Intentionally blank	**17.3.9-12.E** Create a personal budget to allocate current or future income, including estimates for fixed and variable expenses.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
	Intentionally blank	Intentionally blank	Intentionally blank	**17.3.9-12.F** Identify methods for adjusting a budget for unexpected expenses or loss of income.
Payment methods	Intentionally blank	**17.3.3-5.G** Explain various payment methods (e.g., cash, checks, gift cards, debit cards, credit cards).	**17.3.6-8.G** Describe the advantages and disadvantages of using various payment methods (e.g., cash, checks, gift cards, debit cards, credit cards, mobile payments).	**17.3.9-12.G** Compare the effects of using various payment methods when making purchases.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.3.9-12.H** Compare various approaches to paying bills, including making automated payments and ensuring bills are paid on time.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.3.9-12.I** Describe the impact of technology on payment methods and how it influences spending.
Major life purchases	Intentionally blank	Intentionally blank	Intentionally blank	**17.3.9-12.J** Analyze a housing decision, including comparing renting and buying, upfront and ongoing costs, and the process of obtaining a mortgage or a lease.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.3.9-12.K** Justify the purchase or lease of a vehicle and the alternatives considered (e.g., new versus used, total cost of ownership or

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
				use).
Sales and Property Taxes	Intentionally blank	**17.3.3-5.L** Identify taxes applied to certain purchases.	**17.3.6-8.L** Calculate the sales tax for various types of purchases.	**17.3.9-12.L** Analyze the impact of paying sales, excise, and property taxes on financial decisions.
Charitable Giving	**17.3.K-2.M** Identify items people might give to someone else or donate to charity.	**17.3.3-5.M** Identify reasons people participate in fundraising efforts and the causes they support.	**17.3.6-8.M** Describe ways people support charitable organizations by donating money, items, and time.	**17.3.9-12.M** Justify a decision to participate in or forgo a fundraising effort based on the organization and cause.
		4. Saving and Investing		
Asset building	**17.4.K-2.A** Describe things people own (e.g., toys, home, money).	**17.4.3-5.A** Identify items that might increase or decrease in value over time.	**17.4.6-8.A** Explain the change in value over time of various assets.	**17.4.9-12.A** Calculate a person's net worth given their assets and liabilities.
Saving	**17.4.K-2.B** Describe reasons people save money for the future rather than spend it now.	**17.4.3-5.B** Describe a variety of savings goals and the factors that influence them.	**17.4.6-8.B** Describe the process of creating, implementing, and adapting a personal savings plan.	**17.4.9-12.B** Develop a savings plan for accomplishing personal short- and long-term financial goals.
	17.4.K-2.C Identify ways children can keep their money safe and avoid losing it.	**17.4.3-5.C** Identify reasons people deposit money to be saved in accounts at financial institutions and factors they might consider when selecting a financial institution.	**17.4.6-8.C** Use different methods and tools to calculate the growth in savings given various scenarios (e.g., simple versus compound interest, starting age, years to save, and interest rates).	**17.4.9-12.C** Compare the features of various savings vehicles (e.g., savings accounts, certificates of deposit, money market accounts) and the interest rates offered by several institutions.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Investing	Intentionally blank	Intentionally blank	**17.4.6-8.D** Describe similarities and differences between saving and investing.	**17.4.9-12.D** Explain factors that contribute to rates of return for various investments, including risk, inflation, and taxes.
	Intentionally blank	Intentionally blank	**17.4.6-8.E** Identify publicly traded companies, the goods and services they produce, and their current stock prices.	**17.4.9-12.E** Explain the similarities and differences between stocks, bonds, mutual funds, and exchange-traded funds, and the factors that influence price fluctuations for each.
	Intentionally blank	Intentionally blank	**17.4.6-8.F** Identify factors influencing investment planning (e.g., age, income, debt, assets, goals, family size, risk tolerance).	**17.4.9-12.F** Describe factors to consider when selecting sources of investment advice and trading methods (e.g., online trading platforms, financial advisors, robo-advisors).
	Intentionally blank	Intentionally blank	**17.4.6-8.G** Describe how people make money through investing (e.g., buying low and selling high, earning dividends, buy and hold).	**17.4.9-12.G** Explain how popular benchmark indices are used.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.4.9-12.H** Recommend an investment portfolio diversified to meet specific goals, including purpose, starting age, time horizon, and tolerance for risk.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.4.9-12.I** Compare retirement-specific

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
				investment options, including employer-sponsored plans, Roth and traditional individual retirement accounts, and accounts available to people who are self-employed.
Investing risk tolerance	Intentionally blank	Intentionally blank	**17.4.6-8.J** Assess the impact of values and attitudes, including tolerance for risk, on saving and investing decisions.	**17.4.9-12.J** Analyze personal attitudes towards risk and how these might impact future investment decisions and outcomes.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.4.9-12.K** Describe methods to avoid or counteract the potentially negative impacts of behavioral biases (e.g., loss aversion, herding, choice overload) on investment decisions.
			5. Risk and Insurance	
Risk identification and management	**17.5.K-2.A** Identify risks in everyday situations.	**17.5.3-5.A** Describe how unexpected events could impact a person's finances (e.g., floods, automobile accidents, illness).	**17.5.6-8.A** Provide examples of financial risks people face (e.g., losing a job, needing to replace damaged property, paying medical expenses).	**17.5.9-12.A** Evaluate a person's potential for financial risk (e.g., loss of personal property, reduction in income, liability).
	17.5.K-2.B Explain ways to avoid or reduce risks.	**17.5.3-5.B** Explain how emergency savings can be used to offset losses from unexpected events.	**17.5.6-8.B** Describe how people manage the risk of financial loss through risk avoidance, reduction, retention, and transfer.	**17.5.9-12.B** Critique approaches to avoiding, reducing, retaining, and transferring risk given a particular scenario.

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Insurance	Intentionally blank	**17.5.3-5.C** Provide examples of insurance people buy in order to transfer financial risk (e.g., health, auto, flood).	**17.5.6-8.C** Compare various types of insurance and what is typically covered by each.	**17.5.9-12.C** Formulate insurance recommendations based on individual needs, situations, and preferences, including but not limited to automotive, homeowners, renters, health, life, and disability, as justified.
	Intentionally blank	Intentionally blank	**17.5.6-8.D** Describe factors that impact the amount an individual can expect to pay for insurance (e.g., coverage types and amounts, age, location, driving record).	**17.5.9-12.D** Use information from various sources to compare insurance providers, plans, and prices.
	Intentionally blank	Intentionally blank	**17.5.6-8.E** Defend decisions to purchase or forgo insurance, extended warranties, and service contracts.	**17.5.9-12.E** Formulate a process of comparing insurance products, determining out-of-pocket costs, and filing claims.
	Intentionally blank	Intentionally blank	**17.5.6-8.F** Describe how insurance works by pooling the premiums of many policyholders in order to pay for claims.	**17.5.9-12.F** Describe circumstances in which a person may be required to show proof of insurance or obtain a minimum amount of coverage.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.5.9-12.G** Evaluate the impact of public insurance programs for individuals facing financial hardship (e.g., Medicare, Medicaid, and unemployment).

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
Financial fraud and identity theft	**17.5.K-2.H** Give examples of personal information that should be kept private.	**17.5.3-5.H** Predict the financial consequences of sharing personal information.	**17.5.6-8.H** Describe strategies people can use to protect their personally identifiable information.	**17.5.9-12.H** Analyze trends in financial fraud and strategies to avoid becoming a victim.
	Intentionally blank	Intentionally blank	Intentionally blank	**17.5.9-12.I** Research the agencies individuals can contact and steps they can take to address financial fraud and scams, including identity theft.
		6. Credit		
Credit use and benefits	**17.6.K-2.A** Describe the process of borrowing items or money from someone else.	**17.6.3-5.A** Identify traits that could impact a person's ability to borrow items or money.	**17.6.6-8.A** Explain how credit is established and tracked over time for an individual.	**17.6.9-12.A** Evaluate pathways to obtaining credit and what lenders look for in a borrower (e.g., character, capacity, capital, collateral).
	Intentionally blank	Intentionally blank	Intentionally blank	**17.6.9-12.B** Describe how credit reports and scores are determined, used, and improved.
Types of credit	Intentionally blank	**17.6.3-5.C** Identify goods and services people often pay for over time using credit.	**17.6.6-8.C** Discuss various forms of credit and their uses (e.g., credit cards, home loans, auto loans, student loans).	**17.6.9-12.C** Compare various forms of credit and how each is used (e.g., secured and unsecured loans, installment and revolving credit, service credit).
	Intentionally blank	**17.6.3-5.D** Describe how using a credit card is a form of borrowing.	**17.6.6-8.D** Summarize the differences between credit cards and debit cards.	**17.6.9-12.D** Analyze the use of loans to finance higher education and home purchases, how they are

Substrand	Kindergarten to Grade 2	Grades 3 to 5	Grades 6 to 8	Grades 9 to 12
				obtained, and options for paying them back.
Costs of credit	**17.6.K-2.E** Identify potential advantages and disadvantages of borrowing from others.	**17.6.3-5.E** Explain why borrowers might be asked to repay more than they initially borrowed (e.g., interest, fees).	**17.6.6-8.E** Analyze factors that influence the total cost of paying with credit, including interest rates, fees, repayment period, and source of credit.	**17.6.9-12.E** Calculate the total cost of credit given a variety of situations (e.g., making minimum payments, paying fees, using alternative financial service providers).
	Intentionally blank	Intentionally blank	**17.6.6-8.F** Explain why potential borrowers could reach different conclusions about the use of credit, including how much debt constitutes overborrowing.	**17.6.9-12.F** Describe the consequences of failing to repay debts and sources of debt management assistance.
Credit rights and responsibilities	Intentionally blank	Intentionally blank	**17.6.6-8.G** Explain why lenders must provide borrowers with certain information (e.g., interest rates, annual percentage rates, fees, terms, conditions).	**17.6.9-12.G** Evaluate various rights and laws related to credit and their impact on consumers.

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